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Volume 1

D.V. Anosov · V.I. Arnold (Eds.)

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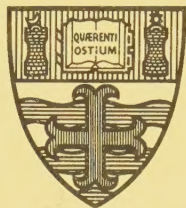
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Volume 1

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D.V. Anosov V.I. Arnold (Eds.)

Dynamical Systems I

Ordinary Differential Equations
and Smooth Dynamical Systems

With 25 Figures



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From the Editorial Board

After a period of intensive "internal" development which led to an unprecedented deepening and interpenetration among the various subfields of mathematics, the last few decades have witnessed a host of new and deep applications of mathematics. These applications have not been limited to fields such as physics and the engineering sciences which have traditionally been heavy users of mathematics, but also have included other endeavours: computer science, information technology, economics, theoretical biology, and medicine. Nor have the types of mathematics that have found application been limited to traditional bread and butter topics such as differential equations and numerical analysis: investigators in other fields have found striking uses for topology, algebra, several complex variables, cohomology, and singularity theory, to name but a few examples. Indeed, a broad mathematical culture has become a necessity for increasingly many specialists in other disciplines.

This kind of "industrialization" of mathematics has resulted in the need for high level mathematical reference books which are, however, accessible to specialists in very different fields. We hope that this series will meet this need.

The books in the series will contain concise expositions of all the main branches of mathematics and their applications. They will be written from the point of view of practicing mathematicians so as to incorporate the achievements, spirit, and perspective of recent decades. In particular, each article will set forth the present state and the prospects for development of a given field. It will enunciate the basic principles of that field from a contemporary point of view, indicate connections with other fields of mathematics, report the main results, and sketch the motivation behind the various concepts which are introduced. Each article will be totally accessible to the interested worker in the other sciences (and not just to professional mathematicians in closely related fields).

In order to enhance the value of the series as a reference tool, each article will contain an annotated bibliography. Hopefully, the books in the series will also serve as a source of material for instructors in colleges and universities in preparing course syllabi and lectures on various subfields of mathematics and their applications.

We intend that the series be considered as a single entity. The choice and content of the articles in each volume will take into consideration the contents of other volumes. However, we have not hesitated to repeat a topic if its treatment benefits from taking different viewpoints.

The volumes will be numbered in the order in which they are published. In addition, they will be grouped into sections according to the main branches of mathematics and the volumes in each section will also be numbered sequentially. The earlier volumes in each section will be devoted to the basic topics. More specialized subjects will appear in the later volumes of each section.

The present volume is the first in the section "Dynamical Systems". An additional five or six volumes in this section will appear in the near future.

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Contents

I. Ordinary Differential Equations

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1

II. Smooth Dynamical Systems

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149

I. Ordinary Differential Equations

V. I. Arnold, Yu. S. Il'yashenko

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Contents

Preface	7
Chapter 1. Basic Concepts	8
§ 1. Definitions	8
1.1. Direction Fields and Their Integral Curves	8
1.2. Vector Fields, Autonomous Differential Equations, Integral Curves and Phase Curves	9
1.3. Direction Fields and Differential Equations	9
1.4. Diffeomorphisms and Phase Flows	10
1.5. Singular Points	11
1.6. The Action of a Diffeomorphism on a Vector Field	12
1.7. First Integrals	12
1.8. Differential Equations with Complex Time	13
1.9. Holomorphic Direction Fields in the Complex Domain	13
1.10. Higher Order Differential Equations	13
1.11. Differential Equations on Manifolds	14
§ 2. Basic Theorems	14
2.1. The Rectifiability Theorem for Vector Fields	14
2.2. The Existence and Uniqueness Theorem	14
2.3. The Rectifiability Theorem for Direction Fields	15
2.4. Methods of Solving Differential Equations	16
2.5. The Extension Theorem	17
2.6. The Theorem on Differentiable and Analytic Dependence of Solutions on Initial Conditions and Parameters	17
2.7. The Variational Equation	17
2.8. The Theorem on Continuous Dependence	18

2.9.	The Local Phase Flow Theorem	18
2.10.	The First Integral Theorem	18
§ 3.	Linear Differential Equations	19
3.1.	The Exponential of a Linear Operator	19
3.2.	The Theorem on the Relation Between Phase Flows of Linear Vector Fields and Exponentials of Linear Operators	19
3.3.	Complexification of the Phase Space	20
3.4.	Saddles, Nodes, Foci, Centers	21
3.5.	The Liouville-Ostrogradsky Formula	21
3.6.	Higher Order Linear Equations	22
§ 4.	Stability	23
4.1.	Lyapunov Stability and Asymptotic Stability	23
4.2.	Lyapunov's Theorem on Stability by Linearization	24
4.3.	Lyapunov and Chetaev Functions	24
4.4.	Generic Singular Points	25
§ 5.	Cycles	26
5.1.	The Structure of the Phase Curves of Real Differential Equations	26
5.2.	The Monodromy Transformation of a Closed Phase Curve. Limit Cycles	26
5.3.	The Multiplicity of a Cycle	27
5.4.	Multipliers	28
5.5.	Limit Sets and the Poincaré-Bendixson Theorem	29
§ 6.	Systems with Symmetries	30
6.1.	The Group of Symmetries of a Differential Equation	30
6.2.	Quotient Systems	30
6.3.	Homogeneous Equations	31
6.4.	Use of Symmetries to Reduce the Order	31
§ 7.	Implicit Differential Equations	33
7.1.	Basic Definitions; the Criminant, Integral Curves	33
7.2.	Regular Singular Points	33
7.3.	Folded Saddles, Nodes, and Foci	34
7.4.	Normal Forms of Folded Singular Points	35
7.5.	Whitney Pleats	35
§ 8.	Attractors	36
8.1.	Definitions	37
8.2.	An Upper Bound for the Dimension of the Maximal Attractor	37
8.3.	Applications	38
	Chapter 2. Differential Equations on Surfaces	39
§ 1.	Structurally Stable Equations on the Circle and on the Sphere	39
1.1.	Definitions	39
1.2.	The One-Dimensional Case	39
1.3.	Structurally Stable Systems on a Two-Dimensional Sphere	40

§ 2. Differential Equations on a Two-Dimensional Torus	40
2.1. The Two-Dimensional Torus and Vector Fields on it	40
2.2. The Monodromy Mapping	41
2.3. The Rotation Number	42
§ 3. Structurally Stable Differential Equations on the Torus	42
3.1. Description of Structurally Stable Equations	42
3.2. A Bound on the Number of Cycles.	43
§ 4. Equations on the Torus with Irrational Rotation Numbers	43
4.1. The Equivalence of a Diffeomorphism of a Circle to a Rotation of the Circle	43
4.2. Diffeomorphisms of a Circle and Vector Fields on S^3	45
§ 5. Remarks on the Rotation Number	45
5.1. The Rotation Number as a Function of the Parameters	46
5.2. Families of Equations on a Torus	46
5.3. Endomorphisms of the Circle	46
Chapter 3. Singular Points of Differential Equations in Higher Dimensional Real Phase Space	47
§ 1. Topological Classification of Hyperbolic Singular Points	47
1.1. The Grobman-Hartman Theorem	47
1.2. Classification of Linear Systems	48
§ 2. Lyapunov Stability and the Problem of Topological Classification	48
2.1. On the Local Problems of Analysis	48
2.2. Algebraic and Analytic Insolubility of the Problem of Lyapunov Stability	49
2.3. Algebraic Solubility up to Degeneracies of Finite Codimension	50
2.4. Topologically Unstabilizable Jets	51
§ 3. Formal Classification of Germs of Vector Fields	52
3.1. Formal Vector Fields and Their Equivalence	52
3.2. Resonances. The Poincaré-Dulac Normal Forms and Their Generalizations	53
3.3. Applications of the Theory of Formal Normal Forms	54
3.4. Polynomial Normal Forms	55
§ 4. Invariant Manifolds and the Reduction Theorem	56
4.1. The Hadamard-Perron Theorem	56
4.2. The Center Manifold Theorem	57
4.3. The Reduction Principle	57
§ 5. Criteria for Stability and the Topological Classification of Singular Points in the Case of Degeneracies of Low Codimension	58
5.1. Structure of the Criteria	58
5.2. Topological Classification of Germs of Smooth Vector Fields up to and Including Degeneracies of Codimension Two	58
5.3. Phase Portraits of Normal Forms	61
5.4. Criteria for Lyapunov Stability of Degeneracies of Codimension up to and Including Three	62

5.5.	The Adjacency Diagram	66
5.6.	Theorems on Algebraic Solubility	66
§ 6.	Smooth Classification of Germs of Vector Fields	66
6.1.	The Relation Between Formal Classification and Smooth Classification	66
6.2.	Germs of Vector Fields with Symmetries	67
6.3.	Quasi-Hyperbolicity	68
6.4.	Finitely Smooth Equivalence of Germs of Vector Fields	68
§ 7.	Normal Forms of Vector Fields in which the Linear Part is a Nilpotent Jordan Block	68
7.1.	Centralized Chains	69
7.2.	Non-Removable Terms	69
7.3.	The Standard Representation of the Group $SL(2)$ and of the Algebra $\mathfrak{sl}(2)$	70
7.4.	Extension of a Nilpotent Operator to a Representation of the Lie Algebra $\mathfrak{sl}(2)$	70
7.5.	Conclusion of the Proof of the Theorem	71
Chapter 4. Singular Points of Differential Equations in Higher Dimensional Complex Phase Space		72
§ 1.	Linear Normal Forms	72
1.1.	Poincaré Domains and Siegel Domains. Small Denominators	72
1.2.	Convergence of the Normalizing Series	72
1.3.	Analytic Theorems on Divergence of the Normalizing Series	73
1.4.	Geometric Theorems on the Divergence of the Normalizing Series	74
§ 2.	The Relation Between Formal and Analytic Classification	74
2.1.	Condition A	75
2.2.	Problems Involving the Smooth and the Analytic Classification	75
§ 3.	Analytic Invariant Manifolds	76
3.1.	The Invariant Manifold Theorem	76
3.2.	Analyticity of the Center Manifold	77
3.3.	Reversible Systems	77
3.4.	Analytic Center Manifolds of Differential Equations in the Plane	78
§ 4.	Topological Classification of Singular Points in the Complex Domain	79
4.1.	Linear Vector Fields	79
4.2.	The Nonlinear Case	79
Chapter 5. Singular Points of Vector Fields in the Real and Complex Planes		80
§ 1.	Resolution of Singularities	80
1.1.	The Polar Blow-up and the σ -Process in the Plane	80

1.2.	Elementary Singular Points	81
1.3.	Good Blow-ups	81
§ 2.	Smooth Orbital Classification of Elementary Singular Points in the Plane	82
2.1.	Table of Normal Forms; the Analytic Case	83
2.2.	Normal Forms in the Smooth Case	83
§ 3.	Topological Classification of Compound Singular Points with a Characteristic Trajectory	83
3.1.	The Fundamental Alternative	84
3.2.	Topological Classification of Differential Equations on the Plane in a Neighbourhood of a Singular Point	85
3.3.	Topological Finite Determinacy. Newton Diagrams of Vector Fields	87
3.4.	Investigation of Vector Fields by Their Principal Part	87
§ 4.	The Problem of Distinguishing Between a Center and a Focus	88
4.1.	Statement of the Problem	88
4.2.	Algebraic Insolubility	89
4.3.	Distinguishing a Center by Linear Terms	89
4.4.	Nilpotent Jordan Block	89
4.5.	Singular Points Without Exclusive Directions	90
4.6.	A Programme for Investigating the General Case	91
4.7.	The Generalized First Focus Number	91
4.8.	Polynomial Vector Fields	91
§ 5.	Analytic Classification of Elementary Singular Points in the Complex Domain	92
5.1.	Germs of Conformal Mappings with the Identity as Linear Part	92
5.2.	Classification of Resonant Mappings and Vector Fields with Generic Nonlinearities	93
5.3.	Degenerate Elementary Singular Points	94
5.4.	Geometric Theory of Analytic Normal Forms	94
5.5.	Embedding in the Flow, Extraction of Roots, Divergence of Normalizing Series, and Holomorphic Center Manifolds	95
5.6.	Taylor Description of the Analytic Equivalence Classes	95
§ 6.	Orbital Topological Classification of Elementary Singular Points in the Complex Plane	96
6.1.	The Non-Resonant Case	96
6.2.	Saddle-Resonant Vector Fields	96
6.3.	Degenerate Elementary Singular Points	96
	Chapter 6. Cycles	97
§ 1.	The Monodromy Mapping	97
1.1.	Definitions	97
1.2.	Realization	98
§ 2.	Local Theory of Diffeomorphisms	99
2.1.	Linear Normal Forms	99

2.2.	The Resonant Case	100
2.3.	Invariant Manifolds of Diffeomorphism Germs	101
2.4.	Invariant Manifolds of a Cycle	101
2.5.	Blow-ups	102
§ 3.	Equations with Periodic Right Hand Side	103
3.1.	Normal Form of a Linear Equation with Periodic Coefficients	103
3.2.	Linear Normal Forms	104
3.3.	Resonant Normal Forms	104
§ 4.	Limit Cycles of Polynomial Vector Fields in the Plane	104
4.1.	The Finiteness Problem and Compound Cycles	105
4.2.	The Monodromy Map of a Compound Cycle	105
4.3.	Finiteness Theorems for Limit Cycles	106
4.4.	Two Particular Finiteness Theorems	107
4.5.	Method of Proving Dulac's Theorem and its Generalization	107
4.6.	Quadratic Vector Fields	108
§ 5.	Limit Cycles of Systems Close to Hamiltonian Systems	108
5.1.	Generation of Real Limit Cycles	109
5.2.	Generation of Complex Cycles	109
5.3.	Investigation of the Variation	110
5.4.	The Weak Hilbert Conjecture	110
5.5.	Abelian Integrals Appearing in Bifurcation Theory	111
§ 6.	Polynomial Differential Equations in the Complex Plane	112
6.1.	Admissible Fields	112
6.2.	Polynomial Fields	113
Chapter 7. Analytic Theory of Differential Equations		114
§ 1.	Equations Without Movable Critical Points	114
1.1.	Definitions	115
1.2.	Movable Critical Points of a First Order Equation	115
1.3.	The Riccati Equation	115
1.4.	Implicit Equations	116
1.5.	Painlevé Equations	116
§ 2.	Local Theory of Linear Equations with Complex Time	117
2.1.	Regular and Irregular Singular Points	117
2.2.	Formal, Holomorphic, and Meromorphic Equivalence	119
2.3.	Monodromy	119
2.4.	Formal Theory of Linear Systems with a Fuchsian Singular Point	120
2.5.	Formal Theory of Linear Systems with a Non-Fuchsian Singular Point	121
2.6.	Asymptotic Series and the Stokes Phenomenon	123
2.7.	Analytic Classification of Nonresonant Systems in a Neighbourhood of an Irregular Singular Point	124
§ 3.	Theory of Linear Equations in the Large	125
3.1.	Equations and Systems of the Fuchsian Class	125

3.2.	Extensibility and Monodromy	125
3.3.	The Riemann-Fuchs Theorem	126
3.4.	Analytic Functions of Matrices	127
3.5.	Connection with the Theory of Kleinian Groups	128
3.6.	Integrability in Quadratures	128
3.7.	Special Equations of Mathematical Physics	129
3.8.	The Monodromy Group of the Gauss Equation	129
§ 4.	The Riemann-Hilbert Problem	130
4.1.	Formulation of the Problem	130
4.2.	The Riemann-Hilbert Problem for a Disk	130
4.3.	The Riemann-Hilbert Problem for a Sphere	132
4.4.	The Riemann-Hilbert Problem for Fuchsian Systems	133
4.5.	Generalizations for Non-Standard Time	133
4.6.	Vector Bundles on the Sphere	134
4.7.	Application to the Riemann-Hilbert Problem	134
4.8.	Isomonodromic Deformations and the Painlevé Equations	135
	Bibliography	135
	Index	142

Preface

This survey is devoted mainly to the local theory of ordinary differential equations. It does not include bifurcation theory, which will be dealt with in a separate article. The averaging method is dealt with in the survey "Mathematical aspects of classical and celestial mechanics" by V. I. Arnold, V. V. Kozlov, and A. I. Neishtadt (volume 3 of this series).

We do not touch on the spectral theory of differential operators with a single independent variable (see, for example, [28]); as regards objectives and methods, this is more closely related to functional analysis. Our survey also does not include the theory of integral transforms and their application to linear differential equations. The asymptotic theory of differential equations is dealt with in M. V. Fedoryuk's survey, "Asymptotic methods in analysis"; however, some general theorems of this theory are presented in Chap. 7. The question of actually integrating particular equations is not touched on at all; the standard book on this subject is E. Kamke's "Differentialgleichungen: Lösungen und Lösungsmethoden", Chelsea (1948).

In recent years there has been a sharp increase in research activity involving classical problems of the theory of differential equations. This is due to the penetration into the theory of other disciplines: algebra (the theory of formal normal forms), algebraic geometry (resolution of singularities), and complex analysis. We have tried, as far as possible, to reflect this research in the present article.

The presentation is carried out from a single point of view using a single terminology. In order to make our survey accessible to non-specialists, we start each chapter by defining the fundamental concepts. In the current literature the terminology is not unique; even the term "singular point" is used with different meanings. Differences in terminology and, even more importantly, in mathematical language have the effect that similar results are formulated entirely differently in different sources. In order that such results be perceived as parts of a single whole, we often express them differently than in the original sources.

We have tried to begin the study of each problem by considering the generic objects; these are the simplest cases and, at the same time, have the most applications because they are the most often encountered. We mention degeneracies of higher codimension in only two cases:

- 1) when all degeneracies of lower codimension in a given problem have already been described;
- 2) when the investigation of problem proceeds in the same way for degeneracies of all codimensions.

The list of references makes no claim to completeness. Some textbooks and general monographs have been included. The greater part of the list consists of papers which contain a more detailed investigation of the results cited in this survey. To shorten the reference list we have used a dual system of references: if a paper is cited in a monograph or article included in our reference list, then a reference to it is of the form $[a; b]$ or $[a, p. b]$. The first indicates the paper $[b]$ cited in the reference list of $[a]$; the second indicates the paper cited on page b of $[a]$.

V. I. Arnold, Yu. S. Il'yashenko

Chapter 1

Basic Concepts

In this chapter (§§ 1–6), we briefly survey some basic concepts and general results in the theory of ordinary differential equations. A more detailed account can be found in the books [6], [40], [49], [87], [93], [106], [110]. Sections 7 and 8 are more specialized and contain some recent results.

§ 1. Definitions

1.1. Direction Fields and Their Integral Curves. We consider a real, finite-dimensional, linear space $V = \mathbb{R}^n$. A *direction field* in a domain U of V is a map which associates to each point $x \in U$ a straight line passing through x .

Definition. An *integral curve* of a direction field is a curve whose tangent line at each point coincides with the image of the direction field at that point.

1.2. Vector Fields, Autonomous Differential Equations, Integral Curves and Phase Curves. A *vector field*, defined in a domain U of the space V , is a map which associates to each point $x \in U$ a vector $v(x)$, based at x , in the space V .

The equation

$$\dot{x} = v(x), \quad x \in U \subset V \quad (1)$$

is called the *differential equation* corresponding to the vector field v ; a dot over a letter denotes differentiation with respect to t : $\dot{x} = \frac{dx}{dt}$.

The domain U is called the *phase space* of this equation, and the direct product $\mathbb{R} \times U$ is called the *extended phase space*. Here $\mathbb{R}$ is the t -axis; the variable t is called “time”.

Equation (1) is also said to be an *autonomous equation* (since the evolution law of a physically autonomous system, i.e., a system which does not interact with other systems does not usually depend on time). An equation whose right-hand side also depends on time

$$\dot{x} = v(t, x) \quad (2)$$

is called a *nonautonomous equation*.

Definition. A *solution* of the differential equation (2) is a differentiable map $\varphi: I \rightarrow U$ of an interval $I = \{t \in \mathbb{R}, a < t < b\}$ of the real t -axis ($a = -\infty$ and $b = +\infty$ are allowed) into the phase space U such that the relation

$$\frac{d}{dt} \varphi(t)|_{t=\tau} = v(\tau, \varphi(\tau))$$

holds for all $\tau \in I$.

Definition. An *integral curve* of a differential equation is the graph of a solution; a *phase curve* is the projection of an integral curve on to the phase space along the t -axis.

1.3. Direction Fields and Differential Equations. Suppose v is a vector field which depends on time, i.e., a mapping which associates to each point (t, x) of a domain in the direct product of the time axis $\mathbb{R}$ and the phase space V a vector (based at the point x) of the phase space.

Definition. The *direction field* of the differential equation

$$\dot{x} = v(t, x), \quad (t, x) \in U \subset \mathbb{R} \times V \quad (2)$$

¹ More precisely, here and elsewhere, $\dot{x}$ or $\frac{dx}{dt}$ denotes the image of the standard tangent vector to the t -axis under the derivative of the mapping $t \mapsto x$.

is the direction field defined in a domain of the extended phase space by associating to each point (t, x) the straight line through (t, x) which contains the vector $(1, v(t, x))$.

The following result is easily proved.

Theorem. *The integral curves of the direction field of equation (2) are the graphs of the solution of this equation, and conversely.*

Not every direction field is a direction field of an equation (2). For example, the field of zeros of the 1-form $f(x, y)dx + g(x, y)dy = 0$ (with the condition $f^2 + g^2 \neq 0$) determines a direction field. But the field of zeros of the form $d(x^2 + y^2)$ on the plane with the origin 0 removed is not a direction field of an equation of the form (2).

1.4. Diffeomorphisms and Phase Flows. A diffeomorphism $f: U \rightarrow V$ of a domain U on to a domain V is a 1-1 mapping such that both f and $f^{-1}: V \rightarrow U$ are differentiable. Henceforth, unless otherwise stated, differentiability means that continuous derivatives of all orders exist.

Definition. A one-parameter group $\{g^t\}$ of diffeomorphisms of a domain U is a map of the direct product $\mathbb{R} \times U$ into U :

$$g: \mathbb{R} \times U \rightarrow U, \quad g(t, x) = g^t x, \quad t \in \mathbb{R}, \quad x \in U,$$

such that

1) g is differentiable,
 2) the family $\{g^t | t \in \mathbb{R}\}$ is a one-parameter group of transformations; that is, $g^t \circ g^s = g^{t+s}$ for any $t, s \in \mathbb{R}$ and $g^0 = \text{id}$ (the identity).

1) and 2) imply

3) for every $t \in \mathbb{R}$ the mapping $g^t: U \rightarrow U$ is a diffeomorphism.

A phase flow defined in a domain U is a one-parameter group of diffeomorphisms of U .

Definition. The phase velocity field of a flow $\{g^t\}$ is the vector field

$$v(x) = \left. \frac{dg^t x}{dt} \right|_{t=0}$$

of velocities with which the flow moves the phase points. This field is also called the *generating field* of the one-parameter group $\{g^t\}$.

Theorem. *The motion of a point under the action of a phase flow $\varphi(t) = g^t(x_0)$ is a solution of the equation (1) defined by the phase velocity field.*

Conversely, given a vector field v , in certain cases (namely, when the solutions of (1) extend over the whole time axis) it is possible to construct a phase flow for which v is the phase-velocity field. However, this is not always possible: an example is the equation $\dot{x} = x^2$. In this case, as can easily be

calculated, the only possible flow is given by the formula $g^t x = \frac{x}{1-tx}$. Although $g^{t+s} = g^t \circ g^s$, this formula does not give a phase flow on the affine line; but rather on the projective line obtained by adjoining the point ∞ to the x -axis.

Associated to every vector field and to each point of phase space there is a *local phase flow*; namely, a one-parameter family of diffeomorphisms defined in a neighbourhood of the point. The maps of the local phase flow correspond to an interval of the time axis and have the group property

$$g^{t+s} = g^t \circ g^s, \quad g^{-t} = (g^t)^{-1}.$$

The first equality holds for all t and s in a neighbourhood of zero and for all x in a neighbourhood of the given point. The phase velocity field of a local phase-flow coincides with the original vector field in this neighbourhood.

The local phase flows $\{g_1^t\}$ and $\{g_2^t\}$ associated to two vector fields at the points x and y are *topologically equivalent* (or *conjugate*) if there is a homeomorphism H of a neighbourhood of each point which carries x into y and conjugates the transformations of the local phase flows:

$$g_2^t \circ H = H \circ g_1^t.$$

1.5. Singular Points

Definition. A *singular point* of a vector field v is a point at which the vector field vanishes.

A *singular point* of a differential equation $\dot{x} = v(x)$ is a singular point of the corresponding vector field v .

Let f be a differentiable mapping of a domain U of $\mathbb{R}^n$ into a domain W of $\mathbb{R}^m$. The *derivative of f at the point x_0* is the “principal linear part” of f at x_0 ; more precisely, it is the linear map $A: \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that

$$f(x) - f(x_0) = A(x - x_0) + o(x - x_0).$$

Let $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_m)$ be coordinates in the source and target respectively. Then f can be written as a vector function $y = (f_1(x), \dots, f_m(x))$. This means that $y_k \circ f = f_k$. The matrix of A with respect to the coordinates x, y is the Jacobian matrix of the vector function $f = (f_1, \dots, f_m)$. We use the same notation for the derivative of a mapping and the matrix representing it:

$$A = \frac{\partial f}{\partial x}(x_0) = f_*(x_0), \quad A = (a_{ij}), \quad a_{ij} = \frac{\partial f_i}{\partial x_j}(x_0).$$

A *differentiable vector field* is a field in which the components are differentiable.

Let x_0 be a singular point of a differentiable vector field v , and let $\frac{\partial v}{\partial x}$ be the derivative of the mapping $v: x \mapsto v(x)$. The equation

$$\dot{x} = Ax, \quad A = \frac{\partial v}{\partial x}(x_0)$$

is called the *linearization of the equation (1) at the singular point x_0* , the field Ax (and, sometimes, the map A associated to it) is called *linear part of the field v at the point x_0* . The map A associated to the linear part is sometimes written as v_* instead of $\frac{\partial v}{\partial x}$.

A singular point of a vector field is said to be *nondegenerate* if the map associated to the linear part of the field is nonsingular at this point.

The singular points of a differential equation are sometimes called *equilibrium points*. The eigenvalues of the map associated to the linear part of the vector field at a singular point a (i.e., the eigenvalues of $\frac{\partial v}{\partial x}(a)$) are sometimes called the *eigenvalues of the singular point*.

1.6. The Action of a Diffeomorphism on a Vector Field

Definition. A diffeomorphism $f: U \rightarrow W$ carries a vector field v on the domain U into a vector field $\tilde{v}$ on the domain W defined as follows: let $\varphi(t)$ be a path in U which passes through the point x at $t=0$ with velocity $v(x)$; define $\tilde{v}(y)$ to be the velocity with which the path $f(\varphi(t))$ passes through the point $y=f(x)$.

This definition of $\tilde{v}$ does not depend on the choice of path φ . This is proved, for example, by using the chain rule: $\tilde{v}(y) = (f_* v)(x)$, $y = f(x)$. In other words, the derivative of the map f at x transforms the vector which the field v assigns to x to the vector which the field $\tilde{v}$ assigns to $f(x)$.

1.7. First Integrals

Definition. A smooth function which is constant on the phase curves of an autonomous differential equation is called a (*time-independent*) *first integral* of the differential equation.

A *time-dependent first integral* of a (generally speaking, nonautonomous) equation is a function on the extended phase-space which is constant on the integral curves.

A *complete system of first integrals* of an (autonomous or nonautonomous) differential equation is a set of first integrals, time-dependent or not, on which any other first integral is functionally dependent.

Example. The system of equations $\dot{x} = x$, $\dot{y} = y$ in the plane has no non-constant, time-independent first integrals. But the same system of equations has a complete system of two time-dependent first integrals, namely: xe^{-t} , ye^{-t} .

1.8. Differential Equations with Complex Time. A vector field defined in a domain of $\mathbb{C}^n$ is said to be *holomorphic* if its components are holomorphic functions. A holomorphic mapping $U \rightarrow W$, $U \in \mathbb{C}^n$, $W \in \mathbb{C}^m$, is defined similarly.

An equation of the form

$$\frac{dz}{dt} = v(z), \quad z \in U \subset \mathbb{C}^n, \quad t \in \mathbb{C},$$

is called a *differential equation with complex time* or a *complex autonomous equation*.

All the definitions of sections 1.1–1.7 can be carried over in an obvious way from real autonomous equations to complex autonomous equations.

1.9. Holomorphic Direction Fields in the Complex Domain. A *holomorphic direction field* is a map which associates to each point of a domain U of $\mathbb{C}^n$ a complex straight line passing through the point and depending analytically on it. This means that each point of U has a neighbourhood on which there exists a holomorphic system of coordinates so that the straight lines passing through the points of the neighbourhood have direction vector $(1, v_2(z), \dots, v_n(z))$, where the v_j are holomorphic functions.

An *integral curve* of a holomorphic direction field is a holomorphic curve whose tangent line at each point coincides with the direction field at that point, and which is connected and maximal with respect to this property; i.e., it is not a proper subset of a connected holomorphic curve which is everywhere tangent to the direction field.

Example 1. Consider the direction field given by the Euler vector field $z_1 \frac{\partial}{\partial z_1} + z_2 \frac{\partial}{\partial z_2}$ on the punctured plane $\mathbb{C}^2 \setminus \{0\}$. Its integral curves are straight lines passing through the origin, with the origin deleted. By definition, the point $\{0\}$ does not belong to the integral curves, because the direction field cannot be continued analytically across it. Notice that it does not belong to the integral curves, even though adding it would leave the curves holomorphic.

Example 2. The integral curves of the direction field $dH=0$, where $H(z, w)$ is a polynomial, are the level curves of H minus any critical points of H . They are Riemann surfaces with points deleted.

Thus the topology of an integral curve of a direction field in the complex domain can be more complicated than that of a real integral curve, which is always homeomorphic to the straight line $\mathbb{R}$. We note that a real phase curve is topologically equivalent to either a straight line, a circle, or a single point.

1.10. Higher Order Differential Equations. The differential equation

$$x^{(n)} = f(t, x, \dot{x}, \dots, x^{(n-1)}), \quad x \in \mathbb{R}^1 \quad (3)$$

reduces to a system of the form (2) under the substitution $y = (x, \dot{x}, \dots, x^{(n-1)})$.

Then

$$\dot{y} = Ay + f(t, y)e_n, \quad e_n = (0, 0, \dots, 0, 1),$$

where the matrix A is a Jordan upper-triangular block with zeros on the diagonal. The solution of equation (3) with the initial conditions

$$x(t_0) = y_0, \quad \dot{x}(t_0) = y_1, \dots, x^{(n-1)}(t_0) = y_{n-1}$$

is the first component of the solution of the corresponding system with the initial condition $y(t_0) = (y_0, \dots, y_{n-1})$. The main theorems for higher order equations reduce to theorems for systems of first order equations.

1.11. Differential Equations on Manifolds. All the definitions of the preceding sections generalize to the case in which a domain in a real or complex linear space is replaced by a real or complex manifold. For example, an autonomous differential equation on a manifold is defined to be a vector field on the manifold (that is, a section of the tangent bundle). The details can be found in [6], [31], [85].

§ 2. Basic Theorems

This section deals with the rectifiability theorem for vector fields, and its consequences. To simplify statements, the vector fields are always assumed to be smooth, *i.e.*, infinitely differentiable, although it would be sufficient everywhere except in section 2.6 to require only that the fields be C^1 ; in section 2.6, the smoothness requirements will be formulated explicitly.

2.1. The Rectifiability Theorem for Vector Fields. The following is the *fundamental theorem* of the theory of ordinary differential equations.

Theorem. *In a sufficiently small neighbourhood of a nonsingular point a differentiable vector field is diffeomorphic to the constant field $e_1 = (1, 0, \dots, 0)$. In other words, in a neighbourhood of a nonsingular point there is a diffeomorphism which carries the original field into the field e_1 .*

The diffeomorphism in this theorem is called the *rectifying diffeomorphism*.

A similar theorem is true for holomorphic vector fields. If the original field is C^r , $1 \leq r \leq \infty$, then the rectifying diffeomorphism can also be chosen to be C^r .

The other theorems in this section are simple consequences of the rectifiability theorem for vector fields.

2.2. The Existence and Uniqueness Theorem

Theorem. *For every point of the extended phase space of a differentiable (or holomorphic) vector field, there is one and only one integral curve of the*

corresponding differential equation with real (or complex) time which passes through that point.

Theorem. For each point of a domain on which a differentiable or complex analytic direction field is defined, there is one and only one integral curve of the field passing through that point.

This theorem becomes false if one replaces the phrase “direction field” by the phrase “plane field”. A *plane field* (or *distribution*) is a map which associates to each point of a domain of a vector space an affine subspace passing through that point. An integral surface of a plane field is a differentiable surface whose tangent space at each point coincides with the subspace determined by the field at that point.

A generic plane field of dimension two or higher has no integral surface. An example is the field in $\mathbf{R}^3$ defined by the zeros of the form $dz - y dx$. Sufficient conditions for the existence of integral surfaces are given by a theorem of Frobenius [7], [85].

The smoothness condition in the preceding theorem can be weakened.

Definition. A mapping $f: U \rightarrow W$, $U \subset \mathbf{R}^n$, $W \subset \mathbf{R}^m$ is said to satisfy a *Lipschitz condition* if there is a positive constant L such that

$$|f(x_1) - f(x_2)| \leq L|x_1 - x_2|$$

for all $x_1, x_2 \in U$.

Theorem. If the right-hand side of the equation

$$\dot{x} = v(t, x) \quad (2)$$

is continuous and satisfies a Lipschitz condition relative to x (i.e., in each hyperplane $t = \text{const}$) with the same constant L in a domain U of $\mathbf{R}^{n+1}$, then one and only one integral curve of (2) passes through each point of U .

If the right-hand side of equation (2) is only continuous, then at least one integral curve still passes through each point. But uniqueness may fail, as the example $\dot{x} = x^{1/3}$ shows (Fig. 1).

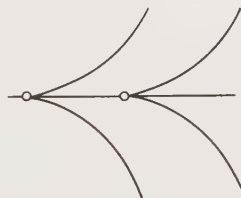


Fig. 1. Lack of uniqueness

2.3. The Rectifiability Theorem for Direction Fields

Theorem. Every point of a domain on which a real differentiable (or a complex analytic) direction field is defined has a neighbourhood on which there

exists a diffeomorphism (resp., a biholomorphic mapping) which transforms the integral curves of the field into parallel straight lines.

A plane field of dimension greater than one is not always rectifiable. An example is the field $dz = y dx$ in $\mathbb{R}^3$.

2.4. Methods of Solving Differential Equations

a) *Integration by means of power series.* A power series solution of a differential equation with an analytic right-hand side can be found by the method of undetermined coefficients. For example, by substituting the series $1 + a_1 t + \dots$ into $\dot{x} = x$, we successively find the coefficients to be $a_1 = 1, a_2 = 1/2, \dots, a_k = 1/k!$ giving the solution with the initial condition $x(0) = 1$.

b) *Euler polygonal curves.* Euler polygonal curves are piecewise-linear curves whose segments arise when the solution is approximated by linear functions in successive small time intervals.

Instead of the differential equation (2) we consider the difference equations

$$\varphi_{k+1} - \varphi_k = v(t_k, \varphi_k) h, \quad t_k = t_0 + kh.$$

From this equation, with the initial condition $\varphi_0 = x_0$ we successively find $\varphi_1, \varphi_2, \dots, \varphi_N$. The piecewise-linear curve with vertices (t_k, φ_k) is called an *Euler polygonal curve*. On a sufficiently small time interval ($Nh \leq a$), the Euler polygonal curves converge uniformly as $h \rightarrow 0$ to the integral curve of the differential equation $\dot{x} = v(t, x)$ passing through the initial point (t_0, x_0) (continuity of v is sufficient to ensure convergence).

Greater accuracy with the small step h is given by difference equations which approximate the solution by polynomials of degree greater than one. For example, one of the most frequently used methods is the *Runge-Kutta method*, which gives as the next vertex after (x, y) , the point $(x + h, y + \Delta y)$, where

$$\Delta y = \frac{1}{8}(k_1 + 3k_2 + 3k_3 + k_4),$$

with

$$\begin{aligned} k_1 &= h v(x, y), & k_2 &= h v\left(x + \frac{h}{3}, y + \frac{k_1}{3}\right), \\ k_3 &= h v\left(x + \frac{2h}{3}, y - \frac{k_1}{3} + k_2\right), & k_4 &= h v(x + h, y + k_1 - k_2 + k_3) \end{aligned}$$

[12, p. 455]. The increment Δy calculated by this formula approximates the increment of the required solution to an accuracy of $O(h^5)$.

c) *Method of successive approximations.* Consider the sequence of mappings $\varphi_k: I \rightarrow \mathbb{R}^n$

$$\varphi_0(t) \equiv x_0, \quad \varphi_{k+1}(t) = x_0 + \int_{t_0}^t v(\tau, \varphi_k(\tau)) d\tau.$$

The mappings φ_k are called *Picard approximations* to the solution φ of equation (2) with the initial condition $\varphi(t_0) = x_0$. If the interval I is sufficiently

small, then the sequence of Picard approximations converges uniformly to the solution φ on I at least as fast as a geometric series with a ratio proportional to the length of the interval. (In fact, Picard approximation converges faster than any geometric series.)

2.5. The Extension Theorem

Definition. An extension of the solution φ is a solution which coincides with φ on the interval on which φ is defined and which is defined on a greater interval of the time axis. An extension is said to be *infinite to the right (or left)* if it is defined on a whole ray to the right (or left) along the time axis. A solution is said to extend to a given subset of phase space (or extended phase space) if it has an extension for which the corresponding phase curve (integral curve) intersects that subset.

Theorem. Suppose that the right-hand side of the equation

$$\dot{x} = v(x), \quad x \in U \subset \mathbb{R}^n, \quad (1)$$

is differentiable, and let K be a subset of U which is the compact closure of a subdomain of the domain U . Then every solution of equation (1) with an initial condition in K can be extended to the right (or left) either up to the boundary of K or infinitely in time. The solution of a nonautonomous equation with an initial condition in a compact subset of the extended phase space can be extended up to the boundary of the subset.

Remark. Differential equations with arbitrarily smooth right-hand sides can have solutions which cannot be extended over the whole time axis. For example, a solution of $\dot{x} = x^2$ goes off to infinity in time $t = 1/x(0)$.

2.6. The Theorem on Differentiable and Analytic Dependence of Solutions on Initial Conditions and Parameters

Theorem. Let

$$\dot{x} = v(t, x, \alpha), \quad x \in U \subset \mathbb{R}^n \quad (\alpha)$$

be a family of differential equations defined in the phase space U by C^r vector fields v which depend differentiably (of class C^r) on a parameter $\alpha \in A$, where A is a domain in a real, linear, finite-dimensional space $\mathbb{R}^a$. Then the solution φ of the equation (α) with the initial condition $\varphi(t_0) = x_0$ depends differentiably (of class C^r) on t, x and α for sufficiently small $|t - t_0|$, $|x - x_0|$ and $|\alpha - \alpha_0|$.

A similar theorem holds for equations with complex time, except that U and A are domains in $\mathbb{C}^n$ and $\mathbb{C}^a$ respectively (a being the complex dimension of the parameter space), and the vector-function v is holomorphic in a domain in the direct product $\mathbb{C} \times U \times A$.

2.7. The Variational Equation. An equation for the first derivative of a solution with respect to the initial conditions can easily be written down

explicitly. Thus, let φ_ξ be the solution of the equation $\dot{x} = v(t, x)$ with the initial condition $\varphi_\xi(0) = \xi$. We fix ξ and put

$$X(t) = \frac{\partial \varphi_\xi(t)}{\partial \xi} \Big|_{\xi = x_0};$$

$X(t)$ is a linear map $\mathbb{R}^n \rightarrow \mathbb{R}^n$ which depends on t . The operator-valued function X satisfies the following variational equation

$$\dot{X}(t) = A(t) X(t), \quad \text{where} \quad A(t) = \frac{\partial v}{\partial x} \Big|_{x = \varphi_{x_0}(t)}.$$

This is a linear, homogeneous, nonautonomous equation.

We now write down the variational equation for the derivative of the solutions with respect to the parameters. Let $\varphi_{\alpha, \xi}$ be the solution of the equation (α) with the initial condition $\varphi_{\alpha, \xi}(0) = \xi$. We fix $\xi = x_0$ and $\alpha = \alpha_0$ and put

$$Y(t) = \frac{\partial \varphi_{\alpha, \xi}(t)}{\partial \alpha} \Big|_{\substack{\xi = x_0 \\ \alpha = \alpha_0}}.$$

$Y(t)$ is a linear map $\mathbb{R}^k \rightarrow \mathbb{R}^n$ which depends on t . The operator-valued function Y satisfies the variational equation:

$$\dot{Y}(t) = A(t) Y(t) + b(t),$$

where

$$A(t) = \frac{\partial v}{\partial x} \Big|_{x = \varphi_{\alpha_0, x_0}(t), \alpha = \alpha_0},$$

$$b(t) = \frac{\partial v}{\partial \alpha} \Big|_{x = \varphi_{\alpha_0, x_0}(t), \alpha = \alpha_0}.$$

This is a linear, inhomogeneous, nonautonomous equation.

2.8. The Theorem on Continuous Dependence. If the dependence of the field v on the parameter α is only continuous in the preceding theorem, then the solutions depend continuously on the parameter α .

But the solutions of an equation with a differentiable right hand side depend differentiably, and not just continuously, on the initial conditions.

2.9. The Local Phase Flow Theorem. It follows from the existence and uniqueness theorem and the theorem on differentiable dependence of solutions on initial conditions that a differentiable vector field determines a local phase flow in a neighbourhood of any point in the phase space.

2.10. The First Integral Theorem. An autonomous equation with a differentiable right-hand side has a complete system of $n-1$ functionally independent, time-independent first integrals $I_1, \dots, I_{n-1}$ in a neighbourhood

of every nonsingular point of an n -dimensional phase space. The phase curves of equation (1) are given in this neighbourhood by the system $I_1 = C_1, \dots, I_{n-1} = C_{n-1}$.

A nonautonomous equation (2) on n -dimensional phase space has a complete system of n time-dependent and locally functionally independent first integrals $I_1, \dots, I_n$. The system $I_1 = C_1, \dots, I_n = C_n$ defines local integral curves of the equation.

§ 3. Linear Differential Equations

Linear autonomous equations constitute almost the only large class of differential equations for which there is a complete theory. This theory, which is essentially a branch of linear algebra, enables all linear autonomous equations to be solved completely.

3.1. The Exponential of a Linear Operator. Let V be an n -dimensional real, or complex, space with a Euclidean, or Hermitian, structure respectively, and let $|\cdot|$ denote the norm in this space.

The *norm* of a linear map $A: V \rightarrow V$ is defined to be the supremum

$$\|A\| = \sup_{|x|=1} |Ax|.$$

This supremum is attained because a finite-dimensional sphere is compact.

A sequence of linear maps $A_k: V \rightarrow V$ is said to *converge to a linear map* $A: V \rightarrow V$ if

$$\|A_k - A\| \rightarrow 0 \quad \text{as } k \rightarrow \infty.$$

Definition. The *exponential of a linear map* $A: V \rightarrow V$ is defined to be the map from V to V given by

$$e^A = E + A + \frac{A^2}{2!} + \dots + \frac{A^k}{k!} + \dots$$

where E is the identity map: $Ex = x$.

An equivalent definition is

$$e^A = \lim_{k \rightarrow \infty} \left(E + \frac{A}{k} \right)^k.$$

3.2. The Theorem on the Relation Between Phase Flows of Linear Vector Fields and Exponentials of Linear Operators

Theorem. Any one-parameter group of diffeomorphisms which are linear transformations $g^t: V \rightarrow V$ is of the form

$$g^t = e^{At},$$

where $A: V \rightarrow V$ is a linear map (the generating transformation of the group). The solution of the linear equation

$$\dot{x} = Ax, \quad x \in V,$$

with the initial condition $\varphi(0) = \xi$ is

$$\varphi(t) = e^{At} \xi.$$

3.3. Complexification of the Phase Space. The complexification of a real, n -dimensional, linear space V is defined to be the n -dimensional complex space ${}^{\mathbb{C}}V$ constructed as follows. The points of ${}^{\mathbb{C}}V$ are pairs (ξ, η) , written $\xi + i\eta$, $\xi \in V$, $\eta \in V$. The operations of addition and multiplication by a complex number are defined in the usual way.

The complexification of a linear map $A: V \rightarrow V$ is the map ${}^{\mathbb{C}}A: {}^{\mathbb{C}}V \rightarrow {}^{\mathbb{C}}V$ defined by the rule

$${}^{\mathbb{C}}A(\xi + i\eta) = A\xi + iA\eta.$$

The solutions of a linear autonomous equation extend without bound. From now on, a solution of a linear equation will be understood to mean a solution defined on the whole time axis.

It follows from the theorem in section 3.2 that the space of solutions of the equation $\dot{x} = Ax$, $x \in V$, is linear and isomorphic to V . A basis of this space is called a *fundamental system of solutions*. The transition from the equation

$$\dot{x} = Ax, \quad x \in V,$$

to the equation

$$\dot{z} = {}^{\mathbb{C}}Az, \quad z \in {}^{\mathbb{C}}V, \tag{4}$$

is called the *complexification of the equation* (the time remains real).

The complexified equation (4) in $z = x + iy$ is equivalent to the system $\dot{x} = Ax$, $\dot{y} = Ay$ ($x, y \in V$).

A linear autonomous equation on a complex phase space, such as equation (4), for example, is easily solved by passing to Jordan normal form.

Let $\xi_0, \dots, \xi_j$ be a Jordan chain of the map ${}^{\mathbb{C}}A$ with eigenvalue λ , i.e., a sequence of vectors such that ${}^{\mathbb{C}}A - \lambda E$ (where λ is a complex number) transforms each vector of the sequence into the next and the last into zero:

$$\xi_0 \mapsto \xi_1 \mapsto \dots \mapsto \xi_j \mapsto 0.$$

The vector ξ_j is called an *eigenvector*, and the other vectors of the chain the *associated vectors*, of the eigenvalue λ .

The vector-function

$$z = e^{\lambda t} \left(\xi_0 + \frac{t}{1!} \xi_1 + \dots + \frac{t^j}{j!} \xi_j \right)$$

is the solution of the equation $\dot{z} = {}^{\mathbb{C}}Az$ with the initial condition ξ_0 .

Every complex linear operator has a basis of Jordan chains. Therefore, a fundamental system of solutions can be made up of solutions of the above type.

3.4. Saddles, Nodes, Foci, Centers. A nondegenerate singular point of a linear vector field on the real plane is of one of the following four types: *saddle, node, focus* or *center* (Fig. 2).

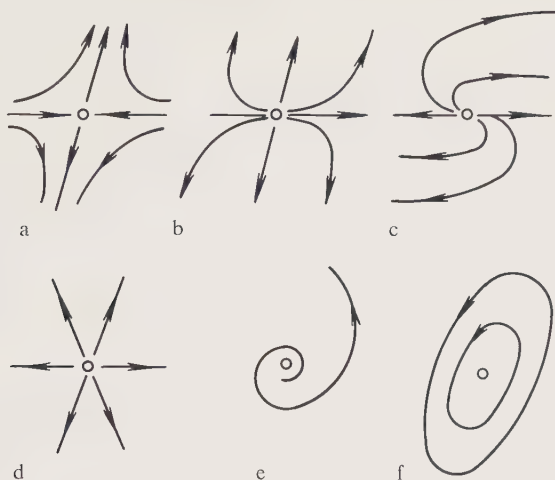


Fig. 2. a) saddle, b) node, c) Jordan node, d) bicritical node, e) focus, f) center

Let λ_1 and λ_2 be the eigenvalues of the linear map A . By the nondegeneracy assumption, they are both nonzero. If $\lambda_1 \lambda_2 < 0$, then the singular point of the equation $\dot{x} = Ax$ is a saddle point (Fig. 2a); if λ_1 and λ_2 are real and have the same sign, it is a node (Fig. 2b, c, d); if λ_1 and λ_2 are complex and not purely imaginary, it is a focus (Fig. 2e); and if λ_1 and λ_2 are purely imaginary, it is a center (Fig. 2f). Among the nodes, we distinguish *Jordan nodes* (traditionally called “degenerate nodes”) in which the matrix A is equivalent to a Jordan block of order two (Fig. 2c), and *bicritical nodes* in which the matrix A is scalar (Fig. 2d). Figs. 2b–2e are drawn for the case where $\operatorname{Re} \lambda_j > 0$.

All these pictures, except the last, remain unchanged under small perturbations, as the Grobman-Hartman and Hadamard-Perron theorems (which are formulated in Chap. 3) prove.

3.5. The Liouville-Ostrogradsky Formula. Suppose an operator-valued function of one variable satisfies the equation

$$\dot{X} = A(t)X, \quad t \in I \subset \mathbb{R}.$$

Then the determinant $W = \det X$ (called the *Wronskian*) satisfies the equation

$$\dot{W} = \operatorname{tr} A(t)W, \quad (5)$$

where $\operatorname{tr} A$ is the *trace* of the operator A (the sum of the eigenvalues).

From (5) we get a formula for W :

$$W(t) = W(t_0) \exp \left\{ \int_{t_0}^t \operatorname{tr} A(\tau) d\tau \right\}.$$

Corollary. Let $V(t)$ denote the Euclidean volume of the image $g^t M$ of a domain M under the phase flow of a vector field v (defined in a domain of Euclidean space). Then

$$\frac{dV}{dt} = \int_{g^t M} \operatorname{div} v \, dx,$$

where dx is the Euclidean volume-element.

From formula (5) and the equation of variation with respect to the initial condition (section 2.7), we obtain a formula for the distortion of the phase

volume: the Jacobian $W(t, x) = \det \left(\frac{\partial g^t}{\partial x} \right) \Big|_{(t, x)}$ is given by

$$W(t, x) = \exp \left\{ \int_0^t (\operatorname{div} v) \circ (g^\tau x) \, d\tau \right\}.$$

In particular, if $\operatorname{div} v \equiv 0$, then the phase flow of the equation $\dot{x} = v(x)$ preserves volume. If $\operatorname{div} v < 0$ everywhere in the phase space, then the phase flow decreases volume as time increases. In this case the equation is said to be *dissipative*.

3.6. Higher Order Linear Equations. Consider the differential equation

$$x^{(n)} + a_1(t) x^{(n-1)} + \dots + a_n(t) x = 0. \quad (6)$$

A maximum interval on which all the coefficients a_j are continuous is called an *interval of continuity of the coefficients* of equation (6). It is not uniquely determined; for example, if $a_i(t) = t^{-1}$, then there are two intervals of continuity of the coefficients: namely, $t > 0$ and $t < 0$.

Theorem. Any solutions of equation (6) at a point in an interval of continuity of the coefficients extends to a solution on the whole interval.

The solutions of equation (6) which are defined on a fixed interval of continuity of the coefficients form an n -dimensional linear space. A basis of this space is called a *fundamental system of solutions*.

Let $\varphi_1, \dots, \varphi_n$ be a system of solutions of equation (6); the function

$$W(t) = \begin{vmatrix} \varphi_1 & \dots & \varphi_n \\ \dot{\varphi}_1 & \dots & \dot{\varphi}_n \\ \vdots & \dots & \vdots \\ \varphi_1^{(n-1)} & \dots & \varphi_n^{(n-1)} \end{vmatrix}$$

is called the *Wronskian* of the system.

The Liouville-Ostrogradsky formula is

$$W(t) = W(0) \exp \left\{ - \int_0^t a_1(\tau) \, d\tau \right\}.$$

§ 4. Stability

Here we set forth the simplest results in stability theory. For a more detailed discussion of the various types of stability, see [86], [26], [74].

4.1. Lyapunov Stability and Asymptotic Stability

Definition. A stationary solution of an autonomous differential equation (i.e., a solution whose image is an equilibrium point) is said to be *Lyapunov stable* if all solutions of the equation, with initial conditions in a sufficiently small neighbourhood of the equilibrium point, are defined on the whole positive time semi-axis and converge uniformly with respect to time to the stationary solution as the initial conditions tend to the equilibrium point.

Definition. A stationary solution is said to be *asymptotically stable* if it is Lyapunov stable and if, in addition, all solutions with initial conditions sufficiently close to the equilibrium point under consideration tend to this equilibrium point as $t \rightarrow +\infty$.

Remark 1. This definition of stability requires that one specify a metric on the phase space in order to define uniform convergence. However, neither stability nor asymptotic stability of an equilibrium point depends on the choice of metric.

Remark 2. Stability (and asymptotic stability) of a stationary solution is a local property of the vector field (giving the differential equation) at the corresponding equilibrium point; it is neither lost nor acquired if the field is changed outside a neighbourhood of the equilibrium point.

Remark 3. The convergence to an equilibrium point as $t \rightarrow \infty$ of all solutions beginning near that equilibrium point is not sufficient for asymptotic stability (as Fig. 3 shows) and is not a local property.

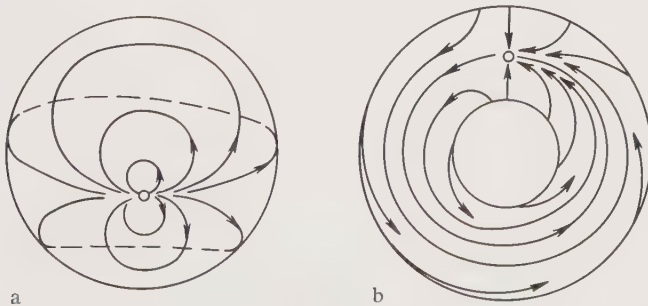


Fig. 3. An unstable singular point, to which all the phase curves which start at nearby points converge

Remark 4. Lyapunov stability of any solution of any differential equation (autonomous or not) is defined in a similar manner. That is, the solutions on the semi-axis $t \geq 0$ are required to converge uniformly to the solution in question as their initial values at $t=0$ tend to the initial value of the solution in question. Here, uniform convergence is defined by means of a metric on the phase (or extended phase) space (or manifold). In contrast to the case of an equilibrium point of an autonomous system, this definition of stability does depend on the choice of metric. For example, a stable solution of an autonomous equation on a euclidean phase space can become unstable after a diffeomorphism of the phase space. Thus, the stability of a motion depends on the coordinates in which the motion is described.

A similar remark pertains to the concept of asymptotic stability.

Stability of a periodic solution of an autonomous equation (like stability of a stationary solution) is a geometric concept which does not depend on the choice of coordinates or metric on the phase space. (In general, this independence always attains when the closure of the phase curve is compact.)

4.2. Lyapunov's Theorem on Stability by Linearization

Theorem. *If all eigenvalues of the linear part of a vector field v at a singular point have negative real part, then the singular point is asymptotically stable. If any eigenvalue has a positive real part, the singular point is not Lyapunov stable.*

4.3. Lyapunov and Chetaev Functions

Definition. A differentiable function f is called a *Lyapunov function* for a singular point x_0 of a vector field v if it satisfies the following conditions:

the function f is defined on a neighbourhood of x_0 and has a strict local minimum at this point;

on a neighbourhood of the point x_0 , the derivative of f along v is nonpositive: $L_v f \leq 0$. (The derivative of a function along a vector field is the rate of change of the function along the phase curves of the field: $L_v f(x) = \frac{d}{dt} f \circ g^t x|_{t=0}$.)

Theorem [87]. *A singular point of a differentiable vector field for which a Lyapunov function exists is stable.*

Definition. A differentiable function f is called a *Chetaev function* for a singular point x_0 of a vector field v if it satisfies the following conditions:

the function f is defined on a domain W whose boundary contains x_0 ; the part of the boundary of W strictly contained in a sufficiently small ball with its center x_0 removed is a piecewise-smooth, C^1 hypersurface along which v points into the interior of the domain (Fig. 4);

$f(x) \rightarrow 0$ as $x \rightarrow x_0$, $x \in W$; $f > 0$ and $L_v f > 0$ everywhere in W .

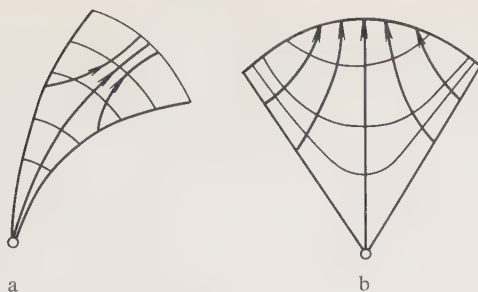


Fig. 4. Domain of definition of the Chetaev function. The thin curves represent the level surfaces of the function, the thick curves the boundary of the domain and the phase curves of the field.

Theorem [26]. *A singular point of a C^1 vector field for which a Chetaev function exists is unstable.*

4.4. Generic Singular Points. Generically, the linear part A of a vector field at a singular point has no eigenvalues on the imaginary axis. In this case Lyapunov's theorem on stability by linearization is applicable. The Lyapunov (or Chetaev) function can be taken to be a quadratic form. Suppose the complexification of the linear part has a basis of eigenvectors. If all the eigenvalues lie in the left half-plane, the Lyapunov function can be taken to be the sum of the squares of the moduli of the coordinates with respect to the eigenbasis (restricted to the real subspace).

Remark. If there is no eigenbasis, then there is an "almost-eigenbasis" for any $\varepsilon > 0$ which is such that the matrix of the linear part is upper-triangular and the moduli of the off-diagonal elements are less than ε . For sufficiently small ε the sum of the squares of the moduli of the coordinates in this basis defines a Lyapunov function. For generic singular points an eigenbasis exists.

If A has at least one eigenvalue in the right half-plane, then we can use the same method to define a Chetaev function on a suitable cone W with vertex at 0 in $\mathbb{R}^n$ (Fig. 4b).

Remark. A stationary solution of the nonautonomous equation

$$\dot{x} = Ax + f(t, x), \quad x \in \mathbb{R}^n,$$

in which all eigenvalues of A lie strictly in the left half-plane is asymptotically stable if

$$|f(t, x)| \leq C|x|^2 \quad \text{for all } t \geq 0 \text{ and } |x| \leq a.$$

Moreover, the solutions converge exponentially to zero:

$$|\varphi(t)| \leq K(\exp(-\alpha t))|\varphi(0)| \quad \text{when } t \geq 0$$

for any $\alpha > 0$ such that $\operatorname{Re} \lambda_j < -\alpha$.

§ 5. Cycles

In physical systems whose law of evolution does not change in time, periodic régimes may be established. A mathematical description of this phenomenon is given by the theory of cycles (closed phase curves) developed by H. Poincaré.

5.1. The Structure of the Phase Curves of Real Differential Equations

Theorem. *A phase curve of a real differential equation $\dot{x} = v(x)$ with smooth right-hand side is either a single point or is diffeomorphic to a circle or a straight line.*

Thus, a phase curve of an equation with real time always has a simple intrinsic geometry; only its disposition in the phase space can be complicated.

A phase curve which is diffeomorphic to a circle is called a *cycle*.

5.2. The Monodromy Transformation of a Closed Phase Curve. Limit Cycles. Suppose that a smooth vector field has a closed phase curve (a cycle). Choose a point on the curve and draw a transversal to the cycle (a smooth transverse hypersurface of dimension $n-1$ if the dimension of the phase space is n). Phase curves which start at points of the transversal sufficiently close to the original point of the cycle return to the transversal. Associate to a point B of the transversal (sufficiently close to the original point A on the cycle) the first point C at which the phase curve starting from B returns to the transversal. The germ at A of this map of the transversal to itself is called the *monodromy transformation* (or the *first-return map*); see Fig. 5.

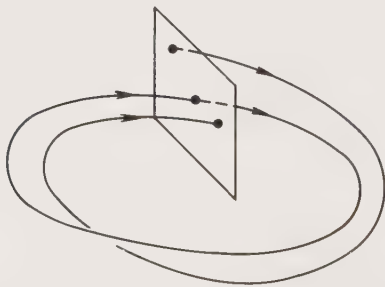


Fig. 5. Monodromy transformation of a cycle

The point on the cycle is a fixed point of the monodromy transformation. The monodromy transformation does not depend on the choice of transversal or even on the choice of the original point, in the sense that the monodromy transformations corresponding to different transversals are conjugate (one is carried into another by the germ of a diffeomorphism of the transversals).

Definition. A *limit cycle* of an autonomous differential equation on the plane is an isolated phase curve (diffeomorphic to a circle) of the equation. In other words, a closed phase curve is called a limit cycle if the fixed point of the corresponding monodromy transformation is isolated.

A neighbourhood of a limit cycle in the plane consists of spirals winding round the cycle as $t \rightarrow +\infty$ or $t \rightarrow -\infty$; the asymptotic behaviour of all phase curves nearby and to one side of the cycle is the same (Fig. 6). In Figs. 6a and 6b generic limit cycles are drawn; Fig. 6c is destroyed by a small perturbation (Fig. 6d).

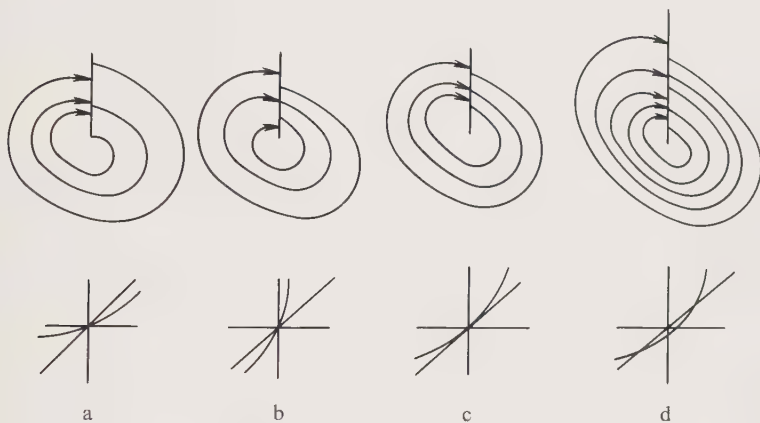


Fig. 6. The graphs of the corresponding monodromy transformations are sketched beneath the limit cycles: a) a stable cycle, b) an unstable cycle, c) a semistable cycle, and d) two cycles obtained by a small perturbation of c).

5.3. The Multiplicity of a Cycle

Definition. The *multiplicity of a cycle* is the local multiplicity of the fixed point of the corresponding monodromy transformation.

If the phase space is 2-dimensional, then the transversal is 1-dimensional, and the monodromy transformation is function of a single variable, $x \mapsto \Delta(x)$, $\Delta(0)=0$. In this case, the multiplicity is the order of the zero of the function $\Delta(x)-x$ at the point 0.

The *multiplicity at the fixed point 0 of a map $x \mapsto \Delta(x)$ in the space $\mathbb{R}^n$ with coordinates $(x_1, \dots, x_n)$* is defined to be

$$\dim_{\mathbb{R}} \mathbb{R}[[x_1, \dots, x_n]] / (\Delta_1(x) - x_1, \dots, \Delta_n(x) - x_n),$$

where the expression in the numerator denotes the algebra of formal power series, and that in the denominator the ideal generated by the Taylor series of the components of the vector $\Delta(x)-x$. For further details on multiplicity, see, for example [9].

Similarly, the *multiplicity of a singular point* 0 of a vector field v is defined to be the multiplicity of the fixed point 0 of the mapping $x \mapsto v(x)$ (see [9, p. 212]).

5.4. Multipliers

Definition. *Multipliers of a cycle* are defined to be the eigenvalues of the linear part of the monodromy transformation at the fixed point corresponding to the cycle.

(Note that multipliers of a cycle are often defined as the eigenvalues of the monodromy operator of the variational equation; this definition differs from that adopted here by the presence of the multiplier one, corresponding to the eigenvector tangent to the cycle. See Chap. 6, § 3.1.)

A cycle is said to be *nondegenerate* if all its multipliers are different from 1. (A nondegenerate cycle is sometimes said to be *robust*.)

A nondegenerate cycle persists under small perturbations. The perturbed differential equation has a nondegenerate cycle close to the original one. Cycles of generic vector fields are nondegenerate.

For equation (1) in the plane the multiplier of a cycle is given by the formula

$$\lambda = \exp \left\{ \int_{\gamma} (\operatorname{div} v) dt \right\},$$

where the cycle γ is parametrized by time t . For a cycle in $\mathbb{R}^n$ the right hand side is equal to the product of all the multipliers.

Definition. A cycle is said to be *orbitally (Lyapunov) stable* if, for an arbitrarily small neighbourhood U of the cycle, all positive semi-trajectories which start in a sufficiently small neighbourhood of the cycle do not emerge from U .

Definition. A cycle is said to be *orbitally asymptotically stable* if it is orbitally Lyapunov stable and if all the phase curves with initial condition sufficiently close to the cycle approach the cycle asymptotically as $t \rightarrow +\infty$.

Remark. A non-constant periodic solution certainly cannot be asymptotically stable because solutions with initial conditions at different points of the cycle do not approach one another as $t \rightarrow \infty$.

If the modulus of every multiplier of a cycle is less than one, then the cycle is orbitally asymptotically stable. The stability follows from the fact that, when all $|\lambda_j| < 1$, the monodromy map is a contraction mapping for a suitable choice of metric on the transversal. This metric is constructed just like the Lyapunov function near a singular point which could be shown to be asymptotically stable by linearization. Because the monodromy map is a contraction, orbital asymptotic stability follows: phase curves near the cycle spiral ever closer to the cycle. It can also be shown that the phase of the motion around the cycle tends to the phase of motion of one of the points along the cycle. Hence, not only do nearby phase curves become

uniformly close to the cycle, but any solution with initial condition close to the cycle becomes uniformly close (on the semi-axis $t \geq 0$) to one of the solutions describing the motion along the cycle.

5.5. Limit Sets and the Poincaré-Bendixson Theorem. The disposition of phase curves in the real plane is fundamentally simpler than in a higher dimensional space (in the theory of differential equations, “higher” means “three or higher”). This difference is essentially due to the fact that a curve locally separates the plane but does not separate space.

Definitions. 1. A *positive semi-trajectory* of an autonomous equation is defined to be the part of a phase curve corresponding to a positively oriented ray along the time axis:

$$\varphi_+ = \{\varphi(t) | t \in [t_0, +\infty)\}, \quad \text{where } \varphi \text{ is a solution of the equation.}$$

2. A *negative semi-trajectory* is defined similarly.

3. The ω -limit set (resp., α -limit set) of a phase curve is defined to be the intersection of the closures of all its positive (resp., negative) semitrajectories. In other words, the ω -limit set Ω consists of the points past which or through which a positive semi-trajectory passes a countable number of times:

$x \in \Omega$ if and only if there is a sequence $\{t_n\}$ such that $t_n \rightarrow \infty$,

$$\varphi(t_n) \rightarrow x \text{ as } n \rightarrow \infty.$$

The choice of notation is motivated by the fact that α is the first, and ω the last, letter of the Greek alphabet.

The Poincaré-Bendixson Theorem. Suppose a positive semitrajectory of a C^1 vector field on the real plane with isolated singular points lies in a bounded domain. Then the ω -limit set of the phase curve must be one of the following types: 1) a singular point, 2) a cycle (a closed phase curve), 3) a union of singular points and phase curves, each of which tends as $t \rightarrow +\infty$ to a singular point and as $t \rightarrow -\infty$ to another, possibly different, singular point of the union (see Fig. 7).

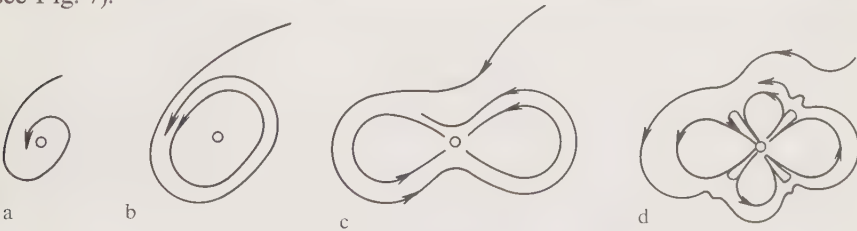


Fig. 7. ω -limit sets

In the last case, the ω -limit set may be pathological. Even an infinitely differentiable vector field with a finite number of singular points can have limit sets with an infinite number of phase curves entering and leaving one singular point (Fig. 7d). Purely topological considerations connected with

the existence and uniqueness theorem do not prohibit this phenomenon. However, this situation is impossible for analytic vector fields and for all smooth fields which do not belong to an exceptional set of infinite codimension. This follows from the theorem on resolution of singularities (Chap. 5). The Poincaré-Bendixson theorem is also true for equations on a sphere, but fails for equations on surfaces of higher genus (see Chap. 2): a closed curve separates a sphere, but need not separate a surface with handles.

§ 6. Systems with Symmetries

In this section, we set out some general considerations which enable one to reduce the order of a differential equation and sometimes allow one to integrate the equation.

6.1. The Group of Symmetries of a Differential Equation

Definition. A *symmetry of a vector field (and the corresponding autonomous differential equation)* is a diffeomorphism of the phase space onto itself which carries the field onto itself. In that case the field is said to be *invariant under the diffeomorphism*.

Definition. A *symmetry of a direction field or of the corresponding nonautonomous equation* is a diffeomorphism of the extended phase space onto itself which carries the field into itself. In that case, the direction field is said to be *invariant under the diffeomorphism*.

The set of all symmetries of a differential equation is a group under the operation of composition. It is called the *symmetry group* of the equation.

Example. The symmetry group of the autonomous equation $\dot{x} = x$, $x \in \mathbb{R}^n$, is the general linear group $GL(n, \mathbb{R})$.

6.2. Quotient Systems

Definition. Let f be a smooth mapping of a domain in a real, linear space on to a domain in a lower dimensional space. A vector field v on the pre-image of this mapping is said to be *f-lowerable* if there is a field $\tilde{v}$ on the image domain which is the image of the original field under the map induced by f . The field $\tilde{v} = f_* v$ is defined by the equation

$$\tilde{v}(y) = f_*(x) v(x), \quad \text{where } y = f(x);$$

the condition that the vector field be *f-lowerable* is that the vector $\tilde{v}(y)$ not depend on the choice of the pre-image of the point y . The system $\dot{y} = \tilde{v}(y)$ is called a *quotient system* of the system $\dot{x} = v(x)$. A vector field which has a one-parameter group of symmetries is often lowerable.

Example. An equation on $\mathbb{R}^2$ which is invariant under rotations can be written, using complex co-ordinates $z = x + iy$, in the form

$$\dot{z} = z g(\rho), \quad (7)$$

where $\rho = z\bar{z}$, and $g: \mathbb{R}^+ \rightarrow \mathbb{C}$ is a complex-valued function on the positive semi-axis. The vector field of this system is f -lowerable where $f: z \mapsto \rho$. The corresponding quotient system is

$$\dot{\rho} = 2\rho \operatorname{Re} g(\rho).$$

Equation (7) with $\operatorname{Re} g(0) = 0$ (a center with respect to linear terms) is used in the theory of stability (Chap. 3, § 5).

We point out below some simple applications of symmetry to the study and integration of equations.

6.3. Homogeneous Equations

Definition. A direction field is said to be *homogeneous* if it is invariant under all dilatations

$$g^\lambda(x) = e^\lambda x, \quad \lambda \in \mathbb{R}$$

(the field must be defined in a cone invariant under dilatations, for example, in $\mathbb{R}^n \setminus \{0\}$). The corresponding differential equation is also said to be *homogeneous*.

Let G denote the group of dilatations $\{g^\lambda | \lambda \in \mathbb{R}\}$ of a domain $U = \mathbb{R}^n \setminus \{0\}$. The quotient space U/G is the sphere S^{n-1} . To each homogeneous equation in $\mathbb{R}^n \setminus \{0\}$ we can associate a direction field on S^{n-1} ; namely, the image of the field in U under the projection $U \rightarrow S^{n-1}$.

Any direction field whatever (possibly with singularities) on S^{n-1} arises from a homogeneous equation in $\mathbb{R}^n \setminus \{0\}$. So all the difficulties which arise in a “global” theory of differential equations on S^{n-1} also appear in the local theory of differential equations in $\mathbb{R}^n$.

6.4. Use of Symmetries to Reduce the Order. A knowledge of a one-parameter group of symmetries of a vector field enables one to reduce the dimension of the phase space by one. To do this, consider the manifold of orbits of the one-parameter group of symmetries. Locally (in the vicinity of a point which is not fixed by the group) these orbits are curves; the original phase space is foliated by the orbits; by the rectification theorem for orbits, they become parallel straight lines after a suitable diffeomorphism. The dimension of the space of orbits is one less than the dimension of the original phase space.

The vector field in the original phase space projects onto a vector field on the space of orbits. The differential equation corresponding to this field is called the *quotient system*. If the quotient system can be integrated, then so can the equation given by the original vector field.

In particular, if the original phase space is two-dimensional, then the phase space of the quotient system is one-dimensional. Therefore an autonomous differential equation with a two-dimensional phase space, for which a one-parameter group of symmetries is known, can be integrated explicitly in quadratures. All elementary integration methods for differential equations of particular types (separable, linear – homogeneous and inhomogeneous, quasihomogeneous, etc.) are based on the fact that there are obvious groups of symmetries in these cases.

Example. The group of quasihomogeneous dilatations of a linear space with co-ordinates $x_1, \dots, x_n$ and weights $\alpha_1, \dots, \alpha_n$ is the one-parameter group of linear transformations

$$g^s: (x_1, \dots, x_n) \mapsto (e^{\alpha_1 s} x_1, \dots, e^{\alpha_n s} x_n).$$

A function on a linear space is said to be *quasihomogeneous of degree r* if it is an eigenvector with the eigenvalue e^{rs} of the action of the quasihomogeneous dilatation on the function space, i.e., if $f(g^s x) \equiv e^{rs} f(x)$.

For example, a polynomial $\sum a_m x^m$ is quasihomogeneous of degree r if $(m, \alpha) = r$ for all exponents m of monomials which appear with non-zero coefficients.

In order that f be a quasihomogeneous function with positive weights α_i it is necessary and sufficient that Euler's identity hold:

$$\sum \alpha_i x_i \frac{\partial f}{\partial x_i} = r \cdot f.$$

This signifies that f is an eigenvector for the operation of differentiation along the direction of the Euler vector field (which is the vector field whose phase flow is the group of quasihomogeneous dilatations).

A vector field is said to be a *quasihomogeneous vector field of degree r* if each quasihomogeneous dilatation of the group multiplies it by e^{sr} . The vector field $v = \sum v_k(x) \frac{\partial}{\partial x_k}$ is quasihomogeneous of degree r if and only if its components are quasihomogeneous functions whose degrees differ from the degree of the corresponding coordinates by r :

$$\deg v_k = \alpha_k + r.$$

Consider the system of differential equations $\dot{x} = a(x, y)$, $\dot{y} = b(x, y)$ defined by a quasihomogeneous vector field of degree 0 on the plane with coordinates of weight $\deg x = \alpha$, $\deg y = \beta$. The orbits of the group have the form $x = x_0 e^{\alpha s}$, $y = y_0 e^{\beta s}$. The coordinate on the quotient space (indexing the orbit) can be taken (when $x > 0, y > 0$) to be $u = y^\alpha / x^\beta$. The quotient system has the form $\dot{u} = f(u)$. The integration of the system is completed by passing to the co-ordinates (x, u) on the phase space.

§ 7. Implicit Differential Equations

This section was written in collaboration with A.A. Davydov. It uses basic concepts from the theory of singularities and from differential geometry in $\mathbb{R}^3$. We formulate the main results about singularities of solutions of first order implicit differential equations.

7.1. Basic Definitions; the Criminant, Integral Curves. By a first order implicit differential equation (*i.e.*, an equation not explicitly solved for the derivative) we mean an equation of the form

$$F(x, y, p) = 0, \quad \text{where } p = \frac{dy}{dx}.$$

We call (x, y, p) -space the *space of 1-jets of functions* $y(x)$. For a generic smooth function F , the equation above defines a smooth surface in this space called the *surface of the equation*. Take the direction of the p -axis to be vertical. Vertical projection of the surface of the equation onto the x, y -plane is called a *folding*. The critical points of the folding are called the *singular points of the equation*.

In a neighbourhood of a nonsingular point, the equation $F=0$ can be solved for p ; *i.e.*, it can be reduced to the usual equation $\frac{dy}{dx} = v(x, y)$ by the implicit function theorem.

The set of singular points of the equation $F=0$ in (x, y, p) -space is called the *criminant* of the equation. For a generic function F the criminant is a smooth curve. The projection of the criminant onto the (x, y) -plane is called the *discriminant curve*. For a generic function, the only singularities of the discriminant curve are semi-cubical cusps and points of transversal self-intersection. Above a cusp of the discriminant curve there is a *Whitney pleat* (normal form $(u, v) \mapsto (u^3 + uv, v)$); above the other points of the discriminant curve there are on the criminant *fold points* (normal form $(u, v) \mapsto (u^2, v)$).

At every point (x, y, p) of the space of 1-jets there is a *contact plane* $dy = p dx$. The *direction field of an equation* is the field cut out by the field of contact planes on the surface of the equation. The integral curves of this field are called the *integral curves of the equation*.

7.2. Regular Singular Points. A singular point of a generic equation is said to be *regular* if the criminant is not tangent to the contact plane at this point.

Two equations are said to be *equivalent* if one can be transformed into the other by a diffeomorphism of the (x, y) -plane.

Cibrario's Theorem (1932) (see the proof in [7, § 4G]). *In a neighbourhood of a regular singular point, the equation $F(x, y, p) = 0$ is equivalent to $p^2 = x$. This equivalence is smooth for smooth equations, and analytic for analytic equations.*

In Fig. 8a the kernel of the derivative of the projection is vertical and the discriminant is horizontal. The projections of the parts of the integral curves from one sheet of the covering are shown by continuous curves, those from the other sheet by dashed curves.

7.3. Folded Saddles, Nodes, and Foci. A contact plane may be tangent to the surface of the equation. For a generic equation, tangency occurs at isolated points. These points necessarily lie on the discriminant. In a neighbourhood of a point of tangency, the direction field is generated by a smooth vector field on the surface of the equation which vanishes at the point of tangency.

For a generic equation:

1) a singular point of the generating vector field is a nondegenerate saddle, node, or focus; and the moduli of the two eigenvalues of nodes or saddles are different.

2) the eigenvectors of the linearization of the field at the singular point are not tangent to either the discriminant or the kernel of the folding.

The singular points which satisfy both these conditions at a fold point are called *folded saddles*, *nodes*, and *foci* respectively. Their integral curves and their projections on the x, y -plane are shown in Figs. 8b, c, d.

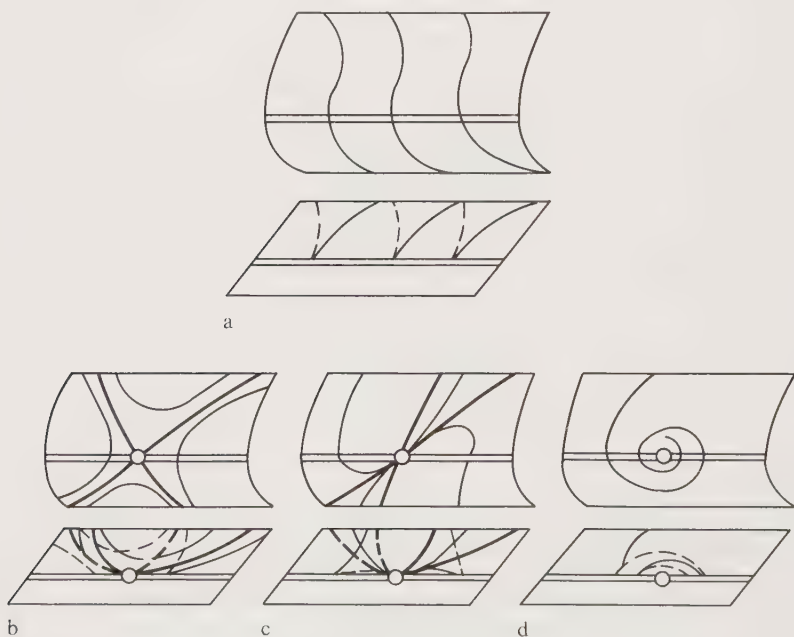


Fig. 8. Singular points of an implicit equation: a) a regular singular point, b) a folded saddle, c) a folded node, d) a folded focus.

7.4. Normal Forms of Folded Singular Points. A diffeomorphism whose square (under composition) is the identity transformation is called an *involution*. Fix a field on the plane with a singular point at the origin. An involution of the plane is said to be *admissible* for this vector field if the fixed points of the involution form a curve going through the origin, if the involution carries the field on this curve into the opposite field, and if neither the invariant nor anti-invariant eigenvector of the linearization of the involution at the origin are eigenvectors of the linear part of the field at the origin.

Theorem (A.A. Davydov, 1984). *Fix a vector field with a singular point at the origin which is a focus or a saddle or a node with eigenvalues of unequal moduli. Then all admissible involutions whose fixed point curves are not separated by the eigenvectors of the linear part of the field at the origin can be locally carried one into the other by diffeomorphisms of the plane which move each point along the phase curve of the field which passes through that point.*

This theorem and its corollaries are true in both the smooth and analytic case.

Corollary 1. *Folded saddles (nodes, foci) are equivalent if the corresponding (non-folded) singular points of the vector fields generating the direction fields on the surface of the equation are orbitally equivalent (i.e., if they have diffeomorphic phase portraits).*

(Note: for precise definitions of orbital equivalence, see the introduction to Chap. 3).

Corollary 2. *In a neighbourhood of a folded saddle (node or focus) a generic implicit equation is equivalent to the normal form $(p + kx)^2 = y$.*

When normalized by the condition $\lambda_1 + \lambda_2 = 2$, the eigenvalues λ_1, λ_2 of the linear part of the vector field which generates the direction field of the equation are equal to $1 \pm \sqrt{1 - 8k}$. (Note that the eigenvalues are defined up to multiplication by the same non-zero constant, because the vector field generating the direction field is defined up to multiplication by a function.) Folded saddles ($k < 0$) are all topologically equivalent to one another: a homeomorphism of the (x, y) -plane transforms the projections of the families of their integral curves into one another. In exactly the same way, all the folded nodes ($0 < k < 1/8$) are topologically equivalent, as are all folded foci ($k > 1/8$).

Example. The net of asymptotic curves on a generic surface in three-dimensional space has regular singularities at general points of a parabolic curve and folded singularities at isolated points of the curve.

7.5. Whitney Pleats. Apart from folded singular points, a generic implicit equation can have only one other type of non-regular singular point, namely, a *Whitney pleat of the folding*. At such a point, the direction field on the

surface of the equation is not singular, but is tangential to the criminant. The criminant is not singular, but is tangential to the kernel of the projection, and the discriminant curve has a cusp.

There are infinitely many topologically different singularities of this type, but all are essentially one or other of only two kinds: a) and b) in Fig. 9.

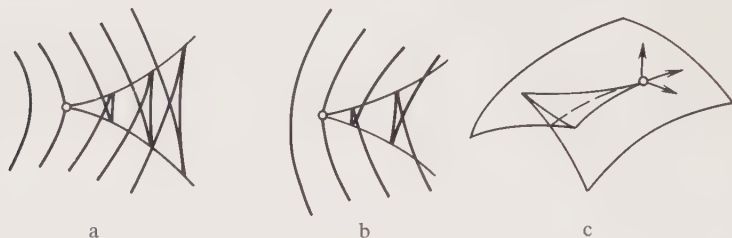


Fig. 9. A singular point of the pleat type

The plane projections (that is, the image of the projection onto the plane) of the integral curves on the surface near a pleat point, as sketched in Fig. 9, can be described as follows (J. Bruce, 1983).

Consider the swallowtail surface (Fig. 9c), i.e., the surface in (a, b, c) -space corresponding to the polynomials $z^4 + az^2 + bz + c$ which have multiple roots. The planes $\{a = \text{const}\}$ cut out curves on the swallowtail surface. In order to obtain a family of plane projections of integral curves of a generic equation close to a pleat point it is sufficient to project these curves to the plane by means of a generic rank two map of (a, b, c) -space into the (x, y) -plane.

In particular, the projection of each individual integral curve is diffeomorphic to the corresponding plane section of the swallowtail surface. For example, the curve passing through the cusp has a singularity of order $4/3$ and is locally diffeomorphic to the curve $u^4 = v^3$.

§ 8. Attractors

In this survey the word “attractor” means “an attracting set”. In this section we mostly discuss upper bounds for the dimension of an attractor.

Suppose that, as $t \rightarrow \infty$, the phase curves of an evolutionary process get closer and closer to a compact set M , the attractor. An observer, equipped with an instrument of finite accuracy, who follows the evolution of a state along a fixed phase curve will, after a certain time, cease to be able to distinguish a point of the phase curve from a point of the attractor. Thus an attractor contains a natural “phase space of the terminal régimes”. In a number of problems in mathematical physics, the phase space is infinite-dimensional, but the attractor is finite-dimensional (see section 8.3).

8.1. Definitions. A domain of the phase space is said to be *absorbing* if it completely contains every positive semi-trajectory which starts in it. It is *globally absorbing* if, in addition, every phase curve falls into it after a finite (non-negative) time.

Definition. Let g^t be the transformation after time t of the phase flow defined by the differential equation

$$\dot{x} = v(x).$$

If B is an absorbing domain with compact closure $\bar{B}$, then the set

$$M = \bigcap_{t > 0} g^t \bar{B}$$

is called an *attractor* of equation (1). (Note: other definitions of an “attractor” which are to be found in the literature are not necessarily equivalent to this one.)

If the domain B is globally absorbing, then the attractor is said to be *maximal*.

Remark. An attractor is always invariant: $g^t M = M$.

Example. Asymptotically stable singular points and orbitally asymptotically stable cycles are attractors.

Definition. An attractor which is not the finite union of submanifolds of phase space is called a *strange attractor* (a term introduced in [96], where it meant an attractor different from a point or a cycle).

Remark. The phase space of stationary régimes may be smaller than the maximal attractor. For example, Fig. 3b shows the phase portrait of a system for which the maximal attractor is a circle, but all the solutions tend to the singular point 0. It seems that an adequate mathematical definition of physically observable attractors is given by the concept, due to Ya. G. Sinai, of a “stochastic attractor” [105]. A “stochastic attractor” is not necessarily a strange attractor.

8.2. An Upper Bound for the Dimension of the Maximal Attractor. An attractor need not be a manifold. The *d-dimensional volume* of a finite covering of a compact set in a metric space by balls is defined to be the sum of the d th powers of the radii of the balls.

Definition. The *Hausdorff dimension* of a compact set is the lower bound of those d for which the compact set admits finite coverings by balls having an arbitrarily small d -dimensional volume.

Examples. 1. The Hausdorff dimension of a smooth submanifold of Euclidean space is equal to its usual dimension.

2. The Hausdorff dimension of the standard Cantor perfect set is equal to $\log_3 2$.

Remark. The Hausdorff dimension of the Cartesian product of two compact sets may be greater than the sum of the dimensions of the factors [58: V. I. Bakhtin [1]].

Definition. A mapping of a domain of Euclidean space into itself is said to be *k-compressive* if it reduces *k*-dimensional volumes; more precisely, if its derivative reduces the volume of any *k*-dimensional parallelepiped in the tangent space and if the ratio of the volumes of the image and pre-image does not exceed some constant less than 1 which is independent of the point and the parallelepiped.

Theorem [30], [58]. *The Hausdorff dimension of a compact invariant set of a k-compressive diffeomorphism of a domain of Euclidean space into itself does not exceed k.*

Suppose system (1) has a globally absorbing domain *B* with a compact closure, and that the phase space is Euclidean.

Definition. System (1) is said to be *k-compressive* if the transformation of the phase flow of the system after a positive time is *k-compressive* in the domain *B*.

Theorem. *The Hausdorff dimension of an attractor of a k-compressive system does not exceed k.*

We formulate a sufficient condition for system (1) to be *k-compressive*. At each point *x* of the domain *B* we define a quadratic form

$$F(x): \xi \mapsto (v_*(x) \xi, \xi), \quad \xi \in T_x B.$$

Let $\lambda_1(x) \geq \dots \geq \lambda_n(x)$ be the eigenvalues of this quadratic form.

Lemma. *A sufficient condition for system (1) to be k-compressive is that the inequality*

$$\lambda_1(x) + \dots + \lambda_k(x) < 0$$

shall hold everywhere in the compact $\bar{B}$.

8.3. Applications. Infinite-dimensional analogues of the theorems in the preceding section have been proved (see [11] and the references therein); they enable one to prove that attractors of a number of evolution equations of mathematical physics are finite-dimensional. For example, the Hausdorff dimension of an attractor for the 2-dimensional Navier-Stokes equation with doubly periodic boundary conditions does not exceed $C \Re^2 \ln \Re$, where $\Re$ is the Reynolds number (a quantity inverse to the dimensionless viscosity) and the constant *C* depends on the lattice of the periods [11], [58], [72].

Chapter 2

Differential Equations on Surfaces

This chapter deals with differential equations on a sphere and those equations on a torus which admit a monodromy transformation. We shall be mostly concerned with the structural stability of such equations and with the theory of diffeomorphisms of a circle.

§ 1. Structurally Stable Equations on the Circle and on the Sphere

According to an idea which goes back to the classical masters, in order that a differential equation adequately describe physical reality it is necessary that small changes in the equation result in only small changes in the qualitative behaviour of the solutions. For the physical parameters which enter an equation are, as a rule, known only approximately; if a small change in the parameters were to result in a sharp change in the properties of the solutions, then conclusions drawn from the equation modeling the phenomenon might be inapplicable to the original physical problem. One of the attempts to formalize this point of view led to the notion of structural stability. This notion was introduced by A.A. Andronov and L.S. Pontryagin [3] who developed Poincaré's work on limit cycles.

1.1. Definitions. Two differential equations are said to be *topologically orbitally equivalent* if there is a homeomorphism of the phase space of the first system onto the phase space of the second which carries the oriented phase curves of the first system onto the oriented phase curves of the second system.

Let M be a compact smooth manifold and v a smooth vector field on M . The system (M, v) is said to be *structurally stable* if there is a neighbourhood of v in the C^1 -topology such that every vector field in the neighbourhood defines a system topologically orbitally equivalent to the original system by a homeomorphism close to the identity homeomorphism.

1.2. The One-Dimensional Case

Theorem. *A vector field on a circle defines a structurally stable system if and only if all its singular points are nondegenerate (a singular point of a field v is said to be nondegenerate if the linear part of the field at the singular point is nonsingular).*

Two vector fields with nondegenerate singular points on a circle are topologically orbitally equivalent if and only if they have the same number of singular points.

The structurally stable vector fields form an open, everywhere dense subset of the space of all vector fields on the circle with the C^1 -topology.

1.3. Structurally Stable Systems on a Two-Dimensional Sphere [3], [7].

A singular point of a vector field is said to be *hyperbolic* if the real parts of the eigenvalues of the linear part of the field at the singular point are nonzero.

The subset of the space of vector fields on a compact manifold, consisting of the fields with the property that all singular points are hyperbolic, is open and everywhere dense in the C^1 -topology [7, Chap. 6]. In the two-dimensional case the hyperbolic singular points are topologically either saddles or nodes. A phase curve which tends to a saddle point as $t \rightarrow +\infty$ is called an *incoming separatrix* of the saddle point, and a phase curve which tends to a saddle point as $t \rightarrow -\infty$ is called an *outgoing separatrix*.

Theorem. *A vector field on a two-dimensional sphere is structurally stable if and only if the following conditions are satisfied:*

1. *the field has a finite number of singular points;*
2. *all the singular points of the field are hyperbolic;*
3. *no outgoing separatrix of a saddle point is also an incoming separatrix;*
4. *the field has a finite number of cycles;*
5. *all the cycles are nondegenerate.*

Remark. Conditions 1 and 4 follow from the other three conditions, but they have been stated separately in order to stress that they are necessary for structural stability of the field.

Theorem. *The structurally stable vector fields form an open, everywhere dense (in the C^1 -topology) subset of space of all vector fields on the sphere.*

Remark. Similar results hold for vector fields on a disc which are not tangent to the boundary circle. Each of these cases can easily be reduced to the other.

§ 2. Differential Equations on a Two-Dimensional Torus

In this section, we investigate differential equations on a torus which admit a monodromy transformation.

2.1. The Two-Dimensional Torus and Vector Fields on it. We shall consider vector fields on the torus $T^2 = \{(x, y) \bmod 2\pi\}$ whose first component is not zero. Every vector field without singular points and cycles on a 2-dimensional torus can be transformed by a suitable change of coordinates into a vector field with a nonzero first component (C.L. Siegel; see [102], § 7). This is not true, in general, for nonsingular vector fields which have cycles (Fig. 10).

An arbitrary equation on a torus can be written in the form

$$\dot{z} = v(z), \quad z \in \mathbb{R}^2,$$

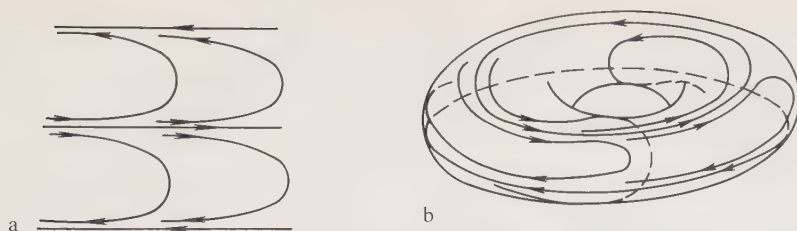


Fig. 10. Equation on a torus without a monodromy transformation. a) The phase portrait of the corresponding doubly-periodic field in the plane; b) the phase curves on the torus.

where

$$v(z + 2\pi e_1) = v(z + 2\pi e_2) = v(z), \quad e_1 = (1, 0), \quad e_2 = (0, 1).$$

The phase curves of an equation defined by a vector field with non-zero first component coincide with the integral curves of the nonautonomous equation

$$\frac{dy}{dx} = f(x, y), \quad \text{where } f(x + 2\pi, y) = f(x, y + 2\pi) = f(x, y). \quad (1)$$

Only such equations on the torus are considered below.

2.2. The Monodromy Mapping. The monodromy mapping (or successor function) of equation (1) is defined to be the mapping of the y -axis onto itself which associates to each point $(0, y)$ the value when $x = 2\pi$ of the solution of the equation with initial condition $(0, y)$. We use the same term for the corresponding mapping of the circle $\mathbb{R}/(2\pi\mathbb{Z})$.

The monodromy mapping and its inverse are differentiable; it differs from the identity by a function a called the *angular function*:

$$A(y) = y + a(y), \quad a(y + 2\pi) = a(y), \quad a'(y) > -1. \quad (2)$$

Definitions. 1. The *trajectory* of a point under the action of a diffeomorphism of a space onto itself is defined to be the set consisting of the point and its images under all iterates of the diffeomorphism and its inverse.

2. A *periodic point* of a diffeomorphism is a point which has a finite trajectory; this trajectory is called a *cycle*, and the number of points in the cycle is its *period*.

3. The *multiplicity* of a cycle with period q of a diffeomorphism A is defined to be the multiplicity of any fixed point of the diffeomorphism A^q which belongs to the cycle (this multiplicity is the same for all points of the cycle). A cycle is *nondegenerate* if the corresponding fixed points are nondegenerate (have no multipliers equal to 1).

The study of equation (1) on the torus reduces to the study of its monodromy map, which is a diffeomorphism of a circle. Thus, for example, the periodic points of the monodromy mapping correspond to closed integral curves of the equation on the torus, and conversely.

2.3. The Rotation Number. Let A be an orientation-preserving homeomorphism of a circle to itself, written in the form (2).

Definition. The *rotation number* of the homeomorphism A is defined to be the limit

$$\mu = \frac{1}{2\pi} \lim_{k \rightarrow \infty} \frac{A^k(y)}{k}.$$

Theorem. The limit in the definition of the rotation number exists and does not depend on the initial point y .

Definition. The *rotation number of the differential equation (1) on the torus* is defined to be the rotation number of the corresponding monodromy mapping.

§ 3. Structurally Stable Differential Equations on the Torus

In this section, we consider differential equations on the torus with a rational rotation number.

3.1. Description of Structurally Stable Equations [7]

Theorem. A differential equation (1) on the torus has a rational rotation number if and only if it has closed integral curves (cycles). If the rotation number is equal to p/q (an irreducible fraction), then the periods of all cycles are equal to $2\pi q$ (the independent variable x plays the rôle of time).

Theorem. A differential equation (1) on the torus is structurally stable if and only if the rotation number is rational and the periodic solutions are nondegenerate.

Similar assertions hold for orientation-preserving diffeomorphisms of a circle.

Definitions. 1. The homeomorphisms $f: M \rightarrow M$ and $g: M \rightarrow M$ are topologically (resp., C^k -, analytically) equivalent if there is a homeomorphism $h: M \rightarrow M$ (resp., C^k or analytic diffeomorphism) conjugating f and g : $f = h \circ g \circ h^{-1}$.

2. A diffeomorphism of a manifold M into itself is said to be *structurally stable* if any C^1 -close diffeomorphism is topologically equivalent to it via a conjugating homeomorphism close to the identity.

Theorem. An orientation-preserving diffeomorphism of a circle is structurally stable if and only if the rotation number is rational and all the cycles are nondegenerate. The structurally stable diffeomorphisms form an open set which is everywhere dense in the space of all twice-differentiable orientation-preserving diffeomorphisms of the circle with the C^2 -topology.

It should not be thought, however, that the rotation numbers of randomly chosen diffeomorphisms of the circle will be preponderantly rational (see § 4).

3.2. A Bound on the Number of Cycles.

Jakobson's Theorem (*Funct. anal. appl.* (Russian), 1985, vol. 19, No. 1). *A diffeomorphism $y \mapsto y + a(y)$ of the circle for which the angular function a is a trigonometrical polynomial of degree n has no more than $2n$ cycles counted with multiplicity.*

Corollary. *Every diffeomorphism in the two-parameter family*

$$f_{\varepsilon, a}: y \mapsto y + a + \varepsilon \sin y$$

has no more than two cycles.

Remark. The rotation number of the diffeomorphisms of the family traverse the whole axis, and the periods of the cycles which occur can be arbitrarily large.

A similar theorem for differential equations on a torus has not been proved.

If $\frac{dy}{dx} = f(x, y)$ is an equation with a trigonometrical polynomial f on the right-hand side, it is not even known whether there is an upper bound depending on the degree of f .

§ 4. Equations on the Torus with Irrational Rotation Numbers

All the integral curves of the standard equation

$$\frac{dy}{dx} = \omega \quad \text{where } \omega \text{ is irrational,}$$

are everywhere dense on the torus and, hence, are not closed. We investigate below the question of when a differential equation on the torus with an irrational rotation number is equivalent to the standard equation. We start with diffeomorphisms of a circle.

4.1. The Equivalence of a Diffeomorphism of a Circle to a Rotation of the Circle

Denjoy's Theorem [7], [106]. *A C^2 orientation-preserving diffeomorphism of a circle with an irrational rotation number is topologically equivalent to a rotation of the circle.*

(Note. For a homeomorphism of a circle with an irrational rotation number the α - and ω -limit sets of all the trajectories coincide. They are either the

whole circle (for homeomorphisms which are C^2) or a Cantor perfect set (the corresponding examples have been constructed for homeomorphisms which are C^1 [106]). In the first case, as has already been mentioned, the homeomorphism of the circle is topologically equivalent to a rotation. A complete topological classification of homeomorphisms of the circle in the second case was obtained in [81].)

The following theorem shows that the conjugating homeomorphism guaranteed by Denjoy's theorem is only slightly less smooth than the original diffeomorphism for almost all (with respect to Lebesgue measure) rotation numbers.

Hermann's Theorem [47]. *There is a set of total measure on the real line such that any C^r diffeomorphism of the circle whose rotation number belongs to the set is C^{r-2} -smoothly equivalent to a rotation. Here, r is ∞ , ω or any integer greater than two and, by convention, $\infty - 2 = \infty$ and $\omega - 2 = \omega$ (where C^ω denotes the class of analytic mappings).*

There is a sufficient condition, known as Hermann's "condition A", on an irrational rotation number μ for μ to belong to the set mentioned in the theorem. To state it, let

$$\mu = a_0 + 1/(a_1 + 1/(a_2 + \dots))$$

be the expansion of μ as a continued fraction. The number μ satisfies condition A if

$$\lim_{B \rightarrow \infty} \limsup_{n \rightarrow \infty} \left(\sum_{a_i \geq B} \ln(1 + a_i) \right) \left(\sum_{1 \leq i \leq n} \ln(1 + a_i) \right)^{-1} = 0.$$

Under the additional assumption that the residual (the difference from a rotation) is small, condition A can be simplified as follows.

Theorem [8: 2]. *Let μ be a number satisfying the condition*

$$\left| \mu - \frac{p}{q} \right| > C q^{-(2+\varepsilon)}$$

for all irreducible fractions p/q and some positive constants C and ε . Then an analytic diffeomorphism of the circle with rotation number μ , which is sufficiently close to a rotation through μ , is analytically equivalent to a rotation.

If $\varepsilon < \sqrt{5} - 2$ in the theorem above, then a C^∞ -diffeomorphism of a circle with a rotation number μ is C^∞ -equivalent to a rotation for any (not necessarily small) angular function (J.C. Yoccoz [42, p. 814]).

The majority of smooth (resp., analytic) diffeomorphisms of a circle which are close to a rotation are smoothly (resp., analytically) equivalent to a rotation. For example, consider the two-parameter family $y \mapsto y + a + \varepsilon C(y)$, where C is a smooth, 2π -periodic function.

The proportion of the values of the parameters (a, ε) in the rectangle $|\varepsilon| \leq \varepsilon_0$, $0 \leq a \leq 2\pi$, for which the diffeomorphism is not smoothly equivalent to a

rotation tends to zero as $\varepsilon_0 \rightarrow 0$. In particular, the total area of all the resonance tongues in Fig. 11 in a small neighbourhood of the α -axis is small compared to the area of the neighbourhood.

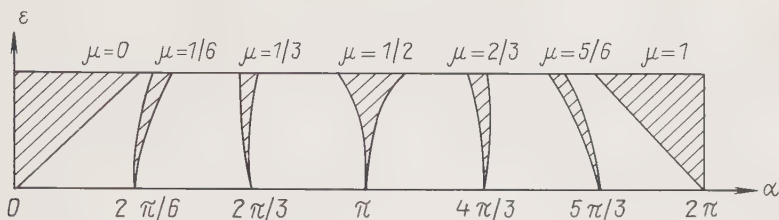


Fig. 11. Resonance zones for a family of diffeomorphism of a circle

In the “exceptional” cases the conjugating homeomorphism in Denjoy’s theorem may not be smooth even if the original diffeomorphism is analytic. This happens when the rotation number is approximated abnormally closely by rational numbers [8: 2].

Similar theorems hold for differential equations on a torus. They follow from the preceding theorems and the following remark.

Remark. A differential equation on a torus which admits a monodromy mapping is topologically (resp., C^r , analytically) equivalent to the standard equation if and only if the corresponding monodromy mapping is topologically (resp., C^r , analytically) equivalent to a rotation.

4.2. Diffeomorphisms of a Circle and Vector Fields on S^3 . Denjoy constructed an example of a C^1 -diffeomorphism of the circle having an irrational rotation number which is not equivalent to a rotation of the circle (see [109]). Using this example, P. Schweitzer a negative answer to the following question posed by Seifert: Is it true that every vector field without singular points on S^3 has a cycle?

Theorem [109]. *There is a C^1 vector field on the three-dimensional sphere without singular points or cycles.*

J. Harrison [45] has recently constructed a C^2 field with the same property. It is not yet known whether it is possible to raise still further the smoothness of the field in Schweitzer’s theorem.

§ 5. Remarks on the Rotation Number

In this section we investigate the rotation number as a function of the parameters, and also the “rotation sets” for endomorphisms of a circle (reminder: an endomorphism is injective, but not necessarily bijective).

5.1. The Rotation Number as a Function of the Parameters. Consider a family of diffeomorphisms of the circle to itself with parameters a and ε

$$y \mapsto y + a + \varepsilon f(y), \quad f(y + 2\pi) = f(y). \quad (3)$$

Definition. A point (a, ε) belongs to a *resonance zone* m/n if the rotation number of the mapping (1) is equal to m/n .

The resonance zones m/n for small n in the case when $f(y) = \sin y$ are shown in Fig. 11. They approach the line $\varepsilon = 0$ along small tongues, with a cusp at the point m/n , the tongue being the narrower the greater n is.

Theorem [8]. Let f be a trigonometric polynomial of degree p of a single variable. Then the width of the resonance zone m/n does not exceed $C\varepsilon^r$, where r is the integral part of the fraction $-n/p$: $r = -[-n/p]$.

5.2. Families of Equations on a Torus. A similar theorem holds for differential equations on a torus:

$$\frac{dy}{dx} = a + \varepsilon F(x, y), \quad F(x + 2\pi, y) = F(x, y + 2\pi) = F(x, y). \quad (4)$$

Theorem (O.G. Galkin). Let F be a trigonometric polynomial of degree p in y . Then the width of the resonance zone m/n (i.e., the set of those (a, ε) for which equation (4) has the rotation number m/n) does not exceed $C\varepsilon^r$, where $r = -[-n/p]$.

5.3. Endomorphisms of the Circle [13]. Let A be an endomorphism of the circle, i.e., let

$$A(y) = y + a(y), \quad a(y + 2\pi) = a(y),$$

but A need not necessarily be bijective, i.e., the condition $a' > -1$ may not hold.

Definitions. 1. The *rotation set* of the endomorphism A of the circle on to itself is defined to be the closure $\mu(A)$ of the set

$$\{\mu^+(A, y) | y \in \mathbb{R}\} \quad \text{where} \quad \mu^+(A, y) = \limsup_{k \rightarrow \infty} \frac{A^k y}{k}.$$

2. The rotation set $\mu(A, y)$ is the set of all limit points of the sequence $\{A^k(y)/k\}$.

The following theorem gives a description of the sets $\mu(A, y)$.

Theorem. Let A be a continuous mapping of degree 1 of the circle on to itself (as described at the beginning of this section). Then:

1. for any y the set $\mu(A, y)$ is an interval which belongs to $\mu(A)$;
2. for each interval σ belonging to $\mu(A)$ there is a y such that $\mu(A, y) = \sigma$.

Chapter 3

Singular Points of Differential Equations in Higher Dimensional Real Phase Space

The local theory of differential equations is concerned to a considerable extent with classification problems. Different branches of the theory arise according to the equivalence relation used in classification.

Definition. Two differential equations (or, what is the same thing, two vector fields) are *topologically equivalent* in neighbourhoods of singular points if there is a homeomorphism carrying the first singular point into the second and conjugating the local phase flows of the equations at these singular points. If the conjugating homeomorphism is C^k (where k is a positive integer or infinity) or analytic, then the differential equations are said to be C^k -*equivalent* or *analytically equivalent*.

Definition. Two differential equations (or two vector fields) are *orbitally topologically equivalent* in neighbourhoods of singular points if there is a homeomorphism of a neighbourhood of the singular point of one field onto a neighbourhood of the singular point of the other field which carries the first singular point into the second and which maps the local phase curves of one equation into the local phase curves of the other, preserving the direction of motion. C^k - and *analytically orbitally equivalent equations* are defined similar to the way in which C^k - and analytically equivalent equations were defined.

These definitions are meaningful in both real and complex domains (in the latter case time is complex). Analytic equivalence of differential equations is most naturally studied in a complex phase space; in this chapter we examine the topological and smooth classification in the real case. From now on, unless otherwise stated, “smooth” means infinitely differentiable, the time is real, and vector fields are smooth.

§ 1. Topological Classification of Hyperbolic Singular Points

The topological type of a differential equation in a neighbourhood of a generic singular point is determined by the linearization of the field at the point (Theorem 1.1 below). The case of more complicated singular points is discussed in sections 2 and 5.

1.1. The Grobman-Hartman Theorem

Definition. A singular point of a differential equation is said to be *hyperbolic* if no eigenvalue of the linear part of the equation at the singular point lies on the imaginary axis.

Theorem [7], [46]. A C^1 vector field is topologically equivalent to its linear part in a neighbourhood of a hyperbolic singular point.

1.2. Classification of Linear Systems. Linear systems with a hyperbolic singular point admit a further simple classification.

Theorem. Let A be a linear map with no eigenvalues on the imaginary axis and suppose that n^+ eigenvalues lie in the right half plane and n^- in the left half plane. Then the differential equation

$$\dot{x} = Ax, \quad x \in \mathbb{R}^n, \quad n = n^+ + n^-,$$

is topologically equivalent to the standard equation

$$\dot{y} = y, \quad \dot{z} = -z, \quad y \in \mathbb{R}^{n^+}, \quad z \in \mathbb{R}^{n^-}.$$

Remark. The topological classification of singular points, hyperbolic or not, of linear systems is the same as the topological classification of linear systems on the whole space $\mathbb{R}^n$, and is as follows.

Theorem [70]. Two linear differential equations $\dot{x} = Ax$ and $\dot{y} = By$, $x, y \in \mathbb{R}^n$, are topologically equivalent if and only if the numbers of eigenvalues with negative (or positive) real parts of the operators A and B are equal, and the restrictions of these operators to their invariant subspaces corresponding to the purely imaginary eigenvalues are linearly equivalent.

§ 2. Lyapunov Stability and the Problem of Topological Classification

In this section, we discuss general approaches to the local problems of analysis and cite theorems which show that in strongly degenerate cases the stability problem and the problem of topological classification of singular points are, in some sense, insoluble. Stability criteria and a classification which apply in cases which are "not too degenerate" have been obtained in recent years and will be given in § 5.

2.1. On the Local Problems of Analysis. We start with the definition of jets of functions and vector fields. We fix a coordinate system in $\mathbb{R}^n$.

Definition. An N -jet of a smooth function at the origin O of the space $\mathbb{R}^n$ is defined to be the class of functions whose Taylor expansions at the point O coincide up to and including terms of degree N .

Hence, in a fixed coordinate system, an N -jet of a function is given by a polynomial of degree less than or equal to N . We give another definition which is clearly independent of the coordinate system.

Definition. The N -jet of a smooth function f at the point O in the space $\mathbb{R}^n$ is the class of all functions which agree with f up to $o(r^N)$ as $r \rightarrow 0$ (r is the distance from the origin); this N -jet is denoted by $f^{(N)}$ or $j_0^N f$.

This definition is clearly equivalent to the first definition and, hence, establishes the independence of the former from the choice of coordinate system.

The N -jet of a function at an arbitrary point x of $\mathbb{R}^n$ is defined similarly; r has merely to be regarded as the distance from x . N -jets of vector fields at an arbitrary point of $\mathbb{R}^n$ are defined similarly. The subspace of jets of vector fields at the singular point O is denoted by $J_0^N(n)$.

A choice of coordinate system in the phase space gives us a coordinate system in $J^N(n)$; each N -jet corresponds to the set of coefficients of the vector polynomial of degree no higher than N which is its *representative*.

Definition. A jet of a vector field at a singular point is *positive (negative)* with respect to some property A if all its representatives have (do not have) property A . A jet is *neutral with respect to property A* if it is neither positive nor negative.

Definition. The problem of determining which vector fields have or do not have property A is said to be *algebraically soluble* if

1. the sets of N -jets positive, negative, or neutral with regard to property A are semi-algebraic subsets of $J^N(n)$ for each N . (A subset of a real space is called *semi-algebraic* if it is the union of a finite number of subsets which are defined by a finite number of algebraic equations and inequalities of the form $P > 0$.)

2. the codimension of the set of neutral N -jets in the space $J^N(n)$ tends to infinity as $N \rightarrow \infty$.

Analytically soluble problems are defined in the same ways as algebraically soluble problems except that in all the definitions *algebraic* is replaced by *analytic*.

In this section, let the property A be Lyapunov stability. Instead of saying "a jet of a vector field is positive (negative, neutral) with regard to the property of Lyapunov stability" we shall say that "the jet is stable (unstable, neutral)".

Example. By Lyapunov's theorem on stability by linearization, a 1-jet Ax of a vector field at the point O is stable if all the eigenvalues of the operator A lie in the left half-plane, unstable if at least one eigenvalue lies in the right half-plane, and neutral if there are no eigenvalues of A in the right half-plane but at least one which lies on the imaginary axis. Hence stable, unstable, and neutral sets of 1-jets are semi-algebraic in a phase space of any dimension.

2.2. Algebraic and Analytic Insolubility of the Problem of Lyapunov Stability. In [7, p. 330] an algebraic subfamily of codimension on the order of 100 in the jet space $J^5(3)$ was found whose intersection with the set of neutral

jets is not semi-algebraic. This shows that the problem of Lyapunov stability is algebraically insoluble.

E.E. Shnol and L.G. Khazin [66:3] discovered a similar phenomenon in a family of jets of small codimension.

Theorem [66:3]. *The subfamily of $J_0^3(4)$ consisting of jets of vector fields at the singular point whose linear part (at this point) has two pairs of purely imaginary eigenvalues of the form $\pm i\omega_1$, $\pm i\omega_2$, with $3\omega_1 = \omega_2$, intersects the set of stable jets in a set which is not semi-algebraic.*

Theorem (G.G. Khazina, L.G. Khazin, 1977). *The subfamily of $J_0^3(4)$ consisting of jets with a diagonalizable linear part whose eigenvalues are double and purely imaginary has the same property.*

The codimensions of these subfamilies in the corresponding jet spaces are equal to 3 and 4 respectively.

In [7, p. 330] the following conjecture was made. "It can be expected that once the boundary between the stable and unstable jets loses its semi-algebraicity, the lack of further restrictions will permit set-theoretic pathologies to arise. For example, the set of stable jets in a finite-dimensional algebraic submanifold of a jet space of fixed order can probably have an infinite number of connected components. Quite likely, such a set could have the property that both it and its complement are everywhere dense." One of the pathologies anticipated there has now been discovered. A one-parameter algebraic family has been constructed in the jet space $J^5(5)$ which intersects the set of stable jets in a countable number of intervals which accumulate to an interior point of the family. The following rather weaker result has been published.

Theorem [54]. *The problem of Lyapunov stability is analytically insoluble. In particular, the set of stable jets in the space $J^5(5)$ is not semi-analytic.*

2.3. Algebraic Solubility up to Degeneracies of Finite Codimension. A natural measure of the extent to which it is possible to investigate the algebraic insolubility of a local problem is the "codimension of the degeneracy".

Definitions. The *germ of a vector field at a singular point* is defined to be the set of all vector fields which coincide with the original field in some neighbourhood (depending on the field) of this point. A field in this set is called a *representative of the germ*. Two germs of vector fields at singular points x and y are said to be *topologically (smoothly, analytically, orbitally topologically, ...) equivalent* if they have topologically (smoothly, analytically, orbitally topologically, ...) equivalent representatives and if the conjugating homeomorphism carries x into y .

Germs of diffeomorphisms at a fixed point, *equivalence* between them, and *germs of functions*, are defined similarly.

Definition. A problem involving germs of vector fields at a singular point O of the space $\mathbb{R}^n$ is *algebraically soluble up to and including codimension k* if, for some N , there is a sequence of embeddings of algebraic manifolds

$$V_0 = J_0^N(n) \supset V_1 \supset V_2 \supset \dots \supset V_k \supset V_{k+1}, \quad \text{co-dim}_{\mathbb{R}} V_j = j,$$

with the following property. Each of the differences $V_j \setminus V_{j+1}$, $j=0, \dots, k$, is the union of a finite number of connected components (strata) and the local problem has the same answer for any two germs whose N -jets belong to the same stratum.

For example, if the problem is the topological classification of germs of vector fields, then all germs belong one stratum must be topologically equivalent.

The problem of Lyapunov stability and the problem of the topological classification of germs of vector fields are algebraically soluble up to and including codimension 2. The algebraic investigation of a local problem can often be extended if one restricts attention to some subset W of the germ space. We say that a problem is algebraically soluble on a subset W up to and including codimension k if the situation of the previous definition holds with $V_0 = W^{(N)}$ and $\text{codim } W_j = j$ in $W^{(N)}$, where $W^{(N)}$ is the set of N -jets of germs in W . Thus, the problem of Lyapunov stability is algebraically soluble up to and including codimension three on the set of germs of vector fields whose linear part has no 4-tuples of eigenvalues of the form $\{\pm i\omega, \pm 3i\omega\}$.

2.4. Topologically Unstabilizable Jets. First we choose a fixed coordinate system. An M -jet of a vector field is said to be an *extension* of an N -jet ($M > N$) if its Taylor polynomial of degree M can be obtained from the Taylor polynomial of the N -jet by adding terms of higher degree.

An Invariant Definition. An M -jet of a vector field is an *extension* of an N -jet if $M > N$ and the vector fields belonging to the M -jet belong to the set of fields representing the N -jet.

Definition. A jet $V^{(N)}$ of a vector field at a singular point is said to be *topologically unstabilizable* in the class of smooth (analytic) germs of vector fields if any higher order jet which is an extension of $V^{(N)}$ contains smooth (analytic) representatives which are not topologically equivalent in any neighbourhood of the singular point.

In other words, if a jet of a vector field is topologically unstabilizable, then no information about a finite number of higher order terms will make it possible to draw, even up to homeomorphism, a phase portrait of the field in a neighbourhood of the singular point.

Takens's Theorem [107]. *There is an open subset of the space of 3-jets of vector fields whose linear parts have 1 zero eigenvalue and 2 pairs of purely imaginary eigenvalues which consists of jets which are topologically unstabilizable in the class of smooth germs of vector fields.*

It is not yet known whether a similar result holds for analytic germs.

The codimension of the set of topologically unstabilizable jets in Takens's theorem (considered as a subset of the space of jets with singular point O) is equal to 3. The topological classification of the germs of the vector fields belonging to a certain codimension 6 subset can even have numerical moduli.

Theorem [108]. *In the space of 5-jets of vector fields whose linear parts have two pairs of purely imaginary eigenvalues $\pm i\omega_1$, $\pm i\omega_2$, $0 < \omega_1 < \omega_2$, there is a codimension 4 submanifold and an open subset of it with the following property. Two topologically equivalent germs of vector fields at the singular point whose 5-jets belong to the subset must have equal ratios ω_1/ω_2 .*

§ 3. Formal Classification of Germs of Vector Fields

According to a general principle which goes back to H. Poincaré, in studying differential equations it is convenient to seek, not a solution, but a substitution which reduces the equation to the simplest possible form. In this section we consider the first steps in this direction — the study of formal substitutions. In what follows we consider germs of vector fields at a singular point, often without stating that fact explicitly.

3.1. Formal Vector Fields and Their Equivalence

Definitions. K denotes one of the fields $\mathbb{R}$ or $\mathbb{C}$, and $x = (x_1, \dots, x_n)$. By a *formal Taylor series* we mean an expression

$$f(x) = \sum_{\mathbb{Z}_+^n} a_k x^k, \quad a_k \in K, \quad x^k = (x_1^{k_1}, \dots, x_n^{k_n});$$

no demands are imposed regarding convergence. The same operations can be carried out on formal Taylor series as can be carried out on convergent Taylor series, namely, addition, multiplication, differentiation, composition (substitution of one series into another), and so on. On convergent series all these operations can be defined as operations on the coefficients of the series without using convergence; these operations can be defined in the formal case by the same formulae. The set of all formal Taylor series of a vector variable x will be denoted by $K[[x]]$; we write $f(0) = a_0$.

By a *formal vector field with a singular point at 0* we mean an expression

$$v = \sum_1^n f_j(x) \frac{\partial}{\partial x_j}, \quad f_j \in K[[x]], \quad f(0) = 0.$$

A *formal substitution with fixed point 0* is an expression

$$H = (H_1(x), \dots, H_n(x)), \quad H_j \in K[[x]], \quad H_j(0) = 0, \quad \det H_*(0) \neq 0.$$

Two formal vector fields v and $\tilde{v}$ with a singular point at 0 are said to be *formally equivalent* if there is a formal substitution H with fixed point 0 for which the relation

$$\frac{\partial H}{\partial x} v = \tilde{v} \circ H$$

holds.

Remarks. 1. This relation holds if v and $\tilde{v}$ are smooth fields and H is a diffeomorphism carrying v into $\tilde{v}$ (see Chap. 1, § 1.6).

2. To every germ of a vector field with a singular point there corresponds a formal vector field, namely, the Taylor series of the germ of v , which we shall denote by $\hat{v}$. A necessary condition for smooth or analytic equivalence of the germs of vector fields is that the corresponding formal Taylor series be formally equivalent. The formal classification therefore precedes the smooth or analytic classification of germs of vector fields.

It turns out that in a neighbourhood of a singular point a *generic* holomorphic vector field is holomorphically equivalent to its linear part, but the situation is considerably more complicated for fields which are not generic.

3.2. Resonances. The Poincaré-Dulac Normal Forms and Their Generalizations. A “generic” formal vector field is formally equivalent at a singular point to its linear part. In order that a field be “generic”, its linear part must be “nonresonant”.

Definition. An n -tuple of numbers $\lambda = (\lambda_1, \dots, \lambda_n) \in \mathbb{C}^n$ is said to be *resonant* if one of the numbers λ_j is an integral linear combination (with nonnegative coefficients whose sum is not less than two) of the other numbers, i.e., if a relation

$$\lambda_j - (\lambda, k) = 0 \quad (1)$$

holds, where $k \in \mathbb{Z}_+^n$, $|k| = k_1 + \dots + k_n \geq 2$, $(\lambda, k) = \sum \lambda_j k_j$, $j \in \{1, \dots, n\}$ and $\mathbb{Z}_+$ is the set of non-negative integers.

A linear vector field is said to be *resonant* if the spectrum of the corresponding linear map is a resonant set; otherwise, it is said to be *nonresonant*.

Poincaré's Theorem [7]. *A formal vector field with a singular point at 0 and a nonresonant linear part is formally equivalent to its linear part. If the field is real, then corresponding formal substitution can also be chosen to be real.*

The formal classification is fundamentally more complicated if there are resonances.

Definition. Let $(z_1, \dots, z_n)$ be coordinates in which the matrix of the linear part of the formal vector field v has Jordan normal form, and let λ be the spectrum of this matrix. The monomial $z^k \frac{\partial}{\partial z_j}$ is called a *resonant term* if the resonance relation (1) holds.

Poincaré-Dulac Theorem [7]. *A formal vector field with a singular point at O and a resonant linear part is formally equivalent to a field in which the linear part has Jordan normal form Jz and the nonlinear terms are resonant. This field has the form*

$$w(z) = Jz + \sum a_{kj} z^k \frac{\partial}{\partial z_j},$$

where the summation is carried out over pairs j, k such that $\lambda_j = (\lambda, k)$, where λ is the spectrum of J . The coefficients a_{kj} may be complex even though the original field is real.

The normal form given by this theorem is called the Poincaré-Dulac normal form; it admits further simplification and is therefore sometimes called the *preliminary normal form*.

If the matrix of the linear part of the field is nilpotent, then the Poincaré-Dulac theorem gives no simplifications. In that case the following theorem due to G. R. Belitsky is useful [15].

Theorem. *A vector field with a singular point at O is formally equivalent to a field in which the linear part has the Jordan normal form Jz , and the nonlinear part commutes with the vector field J^*z (the raised $*$ denotes the Hermitian conjugate $(a_{ij}) \rightarrow (\bar{a}_{ji})$). (Note: Two vector fields v and w commute if the corresponding differential operators commute: $L_v L_w f = L_w L_v f$ for all smooth functions f)*

Example [107]. A real two-component vector field with linear part $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}_x$ is formally equivalent to a field

$$x_2 + f(x_1) \frac{\partial}{\partial x_1} + g(x_1) \frac{\partial}{\partial x_2}.$$

A similar problem for fields in a higher dimensional space is investigated in § 7.

3.3. Applications of the Theory of Formal Normal Forms

1. Differential equations with a resonant linear part, written in Poincaré-Dulac normal form, have, as a rule, an abundant group of symmetries, and this enables the order to be lowered. The order of the resulting system (the so-called quotient system) is equal to the number of linearly independent resonance relations on the spectrum of the linear part. If this number is equal to one, the Poincaré-Dulac normal form can be integrated by quadratures; it is not integrable, generally speaking, if this number is greater than one. Thus, the application of the theory of normal forms to the explicit integration of equations is substantial, but strictly limited.

2. In investigating topologically complicated cases, when the linear part of the equation at the singular point has eigenvalues on the imaginary axis, Poincaré's method is very useful. It is applied to reduce a finite jet (i.e.,

a finite number of terms of the Taylor series of the vector field at the singular point) to normal form. After this, the higher-order terms are discarded, the truncated equation is investigated, and it is then shown that the higher-order terms do not change the qualitative picture. The results in § 5 are not only proved, but actually formulated, by this method. This method is particularly useful in the theory of bifurcations (see [7]).

3. The reduction to formal normal form is particularly effective when it is possible to annihilate all except a finite number of terms. This always happens when all the eigenvalues lie in a half-plane which does not contain the point O . It also happens in certain other cases which will be examined in the next section.

3.4. Polynomial Normal Forms

Definition. We shall say that a formal power series *has a zero of order N at O* if it contains no terms of degree less than N , and we write $f = o(|x|^{N-1})$. Jets of formal vector fields are then defined in the same way as jets of smooth vector fields.

Definition. A formal vector field is said to be *formally N -determined* if all formal vector fields with the same N -jet are formally equivalent to it. A formal vector field is said to be *formally finitely determined* if it is formally N -determined for some N .

Remarks. 1. A formally finitely determined vector field formally admits a polynomial formal normal form.

2. If the spectrum of the linear part of a formal vector field satisfies only a finite number of resonance relations, then the field is formally finitely determined. This follows immediately from the Poincaré-Dulac theorem.

Definition. An n -tuple $\lambda \in \mathbb{C}^n$, and also a vector field with the spectrum λ , is said to be *k -resonant* if the number of generators of the additive group generated by the set of vectors $\{r \in \mathbb{Z}_+^n \mid (r, \lambda) = 0\}$ is equal to k ; here $\mathbb{Z}_+$ is the set of nonnegative integers.

Theorem ([52], a closely related theorem is contained in the book [23]). *Suppose a formal vector field v has a 1-resonant linear part and suppose that all but a finite number of the resonance relations are consequences of a single one: $(r, \lambda) = 0, r \in \mathbb{Z}_+^n$. Let $w(z) = Jz + Zg(u)$ be the Poincaré-Dulac normal form of the field v ; here $Z = \text{diag } z, u = z'$, and $g = (g_1, \dots, g_n)$ is a vector formal series of a single variable. Then the following assertions are equivalent:*

1. *the field v is formally finitely determined;*
2. *$(r, g) \not\equiv 0$.*

Remark. If the relation $(r, g) \equiv 0$ holds for one Poincaré-Dulac normal form of the field v , then it also holds for any other such form.

Theorem [52]. *A formal vector field whose linear part has a k -resonant spectrum with $k \geq 2$ is never formally finitely determined.*

It can be seen by comparing the results of § 3.3 and § 3.4 that the properties of integrability and formal finite determinacy of the Poincaré-Dulac normal form are very closely related.

§ 4. Invariant Manifolds and the Reduction Theorem

A homeomorphism which linearizes a vector field in a neighbourhood of a hyperbolic singular point cannot always be chosen to be smooth. Resonance of the linear part, for example, will prevent this. Nevertheless, considerable similarities between the equation and its linearization at a singular point are retained. This is expressed by the theorems below about invariant manifolds.

4.1. The Hadamard-Perron Theorem

Definition. An *invariant manifold* of a vector field and of the corresponding differential equation is defined to be a submanifold of phase space which is tangent to the vector field at each of its points.

Consider a linear map $A: \mathbb{R}^n \rightarrow \mathbb{R}^n$. The space $\mathbb{R}^n$ can be written as a direct sum of three subspaces:

$$\mathbb{R}^n = T^s \oplus T^u \oplus T^c$$

(the superscripts s, u, c correspond to “stable”, “unstable”, and “center”). This splitting is determined by requiring that: all three subspaces be invariant under A ; the spectrum of the restriction $A|_{T^s}$ lie in the open left half-plane, that of $A|_{T^u}$ in the open right half-plane, and that of $A|_{T^c}$ on the imaginary axis.

We consider first the case when O is a hyperbolic singular point of the equation $\dot{x} = Ax$, i.e., $T^c = \{O\}$. The following theorem generalizes the results of Hadamard and Perron, and by tradition it is called the Hadamard-Perron theorem. The following formulation is given in the book [77], which gives references to the original papers. In the next two theorems r is either a natural number or infinity.

Theorem. Let v be a C^r vector field with a hyperbolic singular point O and with linear part Ax at O ; let T^s and T^u be the subspaces of the splitting determined by A . Then the differential equation $\dot{x} = v(x)$ has two invariant C^r manifolds W^s and W^u passing through O and tangent at O to the planes T^s and T^u respectively. Solutions with initial conditions on W^s (resp. W^u) tend exponentially to O as $t \rightarrow +\infty$ (resp. $-\infty$).

W^s is called the *stable manifold*, W^u the *unstable manifold* of the singular point O .

The following two theorems form the basis of the local theory of stability and the local theory of bifurcations.

4.2. The Center Manifold Theorem

Theorem. Let v be a C^{r+1} vector field with singular point O and a linear part Ax ($r < \infty$) at O . Let T^s, T^u, T^c be the subspaces of the splitting corresponding to the operator A , as described in § 4.1.

Then the differential equation $\dot{x} = v(x)$ has invariant manifolds W^s, W^u , and W^c of class C^{r+1}, C^{r+1} and C^r , respectively, which go through O and are tangent to T^s, T^u and T^c , respectively, at O . The phase curves of the equation on the manifolds W^s, W^u behave as stated in the Hadamard-Perron theorem. The behaviour of the phase curves on the center manifold W^c is determined by the nonlinear terms.

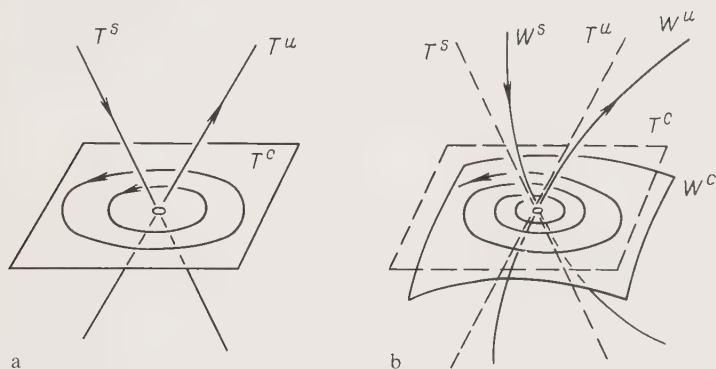


Fig. 12. Stable, unstable, and central manifolds: a) of a linear system, b) of a non-linear system

Under the same conditions the equation $\dot{x} = v(x)$ has two more invariant manifolds, a stable-center manifold W^{sc} and an unstable-center manifold W^{uc} , both of which are C^r . The first is central and stable, the second central and unstable. Like the center manifold, these manifolds, are C^r , in general, only when $r < \infty$.

As before, the manifolds W^s and W^u in the above theorem are said to be stable and unstable. If the germ of v is C^∞ or analytic, then the stable and unstable manifolds in both previous theorems are C^∞ or analytic; the center manifold is only finitely smooth.

The manifold W^c is called the center manifold, and the subspace $T^s \oplus T^u$ the subspace of hyperbolic variables. The following theorem asserts that to investigate the topology of the germ of a nonlinear vector field one need only consider the restriction of the germ to the corresponding center manifold; the hyperbolic variables need not be considered.

4.3. The Reduction Principle

Reduction Theorem [7], [101]. Suppose a differential equation with C^2 right hand side has singular point O and a linear part Ax . Let T^s, T^u, T^c be the

invariant subspaces corresponding to the map A in the notation of § 4.1. Then, in a neighbourhood of the singular point O , the equation under consideration is topologically equivalent to the direct product of two equations: the restriction of the original equation to the center manifold, and the "standard saddle"

$$\dot{x} = -x, \quad \dot{y} = y, \quad x \in T^s, \quad y \in T^u.$$

This theorem is used to study both individual equations and families of equations; a family $\dot{x} = f(x, \varepsilon)$ being equivalent to the equation $\dot{x} = f(x, \varepsilon)$, $\varepsilon = 0$. These applications are discussed in § 5 and in [7]. Other variants of the last two theorems were obtained by V.A. Pliss [90] and by M.W. Hirsch, C. Pugh, and M. Shub [50]. Infinite-dimensional analogues of the reduction theorem are useful in mathematical physics and, in particular, in the hydrodynamics of a viscous fluid. The present state of the problem is discussed by V.I. Yudovich in another volume of the present work. The theory of invariant manifolds for the analytic case is discussed in Chap. 4, § 3.

§ 5. Criteria for Stability and the Topological Classification of Singular Points in the Case of Degeneracies of Low Codimension

In this section we describe the topology of germs of vector fields at all degenerate singular points off a manifold of codimension three in the space of vector fields. We also give the corresponding stability criteria.

5.1. Structure of the Criteria. In order to apply the criteria of this section to a fixed germ of a vector field, it is necessary to carry out the following steps.

1. Determine the class enumerated below to which the linear part of the germ belongs and reduce the N -jet of the germ to Poincaré-Dulac normal form; the number N is indicated in the tables below. (Note: the normal form described in § 3.3 is used in the third row of Table 4.)

2. Equate the hyperbolic terms to zero.

These two steps effect the normalization of the N -jet of the restriction of the original germ to its center manifold.

3. Verify that the Taylor coefficients of the normalized jet do not satisfy the conditions defining an exceptional submanifold in the space of N -jets corresponding to a degeneracy of higher codimension.

4. Find the class in the appropriate table to which the normalized jet belongs; to do this a preliminary linear substitution may be needed, e.g., $(x, z) \mapsto (-x, z)$, or $(z_1, z_2) \mapsto (z_2, z_1)$.

5.2. Topological Classification of Germs of Smooth Vector Fields up to and Including Degeneracies of Codimension Two. The space of germs of

smooth vector fields on $\mathbb{R}^n$ with singular point O whose 1-jets do not belong to a prescribed codimension three algebraic submanifold fall into five classes, enumerated in Table 1.

These classes differ in the dimension of the center manifold W^c or the Jordan normal form A of the linear part of the restriction of the germ to W^c .

Table 1

Class	$\dim W^c$	A
W_1^0	1	0
W_2^I	2	$I = \mathbb{R}i\omega, \omega \neq 0$
W_2^J	2	$J_2 - \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$
$W_3^{0,I}$	3	$\text{diag}(0, I)$
$W_4^{I,I}$	4	$\text{diag}(I_1, I_2), I_j = \mathbb{R}i\omega_j, 0 < \omega_1 \leq \omega_2$

The differential equation on the real vector space $\mathbb{R}^n$ with singular point O corresponding to the Poincaré-Dulac normal form can be written as

$$\dot{x} = f(x, z, w), \quad \dot{z} = g(x, z, w), \quad \dot{w} = h(x, z, w).$$

Here $(x, z, w) \in \mathbb{C}\mathbb{R}^n$, x is real on $\mathbb{R}^n$, and $w = \bar{z}$ on $\mathbb{R}^n$. Also $h(x, z, w) = \bar{g}(x, \bar{w}, \bar{z})$. Thus the equation for w is uniquely determined by the equation for z , and will not be written out.

Tables 2–5 give the differential equations corresponding to the normalized jets; for brevity the corresponding column is headed “normalized N -jet”. We omit terms in the expressions of the normalized jets which do not affect the answer.

Notation: $z_j = x_j + iy_j, z = (z_1, \dots, z_k), k \in \{1; 2; 3\}$,

$$r_j = |z_j|, \quad \rho_j = r_j^2, \quad \mathbb{R}^+ = \{x \in \mathbb{R} \mid x \geq 0\}.$$

Theorem. *The restriction to the center manifold of a germ of the class W_1^0 , W_2^I , W_2^J is topologically equivalent to a germ whose N -jet is listed in Table 2, provided only that the normalized N -jet of the germ in question does not belong to the exceptional variety; this variety and the number N are also shown in the table.*

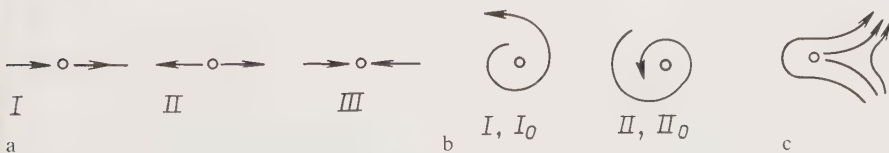


Fig. 13. Phase portraits of vector fields, a) of the class W_1^0 (cases I, II, III, b) of the class W_2^I (cases I and I^0 , II and II^0), c) of the class W_2^J ($a \neq 0$)

Table 2. Topological normal forms

Class	N	Normalized N-jet	Degeneracy of codimension 3	Class of topological equivalence	
				Definition	Notation
W_1^0	3	$\dot{x} = ax^2 + bx^3$	$a = b = 0$	I. $a \neq 0$	$W_1^{0,*}$
				II. $a = 0, b > 0$	$W_1^{0:0}$
				III. $a = 0, b < 0$	
W_2^I	5	$\dot{z} = z(i\omega + \alpha\rho + \beta\rho^2), \omega > 0$ $a = \operatorname{Re} \alpha, b = \operatorname{Re} \beta$	$a = b = 0$	I. $a > 0$	$W_2^{I,*}$
				II. $a < 0$	
				I ₀ . $a = 0, b > 0$ II ₀ . $a = 0, b < 0$	$W_2^{I:0}$
W_2^J	2	$\dot{x} = y, \dot{y} = ax^2$	$a = 0$	$a \neq 0$	$W_2^{J,*}$

For 1-dimensional and 2-dimensional systems, this result was known to Lyapunov.

Theorem [107], [35]. *The restriction to the center manifold of a germ of class $W_3^{0,I}$ or $W_4^{I,I}$ is, up to a change of sign (multiplication of the vector field by -1), topologically equivalent to a germ whose N-jet is listed in Table 3, provided that the normalized N-jet of the germ in question does not belong to the exceptional variety; this variety and the number N are shown in the table.*

Table 3. Topological normal forms

Class	N	Normalized N-jet	Degeneracy of codimension 3	Classes of topological equivalence	
				Definition	Notation
$W_3^{0,I}$	2	$\dot{x} = ar^2 + bx^2$ $\dot{z} = z(i\omega + \alpha x)$ $c = \operatorname{Re} \alpha$ $a > 0 \quad (*)$	$ab(b - c) = 0$ or $\begin{cases} c = 0 \\ b - c < 0 \end{cases}$	I. $a > 0, b - c > 0, b > 0$ II. $a > 0, b - c > 0, b < 0$ III. $a > 0, b - c < 0, b > 0, c > 0$ IV. $a > 0, b - c < 0, b < 0, c > 0$ V. $a > 0, b - c < 0, b < 0, c < 0$	$W_3^{0,I,*}$
$W_4^{I,I}$	3	$\dot{z} = Z(i\omega + \mathcal{A}\rho)$ $z = (z_1, z_2)$ $\rho = (\rho_1, \rho_2)$ $\omega = (\omega_1, \omega_2)$ $0 < \omega_1 < \omega_2$ $A = \operatorname{Re} \mathcal{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ $\det A = \Delta$ $b_1 = a_{11} - a_{21}$ $b_2 = a_{12} - a_{22}$ $a_{11} \geq a_{22} \quad (*)$	$a_{11}a_{22}b_1b_2 = 0$ or $\Delta = 0, b_1b_2 \leq 0$ or $k\omega_1 = \omega_2,$ $k \in \{1, 2, 3\}$	I. $a_{22} < 0 \quad (**), b_1b_2 > 0$ a) $a_{11} > 0, b_1 > 0$ b) $a_{11} > 0, b_1 < 0$ c) $a_{11} < 0$ II. $b_1 < 0 \quad (**), b_1b_2 < 0$ a) $a_{11} > 0, a_{22} > 0, \Delta < 0$ b) $a_{11} > 0, a_{22} < 0, \Delta < 0$ c) $a_{11} > 0, a_{22} < 0, \Delta > 0$ d) $a_{11} < 0, a_{22} < 0, \Delta < 0$ e) $a_{11} < 0, a_{22} < 0, \Delta > 0$	$W_4^{I,I,*}$

Remark. Throughout this section, a condition marked (*) will be deemed to be satisfied if it is satisfied after making the substitution $(x, r) \mapsto (-x, r)$ or $(z_1, z_2) \mapsto (z_2, z_1)$. An inequality (**) will be deemed satisfied, if it can be satisfied by reversing time. In Table 3 the condition I is imposed on all germs in the subclasses I_a, I_b, I_c and condition II on all germs in $II_a, \dots, II_e$.

In the last 5 rows of Table 3, we list all combinations of signs of the nonzero quantities a_{11}, a_{22}, Δ which are compatible with the inequalities $a_{11} \geq a_{22}, b_1 < 0, b_1 b_2 < 0$. The combinations $-++$ and $-+-$ drop out for obvious reasons and the combination $+++$ drops out because the inequalities $a_{11} > 0, a_{22} > 0, b_1 < 0$ and $b_2 > 0$ imply $a_{21} > a_{11} > 0$ and $a_{12} > a_{22} > 0$ which, in turn, mean that $\Delta < 0$.

5.3. Phase Portraits of Normal Forms

1. All the equations in Table 3 can be integrated by quadratures. For germs in $W_3^{0,I}$, they are conveniently investigated by passing to the quotient system relative to the variables (x, r) and, for germs in $W_4^{I,I}$, by passing to the quotient system relative to (ρ_1, ρ_2) . In the first case, the quotient system is defined in the half-plane $r \geq 0$, and in the second case in the quadrant

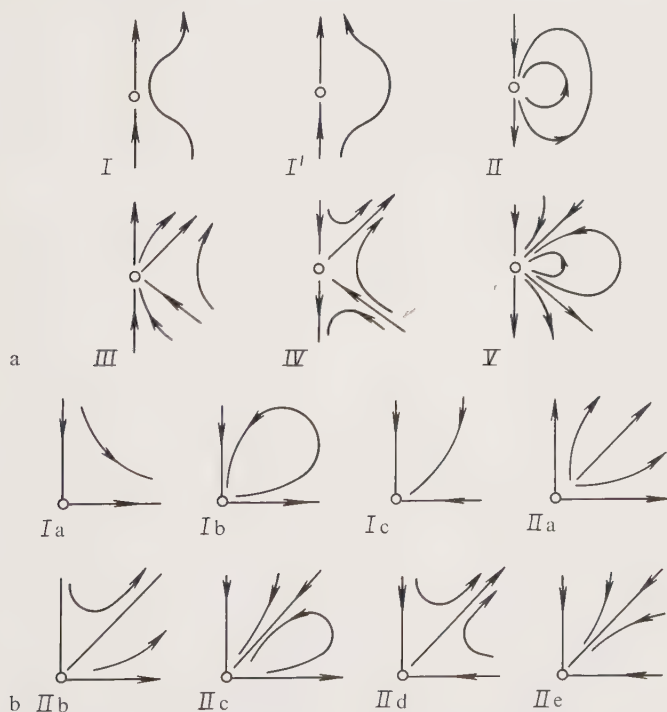


Fig. 14. a) Phase portraits of quotient systems corresponding to nondegenerate equations of the class $W_3^{0,I}$. Figs. I and I' represent the phase portraits of topologically equivalent quotient systems for $c > 0$ and $c < 0$ respectively; the difference between them shows up at bifurcations. b) Phase portraits of quotient systems corresponding to nondegenerate equations of class $W_4^{I,I}$.

$\rho_j \geq 0$. Their phase portraits are shown in Figs. 14a and 14b. These diagrams and the preceding theorems allow us to obtain a lot of information about the behaviour of the phase curves of the original germ. For example, for a suitable representative of any germ of the fourth sub-class $W_3^{0,I;*}$, the union of the O^+ -curves and the O^- -curves (these are the curves which enter the point O as $t \rightarrow +\infty$ and $t \rightarrow -\infty$ respectively; together they are called O -curves) is homeomorphic to the union of the x -axis of the space $\mathbb{R}^3 = \{(x, z)\}$ and the cone $x^2 - |z|^2 = 0$ with the origin removed.

2. *Problem*: Classify the germs listed in Table 3 up to (finite) smooth equivalence.

5.4. Criteria for Lyapunov Stability of Degeneracies of Codimension up to and Including Three. The preceding theorems allow us to easily establish criteria for Lyapunov stability of degeneracies of codimension up to and including two. These criteria were essentially obtained by Lyapunov himself (cf. [66] and the references therein). Table 4 shows the subclasses in Tables 2 and 3 whose representatives are stable at the point O . The third column gives the conditions which signify codimension three degeneracies in the stability problem. The notation is the same as in Tables 2 and 3.

The criteria below are due to many authors (references in [66]). We set these out in Table 5, where we have given the quotient system instead of

Table 4. Stability criteria in the case of degeneracies of codimension 1 and 2

Class	Stable subclasses	Classes of germs having degeneracies of codimension three	
		Definition ^a	Notation
W_1^0	III	$a = b = 0$	$W_1^{0;0,0}$
W_2^I	II and II ₀	$a = b = 0$	$W_2^{I;0,0}$
W_2^J	$\emptyset$	$a = 0$	$W_2^{J;0}$
$W_3^{0,I}$	$\emptyset$	$a = 0$	$W_3^{0,I;0}$
$W_4^{I,I}$	Ic, IIe and -IIa (obtained from IIa by reversing the time)	$a_{11} a_{22} = 0$	A_1
		$b_1 b_2 < 0, \Delta = 0$	A_2
		$\omega_1 = \omega_2$	A_3
		$2\omega_1 = \omega_2$	A_4
		$3\omega_1 = \omega_2$	$A_5; A = \bigcup_1^5 A_j$

^a The nonzero coefficient a_{ii} must be negative in the definition of the class A_1 . In the definition of the classes A_2 - A_5 , we must have $a_{11} < 0$ and $a_{22} < 0$.

the normalized jet whenever convenient. Any germ which does not satisfy the stability criteria and which does not belong to the exceptional variety is unstable. The notation is the same as in Tables 2 and 3.

In Table 5, $\mathbf{1}$ denotes the vector with each component equal to 1 and A is the Jordan normal form of the linear part of the restriction of the germ to the center manifold. The symbol $\emptyset$ in the column headed "stability criterion" means that the nondegenerate jet is unstable.

Remark. The last four lines show the codimension 4 degeneracies which are distinguished only by conditions on the nonlinear terms. One must add the classes which correspond to additional degeneracies of the linear part and to which the classes $W_3^J, \dots, W_6^{I,I,I}$ are adjacent, in the sense of § 5.5 below; for example, the class $W_3^{J,0}$ is adjacent to the W_3^J :

$$\dim W^C = 3, \quad A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix};$$

the class W_4^J : $\dim W^C = 4$, with A a nilpotent Jordan block of order 4, is adjacent to $W_4^{I,J}$, and so on. In the last row, and in the rows corresponding to the classes A_1 and A_2 , the additional degeneracies of the linear part relate, among other things, to "internal resonances of order no higher than 5", i.e., relations of the form $\omega_j = (k, \omega)$, $2 \leq |k| \leq 5$, $k \in \mathbb{Z}_+^3$;

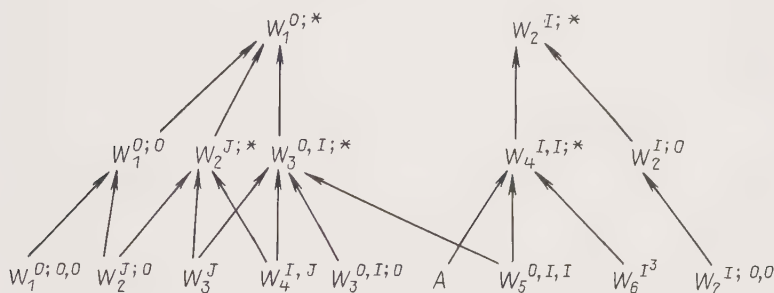


Fig. 15. The adjacency diagram for the Lyapunov stability problem. When it is not too complicated, the notation encodes the definition of the class. The subscript gives the dimension of the center manifold. The number preceding the semicolon in the superscript indicates the eigenvalues of the linear part on the imaginary axis and the corresponding Jordan blocks; for example, $W_3^{0,I,I}$ is the class of vector fields whose linear part has one zero and two pairs of purely imaginary eigenvalues. The part of the superscript after the semicolon refers to the nonlinear terms: an asterisk after the semicolon in the superscript of a class of codimension two indicates the absence of the codimension 3 degeneracies listed in Tables 2 and 3; the number of zeros after the semicolon indicates the number of degeneracies in the nonlinear terms. Tables 2–5 completely detail the notation.

Table 5. Stability criteria for codimension three degeneracies (the subclasses A_j of $W_4^{I,I}$ have been labelled as in Table 4.)

Class ^a	N	Normalized jet	Stability criterion	Degeneracy of codimension 4
$W_1^{0,0,0}$	4	$\dot{x} = cx^4$	$\emptyset$	$c = 0$
$W_2^{1,0,0}$	7	$\dot{z} = z(i\omega + \gamma\rho^3)$	$\operatorname{Re} \gamma < 0$	$\operatorname{Re} \gamma = 0$
$W_2^{2,0}$	4	$\dot{x} = y + x^2(b + cx)$ $\dot{y} = -x^3(d + ex)$	$b^2 + 2d < 0$ $2be - 5cd < 0$	Linear part of the restriction of the germ to W^c is equal to 0, or $(b^2 + 2d)(5cd - 2be) = 0$
$W_3^{0,1,0}$	3	$\dot{x} = br^2 + dx^3$, $\dot{z} = i\omega z + \alpha xz$, $c = \operatorname{Re} \alpha$	$\begin{cases} bc < 0 \\ d < 0 \end{cases}$	$bcd = 0$
A_1	5	$\dot{\rho}_1 = \rho_1(a_{12}\rho_2 + a_{31}\rho_2^2)$, $\dot{\rho}_2 = \rho_2(a_{21}\rho_1 + a_{22}\rho_2)$	$\begin{cases} a_3 < 0 \\ b_1 b_2 < 0 \end{cases}$ or $\begin{cases} a_3 < 0, \\ b_1 b_2 > 0 \\ b_1 \Delta < 0 \end{cases}$	$a_{22} a_{31} b_1 b_2 = 0$ or $b_1 b_2 < 0, \Delta = 0$
A_2	5	$\dot{\rho}_1 = \rho_1(a_{11}\rho_1 + a_{12}\rho_2 + f_2(\rho))$ $\dot{\rho}_2 = \rho_2(a_{21}\rho_1 + a_{22}\rho_2 + g_2(\rho))$ f_2, g_2 are homogeneous, 2nd-degree polynomials	$K < 0$ $K = a_{12}g_2(a, 1) \cdot a_{22}f_2(a, 1)$ $a = \frac{a_{12}}{a_{11}} - \frac{a_{22}}{a_{21}}$	$a_{11} a_{22} K = 0$
A_3	3	$\dot{z}_1 = i\omega z_1 + z_2$ $\dot{z}_2 = i\omega z_2 + \alpha z_1 \rho_1$	$\emptyset$	$\operatorname{Im} \alpha = 0$ or $\Delta = \operatorname{diag}(\pm i\omega, \pm i\omega)$
A_4	4	$\dot{z}_1 = i\omega z_1 + \alpha \bar{z}_1 z_2$ $\dot{z}_2 = 2i\omega z_2 + \beta z_1^2$	$\emptyset$	$\alpha\beta = 0$ or $\operatorname{Im}(\beta/\alpha) = 0, \operatorname{Re}(\beta/\alpha) < 0$
A_5		There is no algebraic criterion for stability		

W_3^J $\dim W^c = 3$ $A = J_3 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$	2	$\dot{x}_1 = x_2$ $\dot{x}_2 = x_3$ $\dot{x}_3 = ax_1^2$	$\emptyset$	$a = 0$
$W_4^{I,J}$ $\dim W^c = 4$ $A = \begin{pmatrix} I & \\ & J_2 \end{pmatrix}$	2	$\dot{z} = i\omega z$ $\dot{x}_1 = x_2$ $\dot{x}_2 = ax_1^2$	$\emptyset$	$a = 0$
$W_5^{0,I,I}$ $\dim W^c = 5$ $A = \begin{pmatrix} 0 & & \\ & I_1 & \\ & & I_2 \end{pmatrix}$	2	$\dot{x} = ax^2$ $\dot{z}_1 = i\omega_1 z_1$ $\dot{z}_2 = i\omega_2 z_2$	$\%_0$	$a = 0$
$W_6^{I,I,I}$ $\dim W^c = 6$ $A = \mathbb{R}\text{-diag}(i\omega)$ $\omega = (\omega_1, \omega_2, \omega_3)$ $\omega_j \neq 0$	3	$\dot{z} = Z(i\omega + \mathcal{A}\rho), \mathcal{A} \in \text{Hom}(\mathbb{C}^3, \mathbb{C}^3)$ $A = \text{Re } \mathcal{A}, \rho = (\rho_1, \rho_2, \rho_3)$ $\mathcal{B}$ is the set of principal minors of the matrix A	For every $B \in \mathcal{B}$ the equation $B\xi = \mathbf{1}$ has no solution with all its components > 0	There is a $B \in \mathcal{B}$ such that the equation $B\xi = 0$ has a solution with all its components > 0

^a The first nine classes are defined in Table 4

5.5. The Adjacency Diagram. We describe the adjacencies between the classes of germs listed in Tables 4 and 5.

Definition. Let A and B be two non-intersecting classes of germs of vector fields at the singular point O . We shall say that B is *adjacent to* A , and write $B \rightarrow A$, if, for every germ v of B , there is a continuous deformation joining v with germs of the class A ; more precisely, a continuous family of germs $\{v_t | t \in [0, 1]\}$ such that $v_0 = v$ and v_t is a germ in A for all $t \in (0, 1]$.

5.6. Theorems on Algebraic Solubility

Theorem. *The stability problem and the problem of topological classification of germs of vector fields are algebraically soluble for the classes W_1^0 , W_2^I , W_2^J .*

This theorem uses the reduction theorem in § 4. For the class W_1^0 it is almost obvious; and for the classes W_2^I and W_2^J it follows from the Poincaré-Lyapunov results (Chap. 5, § 4); see also [75, Chap. 5].

Theorem (Yu.S. Il'yashenko, Proc. of I.G. Petrovsky Sem., v. 12 (1981) 119–136). *The Lyapunov stability problem for germs of vector fields is algebraically soluble, up to degeneracies of arbitrary codimension, for the classes $W_3^{0,I}$ and $W_4^{I,I}$, under the additional condition that the ratio ω_1/ω_2 is irrational and fixed. The jet of the restriction to the center manifold of a vector field in either of these classes, when written in Poincaré-Dulac normal form (§ 5.2) in the variables x, z or z_1, z_2 , respectively, is stable if and only if the corresponding quotient system relative to the variables x, r or ρ_1, ρ_2 is stable.*

Problem. Does a similar theorem hold for the problem of the topological classification of germs of these classes? (See F. Dumortier in Lect. Notes in Math., 1985, No. 1125.)

§ 6. Smooth Classification of Germs of Vector Fields

6.1. The Relation Between Formal Classification and Smooth Classification. The generic case has been investigated by K.T. Chen [46: 1].

Theorem. *If two germs of smooth vector fields at a hyperbolic singular point are formally equivalent, then they are smoothly equivalent.*

Problem. Does a similar theorem hold for nonhyperbolic germs with one-resonant linear parts?

Germs of this type belong to the classes W_1^0 and W_2^I (§ 5.2). Let us assume in addition, that the restriction of the germ to the center manifold has nonzero nonlinear terms when written in Poincaré-Dulac formal normal form. In this case, F. Takens proved in 1973 [15: 90] that the answer is affirmative for germs of class W_1^0 on the line and for germs of class W_2^I on the plane.

In the higher dimensional case, an affirmative answer for these classes has recently been obtained by G.P. Belitskiy, "Smooth equivalence of germs of vector fields with one zero or an imaginary pair of eigenvalues", *Funct. Anal. Appl.* **20** (1986), 1–8.

6.2. Germs of Vector Fields with Symmetries

Theorem [15]. *For every germ of a smooth vector field at the point O in the space $\mathbb{R}^n$, and for every Euclidean structure on $\mathbb{R}^n$, there is a formally equivalent germ with linear part Ax and nonlinear part which commutes with the vector field A^*x (where $*$ denotes the conjugation with respect to the chosen Euclidean structure).*

Consider a Hermitian scalar product on $\mathbb{C}\mathbb{R}^n$ in which the Jordan basis of A invariant under complex conjugation is orthonormal. To apply the previous theorem it is convenient to take this scalar product as the Euclidean structure on $\mathbb{R}^n$.

Corollary 1. In the absence of internal resonances $\omega_2 = k\omega_1$, $k = 1, 2, \dots$, $0 < \omega_1 < \omega_2$, the germ at O of a smooth vector field on $\mathbb{R}^n$ of class $W_4^{I,I}$ is formally equivalent to the germ defined by the differential equation

$$\dot{z} = Z(i\omega + f(\rho)), \quad \omega = (\omega_1, \omega_2), \quad (2)$$

where f is the germ at O of a smooth, complex-valued vector field and $Z = \text{diag}(z_1, z_2)$.

Corollary 2. *The germ of a smooth vector field of class W_2^J on the plane is formally equivalent to the germ of the vector field defined by the differential equation*

$$\dot{x} = y + x^2 f(x), \quad \dot{y} = x^2 g(x),$$

where f and g are germs at O of smooth functions.

In the case of equation (2), it is easy to investigate the asymptotic behaviour of phase curves tending to O and of the invariant manifolds through O . This is because equation (2) is symmetric relative to rotations

$$(z_1, z_2) \mapsto (\lambda z_1, \mu z_2), \quad |\lambda| = |\mu| = 1,$$

and, thus, defines a quotient system on the plane $\rho = (\rho_1, \rho_2)$. The investigation of the O -curves of the original system reduces to the investigation of the O -curves of the quotient system, for which there are well-established techniques (cf. Chap. V).

So the implication

$$\text{"formal equivalence"} \Rightarrow \text{"smooth equivalence"}$$

is meaningful even when the formal normal form depends on an infinite number of nonlinear terms.

6.3. Quasi-Hyperbolicity. The implication “formal equivalence $\Rightarrow$ smooth equivalence” holds not only for germs of vector fields with a hyperbolic linear part (Chen’s theorem), but also for the so-called “quasi-hyperbolic” germs (see [15], in which germs of diffeomorphisms are considered). Like hyperbolic systems, quasi-hyperbolic germs of vector fields have stable and unstable sets; but these sets are not necessarily manifolds, and the phase curves do not approach the singular point along the stable set exponentially, but only as a power, *i.e.*, the modulus of the solution decreases as a power of t when $t \rightarrow +\infty$. Solutions whose phase curves lie on the unstable set behave similarly when $t \rightarrow -\infty$.

6.4. Finitely Smooth Equivalence of Germs of Vector Fields. Generally speaking, a vector field with a resonant linear part can be reduced to linear normal form by a C^r substitution, but r will be less than the smallest degree of the resonant terms. The first result in this direction is due to S. Sternberg [46: 3]. Various theorems devoted to increasing the smoothness of the conjugating homeomorphism have been obtained by V.A. Kondrat’ev, V.S. Samovol [15: 44], and G.R. Belitskiy [15]. The strongest result is due to V.S. Samovol (Linearization of systems of ordinary differential equations in a neighbourhood of invariant toroidal manifolds, *Proc. Moscow Math. Soc.*, **38** (1979), 187–219) and is the following.

Theorem. Consider a smooth hyperbolic germ of a vector field at a singular point. Let $\{\lambda_j, \mu_k\}$ be the eigenvalues at this point and order them as follows:

$$\operatorname{Re} \lambda_s \leq \dots \leq \operatorname{Re} \lambda_1 < 0 < \operatorname{Re} \mu_1 \leq \dots \leq \operatorname{Re} \mu_u.$$

Suppose that the following condition $S(k)$ holds for some positive integer k . For each resonance of the form

$$\lambda_j = \sum_{m=1}^s \lambda_m r_m^- + \sum_{l=1}^u \mu_l r_l^+ \quad \text{or} \quad \mu_j = \sum_{m=1}^s \lambda_m r_m^- + \sum_{l=1}^u \mu_l r_l^+$$

there is an $m \leq s$ or an $l \leq u$ such that

$$k \operatorname{Re} \mu_l < \operatorname{Re}(\mu_1 r_1^+ + \dots + \mu_l r_l^+)$$

or

$$-k \operatorname{Re} \lambda_m < -\operatorname{Re}(\lambda_1 r_1^- + \dots + \lambda_m r_m^-).$$

Then the germ is C^k -equivalent to its linear part.

Sell’s theorem [99, p. 1292] is a consequence of the above.

§ 7. Normal Forms of Vector Fields in which the Linear Part is a Nilpotent Jordan Block

The study of normal forms whose linear part is a nilpotent Jordan block of any order is facilitated by the theory of representations of the Lie algebra

sl (2). The heart of the matter is the following algebraic problem. Consider the phase flow of a linear vector field defined by a nilpotent Jordan block. The corresponding one-parameter group of linear transformations also acts on various tensor spaces. For example, it acts on the space of vector fields whose coefficients are homogeneous polynomials of fixed degree. The problem is to determine the Jordan structure of these transformations on the tensor spaces.

7.1. Centralized Chains. The representation theory of the Lie algebra sl (2) enables us to find the number of blocks and say something about the Jordan basis (V.N. Bogaevskiy, A.Ya. Povzner, A.B. Givental). We index the basis vectors of the original $(d+1)$ -dimensional Jordan block by the integers going from $-d$ to $+d$ in steps of 2 (the convenience of this enumeration will become clear later). Thus, the Jordan block J acts on the basis by the rule

$$e_{-d} \mapsto e_{-d+2} \mapsto \dots \mapsto e_d \mapsto 0. \quad (3)$$

Let the corresponding coordinate functions be $x_{-d}, \dots, x_d$ and set $\deg x_a = a$ and $\deg \left(\frac{\partial}{\partial x_a} \right) = -a$. The weight of a monomial is defined to be the sum of the weights of its components, e.g.,

$$\deg x_a^p = p a, \quad \deg \left(x_b \frac{\partial}{\partial x_a} \right) = b - a, \quad \left(\frac{\partial}{\partial x_a} = e_a \right).$$

The weight of a sum of elements of the same weight is that of any element.

Theorem. Consider the action of the group e^{Jt} on a vector space V and the corresponding group of transformations on a space L of tensors on V . The Jordan chains of the operator generating the group on the tensor space can be chosen to be centralized (that is, the extreme elements of each chain have opposite weights and moving along each chain decreases the weight in steps of 2 as in formula (3)).

7.2. Non-Removable Terms. The dimension of the space of homogeneous vector polynomials of degree N which enter the residual of a formal normal form of a vector field with the linear part Jx does not exceed the number of Jordan blocks of the map $\text{ad } J$. Here, $\text{ad } J$ is the map on the space L of all homogeneous vector polynomials of degree N given by commutation with J . By the previous theorem, the number of these blocks is equal to the number of centralized chains. The dimension of a space of tensors of fixed weight can be calculated without difficulty. Therefore, after some calculation, we obtain the following corollary from the theorem.

Corollary. A vector field with a nilpotent linear part consisting of one Jordan block of order n is formally equivalent to a vector field whose terms of weight

μ and homogeneous degree $N-1$ belong to a linear space of dimension

$$\text{res}_{x=0} \left[x^{\mu/2-s} (1-x)^n \text{res}_{t=0} \left(t^{-N} \prod_{k=1}^n (1-tx^k)^{-1} \right) \right]$$

where $S = N(n+1)/2$.

In particular, all chains with a fixed N have the same parity: if n is odd, then the length of all chains is odd; if n is even, then the parity of the length of a chain is equal to the parity of N .

§§ 7.3-7.5 are devoted to proving the theorem for the space L defined at the beginning of § 7.2.

7.3. The Standard Representation of the Group SL (2) and of the Algebra sl (2). The group SL (2) of area-preserving, linear transformations of the plane naturally acts on the space P^{m+1} of homogeneous polynomials of degree m in two variables. Consider the action of the "Cartan" subgroup of hyperbolic rotations and the "Borel" subgroup of shifts:

$$H^t(x, y) = (e^t x, e^{-t} y), \quad S^t(x, y) = (x + t y, y).$$

Under the action of the hyperbolic rotations the monomials $x^m, x^{m-1}y, \dots, y^m$ are multiplied by $e^{mt}, e^{(m-2)t}, \dots, e^{-mt}$. Therefore, the eigenvalues of the generating transformation A_H of the action of the Cartan subgroup are equal to $m, m-2, \dots, -m$. We index the eigenvectors of A_H by the corresponding eigenvalues.

Under a shift S^t , the restriction of a polynomial to the straight line $y = y_0$ is shifted through $y_0 t$. Therefore the generating operator A_S of the action of the Borel subgroup reduces to differentiation with respect to x followed by multiplication by y . That is, the operator carries each eigenvector of the Cartan subgroup into the one following (thereby reducing the eigenvalues of the generating operator A_H by 2 units).

We call this the *standard representation* of the group SL (2) (and the corresponding algebra representation is called the *standard representation* of the algebra sl (2)).

7.4. Extension of a Nilpotent Operator to a Representation of the Lie Algebra sl (2). A linear map J which acts on a $(d+1)$ -dimensional linear space V in accordance with formula (3) can be interpreted as an element of a representation V of the Lie algebra sl (2) of the group of area-preserving transformations of the plane. To do this, it suffices to carry the standard representation of the group SL (2) (and the algebra sl (2)) over to V by the isomorphism $C: V \rightarrow P^{d+1}$, putting $m = d$ in the formulae of the previous section:

$$e_{-d} \mapsto \frac{x^d}{d!}, \quad e_{-d+2} \mapsto \frac{x^{d-1}y}{(d-1)!}, \dots, e_d \mapsto y^d.$$

This isomorphism converts the action of J into the action of the generating operator of the Borel subgroup defined in § 7.3, and A_H into the operator $\tilde{A}_H: e_d \mapsto -de_d, \dots, e_{-d} \mapsto de_{-d}$. The representations of the Lie group $SL(2)$ and the Lie algebra $\mathfrak{sl}(2)$ on V obtained in this way are called the *representations associated with the operator J* . The following theorem asserts that every irreducible representation of the Lie algebra $\mathfrak{sl}(2)$ can be obtained in a similar way.

Theorem. *Every reducible representation of the Lie algebra $\mathfrak{sl}(2)$ splits into irreducible ones, and every irreducible representation is conjugate to a standard representation.*

7.5. Conclusion of the Proof of the Theorem. Let L be the space of vector fields on V whose components are homogeneous polynomials of degree N in $d+1$ variables (or some other space of tensors over V). Let $J: V \rightarrow V$ be the operator above. Consider the representation on V of the algebra $\mathfrak{sl}(2)$ associated with J , and the corresponding representation T of $\mathfrak{sl}(2)$ on L . The eigenvectors of the map T_{A_H} corresponding to the generating operator of the Cartan subgroup are vector polynomials, consisting of monomials of a single weight, with eigenvalue equal to the weight. We decompose the representation T into irreducible representations. Let V^{m+1} be the space of one of the irreducible representations with dimension $m+1$. By the previous theorem, the irreducible representation $T|_{V^{m+1}}$ is conjugate to a standard representation. Therefore, the eigenvectors of the restriction of T_{A_H} to V^{m+1} form a centralized chain

$$v_m, v_{m-2}, \dots, v_{-m},$$

and the weight of the vector polynomial v_a is equal to a . Now, we consider the operator T_{A_S} corresponding to the generating operator A_S of the Borel subgroup. Its restriction to V^{m+1} acts by the formula

$$v_m \mapsto v_{m-2} \mapsto \dots \mapsto v_{-m} \mapsto 0 \quad (4)$$

(if necessary, we replace the vectors v_a by collinear ones).

In conclusion, we note that T_{A_S} is indeed the generating operator mentioned in the theorem of § 7.1. Namely, it corresponds to commutation with the vector field Jx , where x is a column with the components $x_d, \dots, x_{-d}$. This is because the Borel subgroup $\{S^t\}$ acts on v by the rule $x \mapsto (\exp Jt)x$, and on L by sending the field v to the field $\tilde{v} = e^{Jt} v \circ e^{-Jt}$. The generating operator of the latter is the operator corresponding to commutation with the field Jx .

By the same token, (4) is the required centralized chain of the operator $\text{ad}(Jx)$.

Chapter 4

Singular Points of Differential Equations in Higher Dimensional Complex Phase Space

The local theory of analytic differential equations is the oldest and best developed branch of the local theory of differential equations.

§ 1. Linear Normal Forms

A germ of a generic analytic vector field at a singular point is analytically equivalent to its linear part, as the theorems below show.

1.1. Poincaré Domains and Siegel Domains. Small Denominators

Definition. An n -tuple $\lambda \in \mathbb{C}^n$ belongs to the *Poincaré domain* if the convex hull of the vectors $\lambda_1, \dots, \lambda_n$ in the complex plane does not contain the origin; otherwise it belongs to the *Siegel domain*. An n -tuple $\lambda \in \mathbb{C}^n$ is a *strict Siegel n -tuple* if the convex hull of $\lambda_1, \dots, \lambda_n$ contains 0 and a neighbourhood of 0.

Definition. A linear vector field is of *Poincaré type*, *Siegel type*, or *strict Siegel type* if the spectrum of the corresponding operator belongs to the Poincaré domain, the Siegel domain, or the strictly Siegel domain.

The set of eigenvalues is said to be *resonant* if one of them is the sum of not less than two eigenvalues (not necessarily distinct).

In this section, the formal substitutions which reduce germs of analytic vector fields with nonresonant linear part to linear normal forms will be called *normalizing series*. In calculating them one has to divide by the expressions

$$(\lambda, k) - \lambda_j, \quad (k, j) \in J, \quad (*)$$

where

$$J = \{(k, j) \mid k \in \mathbb{Z}_+^n, \quad |k| = \sum k_i \geq 2, \quad j \in \{1, \dots, n\}\}.$$

Some of these expressions vanish for resonant n -tuples. For nonresonant n -tuples λ the set of numbers $\{(\lambda, k) - \lambda_j \mid (k, j) \in J\}$ has 0 as a limit point if and only if λ belongs to the Siegel domain. The numbers in this set are called *small denominators*; their appearance impedes the convergence of the normalizing series.

1.2. Convergence of the Normalizing Series

Poincaré's Theorem [7], [86]. *A germ of an analytic vector field with nonresonant linear part of Poincaré type is analytically equivalent to its linear part.*

Siegel's Theorem [7]. *For almost all (in the sense of Lebesgue measure) sets of eigenvalues of linear parts of germs of holomorphic vector fields at a singular point, the germ of a holomorphic vector field is biholomorphically equivalent to its linear part.*

There is a condition on the eigenvalues which is almost always satisfied and is sufficient for a germ to be holomorphically equivalent to its linear part. It demands that the small denominators be bounded from below by the order¹ of the corresponding resonance and is formulated precisely as follows.

Definition. A set $\lambda \in \mathbb{C}^n$ is *Bryuno incommensurable* if there are positive constants C and ε such that

$$|(\lambda, k) - \lambda_j| > C \exp(-|k|^{1-\varepsilon}) \quad \text{for all } (k, j) \in J$$

for which $(\lambda, k) - \lambda_j \neq 0$ (the analogous estimate with right-hand side $C|k|^{-(n+1)}$ even holds for almost all $\lambda \in \mathbb{C}^n$).

Theorem [23:31]. *A germ at a singular point of an analytic vector field whose linear part has a nonresonant and Bryuno incommensurable spectrum is biholomorphically equivalent to its linear part.*

In [23:31] equivalence of a germ and its linearization is established under even weaker restrictions on λ .

Similar theorems hold when λ is a resonant set if the germ of the vector field is formally equivalent to its linear part (as, for example, in the case of a center in the linear terms when there is a first integral $x^2 + y^2 + \dots$). General theorems of this kind are to be found in the papers of A.D. Bryuno [23:31] and V.A. Pliss [15:54].

1.3. Analytic Theorems on Divergence of the Normalizing Series. If the small denominators are pathologically small, then the normalizing series will, as a rule, diverge.

Definition. An n -tuple $\lambda \in \mathbb{C}^n$ is *almost resonant* if the series

$$\sum z^k [(\lambda, k) - \lambda_j]^{-1} \frac{\partial}{\partial z_j}$$

does not converge in any ball centred at 0.

Theorem [57]. *Let λ be an almost resonant n -tuple and let the coefficients of the convergent series*

$$f(z) = \sum_j f_{kj} z^k \frac{\partial}{\partial z_j}$$

be bounded from below by a geometric progression:

$$|f_{kj}| > C q^{|k|}.$$

¹ The order of resonance is $|k|$ in the corresponding relation (*).

Then, for almost all (in the sense of Lebesgue measure) complex α , the equation

$$\dot{z} = (\text{diag } \lambda) z + \alpha f(z)$$

is not analytically equivalent to the linear equation in any neighbourhood of the point 0.

1.4. Geometric Theorems on the Divergence of the Normalizing Series. The geometric reason for the divergence of the normalizing series in the case of pathologically small denominators is the phenomenon of materialization of resonances; it consists in the following (see [7]).

Consider a one-parameter family of vector fields whose linear parts pass transversally through resonance when the value of the parameter is zero (more precisely, the spectra of the linear parts intersect the resonance plane transversely when the parameter passes through zero). When the parameter passes through zero, an analytic manifold separates from the coordinate planes of a chart. This manifold depends on the parameter and is an invariant manifold of the equation corresponding to the parameter value. (The simplest geometric example of this sort is the family of manifolds $xy = \varepsilon$; when $\varepsilon = 0$ the member of the family is the union of the coordinate axes.) The topology of this manifold is determined by the arithmetic of the resonance. The presence of a "large piece" of this manifold in a neighbourhood of the singular point prevents the reduction of the equation to linear normal form in this neighbourhood. If an n -tuple λ is pathologically close to a countable number of resonances, then every neighbourhood of the singular point of the equation

$$\dot{z} = (\text{diag } \lambda) z + f(z)$$

has a countable number of invariant manifolds, provided that the nonlinear part f is generic. Each arises when one of the resonances close to λ materializes; taken together, these manifolds prevent the convergence of the normalizing series in any neighbourhood of zero.

The geometric theorems proved in this way about the divergence [62], [62:6] assume that the small denominators decrease even more rapidly than in the previous section; namely, that for some $\sigma > 0$ and for any $C > 0$ there is a $k \in \mathbb{Z}_+^n$ such that

$$0 < |(\lambda, k) - \lambda_j| < C|k|^{-\sigma|k|}.$$

§ 2. The Relation Between Formal and Analytic Classification

In this section, we discuss the relation between the formal and analytic classification of the germs of vector fields whose linear parts have resonant, Bruno incommensurable spectra.

2.1. Condition A. Fix a germ of an analytic vector field whose linear part has Bryuno incommensurable spectrum. A.D. Bryuno [23:31] found a necessary and sufficient condition (called condition A) under which every analytic vector field which is formally equivalent to a fixed germ is analytically equivalent to the germ.

Condition A pertains to the nonlinear terms of the preliminary normal form of a germ v in the case when the linear part of the germ is resonant. All formally equivalent preliminary normal forms either satisfy or do not satisfy condition A simultaneously; therefore condition A is a condition on the germ itself and not merely on its normal form. The actual formulation of the condition is complicated and will be omitted. We note by way of an example that a germ of a vector field on the plane with linear part

$$p x \frac{\partial}{\partial x} - q y \frac{\partial}{\partial y} \quad (\text{where } p, q \text{ are natural numbers})$$

satisfies condition A if and only if the germ is orbitally analytically equivalent to its linear part.

Furthermore, condition A is automatically satisfied for all germs with linear part of Poincaré type. The convergence of the normalizing series in this case was proved by Dulac (see [23]). Condition A is rarely satisfied outside the Poincaré domain.

2.2. Problems Involving the Smooth and the Analytic Classification. The problem of effecting a smooth analytic classification of germs of vector fields with a singular point naturally divides into three parts.

I. Formal classification (see Chap. 3, § 3).

II. Investigation of the connection between the formal and smooth (or analytic) classifications. In the nonresonant case the formal normal form is linear. The connection between the smooth and analytic variants of the classification of nonresonant germs was described in Chap. 3, § 6 and Chap. 4, § 1. In the context of holomorphic germs, the resonant case has been investigated in as much detail as nonresonant case (see § 2.1). Because condition A is so stringent, the class of formally equivalent analytic germs of vector fields with a resonant linear part at a singular point hardly ever coincides with the class of analytically equivalent germs. With regard to the smooth case, see Chen's theorem and the other theorems in Chap. 3, § 6 and Chap. 6, § 2.

III. The problem of obtaining a (complete) system of invariants of the analytic classification of germs of vector fields. (As far as we know, the smooth analogue of this problem has been investigated only at the formal level; the only exception is a result obtained by S.M. Voronin [64, p. 143] and Martinet-Ramis [79, Chap. III, § 5].)

Up to now, attempts to solve this problem have been confined to seeking invariants of the formal classification. The convergence of the corresponding

formal substitutions was studied. If these converged, the problem was solved; if they diverged, the investigation was abandoned. Papers devoted to finding invariants of the analytic classification, as distinct from the formal classification, are discussed in Chaps. 5 and 7.

§ 3. Analytic Invariant Manifolds

An analytic system may have fewer analytic invariant manifolds than smooth invariant manifolds. (For example, a germ of an analytic vector field on $\mathbb{R}^2$ whose linear part has spectrum $(\lambda, 0)$, $\lambda \neq 0$, always has a C^r center manifold (Chap. 3, § 4), but there is no analytic invariant manifold as a rule (Chap. 5, § 5).

Theorems about stable (resp., unstable), analytic invariant manifolds were formulated in Chap. 3, § 4. Similar theorems hold for holomorphic vector fields [23:31]. The theorems formulated below about local invariant manifolds of holomorphic vector fields make it possible to find analytic invariant manifolds through a singular point of a real-analytic field which belongs to neither the stable nor unstable manifold of the point.

3.1. The Invariant Manifold Theorem. Consider a germ of an analytic vector field at the singular point 0 whose linear part has an invariant subspace. We investigate the question of whether the germ has an analytic invariant manifold tangent to the subspace at 0.

Consider the formal differential equation

$$\begin{aligned} \dot{x} &= Ax + f(x, y), & \dot{y} &= My + g(x, y), \\ f(0) &= 0, & g(0) &= 0, & f_*(0) &= 0, & g_*(0) &= 0, \\ x &= (x_1, \dots, x_n), & y &= (y_1, \dots, y_m). \end{aligned} \quad (1)$$

Definition. The subspace $y=0$ is called a *formal invariant manifold* of equation (1) if $g(x, 0)=0$. The equation $\dot{x}=Ax+f(x, 0)$ is called the *restriction of equation (1) to the formal invariant manifold*.

Definition. We shall say that a formal vector field belongs to the class *NFIM* (Normal Form on the Invariant Manifold) with linear part (Ax, My) if it corresponds to an equation (1) in which the subspace $y=0$ is a formal invariant manifold and if the restriction of (1) to this plane is in preliminary normal form.

Theorem (Yu.N. Bibikov) [19]. *Let v be the germ of an analytic vector field at a singular point which is formally equivalent to a formal vector field w in the class NFIM with linear part (Ax, My) . Let $\lambda=(\lambda_1, \dots, \lambda_n)$ be the spectrum of the matrix A , and $\mu=(\mu_1, \dots, \mu_m)$ be that of the matrix M . Suppose that following conditions hold:*

1: There are no "cross-resonances"

$$\mu_j \neq (\lambda, k) \quad \text{for all } k \in \mathbb{Z}_+^n, \quad |k| \geq 2, \quad j = 1, \dots, m.$$

2: λ is a resonant n -tuple without small denominators: 0 is an isolated point of the set of numbers $\{(\lambda, k) - \lambda_j | (k, j) \in J\}$.

3: The restriction of the equation (1), corresponding to the field w , to the subspace $y=0$, satisfies Bryuno's condition A (see [23:31]).

Then the germ v is analytically equivalent to a germ of the class NFIM with linear part $(\lambda x, My)$.

Some extremely general theorems about invariant manifolds and sets have been stated by A.D. Bryuno [23:40, 42, 43, 46], but the proofs have not yet been published.

3.2. Analyticity of the Center Manifold

Definition. Suppose that a formal vector field is written in preliminary normal form. The plane on which the hyperbolic variables vanish is called the *formal center manifold* of the field.

Remark. If the preliminary normal form is analytic, this definition gives the usual center manifold.

Theorem. If the restriction of the preliminary normal form to the center manifold has the form

$$\dot{z} = iz \sum_{k=0}^{\alpha} a_k z^k \bar{z}^k, \quad z \in \mathbb{C}, \quad \operatorname{Im} a_k = 0, \quad a_0 = \omega \neq 0,$$

or the form

$$\dot{x} = 0, \quad \dot{x} \in \mathbb{R},$$

then the center manifold is analytic and consists of closed phase curves (in the first case) or equilibrium points (in the second case).

This is an analogue of the Poincaré-Lyapunov theorem (Chap. 5, § 4). For smooth germs there are, generally speaking, no manifolds of fixed points or closed curves.

3.3. Reversible Systems. Differential equations which are invariant under time-reversal are called *reversible systems*.

Consider the equation $\ddot{x} + Ax = 0$ for small oscillations, where $x \in \mathbb{R}^n$ and the matrix A is symmetric and positive-definite. The corresponding phase space $\mathbb{R}^{2n}$ splits into a direct sum of 2-dimensional invariant planes L_j , $j = 1, \dots, n$. Each of these planes is filled up by closed phase curves, along which motion has frequency ω_j , where the ω_j^2 , $j = 1, \dots, n$, are the eigenvalues of A . The following theorem shows that if the equation is perturbed by nonlinear terms for which the resulting system remains invariant under time-

reversal, then the perturbed system will have, as in the linear case, n one-parameter families of closed phase curves. A weak restriction is imposed on the frequencies ω_j .

Theorem. *Let*

$$\ddot{x} + Ax = \Phi(x, \dot{x}), \quad \Phi(0, 0) = 0, \quad \Phi_*(0, 0) = 0 \quad (2)$$

be an analytic system invariant under time-reversal, i.e., suppose that $\Phi(x, p) = \Phi(x, -p)$ and that the matrix A , the planes L_j , and the frequencies ω_j are as above. Suppose, further, that $\omega_j \neq k\omega_l$ for any $k \in \mathbb{Z}$, $j \neq l$. Then there are n two-dimensional analytic surfaces in the phase space $\mathbb{R}^{2n}$ of (2) which are tangent to the planes L_j at zero and which are filled up by closed phase curves of the system.

A stronger theorem is proved in [19].

Reversible systems have many properties in common with Hamiltonian systems. (See V.I. Arnold, *Reversible Systems*, in *Nonlinear and Turbulent Processes*. Acad. Pr., New York, 1984, pp. 1161–1174.)

3.4. Analytic Center Manifolds of Differential Equations in the Plane. In this section we investigate differential equations in a neighbourhood of a singular point in $\mathbb{R}^2$ with the property that one of the eigenvalues of the linear part is equal to zero, and the other is different from zero:

$$\begin{aligned} \dot{x} &= f_1(x, y), & \dot{y} &= \lambda y + f_2(x, y) \\ \lambda &\neq 0, & f_j(0, 0) &= 0, \quad df_j(0, 0) = 0, \quad j = 1, 2. \end{aligned} \quad (3)$$

Equations of this sort always have a C^∞ center manifold (a two-dimensional phenomenon), but do not, in general have an analytic center manifold. The latter circumstance was noted by Euler in the case of the equation

$$\dot{x} = x^2, \quad \dot{y} = y - x$$

(which is reducible to the form (3) by a linear substitution). It turns out that the question of whether there is an analytic center manifold of equation (3) cannot be decided from the normal form of the equation at an isolated singular point 0: in every class of formally equivalent equations of this type there are some equations which have, and some equations which do not have, an analytic center manifold [78]. In order that all equations of the form (3) which are formally equivalent to one another should have an analytic center manifold it is necessary and sufficient that the singular point 0 should not be an isolated point; in that case the center manifold consists of the singular points. This condition defines a set of equations of infinite codimension. Meanwhile, we note that the set of equations (3) which have an analytic center manifold has complex codimension one in the class of all equations of the form (3) [78].

§ 4. Topological Classification of Singular Points in the Complex Domain

The theorems formulated below show that, in contrast to the real case, the topological classification of vector fields in a complex space is not even discrete (it has continuous moduli) for generic linear vector fields.

4.1. Linear Vector Fields

Definition. A linear vector field in a complex phase space is said to be $\mathbb{C}$ -hyperbolic (resp. weakly hyperbolic) if no two eigenvalues of the corresponding linear map have a real ratio (resp., a real non-positive ratio).

Guckenheimer's Theorem. *Germs of holomorphic vector fields on $\mathbb{C}^n$ at a singular point 0 with a hyperbolic linear part of Poincaré-type are orbitally topologically equivalent to one another.*

The situation changes abruptly for linear parts of Siegel type.

Theorem (N.N. Ladis, C. Camacho, N.M. Kuiper, R. Palis, Yu.S. Il'yashenko; see the references in § 36B of [8]). *Diagonalizable linear vector fields whose spectra lie strictly within the Siegel domain are orbitally topologically equivalent in a neighbourhood of 0 of the space $\mathbb{C}^n$ if and only if the sets of reciprocals of their eigenvalues can be carried into one another by an $\mathbb{R}$ -linear mapping of $\mathbb{C}$ into $\mathbb{C}$.*

Theorem (Yu.S. Il'yashenko, N.N. Ladis) [52]. *Two strictly Siegel, nondegenerate vector fields which have a nontrivial Jordan block are orbitally topologically equivalent if and only if they are affinely equivalent.*

Problems. To obtain a complete orbital topological classification of linear vector fields on $\mathbb{C}^n$ it is necessary to analyze the following remaining cases:

1. linear vector fields whose spectra lie on the boundary of the Siegel domain;
2. linear vector fields which have a nontrivial Jordan block with a zero eigenvalue.

4.2. The Nonlinear Case. A germ of a holomorphic nonresonant vector field at a singular point with spectrum in the Poincaré domain is topologically (and even analytically) equivalent to its linear part. This is also the case for germs with spectrum in the Siegel domain for almost all (in the sense of Lebesgue measure) sets of eigenvalues. This follows from the Poincaré and Siegel theorems. However, a germ is often topologically equivalent to its linear part when Siegel's theorem is inapplicable; small denominators do not prevent this.

M. Chaperon [42, p. 96] has proved that if the spectrum of the linear part of the germ of a holomorphic vector field at a singular point is a weakly hyperbolic set, then the germ is topologically equivalent to its linear part and the conjugating homeomorphism can be chosen to satisfy a Hölder condition.

Chapter 5

Singular Points of Vector Fields in the Real and Complex Planes

In this chapter, we describe a method for resolving singularities. This enables us to replace an arbitrarily degenerate singular point of a vector field on the plane by so-called elementary singularities. The latter have been thoroughly investigated for both the real and the complex plane.

§ 1. Resolution of Singularities

In this section we present a classical theorem of Bendixson and its modifications.

1.1. The Polar Blow-up and the σ -Process in the Plane. We start by defining a polar blow-up. First, we shall say that a smooth vector field is *flat* at a point if the vector field and all its derivatives vanish at the point.

Let v be a smooth vector field, which is not flat at 0, defined on a neighbourhood U of the singular point 0 in the plane $\mathbb{R}^2$. Let (r, φ) be the point with polar coordinates r and φ . Consider the mapping

$$U \setminus \{0\} \rightarrow \mathbb{R}^2, \quad (r, \varphi) \mapsto (r+1, \varphi),$$

which carries the domain $U \setminus \{0\}$ into a ring with interior boundary $r=1$. The vector field obtained from v under this mapping can be smoothly extended over a full neighbourhood of the circle $r=1$ so that, after division by a suitable power of $(r-1)$, the extended field is smooth on a neighbourhood of the circle $r=1$, has only a finite number of singular points on the circle, and is not flat at any of these points.

The construction is called the *polar blow-up* of the singular point of the vector field. The circle $r=1$ arising from the singular point is called the *pasted circle*.

Each of the singular points obtained after blowing-up is, in general, simpler than the original one. One can perform more polar blow-ups to further simplify each of the new singular points. Before describing the simplest points which can be obtained by this process, we define an algebraic variant of the blow-up, which is called the σ -process.

We consider the natural mapping of the punctured real plane $\mathbb{R}^2 \setminus \{0\}$ to the projective line $\mathbb{R}P^1$: to each point of the punctured plane, we assign the straight line joining that point to the origin. Let M denote the graph of the mapping; its closure $\bar{M}$ in the direct product $\mathbb{R}^2 \times \mathbb{R}P^1$ is diffeomorphic to the Möbius strip. The projection $\pi: \mathbb{R}^2 \times \mathbb{R}P^1 \rightarrow \mathbb{R}^2$ along the second

component carries $\bar{M}$ into $\mathbb{R}^2$; the pre-image of 0 under this mapping is the projective line $L = \mathbb{R}P^1$ (henceforth, called the *pasted projective line*); the projection $\pi: \bar{M} \setminus L \rightarrow \mathbb{R}^2 \setminus \{0\}$ is a diffeomorphism.

Lemma. *To each smooth vector field v which is not flat at 0 and which is defined in a neighbourhood of the point 0 of $\mathbb{R}^2$ there corresponds a smooth direction-field α on a neighbourhood of the pasted projective line L on the surface $\bar{M}$ which is defined everywhere except for a finite number of points on L which are called singular points. The field α is carried by the projection $\pi: \bar{M} \setminus L \rightarrow \mathbb{R}^2 \setminus \{0\}$, onto the direction field generated by v . The field α is generated in a neighbourhood of each singular point by a vector field $\tilde{v}$ which is not flat at the singular point.*

This result allows the σ -process to be iterated inductively.

This construction, in contrast to the polar blow-up, can be carried over word for word to the complex domain except that $\mathbb{R}^2$ and $\mathbb{R}P^1$ must be replaced by $\mathbb{C}^2$ and $\mathbb{C}P^1$.

1.2. Elementary Singular Points

Definition. A singular point of a vector field on the plane is said to be *elementary* if at least one eigenvalue of the linear part of the field at that point is different from zero.

The topological classification of isolated elementary singular points in the real plane is easy; apart from nodes, saddle-points, foci, and centers, there are only saddle-nodes (Fig. 16).

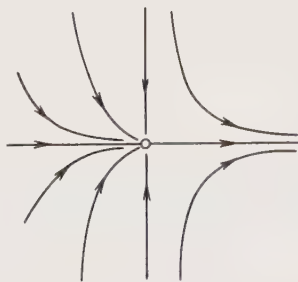


Fig. 16. A saddle-node

1.3. Good Blow-ups. By performing a finite number of polar blow-ups, or σ -processes, a degenerate singular point of a vector field on the plane can be split into a finite number of elementary singular points under very weak conditions on the vector field.

Bendixson's Theorem [16], [33]. *A real-analytic vector field defined in a neighbourhood of an isolated singular point in $\mathbb{R}^2$ can be transformed by finite*

number of σ -processes into an analytic direction field which is defined on a neighbourhood of pasted projective straight lines and which has only a finite number of singular points, each of which is elementary.

The blow-up mentioned in Bendixson's theorem is called a *good blow-up*. (In the original paper [33] a "good blow-up" satisfies stricter conditions.)

Seidenberg's Theorem [98]. *A good blow-up exists for any holomorphic vector field defined in a neighbourhood of an isolated singular point in $\mathbb{C}^2$.*

A similar theorem holds for smooth vector fields under slightly more restrictive conditions.

Definition. A vector field v satisfies a *Lojasiewicz condition* at a singular point x_0 if $|v(x)|$ does not decrease as $x \rightarrow x_0$ more rapidly than some power of the distance from the singular point *i.e.*,

$$|v(x)| \geq C|x - x_0|^p$$

for some positive constants C and p .

The set of germs of vector fields which do not satisfy a Lojasiewicz condition have infinite codimension in the space of all germs. For example, a Lojasiewicz condition is always satisfied for a singular point of finite multiplicity.

Dumortier's Theorem [33], [34]. *A good blow-up exists for any smooth vector field which is defined on a neighbourhood of a singular point in $\mathbb{R}^2$ and which satisfies a Lojasiewicz condition. Moreover, every vector field of this sort can be converted by a finite number of polar blow-ups into a smooth vector field on a planar domain with finitely many singular points, each of which is elementary.*

Remark. In the paper [33] some stronger formulations are given.

§ 2. Smooth Orbital Classification of Elementary Singular Points in the Plane

The word "elementary" applied to singular points has a double significance:

- 1) a complicated singular point scatters into elementary ones, as if into atoms;
- 2) the elementary singular points have a comparatively simple structure (see § 2 and § 5).

The classification mentioned in the heading is obtained for all germs of smooth vector fields with the exception of a subvariety of infinite codimension. In particular, it applies to all germs of analytic vector fields with an isolated singular point. The classification is given by the following two theorems.

2.1. Table of Normal Forms; the Analytic Case

Theorem. *A germ of an analytic vector field with an isolated singular point on the real plane is smoothly orbitally equivalent to one of the vector fields listed in the following table; (here, and throughout this chapter, p, q , and k are natural numbers, the fraction p/q is in lowest terms, a is real, $\mathbf{x}=(x, y)\in\mathbb{R}^2$, $r^2=x^2+y^2$ and $\varepsilon\in\{0; 1; -1\}$. I is the operator given by rotation through $\pi/2$).*

Type of singular point	Normal form
1. Field with a nonresonant linear part	$w(x)=Ax$
2. Center by linear terms	$w(x)=Ix+\varepsilon(r^{2k}+cr^{4k})x$
3. Resonant node	$w(x, y)=x\frac{\partial}{\partial x}+(ny+\varepsilon x^n)\frac{\partial}{\partial y}$
4. Resonant saddle-point with ratio of eigenvalues $\lambda=-p/q$	$w(x, y)=x[1+\varepsilon(u^k+au^{2k})]\frac{\partial}{\partial x}+\lambda y\frac{\partial}{\partial y},$ $u=x^p y^q$ is the resonant monomial
5. Degenerate elementary singular point	$w(x, y)=x(\pm x^k+ax^{2k})\frac{\partial}{\partial x}\pm y\frac{\partial}{\partial y}$ (the two signs $\pm$ are independent)

2.2. Normal Forms in the Smooth Case

Definition. A formal first integral of a germ of a vector field v is a formal series F for which $(\hat{v}, \text{grad } F)=0$.

Theorem. *Germs of smooth vector fields at elementary singular points which satisfy a Lojasiewicz condition, and which do not have a formal first integral whose Taylor series begins with a positive-definite quadratic form, reduce to the same normal forms as above.*

These theorems are proved by parts in [51], [46], [23], [21], [59], [15:90] and in a paper by Yu.S. Il'yashenko entitled "Dulac's memoir 'On limit cycles' and related problems of the local theory of differential equations" (*Uspekhi Mat. Nauk.* **40**:6 (1985), 41–78; English translation: *Russian Math. Surveys* **40**:6 (1985), 1–49.)

§ 3. Topological Classification of Compound Singular Points with a Characteristic Trajectory

The problem of topologically classifying compound singular points with a characteristic trajectory is algebraically soluble. In fact, there is a simple algorithm for classifying such singularities. Throughout this section, we shall be referring to the real plane.

3.1. The Fundamental Alternative

Definition. A phase curve of a differential equation on the plane is called a *characteristic trajectory of the singular point* if, as $t \rightarrow +\infty$ or $t \rightarrow -\infty$, it approaches the singular point and becomes tangent to a fixed straight line.

Definition. A singular point of a vector field is said to be *monodromic* if there is a neighbourhood of it and a *smooth arc* (we say that an arc is smooth at its initial-point if the arc can be continued smoothly in both directions from the initial-point) originating at the singular point which is transversal to the field at every point, except the origin, with the property that the direction field on the neighbourhood with the arc removed is diffeomorphic to the standard field (see Fig. 17). More precisely there must be continuous map of a closed rectangle onto the closure of the neighbourhood of the singular point which carries the interior of the rectangle diffeomorphically onto the complement of the transversal arc in the neighbourhood, transforms the horizontal field into the original field, and maps the vertical sides of the rectangle onto the transversal and the lower horizontal side onto the singular point. Each phase curve of the original field with initial point on the transversal sufficiently close to the singular point returns onto the transversal after making one loop around the singular point. The mapping which carries the initial point of each such arc into its endpoint (the point of first return on the transversal) is called the *monodromy transformation of the singular point* (Fig. 17).

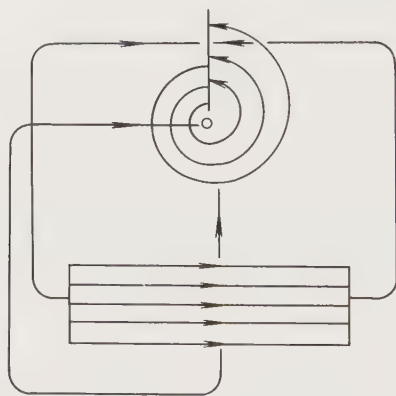


Fig. 17. Monodromic singular point

Remark. At interior points of the transversal the monodromy transformation is as differentiable as the vector field and is analytic when the field is analytic. However, even for analytic fields, it cannot be extended smoothly beyond the origin of the transversal.

It is easy to give an example of a smooth vector field with a singular point which does not have a characteristic trajectory and which is not monodromic (see Fig. 18). However, the corresponding vector field will necessarily be flat at the singular point:

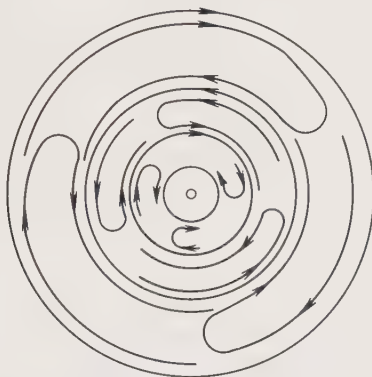


Fig. 18. A non-monodromic singular point without a characteristic trajectory

Theorem. *A singular point of a smooth vector field on the plane at which the field is not flat will either have a characteristic trajectory or be monodromic.*

3.2. Topological Classification of Differential Equations on the Plane in a Neighbourhood of a Singular Point. As shown in [2], under relatively mild restrictions on the field, a neighbourhood of a singular point of a vector field with a characteristic trajectory will split into a finite union of so-called hyperbolic, parabolic, and elliptic sectors.

Definition. The *standard hyperbolic* (resp., *parabolic, elliptic*) *sector* is defined to be the set S_h (resp., S_p, S_e) with vector field v_h (resp., v_p, v_e) where

$$v_h = x \frac{\partial}{\partial x} - y \frac{\partial}{\partial y},$$

and S_h (the hyperbolic sector) is bounded by the curves

$$x=0, \quad x=1, \quad y=0, \quad y=1, \quad xy=1/2;$$

$$v_p = x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y},$$

and S_p (the parabolic sector) is bounded by the curves

$$x=0, \quad y=0, \quad x^2 + y^2 = 1;$$

$$v_e = z^3 \frac{\partial}{\partial z}, \quad z = x + iy,$$

and S_e (an elliptic sector) is bounded by a phase curve of the field v_e in the domain $x > 0, y > 0$ together with the point 0 (see Fig. 19).

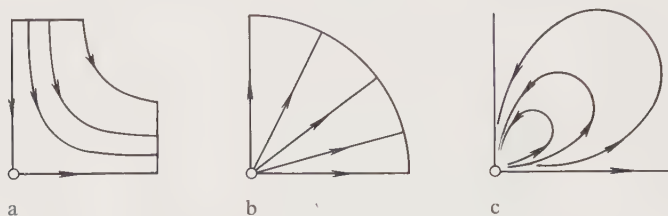


Fig. 19. a) hyperbolic sector; b) parabolic sector; c) elliptic sector

Definition. The closure S of an open set whose boundary contains the origin is said to be a *hyperbolic* (resp., *parabolic*, *elliptic*) sector of a vector field with the singular point 0 there is a homeomorphism $S_h \rightarrow S$ (resp., $S_p \rightarrow S$, $S_e \rightarrow S$) which carries the phase curves of the standard field v_h (resp., v_p, v_e) onto the characteristic trajectories of the field v .

Theorem 1 [2]. Let v be a smooth vector field with an isolated singular point which is not flat. If the singular point has a characteristic trajectory and a neighbourhood which does not contain a countable number of elliptic sectors without common interior points, then it has a neighbourhood which splits into a finite union of parabolic, elliptic, and hyperbolic sectors of v which have no common interior points.

The following theorem follows from Dumortier's theorem (§ 1.3).

Theorem 2. The conclusion of the previous theorem holds for a singular point of a smooth vector field which satisfies a Lojasiewicz condition at the point. In particular, it holds for an isolated singular point of an analytic vector field.

By the same token, a pathological phase portrait with a countable number of elliptic sectors, which is not prohibited by purely geometrical considerations (the existence, uniqueness, and continuous dependence theorems), is not possible in the analytic case.

The book [2] gives a complete collection of topological invariants of vector fields on the sphere with a finite number of singular points, all of which satisfy the conditions of Theorem 1. In particular, it gives a topological classification of analytic vector fields with isolated singular points on a sphere. The realization problem, that is, the question of which of the phase portraits listed in [2] are actually realized by analytic vector fields on a sphere, remains open: it is conjectured that every portrait on the list which has a finite number of limit cycles can be realized.

3.3. Topological Finite Determinacy. Newton Diagrams of Vector Fields

Theorem [33], [34]. *If a germ of a smooth vector field on the plane satisfies a Lojasiewicz condition, then it is possible to determine whether or not the germ has a characteristic trajectory from a finite jet. If there is a characteristic trajectory, then the germ has a finite jet all of whose representatives are topologically equivalent. In this case, the phase portrait of any representative of this jet can be determined up to homeomorphism by performing a finite number of algebraic operations on the Taylor coefficients of the jet.*

This construction is carried out as follows. First, we make a good polar blow-up. We then draw the phase portraits in a neighbourhood of each of the elementary singular points obtained after blowing up. If there is a characteristic trajectory, this enables us to construct the phase portrait, up to homeomorphism, of the blown-up vector field in a neighbourhood of the pasted curve (the union of the pasted circles). This portrait is then projected onto the original neighbourhood by the map inverse to the blow-up.

A substantially quicker method of investigation using Newton diagrams and normal forms was proposed by A.D. Bryuno [23]; a suitable algorithm has been implemented on the computer [23:16].

Definitions. 1. The *support* of a vector monomial $\mathbf{x}^k \frac{\partial}{\partial x}$ or $\mathbf{x}^k \frac{\partial}{\partial y}$, $\mathbf{x}=(x, y) \in \mathbb{R}^2$ is defined to be the point $k+e_2$ or $k+e_1$ of the lattice $\mathbb{Z}^2$ where $e_1=(1, 0)$ and $e_2=(0, 1)$.

2. The *support* of an analytic vector field v on the plane with a singular point at 0 is defined to be the union of the supports of all monomials in the Taylor expansion of v which appear with nonzero coefficients.

3. The *Newton diagram* of a vector field v is defined to be the polygon constructed as follows. Consider the union of all the quadrants with vertices at the points of the support of the field v and whose sides are parallel to and oriented like the positive semi-coordinate axes. The boundary of the convex hull of this set consists of two open rays and a polygon no side of which is parallel to the coordinate axes (the polygon may consist of a single point). This polygon is called the *Newton diagram* of the field v .

4. The *principal part* of a vector field v is defined to be the sum of all terms of the Taylor expansion of v whose exponents belong to the Newton diagram.

We shall say that a property holds for Γ -nondegenerate vector fields if the subset of vector fields with fixed Newton diagram Γ which does not have this property is distinguished by a finite number of nontrivial algebraic equations in the coefficients of the terms of the Taylor expansion with exponents belonging to Γ .

3.4. Investigation of Vector Fields by Their Principal Part

Theorem (F.S. Berezovskaya) [17]. *Let Γ be the Newton diagram of a vector field on the plane with an isolated singular point at 0. In the space of all*

analytic vector fields with Newton diagram Γ there is an everywhere dense, open set U , of " Γ -nondegenerate vector fields" with the following property. The domain U splits into two domains S and M ; vector fields with their principal part in S have characteristic trajectories, those with their principal part in M have no characteristic trajectories. The domain S consists of a finite number of connected components S_i ; fields with their principal part in the same component S_i are orbitally topologically equivalent in a neighbourhood of 0. The boundaries of M and S are semi-algebraic sets.

Remarks. 1. The criteria for belonging to M or S have been formulated as explicit conditions on the principal parts. There is a computer program for verifying these conditions (which is an awkward calculation by hand). There is also a program for calculating the principal terms of the asymptotics of the characteristic trajectories.

2. The preceding theorem says nothing about distinguishing between a center and a focus. Even the answer to the following question is unknown:

Suppose that, for a given Newton diagram, there are vector fields without characteristic trajectories. Is it true that at least one such field must have a singular point of focus type?

The answer is conjectured to be yes; this conjecture was recently proved for "most" diagrams by F.S. Berezovskaya and N.B. Medvedeva.

§ 4. The Problem of Distinguishing Between a Center and a Focus

In this section, we discuss one of the oldest problems in the qualitative theory of differential equations.

4.1. Statement of the Problem. A germ of an analytic vector field on the plane will have a singular point of the type of a center if a countable number of conditions on the nonlinear terms are satisfied. It is therefore impossible to find a necessary and sufficient condition for the existence of a center in general infinite-dimensional classes of equations. It seems realistic to state the problem of distinguishing between a center and a focus as follows. Find an algorithm with the following properties:

1. Each step of the algorithm uses only a finite jet of the field under consideration and is achieved by means of algebraic operations and integration.

2. For all germs of smooth or analytic vector fields at a singular point in the plane which do not belong to an exceptional set of infinite codimension in the corresponding function space, the algorithm stops after a finite number of steps, and in this case the singular point is a focus.

Such an algorithm has been found for the classes of vector fields enumerated in §§ 4.3–4.5 below. The status of the center-focus problem for polynomial vector fields of fixed degree is discussed in § 4.8.

4.2. Algebraic Insolubility. The difficulty of the problem of distinguishing between a center and a focus resides in the fact that the problem is algebraically insoluble in the sense of the definition given in Chap. 3, § 2.1 [7, p. 330]. It is not known whether it is analytically soluble (the conjectured answer is yes). It has been proved that the problem is algebraically soluble up and including to codimension 10 (see sections 4.3, 4.4) and is insoluble up to codimension 11 (B.V. Alekseev). Analytic solubility has been proved only up to and including codimension 11.

We enumerate the classes of vector field for which the problem of distinguishing between a center and a focus is algebraically soluble.

4.3. Distinguishing a Center by Linear Terms

Theorem. *The problem of distinguishing between a center and a focus in the class of equations with linear part $\dot{z} = i\omega z$, $\omega > 0$, is analytically soluble.*

This is a modification of the theorem of A.M. Lyapunov [75] and H. Poincaré [91]. We outline the proof because it contains concepts needed subsequently.

Let $\hat{v}$ be the formal Taylor series for the field under consideration. The equation

$$(\hat{v}, \nabla F) = g(r^2) \quad (1)$$

in the formal series F of two variables and g of one variable is always soluble. There is a unique solution of this equation for which the expansion of F in powers of $z = x + iy$ and $\bar{z} = x - iy$ starts with the monomial $z\bar{z}$ and has no terms of the form $z^n \bar{z}^n$ for $n > 1$. We shall call this the *distinguished solution*.

The coefficients (called the *Lyapunov focus numbers*) in the expansion of g for the distinguished solution are polynomials in the coefficients of the series $\hat{v}$. The singular point 0 of v is a center if and only if all these polynomials vanish. This proves the theorem.

Theorem [75], [91]. *If an analytic equation with linear part $\dot{z} = i\omega z$, $\omega > 0$, has a formal first integral (i.e., if there is a solution of equation (1) for which $g \equiv 0$, $F \not\equiv \text{const.}$), then the field v has a center at 0.*

For a generalization to the higher dimensional case, see Chap. 4, § 3.2.

4.4. Nilpotent Jordan Block. The problem of distinguishing between a center and a focus has been solved for fields whose linear part is a nilpotent Jordan block [75], [83]. The solution is carried out in two steps.

1. By a formal change of variables and multiplication by a formal series with a nonzero constant term, the original equation can be put in the form

$$\dot{x} = y + f(x), \quad \dot{y} = \varepsilon x^l, \quad \varepsilon = \pm 1;$$

the Taylor coefficients of the formal series f are polynomials in the Taylor coefficients of the original series.

2. Theorem. Suppose that

$$f(x) = x^m \tilde{f}(x) \quad \text{and} \quad \tilde{f}(0) = a \neq 0$$

in the previous equation. The singular point 0 of the original equation is monodromic if and only if one of the following conditions holds:

$$\varepsilon = -1, \quad l = 2n - 1, \quad m > n$$

or

$$\varepsilon = -1, \quad l = 2n - 1, \quad m = n, \quad a^2 n^2 < 4m.$$

If, in addition, $f(x) \neq f(-x)$, then the singular point 0 of the original field is a focus; it is stable if the coefficients of the smallest term of odd degree in the expansion of f is negative, and unstable otherwise.

Remark. It easily follows that the problem of distinguishing between a center and focus is algebraically soluble in the class under consideration (see also Amelkin, V.V., Lukashevich, H.A., Sadovsky, A.P., "Non-linear oscillations in systems of the second order", Minsk, BGU edition, 1982, p. 208).

Theorem. If the original germ is analytic, and if the formal series constructed above for f contains no odd powers (so that $f(x) = f(-x)$), then the singular point 0 is a center.

The theorems above and those of the previous section show that the problem of distinguishing between a center and a focus is algebraically soluble in the class of vector fields on the plane with a nonzero 1-jet (linear part) at the singular point.

For smooth germs with a zero linear part and a nonzero 2-jet, the problem of distinguishing between center and focus does not arise. Every such germ which satisfies a Lojasiewicz condition always has a characteristic trajectory. The space of germs with zero 2-jet has codimension 10. The problem of distinguishing between a center and a focus is algebraically soluble in this space up to and including codimension 0 [86]. Therefore, the problem of distinguishing between a center and focus is algebraically soluble up to and including codimension 10 in the class of all germs of vector fields at a singular point on the plane.

4.5. Singular Points Without Exclusive Directions

Definition. We say that a smooth vector field on a neighbourhood of a singular point which is not flat has no exclusive directions if it can be transformed by a single polar blow-up into a vector field with no singular points on the pasted circle.

Remark. Polar blowing-up reduces the investigation of a neighbourhood of a singular point without exclusive directions to the investigation of a neighbourhood of a closed phase curve (the pasted circle). This in turn reduces to the calculation of the corresponding monodromy map. If the latter is

not the identity, then the original singular point is a focus. The Taylor coefficients of the monodromy map are calculated recursively from the Taylor coefficients of the field. However, the corresponding functions turn out to be non-algebraic and their zero sets are not algebraic. Thus, an algorithm exists for distinguishing between a center and a focus in the class of fields without exclusive directions, but the actual problem of making the distinction in this class is algebraically insoluble.

4.6. A Programme for Investigating the General Case. We shall only outline an algorithm for distinguishing between a center and a focus in the general case. A sequence of σ -processes leading to a good blow-up replaces the singular point of the vector field by a “compound cycle”. This cycle is a union of the singular points obtained by blowing-up the direction field, together with the integral curves lying on the pasted projective lines joining these singular points. All the singular points obtained in this way are elementary. A neighbourhood of a compound cycle which contains only elementary singular points can be investigated by Dulac’s method (Chap. 6, § 4). This also allows us to calculate the asymptotics of the monodromy map of the original field. If any segment of the asymptotic series obtained in this way defines a mapping distinct from the identity, then the original singular point is a focus. The asymptotic series of the monodromy map coincides with the identity for a set of vector fields of infinite codimension.

4.7. The Generalized First Focus Number

Theorem [59]. *The leading term of the monodromy map of a monodromic singular point of a smooth vector field satisfying a Lojasiewicz condition is always linear.*

Definition. Suppose that the principal term in the theorem above is $x \mapsto cx$. The number $\ln c$ is called the *generalized first focus number* of the corresponding compound singular point.

Remark. The generalized first focus number depends on the choice of transversal; its sign depends only on the vector field. If the first focus number is not zero, then the singular point is a focus.

The preceding theorem, like the actual calculation of the first focus number, uses the geometry of the surface obtained from a neighbourhood of the singular point by repeatedly applying σ -processes. This surface is a union of circular rings and Möbius strips.

4.8. Polynomial Vector Fields. In the following, we only consider vector fields whose linearization has a center. The original field has a center if and only if all the Lyapunov focus numbers vanish. Therefore, the condition that a polynomial field of degree n have a center is given by an infinite number of algebraic equations in the (finite number of) coefficients of the polynomial vector field. By Hilbert’s basis theorem, this infinite system of

algebraic equations is equivalent to a finite system. The problem of distinguishing between a center and a focus of a polynomial field of degree n reduces to calculating this system.

The Lyapunov focus numbers can be calculated recursively. There are computer programs for performing the calculation. The unsolved part of the problem is to find an $N(n)$ for each n with the property that the vanishing of the first $N(n)$ focus numbers guarantees the existence of a center. It is not even known whether $N(n)$ is less than the dimension of the space of coefficients of polynomial vector fields of degree n .

An unexpectedly simple answer was obtained by H. Dulac [103:9] for the case $n=2$. It turns out that $N(2)=3$; when the first three focus numbers vanish, the equation can be integrated by quadratures and it has a center. For the fields $Ix + f_3(x)$ where $x \in \mathbb{R}^2$ and f_3 is a homogeneous vector polynomial of degree 3, we have $N=6$. There is a detailed exposition of this problem in K.S. Sibirskiy's book [103].

§ 5. Analytic Classification of Elementary Singular Points in the Complex Domain

This section deals with the orbital analytic classification of germs of vector fields at an elementary singular point with a resonant linear part. This classification has functional moduli. The first step is to investigate germs of one-dimensional mappings.

5.1. Germs of Conformal Mappings with the Identity as Linear Part

Definition. Two germs f and g of conformal mappings $(\mathbb{C}, 0) \rightarrow (\mathbb{C}, 0)$ are said to be *analytically equivalent* if there is a germ of a conformal mapping $h: (\mathbb{C}, 0) \rightarrow (\mathbb{C}, 0)$ which conjugates the germs f and g : $f = h \circ g \circ h^{-1}$.

Let A_2 denote the space of germs $f: (\mathbb{C}, 0) \rightarrow (\mathbb{C}, 0)$ of the form

$$z \mapsto z + az^2 + \dots, \quad a \neq 0.$$

All germs in A_2 with the same 3-jet are formally equivalent, but far from all such germs are analytically equivalent.

To each germ in A_2 we associate an equivalence class of pairs of germs of conformal functions (which will be an invariant of the analytic classification); these classes coincide for two formally equivalent germs if and only if the corresponding germs are analytically equivalent. We describe this "space of invariants". Let 0 and ∞ be points on the Riemann sphere and let

$$\varphi = (\varphi_+, \varphi_-); \quad \varphi_+: (\mathbb{C}, 0) \rightarrow (\mathbb{C}, 0), \quad \varphi_-: (\mathbb{C}P, \infty) \rightarrow (\mathbb{C}P, \infty), \quad \varphi'_+(0) = 1,$$

be a pair of conformal mappings.

Definition. The pairs φ and ψ are said to be *equivalent* if there is a linear map $\lambda: \mathbb{C} \rightarrow \mathbb{C}$ such that $\varphi \circ \lambda = \lambda \circ \psi$.

The equivalence classes of such pairs will be invariants.

The space of such classes is called the *space of invariants*, and is denoted by M_2 .

Theorem [112], [59], [37], [38], [76]. 1. The invariant. *A set of analytically equivalent germs in A_2 corresponds to a unique equivalence class $[\varphi]$ in M_2 of pairs (called the functional invariant).*

2. Equimodality and equivalence. *Formally equivalent germs in A_2 with the same functional invariant are analytically equivalent.*

3. Realization. *Each class $[\varphi]$ in M_2 of pairs is the invariant of some map germ in A_2 .*

4. Analytic dependence. *If a germ in A_2 depends analytically on a finite number of complex parameters, then the corresponding invariant also depends analytically on the same parameters (to say that a class of pairs depends analytically on a parameter means that it is possible to select representatives in each class of pairs so as to obtain an analytic family of pairs of germs).*

5.2. Classification of Resonant Mappings and Vector Fields with Generic Nonlinearities. Consider the set of germs of mappings $(\mathbb{C}, 0) \rightarrow (\mathbb{C}, 0)$ with a resonant linear part of the form

$$f: z \mapsto vz + \dots, \quad v = \exp(-2\pi i p/q), \quad z \in \mathbb{C}, \quad (2)$$

where, here and in what follows, p and q are relatively prime natural numbers. The germ f^q has the form $z \mapsto z + \alpha z^{q+1} + \dots$, where f^q denotes the iterate $f \circ \dots \circ f$ of f with itself q times. Let B_q denote the set of germs f of the form (1) for which $\alpha \neq 0$ in the expansion of f^q .

Theorem (S.M. Voronin, Yu.S. Il'yashenko) [59]; J. Martinet, J.P. Ramis [79]). *Theorem 5.1 holds for germs in B_q , if A_2 is replaced by B_q .*

Consider the set of germs of *saddle resonant* vector fields:

$$\begin{aligned} v(z) &= Az + f(z), \quad A = \text{diag}(\lambda_1, \lambda_2), \quad f(0) = 0 \\ \frac{\partial f}{\partial z}(0) &= 0, \quad \lambda_2/\lambda_1 = -p/q, \quad z \in \mathbb{C}^2. \end{aligned} \quad (3)$$

There are just two phase curves of the equation $\dot{z} = v(z)$ which can be extended holomorphically across the point 0; they are tangent to the coordinate axes at the origin and, by analogy with the real case, are called *separatrices*. Consider a positively oriented loop encircling 0 on the separatrix tangent to the axis $0z_j$ at 0. Let $\Delta_j: (\mathbb{C}, 0) \rightarrow (\mathbb{C}, 0)$ be the monodromy mapping corresponding to this loop (see Chap. 6, §1 below). It turns out that if v has the form (3), then Δ_1 has the form (2). We let $\tilde{B}_q$ denote the set of germs of saddle-resonant vector fields (3) for which Δ_1 is in B_q .

Definition. Two germs v and w of holomorphic vector fields are said to be *formally orbitally equivalent* if there is a formal substitution H and a formal series T with a nonzero constant term such that

$$H_* \hat{v} = T \hat{w} \circ H.$$

Remark. If both the series H and T converge, this equality implies that v and w are orbitally analytically equivalent.

Theorem [59], [79]. *Theorem 5.1 holds for germs in $\tilde{B}_q$ with $p/q \neq 1$ (i.e., if $\lambda_1 \neq -\lambda_2$ in formula (3)), if we replace B_q by $\tilde{B}_q$ and “formally equivalent” (resp., “analytically equivalent”) by “formally orbitally equivalent” (resp., “analytically orbitally equivalent”).*

5.3. Degenerate Elementary Singular Points. Let $\tilde{A}$ be the set of germs of holomorphic vector fields with a degenerate elementary singular point. Each such germ has a phase curve which can be extended holomorphically through 0. The extended curve is tangent to the eigenvector corresponding to the zero eigenvalue of the linear part of the germ. As above, we obtain a corresponding monodromy mapping $\Delta: (\mathbb{C}, 0) \rightarrow (\mathbb{C}, 0)$. It turns out that $\Delta(z) = z + \alpha z^2 + \dots$. Let $\tilde{A}_2$ denote the class of germs for which Δ is in A_2 . The space of functional invariants of germs in $\tilde{A}_2$ is narrower than that of A_2 (not all germs in A_2 are realized as monodromy mappings of germs in $\tilde{A}_2$).

Let M_+ denote the subset of classes in M_2 consisting of pairs (φ_+, φ_-) for which

$$\varphi_-(z) = z + c$$

(this is the germ of a holomorphic mapping $(\mathbb{C}P, \infty) \rightarrow (\mathbb{C}P, \infty)$).

Theorem [78]. *Theorem 5.1 holds for germs in $\tilde{A}_2$ with “formally equivalent” replaced by “formally orbitally equivalent” and A_2 and M_2 replaced by $\tilde{A}_2$ and M_+ , respectively.*

Remark. The theorems of the last two sections were formulated for germs with generic nonlinearities; similar theorems are true without any restrictions on the nonlinearities (one need only require that the germs under consideration should not be formally, or formally orbitally, equivalent to linear germs); the restriction $\lambda_1 \neq -\lambda_2$ is also superfluous, but without it the formulations become more awkward.

5.4. Geometric Theory of Analytic Normal Forms. All germs in the sets A_2 , $\tilde{A}_2$, B_q and $\tilde{B}_q$ above (§§ 5.1–5.3) have the following property in common. Each such germ has a representative which is analytically, or orbitally analytically, equivalent to a very simple normal form in a domain which contains 0 on its boundary (but is not equivalent to such a form in a neighbourhood of the singular point 0). These normal forms for germs of vector fields in $\tilde{B}_q$ or $\tilde{A}_2$ are given in the Table in § 2.1 (the cases 4 and 5), except that

we have to take $\varepsilon=1$, $k=1$, and $a \in \mathbb{C}$. For any of the germs above it is possible to choose a pair of intersecting domains which cover a punctured neighbourhood of 0 in the 1-dimensional case or a neighbourhood with the w -axis removed in the 2-dimensional case. In each of these domains the germ is analytically (resp. orbitally analytically) equivalent to one and the same normal form. The existence of this pair is guaranteed by some theorems on "sectorial normalization" [112], [51]. But the conjugating holomorphisms in these domains are different. On the intersection of the domains, we obtain a transition function; that is, a biholomorphic mapping which preserves the normal form of the germ. The last requirement is very severe; it enables the transition function to be described by means of the pair of germs (φ_+, φ_-) .

The collection consisting of: the formal or orbital formal normal form, the pair of domains with 0 on the boundary in which the germ has been reduced to the normal form, and, finally, the transition function, makes up the analytic normal form of the germ. If this collection is known, much can be said about the properties of the germ. It is in this way that the theorems of the next section were obtained. A similar approach can be used in the theory of linear systems with an irregular singular point (Chap. 7, § 2).

5.5. Embedding in the Flow, Extraction of Roots, Divergence of Normalizing Series, and Holomorphic Center Manifolds. The functional invariants of germs $f, v, \dots$ are denoted by $\mu_f, \mu_v, \dots$.

Theorem [112], [38]. *A germ f in A_2 is the n th power of a germ g in A_2 (the operation being composition) if and only if the invariant μ_f consists of germs of mappings which commute with multiplication by all the n th roots of unity.*

Theorem [112], [38]. *A germ f in A_2 can be embedded in a flow (i.e., can be expressed as a unit time shift along a phase curve of a holomorphic vector field) if and only if the corresponding functional invariant is trivial, i.e., if the mappings φ_- and φ_+ are linear.*

Theorem [59]. *A germ of a vector field in $\tilde{B}_q$ is analytically equivalent to its preliminary normal form if and only if the corresponding functional invariant is trivial.*

Theorem [78]. *A germ v of a vector field in $\tilde{A}_2$ has an analytic center manifold (a phase curve which extends holomorphically through zero and which is tangent to the kernel of the linear part at zero) if and only if $\mu_v = (\text{id}, \varphi_+)$.*

5.6. Taylor Description of the Analytic Equivalence Classes. A.D. Bryuno and P.M. Elizarov [64, pp. 165 & 144] have announced theorems about analytic normal forms of germs of resonant vector fields similar to the Poincaré-Dulac theorem (see also A.D. Bryuno, *Doklady AN SSSR*, 263 (1983), pp. 781

784; P.M. Elizarov, A.A. Scherbakov. Functional moduli of analytic vector fields and mappings and divergence of normalizing series. *Russ. Math. Surv.*, v. 38 (1983), p. 144–145). These theorems show which monomials of the Taylor expansion of a vector field germ can be removed by an analytic change of coordinates. The normal forms so obtained contain so few terms that two germs with different forms are not analytically equivalent.

§ 6. Orbital Topological Classification of Elementary Singular Points in the Complex Plane

6.1. The Non-Resonant Case. A vector field germ with nonresonant linear part in the complex plane is either analytically equivalent to its linear part (a sufficient condition for which is given in Chap. 4, § 1), or it is not even topologically equivalent to it (V.A. Nayshul). In the first case the orbital topological classification is simple; in the second case it is still almost totally unexplored.

6.2. Saddle-Resonant Vector Fields. As mentioned in § 2.1, a holomorphic saddle-resonant vector field at a singular point in the plane $\mathbb{C}^2$ is formally orbitally equivalent to the field

$$z(1+u^k+au^{2k})\frac{\partial}{\partial z}+\lambda w\frac{\partial}{\partial w}, \quad \lambda=-p/q, \quad u=z^pw^q, \quad a\in\mathbb{C}, a\neq 0. \quad (3)$$

Theorem (C. Camacho, P. Sad) [24]. *The numbers p, q and k are a complete set of topological invariants of germs of saddle-resonant vector fields.*

6.3. Degenerate Elementary Singular Points. A germ in $\tilde{A}_2$ is formally orbitally equivalent to a germ

$$(z^2+az^3)\frac{\partial}{\partial z}+w\frac{\partial}{\partial w}, \quad a\in\mathbb{C} \quad (4)$$

and has functional modulus (φ_-, φ_+) , where $\varphi_- = \text{id} + c$.

P.M. Elizarov classified the germs (4) up to orbital topological equivalence for all a, φ_-, φ_+ except those in which $\varphi_- = \text{id}$ and a is an irrational number abnormally closely approximated by rational numbers. In particular, the following theorem holds.

Theorem (P.M. Elizarov, Orbital topological classification of analytic differential equations near a degenerated elementary singular point, *Russ. Math. Surv.*, v. 40 (1985), pp. 253–254). *Let v be a germ in $\tilde{A}_2$ with functional modulus (φ_-, φ_+) where $\varphi_- = \text{id}$. Suppose the number a in formula (4) is rational, $a = p/q$, and suppose the map $z \mapsto v\varphi_+(z)$ with $v = \exp(2\pi ia)$ is formally equivalent to the map $z \mapsto v z(1+z^k+\alpha z^{2k})$. Then the numbers p, q and k form a complete*

set of invariants of the orbital topological classification of the germs for which $\varphi_- = \text{id}$ and $a \in \mathbb{Q}$.

We note that the number k is connected with the modulus of the analytic classification of the original germ, and not with its formal orbital normal form. It cannot be calculated from any finite jet of the germ v .

Chapter 6

Cycles

This chapter deals with (limit) cycles of differential equations in a real or complex domain and with general properties of polynomial differential equations on the complex plane.

§ 1. The Monodromy Mapping

In this section, we define the monodromy map which connects cycles of differential equations in the complex domain with holomorphic mappings on transversals to the cycles.

1.1. Definitions. A phase curve of a holomorphic vector field is a connected one-dimensional complex manifold (of real dimension two). In a neighbourhood of a nonsingular point of an equation, the phase space is fibered by the phase curves. But, in the large, the phase curves will not, in general, define a fibration, but only a foliation.

A foliation with k -dimensional leaves on an n -dimensional manifold is a partition of the n -dimensional manifold into smooth, embedded, k -dimensional submanifolds (the leaves of the foliation) with the property that the partition of some neighbourhood of each point of the manifold by the connected components of the intersections of the leaves with the neighbourhood is rectifiable (that is, diffeomorphic to the partition of an n -dimensional cube by planes parallel to a k -dimensional face).

A foliation is said to be *holomorphic* if the manifold, leaves, and rectifications are holomorphic.

To each closed path on a leaf of a foliation, we can associate a map germ called the *monodromy of the path*. This will be the germ at an initial point of a mapping of a transversal (to the leaf) into itself.

The precise definition is as follows. A rectification of the foliation in a neighbourhood of a point of the path defines diffeomorphisms between the local transversals at nearby points of the path by carrying a point of a

leaf into a point of the same leaf. Choosing a finite number of neighbourhoods which cover the path, we obtain a diffeomorphism of the local transversal at the initial point of the path into itself which carries a point of each leaf into a point of the same leaf and fixes the initial point.

The monodromy does not depend on the choice of transversal: mappings defined on different transversals are conjugate (one being carried into another by a diffeomorphism of the transversals). In fact, the monodromy can be considered as a mapping of the local manifold of leaves close to the initial point of the path into itself, and it is possible to define it without even mentioning transversals.

The monodromy of a path does not change if the path is replaced by a path homotopic to it on the same leaf.

This gives a representation of the fundamental group of the leaf on the group of germs of diffeomorphisms of a transversal which is called the *monodromy group* (or the *holonomy group*) of the foliation.

The monodromy group of a holomorphic foliation consists of germs of holomorphic mappings (with respect to the natural complex structure on the local transversal, i.e., on the local manifold of leaves).

When the initial point of a closed path on a leaf is replaced by another point on the same leaf the monodromy mapping obtained is conjugate to the original one. Therefore the monodromy mapping is defined up to conjugation in the free homotopy class of closed paths on the leaf. (Two closed paths on a leaf are said to be *freely homotopic* if the corresponding maps of the circle into the leaf are homotopic.)

Definition. A *complex cycle* of a holomorphic vector field is defined to be a free homotopy class of closed paths on a phase curve.

The *monodromy map of a complex cycle* is defined to be a monodromy map of the foliation by phase curves. It is the germ of a map of a transversal to the phase curve into itself. The monodromy of a complex cycle is holomorphic and does not depend (up to biholomorphic conjugation) on the choice of transversal, initial point, or representative of the free homotopy class of closed paths.

1.2. Realization. Every germ of a C^∞ -diffeomorphism $(\mathbb{R}^{n-1}, 0) \rightarrow (\mathbb{R}^{n-1}, 0)$ can be realized as the monodromy map of a closed phase curve of a smooth vector field on a real n -dimensional manifold (but not in $\mathbb{R}^n$ as, for example, the mapping $x \mapsto -x$, $x \in \mathbb{R}$ shows).

Theorem [61:6]. *Any germ of an analytic diffeomorphism $(\mathbb{C}^{n-1}, 0) \rightarrow (\mathbb{C}^{n-1}, 0)$ can be realized as the monodromy map of a complex cycle of a holomorphic vector field defined on a subdomain of $\mathbb{C}^n$.*

The proof uses a result due to Y.T. Sue (*Invent. Math.*, **38**, No. 1 (1976), pp. 89-100), viz., a Stein submanifold of an analytic manifold has a neighbourhood which also is a Stein manifold.

§ 2. Local Theory of Diffeomorphisms

The local theory of diffeomorphisms and the local theory of differential equations are almost identical. In this section a brief survey of the first of these theories is given.

2.1. Linear Normal Forms. In this section we consider analytic, smooth, and topological equivalence of a diffeomorphism germ at a fixed point to its linear part.

Definition. An n -tuple $\lambda = (\lambda_1, \dots, \lambda_n) \in \mathbb{C}^n$ is called *multiplicatively resonant* if there exists a vector $m = (m_1, \dots, m_n)$ with nonnegative integral components, the sum of which is no less than 2, such that

$$\lambda_j = \lambda^m$$

for some $j \in \{1, \dots, n\}$ where $\lambda^m = \lambda_1^{m_1} \dots \lambda_n^{m_n}$.

Definition. An n -tuple $\lambda \in \mathbb{C}^n$ belongs to the *Poincaré domain* if the moduli of its components are either all less than 1 or all greater than 1. The *Siegel domain* is the complement of the Poincaré domain.

Theorem. A germ of a holomorphic diffeomorphism at a fixed point is biholomorphically equivalent to its linear part (conjugate to its linear part by a biholomorphic change of coordinates in a neighbourhood of the fixed point) for almost all (in the sense of Lebesgue measure) sets of eigenvalues of the linear part.

The following theorem makes this more precise.

Poincaré's Theorem. If the spectrum of the linear part of the germ of an analytic diffeomorphism at a fixed point belongs to the Poincaré domain and is multiplicatively nonresonant, then the germ is biholomorphically equivalent to its linear part.

Definition. An n -tuple $\lambda \in \mathbb{C}^n$ is said to have *multiplicative type* (C, ν) if the inequality

$$|\lambda_j - \lambda^m| \geq C|m|^{-\nu}$$

holds for all $j \in \{1, \dots, n\}$, $m \in \mathbb{Z}_+^n$; $|m| = \sum m_j \geq 2$, $m_j \geq 0$.

Siegel's Theorem. If the spectrum of the linear part of the germ of an analytic diffeomorphism at a fixed point has multiplicative type (C, ν) for some positive C and ν , then the germ is biholomorphically equivalent to its linear part.

The arithmetic condition on λ in Siegel's theorem can be weakened [23:31]. In the case that λ is pathologically close to a countable number of resonant n -tuples [61:6], there are analytic and geometric theorems about the diver-

gence of formal substitutions reducing an analytic diffeomorphism to linear normal form which are analogous to the theorems formulated in Chap. 4, § 1 for vector fields.

In the smooth theory it is not necessary to impose arithmetic requirements on the spectrum of the linear part.

Sternberg's Theorem. *If the spectrum of the linear part of the germ of a smooth diffeomorphism at a fixed point in $\mathbb{R}^n$ is multiplicatively nonresonant, then the germ is smoothly equivalent to the germ of its linearization at the fixed point.*

Remark. A real nonsingular linear map with multiplicatively nonresonant spectrum has no eigenvalues of modulus equal to 1. Hence the fixed point in Sternberg's theorem is hyperbolic in the following sense.

Definition. A fixed point of a diffeomorphism is said to be *hyperbolic* if no eigenvalue of the linear part of the diffeomorphism at the point has modulus equal to 1.

In the topological theory it is not necessary to require that there be no resonances.

The Grobman-Hartman Theorem [46]. *At a hyperbolic fixed point, the germ of a C^1 diffeomorphism is topologically equivalent to its linear part.*

Generic mappings are hyperbolic and nonresonant.

2.2. The Resonant Case. In the resonant case, the formal normal form of diffeomorphism germs is given by the Poincaré-Dulac theorem formulated below.

Definition. Let $z_1, \dots, z_n$ be coordinates in which the matrix of the linear part of the germ of the diffeomorphism at a fixed point is in Jordan normal form. Let λ_j be the eigenvalue corresponding to z_j (the numbers λ_j are not necessarily distinct). The monomial $z^m \frac{\partial}{\partial z_j}$ is called a *multiplicatively resonant term* if $\lambda_j = \lambda^m$.

The Poincaré-Dulac Theorem. *A formal mapping with resonant linear part is formally equivalent to a mapping whose linear part is in Jordan normal form and whose nonlinear part consists only of multiplicatively resonant terms.*

If the spectrum of the linear part of a germ of an analytic diffeomorphism at a fixed point belongs to the Poincaré domain, then the germ is analytically equivalent to the formal normal form described in the Poincaré-Dulac theorem. If the spectrum of the linear part belongs to the Siegel domain, the formal series reducing the germ to Poincaré-Dulac normal form are, with rare exceptions, divergent (a theorem due to A.D. Bryuno [23:31]). In the smooth case the following theorem holds.

Chen's Theorem [46]. *If two germs of diffeomorphisms at a hyperbolic fixed point are formally equivalent, then they are smoothly equivalent.*

2.3. Invariant Manifolds of Diffeomorphism Germs. The Hadamard-Perron theorem, the center manifold theorem, and the reduction principle (see Chap. 3, § 4) all hold for mappings.

Let f be a diffeomorphism. We recall that f^q , $q > 0$, denotes the q^{th} iterate $f \circ f \circ \dots \circ f$ (q times), and $f^{-q} = (f^q)^{-1}$ its inverse, where defined.

Definition. The *trajectory of a point p under the action of the diffeomorphism f* is the set of points $f^q(p)$ for all q at which the expression is defined.

Let $A: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a nonsingular linear map. The space $\mathbb{R}^n$ splits as a direct sum of three invariant subspaces

$$\mathbb{R}^n = T^s \oplus T^u \oplus T^c$$

(as in Chap. 3, § 4, the letters s, u, c stand for “stable”, “unstable”, and “center”, respectively). This splitting is determined by requiring that: the spectrum of the restriction $A|_{T^s}$ (resp., $A|_{T^u}$, $A|_{T^c}$) lie inside (resp., outside, on) the unit circle. The invariant stable manifold of the diffeomorphism germ $x \mapsto Ax + \dots$ is tangent to T^s at 0, the invariant unstable manifold is tangent to T^u , and the invariant center manifold is tangent to T^c . The theorems in Chap. 4, § 3 apply to mappings if we replace the words “vector field” and “differential equation” by “diffeomorphism”, the word “solution” by “trajectory of the diffeomorphism”, and if we replace continuous time by discrete time (t by q). In the discrete case, the role of the standard saddle is played by the mapping

$$(x, y, u, v) \mapsto (2x, -2y, u/2, -v/2),$$

where x, y, u, v are points of four subspaces with the property that the sum of the first two is equal to T^u , and the sum of the second two to T^s .

2.4. Invariant Manifolds of a Cycle. Consider a periodic solution of a differential equation and fix a neighbourhood U of the corresponding closed phase curve γ in phase space. Let Δ be the monodromy mapping which corresponds to this cycle; and let W^s , W^u and W^c be the stable, unstable and center manifolds of the germ of Δ at the fixed point corresponding to the cycle. We consider the union of all arcs of phase curves with initial point on W^s ; each arc being considered over the maximum time interval for which it remains in the neighbourhood U . The neighbourhood U can be chosen so that this union is a manifold, called the *stable manifold of the cycle γ* . This manifold is invariant with respect to the original equation, and all the solutions on it approach γ exponentially as $t \rightarrow +\infty$. The *unstable manifold of the cycle* is defined similarly; it consists of arcs of phase curves which

approach γ exponentially as $t \rightarrow -\infty$. The *center manifold of the cycle* γ is also defined similarly; in this case, the behaviour of the phase curves not only depends on the linear terms of the monodromy map, but on the nonlinear terms as well.

If the fixed point of the germ of the monodromy mapping Δ is hyperbolic, then the cycle γ is said to be *hyperbolic*. All cycles of a generic equation are hyperbolic.

Theorems about invariant manifolds in a neighbourhood of closed phase curves of analytic vector fields have been announced by A.D. Bryuno [23:42]. By the theorem of § 1.2, these carry over to the local theory of analytic diffeomorphisms.

2.5. Blow-ups. In conclusion we will mention a class of diffeomorphism germs for which the topological classification problem is algebraically soluble up to degeneracies of arbitrary finite codimension. This class is a class of diffeomorphism germs at a fixed point of the plane for which the classification problem is simultaneously nontrivial and soluble. But first we list classes of germs for which the problem is trivial.

Germes with hyperbolic singular points are studied using the Grobman-Hartman theorem. Germes whose linear part is a rotation through an irrational angle are either attractive or repulsive, except for a set of degenerate germes of infinite of codimension (this set, however, includes important classes such as the conformal and area-preserving mappings whose linear part at the fixed point is a rotation).

The remaining germes are those whose linear part is:

a) a rotation through a rational angle (the linear part of some iterate of such a germ is the identity);

b) a unipotent Jordan block $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

Definition. Suppose that f fixes the origin 0 and is a diffeomorphism of a neighbourhood of 0 in the real plane. Suppose the linear part of f at the origin is unipotent (possibly the identity). The trajectory of a point p under f is called a *characteristic trajectory* if it tends to 0 with a well-defined tangent. This means that $f^q(p)$ is defined for all $q \geq 0$ or all $q \leq 0$, that $\lim f^q(p) = 0$ (as $q \rightarrow +\infty$ or $q \rightarrow -\infty$), and that there exists a line l such that the distance from $f^q(p)$ to l vanishes to higher order than the distance from $f^q(p)$ to 0. A diffeomorphism germ at a fixed point is said to have a characteristic trajectory if any representative has a characteristic trajectory.

The existence of a characteristic trajectory of a diffeomorphism germ whose linear part is unipotent can be established from a finite jet of the germ. The topological type of a germ with a characteristic trajectory can be determined by resolving singularities [35] which involves performing a finite number of algebraic operations on a finite jet.

§ 3. Equations with Periodic Right Hand Side

In this section, we consider equations with a periodic right hand side in a neighbourhood of a constant solution, *viz.*, equations of the form

$$\frac{dx}{dt} = v(x, t), \quad \text{where } v(0, t) = 0 \quad \text{and} \quad v(x, t + 2\pi) \equiv v(x, t), \quad (1)$$

and x belongs to a neighbourhood of 0 in the space V with $V = \mathbb{R}^n$ or $V = \mathbb{C}^n$, and t is real. We shall suppose that t runs over the circle $S^1 = \mathbb{R}/(2\pi\mathbb{Z})$. By choosing a new time-variable τ and adding the equation

$$\dot{t} = 1, \quad \text{where the dot denotes } d/d\tau,$$

we make the original equation autonomous:

$$\dot{x} = v(x, t), \quad \dot{t} = 1, \quad v(0, t) = 0, \quad t \in S^1.$$

The monodromy mapping corresponding to the closed phase curve $x=0$ of this autonomous equation is called the *monodromy mapping of the (original) periodic equation*. This construction, together with the realization theorem in § 1, reduces the theory of periodic equations to the local theory of diffeomorphisms: all effects observed in the one theory are observed in the other. However, calculating the asymptotics of the monodromy map is, as a rule, impossible without reducing the periodic differential equation to normal form. The first step is to investigate the linear case.

3.1. Normal Form of a Linear Equation with Periodic Coefficients. Consider the linear periodic equation

$$\dot{x} = A(t)x, \quad A(t + 2\pi) \equiv A(t), \quad x \in \mathbb{C}^n.$$

The monodromy mapping of this equation is a nonsingular linear map $C: \mathbb{C}^n \rightarrow \mathbb{C}^n$, called the *monodromy operator*.

Floquet's Theorem. *There exists a change of variables $x = B(t)y$, linear in x and 2π -periodic in t , which transforms the original equation into an equation $\dot{y} = Ay$ with constant coefficients, where $C = \exp(2\pi A)$.*

Remark. A linear equation on a real phase space which is periodic in t can be reduced to an equation with constant coefficients by a real linear substitution $x = B(t)y$, but the period in t of the latter may necessarily be twice that of the right hand side. The reason: a nonsingular linear map has a complex logarithm which need not be real. However, the square of a real operator has a real logarithm.

Floquet's theorem allows us to change variables so that the linear part (with respect to x when $x=0$) of equation (1) becomes autonomous.

3.2. Linear Normal Forms. The eigenvalues of the linearization of a periodic differential equation $\dot{x} = Ax + \dots$ are said to belong to the Poincaré domain if they all lie in the left half plane $\operatorname{Re} \lambda < 0$ (or all lie in the right half plane).

Definition. An n -tuple $\lambda \in \mathbb{C}^n$ of eigenvalues of A is said to be *resonant* for the 2π -periodic equation $\dot{x} = Ax + \dots$ if

$$\lambda_j = (\lambda_0 m) + ik_1 \quad m \in \mathbb{Z}_+^n, \quad |m| = \sum m_j \geq 2, \quad k \in \mathbb{Z}, \quad 1 \leq j \leq n.$$

Poincaré's Theorem. A periodic differential equation whose linear part has a nonresonant spectrum belonging to the Poincaré domain can be reduced, in a neighbourhood of the zero solution, to the autonomous, linear, normal form $\dot{x} = Ax$ by a biholomorphic transformation which is 2π -periodic in t .

Siegel's Theorem [23:31]. A biholomorphic reduction to autonomous linear normal form exists for all equations whose linear part has spectrum belonging to a set of full Lebesgue measure in $\mathbb{C}^n$ (i.e., the set of λ for which there exist positive constants C and σ such that

$$|\lambda_j - (\lambda, m) - ik| > C(|m| + |k|)^{-\sigma} \quad \text{for all } m \in \mathbb{Z}_+^n, \quad k \in \mathbb{Z}, \quad |m| \geq 2).$$

3.3. Resonant Normal Forms. Let A be a linear map in Jordan normal form. Let $z = (z_1, \dots, z_n)$ be the coordinates with respect to a Jordan basis and suppose that z_j corresponds to the eigenvalue λ_j .

A monomial $z^m e^{ikt} \frac{\partial}{\partial z_j}$ is said to be a *resonant term* if the relation $\lambda_j = (\lambda, m) + ik$ holds.

In the resonant case, a periodic differential equation can, by means of a substitution which is formal in x and periodic in t , be reduced to the form

$$\dot{y} = Jy + w(y, t),$$

where J is a matrix in Jordan normal form, and w is a formal series which consists of resonant terms and which is a Fourier series in t and a Taylor series in y .

In the local theory of autonomous differential equations and the local theory of diffeomorphisms, we can calculate any finite number of terms of the Poincaré-Dulac normal form by a finite number of algebraic operations. In the case of periodic differential equations, even calculating the monodromy operator of the linearized system requires solving a linear system with periodic coefficients in $\mathbb{R}^n$; when $n > 1$, such equations cannot, as a rule, be solved by quadratures (see Chap. 7, § 3).

§ 4. Limit Cycles of Polynomial Vector Fields in the Plane

The so-called finiteness problem is to determine whether the number of limit cycles of every polynomial vector field on the real plane is finite. The finiteness proof advanced by Dulac [32] has a gap [59].

4.1. The Finiteness Problem and Compound Cycles. Poincaré reduced the finiteness problem to the study of neighbourhoods of so-called “compound cycles”.

Definition. A union of a finite (nonzero) number of singular points and finitely many phase curves which are not points is called a *compound cycle* of a vector field if: the solutions corresponding to the phase curves which are not points tend to the singular points as $t \rightarrow +\infty$ and $t \rightarrow -\infty$; if the cycle is connected; and if it cannot be contracted onto a proper subset of itself.

Remark. Compound cycles can consist of a single point. Compound cycles often result from resolving singularities. Examples of compound cycles are shown in Fig. 20.

The finiteness problem is equivalent to the following: can countably many limit cycles of a polynomial vector field accumulate onto a compound cycle of the field?

The answer is “no”, both for polynomial and analytic vector fields (see § 4.3 below).

4.2. The Monodromy Map of a Compound Cycle. The *monodromy map of a compound cycle* is defined in the same way as the monodromy of a closed phase curve, except that a *semi-transversal* is used instead of a transversal. A *semi-transversal* is the homeomorphic image of a semi-interval with vertex on the cycle (Fig. 20a); it is required to be transverse to the vector field at its interior points. Not all compound cycles have a monodromy map (Fig. 20b).

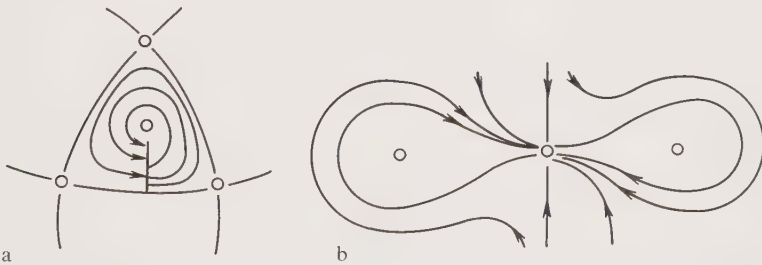


Fig. 20. A compound cycle which: a) has a monodromy map, b) does not have a monodromy map

If a semi-transversal belongs to an analytic curve which is transverse to the compound cycle, then the monodromy mapping is analytic at interior points of the semi-transversal; but it does not even have a C^1 extension to a neighbourhood of the vertex. This fact is connected with the nature of monodromy maps; these maps often belong to the class of “semi-regular mappings” introduced by Dulac [32].

Definition. A germ of a map $f: (\mathbb{R}^+, 0) \rightarrow (\mathbb{R}^+, 0)$ is said to be *semi-regular* if, for any sufficiently large integer N , there is a k such that

$$f(x) = Cx^{v_0} + \sum_1^k P_j(\ln x) x^{v_j} + o(x^N),$$

where $C > 0$, $0 < v_0 < v_1 < \dots < v_k \leq N$, and the P_j are polynomials. A representative of a semi-regular germ is called a *semi-regular mapping*.

Definition. A germ of a mapping at a point is said to be *flat* if all its derivatives at the point are equal to 0.

Dulac's Theorem [32]. *If a compound cycle of a polynomial vector field has a monodromy map, then there is a semi-transversal for which the monodromy map is either flat, semi-regular, or the inverse of a flat map.*

Generalization [59]. The conclusion of Dulac's theorem holds for smooth vector fields which satisfy a Lojasiewicz condition at each singular point.

If the closed phase curves of a vector field which satisfies Lojasiewicz conditions at its singular points accumulate to a compound cycle, then the compound cycle has a monodromy map with an isolated fixed point corresponding to each limit cycle situated close to the compound cycle.

Dulac uses the following lemma to deduce from his theorem that there is a neighbourhood of a compound cycle which is free from limit cycles:

A semi-regular map is either the identity or it has an isolated fixed point at 0.

But, here is a counter-example: $f = x + e^{-1/x} \sin(1/x)$.

There are examples of analytic vector fields with a compound cycle for which the monodromy map is the identity plus an additional nonzero, flat term [60].

4.3. Finiteness Theorems for Limit Cycles

Theorem 1. *A polynomial vector field on the real plane has only a finite number of limit cycles.*

Theorem 2. *An analytic vector field on a closed (real) two-dimensional surface has only a finite number of limit cycles.*

Theorem 3. *The germ at a singular point of an analytic vector field on the plane has a representative with no limit cycles.*

All the theorems above are consequences of the following result.

Theorem 4. *Limit cycles of an analytic vector field on a two-dimensional surface cannot accumulate to a compound cycle of the field.*

These theorems were announced by Yu.S. Il'yashenko in June, 1986 at the USSR Conference in Irkutsk on the Qualitative Theory of Differential Equations and by J.P. Martinet in May, 1987 at a conference in Oberwolfach. Citations for the published announcements are: J. Ecalle, J. Martinet, R.

Moussu, J.-P. Ramis, Non-accumulation des cycles limites. *C.R. Acad. Sci. Paris*, **304**, s. I Nr. 13 (1987) pp. 375–377; Nr. 14, pp. 431–434; and Yu.S. Il'yashenko, Finiteness theorems for limit cycles, *Russ. Math. Surv.*, v. **42**, no. 3 (1987) p. 223.

The proofs use the geometric theory of analytic normal forms discussed in § 5 of Chap. 5 below and the first result of section 4.4 below.

Theorem 3 implies that a singular point of an analytic vector field without characteristic trajectories is either a center or a focus. Before the proof of Theorem 3, the latter had been a widely disseminated “result” in the mathematical literature and folklore which had not, in fact, been proved.

Conjecture. For every n , there is an N such that an equation on the plane whose right hand side is a polynomial of degree n cannot have:

- a) a complex limit cycle of multiplicity greater than N [52];
- b) more than N real limit cycles (this is Hilbert's sixteenth problem, which has not even been solved for $n=2$).

4.4. Two Particular Finiteness Theorems. Some finiteness theorems have been published for vector fields satisfying additional restrictions.

Theorem [60]. *A polynomial vector field on the real plane, all of whose singular points (including those at infinity) are nondegenerate, has a finite number of limit cycles.* (The definition of a “singular point at infinity” will be given later in § 6.1.)

This theorem also holds for nondegenerate analytic vector fields on a closed two-dimensional surface.

Theorem (R. Bamon, *Publ. Math. I.H.E.S.*, **64** (1986) 111–142). *A quadratic vector field (that is, a polynomial vector field whose components have degree two) on the plane has only a finite number of limit cycles.*

4.5. Method of Proving Dulac's Theorem and its Generalization

Definition. A compound cycle is said to be *elementary* if all singular points on it are elementary.

Step 1. A compound cycle in the generalized Dulac theorem can be replaced by an elementary compound cycle. This is done by a good blow-up of the non-elementary singular points; such a blow-up exists by virtue of the Bendixson-Dumortier theorem (Chap. 5, § 1).

Step 2. The monodromy mapping of an elementary compound cycle is written as a composition of the correspondence mappings defined below.

The *correspondence mapping for a hyperbolic sector* of a singular point of a vector field on the plane is the map, obtained by following phase curves, of the semi-transversal through which the phase curves enter the sector to the semi-transversal through which the phase curves leave the sector (Fig. 21); we define the map at the vertex by demanding that the vertex of one semi-transversal be carried into the vertex of the other.

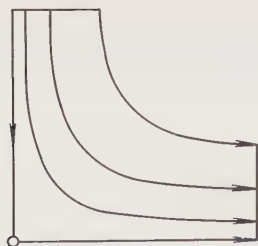


Fig. 21. Correspondence mapping of a hyperbolic sector

If an elementary compound cycle has a monodromy mapping, then all the singular points on it belong to classes 1, 4, and 5 in the table in Chap. 5, § 2.1. The classification theorem in Chap. 5, § 2.1 and § 2.2 allows us to obtain the following result.

Corollary. *The correspondence mapping for a hyperbolic sector of a smooth vector field on the real plane with a singular point satisfying a Lojasiewicz condition is semi-regular when the singular point is nondegenerate; when the singular point is degenerate it has the form $\exp(-1/h)$ or $h(-1/(\ln x))$ where h is a semi-regular mapping.*

The classification theorem reduces the proof of this assertion to the case where the vector field has the normal form shown in the table. In this case the correspondence mapping is easily calculated.

Step 3. It is proved that a composition of the correspondence mappings given in the preceding corollary belongs to one of the classes mentioned in Dulac's theorem (the germs of semi-regular mappings form a group).

For details, see Ju.S. Il'yashenko: "Dulac's memoir "On limit cycles" and related questions in the local theory of differential equations," *Russ. Math. Surv.*, **40**, No. 6 (1985) 41–78.

4.6. Quadratic Vector Fields. It has been proved that the number of cycles is finite in the case of a polynomial vector field of degree two (see § 4.4 above and also R. Bamon: "The solution of Dulac's problem for quadratic vector fields", *Ann. Acad. Bras. Ciênc.*, **57**, No. 3 (1985)). Some information about the form and disposition of the limit cycles has been obtained [27:3]. For example, they are always convex and each encloses only one singular point. If a monodromic compound cycle contains a saddle connection, then it contains a straight line segment, and so on [27]. Quadratic fields have been constructed with four limit cycles [25], [10].

§ 5. Limit Cycles of Systems Close to Hamiltonian Systems

In this section we consider perturbations of hamiltonian systems on the real and complex planes.

5.1. Generation of Real Limit Cycles. Suppose that a vector field on the real plane has a family of closed phase curves dependent on a parameter. Perturbing such a (highly degenerate) field will, generally speaking, give rise to a field with isolated nondegenerate cycles.

Example. Consider a small perturbation of a hamiltonian field:

$$\dot{x} = \frac{\partial H}{\partial y} + \varepsilon A(x, y, \varepsilon), \quad \dot{y} = -\frac{\partial H}{\partial x} + \varepsilon B(x, y, \varepsilon).$$

Definition. A cycle $\gamma(c)$ lying on a level curve $H=c$ is called a *generating cycle* if the perturbed equation has a cycle which depends continuously on ε for small $|\varepsilon|$ and which tends to $\gamma(c)$ as $\varepsilon \rightarrow 0$.

Definition. The *variation (of the monodromy)* is the function

$$I(c) = \oint (B dx - A dy)$$

(defined by integrating along $\gamma(c)$ when $\varepsilon=0$).

Theorem [94]. *If $\gamma(c)$ is a generating cycle, then $I(c)=0$. Conversely, if c is a simple zero of the variation, then $\gamma(c)$ is a generating cycle, and the cycle of the perturbed system that it generates is nondegenerate for small $|\varepsilon|$ and depends smoothly on ε .*

The proof is based on the fact that I is the derivative of the increment of H along a loop of the phase curve with respect to ε when $\varepsilon=0$.

5.2. Generation of Complex Cycles. We consider the family of equations $\omega_\varepsilon=0$ in the complex case. We suppose that ω_ε viewed as a 1-form on a two-dimensional complex manifold is holomorphic and holomorphically dependent on the (small) one-dimensional, complex valued parameter ε . Suppose, further, that the unperturbed form (corresponding to $\varepsilon=0$) is exact: $\omega_0=dH$, where H is a holomorphic function. Suppose that the equation $\omega_0=0$ has a family of integral curves which are not simply connected. The role of $\gamma(c)$ is played by a complex cycle represented by a closed path on a non-simply connected integral curve of the form ω_0 ; the path depends continuously (and the curve depends holomorphically) on the complex parameter ε .

The variation I is a holomorphic function of c :

$$I(c) = \int_{\gamma(c)} \omega, \quad \omega = \left. \frac{d\omega_\varepsilon}{d\varepsilon} \right|_{\varepsilon=0}.$$

The functions I so defined in this way are called *Abelian integrals* dependent on a parameter.

Theorem [88], [53:2]. *If a complex cycle $\gamma(c)$ is a generating cycle, then $I(c)=0$. An isolated zero of I corresponds to a complex generating cycle; the cycles it generates depend holomorphically on ε for small $|\varepsilon|$, $\varepsilon \neq 0$.*

Cycles corresponding to simple zeros of I also depend holomorphically on ε at $\varepsilon=0$.

5.3. Investigation of the Variation. Verifying that the zeros of the variation are isolated is simpler in the algebraic situation than otherwise.

Definition. A polynomial of degree $n+1$ in two complex variables is said to be *regular* if it has n^2 critical points with distinct critical values and its level curves have $n+1$ distinct asymptotic directions.

A generic polynomial is regular; the critical points of a regular polynomial are nondegenerate.

In a neighbourhood of a nondegenerate critical point, a polynomial can be reduced, by a biholomorphic change of coordinates, to the normal form $H = x^2 + y^2 + \text{const}$. The cycle on a nonsingular level curve of the function H which is represented in this coordinate system by the real circle $x^2 + y^2 = c$ when c is real and the circle

$$x + iy = \sqrt{c} e^{i\varphi}, \quad x - iy = \sqrt{c} e^{-i\varphi}, \quad \varphi \in [0, 2\pi], \quad (x, y) \in \mathbb{C}^2,$$

when c is complex, is called a *vanishing cycle of the critical point*. We let $\delta(c)$ denote the vanishing cycle on the level curve $H=c$ (this cycle is defined up to a sign, depending on the orientation, for values of c close to the critical value; it vanishes by shrinking into the critical point as c tends to the critical value of the function H).

Theorem [53]. Let H be a regular polynomial of degree $n+1$ and let ω be a 1-form in the plane with polynomial coefficients of degree no higher than n . If

$$I(c) = \oint_{\delta(c)} \omega \tag{2}$$

is identically zero (as a function of c), then the form ω is exact.

Remark. Thus, under the conditions of the theorem, a perturbation $dH + \varepsilon\omega = 0$ of the family of level curves of the function H has variation of the monodromy identically zero only in the obvious case where the perturbed integral curves are algebraic and, more precisely, level curves of a polynomial close to H and of the same degree as H .

5.4. The Weak Hilbert Conjecture. In his 16th problem, Hilbert conjectured that the number of limit cycles of a polynomial vector field in the real plane is bounded by a number depending only on the degree of the field.

Hilbert's conjecture implies that the number of generating cycles for polynomial perturbations of polynomial hamiltonian fields is bounded. In contrast to Hilbert's conjecture, this weaker statement has been proved.

Varchenko's Theorem [111]. *There is a number $N(n, m)$ such that, for any degree n real polynomial of two variables, for any continuous family of closed connected components of its level curves $\gamma(c)$, and for any polynomial 1-form ω with components of degree no higher than m , the integral of ω along $\gamma(c)$ is either identically equal to 0 in c , or it has no more than $N(n, m)$ real zeros (taking multiplicity into account).*

Remarks. 1. It has not been proved, even for polynomial perturbations of a polynomial hamiltonian system, that there are only finitely many generating limit cycles. The point is that limit cycles may arise, not only from closed phase curves, but also from compound cycles formed by the separatrices of a hamiltonian equation.

2. There are generalizations in [111] and [68] of the theorem to the higher dimensional and complex cases, but the proofs do not give an explicit estimate of the number N , even in principle.

5.5. Abelian Integrals Appearing in Bifurcation Theory. The particular integrals given below arise in the application of the theorem of §5.1 to the standard equations of the theory of bifurcations of general dynamical systems. In these particular cases the number of zeros of the variation usually turns out to be equal to the minimum allowed by dimensional considerations (which is related to a nonoscillation property of the corresponding Picard-Fuchs linear differential equation).

Example 1. A real n -dimensional space is called a *Chebyshev space* if non-zero functions on the space have less than n zeros counting multiplicity.

We let Ω_m denote the space of all polynomial 1-forms

$$\omega = A(x, y) dx + B(x, y) dy,$$

where A and B are real polynomials of degree no higher than m .

Theorem (G.S. Petrov, 1984–85).

1. *The space of integrals (2) of forms $\omega \in \Omega_m$ over vanishing cycles of the polynomial H is a Chebyshev space for any m . Here $H = y^2 - x^3 + 3x$ and the vanishing cycles correspond to the interval $|c| < 2$ between the two critical values.*

2. *The space of functions obtained by analytic continuation of the integrals above from the interval $|c| < 2$ to the plane $\mathbb{C}$ cut along the ray $c < -2$ and having the point $c = 2$ deleted (for integration over vanishing cycles at $c = 2$), or cut along the ray $c > 2$ and having the point $c = -2$ deleted (for integration over vanishing cycles at $c = -2$), is a Chebyshev space.*

3. *If $H = y^2 - x^3 + 3x$, the dimension of the spaces of integrals described in 1 and 2 above is equal to m .*

4. *The analogue of assertion 1 is true for the Hamiltonians $H_1 = y^2 + x^2 + x^4$, $H_2 = y^2 + x^2 - x^4$ and $H_3 = x^3 + y^3 + \lambda xy$. The dimension of the corresponding space for the Hamiltonians H_1 and H_2 is equal to $2[(m+1)/2] - 1$.*

A weaker variant of this theorem has been published: G.S. Petrov, *Funct. Anal. Appl.* **18**, No. 2 (1984) 73–74.

The proofs use an explicit form of the Picard-Fuchs theorem for the integrals above together with the theory of fewnomials [68].

Example 2. Algebraic-geometric methods of investigating the zeros of integrals arising in the theory of loss of stability of self-oscillations are described in [7] (see the references therein). In these cases, H is one of the polynomials

$$y^2 + 3x - x^3, \quad \pm y^2 \pm 2x^2 + x^4 \quad (\text{the signs } + \text{ and } - \text{ are independent}), \\ x^3 - 3xy^2 + 2(x^2 + y^2).$$

Example 3. The investigation of loss of stability in self-oscillations with two pairs of purely imaginary eigenvalues, and also the investigation of bifurcations of cycles in the generalized Lotka-Volterra model (in which the plane vector fields are permitted to be tangent to the coordinate axes) are based on an estimate of the number of zeros of the integral

$$I(c) = \oint \frac{\omega_1 + \lambda \omega_2}{x y z}, \quad \text{where } \omega_1 = \beta y dx - \alpha x dy, \quad \omega_2 = x y (x dy - y dx)$$

along a closed component of a level curve $H = c$ of the function

$$H(x, y) = x^\alpha y^\beta z^\gamma \quad \text{in the triangle } x > 0, y > 0, z > 0,$$

where $z = 1 - x - y$, $\alpha\beta = 1$, $\alpha > 0$, $\gamma > 0$. This number is equal to 1 (H. Zoladek, *J. Diff. Eq.*, **67**, No. 1 (1987) 1–55).

§ 6. Polynomial Differential Equations in the Complex Plane

Holomorphic vector fields on the complex projective plane are the fields of the Lie algebra of the projective group; they are linear in homogeneous coordinates. Other direction fields with a finite number of singular points also turn out to be algebraic. They are structurally unstable.

6.1. Admissible Fields

Definition. An *admissible direction field in the complex plane* is a field of lines which is holomorphic on the complement of an analytic subset of complex codimension at least two. The direction field of a polynomial vector field on $\mathbb{C}^2$ extends to an admissible direction field on $\mathbb{C}P^2$.

Theorem [56]. *An admissible direction field on the complex projective plane (or on a complex projective space of any dimension) is generated in any affine neighbourhood by a polynomial vector field.*

Definition. Two admissible fields on a complex manifold are said to be *topologically (analytically) equivalent* if there is a homeomorphism (biholomorphism) of the manifold on to itself which carries the integral curves of one field into the integral curves of the other.

Let $\mathcal{A}$ be a family, depending on a finite number of parameters, of admissible direction fields in $\mathbb{C}P^2$. We will say that a *generic field in class $\mathcal{A}$ has property A* if the set of parameter values corresponding to fields which do not have the property has Lebesgue measure zero.

We say that property A is Petrovsky-Landis generic if the set of parameter values corresponding to fields not having property A is nowhere dense and does not separate the parameter space $\mathcal{A}$ (i.e., its complement is path connected).

6.2. Polynomial Fields. Consider the admissible direction fields on $\mathbb{C}P^2$ corresponding to all polynomial vector fields of degree n in a fixed affine chart. We let $\mathcal{A}_n$ denote both the set of such admissible fields and the corresponding set of equations. The following property is Petrovsky-Landis generic for fields in $\mathcal{A}_n$: “the field has $n+1$ singular points on the line at infinity which is the complement of the affine chart”. These singular points are called the *singular points at infinity*. In a neighbourhood of such a singular point, the admissible direction field is the direction field of a vector field with a nondegenerate singular point. The ratio of the eigenvalues of the linear part of the field at the singular point is called the *characteristic number of the singular point* (where the eigenvalue in the denominator is the one corresponding to the eigenvector directed along the line at infinity).

After removing these points, the line at infinity becomes an integral curve. In what follows, we take $n \geq 2$.

Theorem [69]. *All but finitely many integral curves of a generic field in $\mathcal{A}_n$ are dense in $\mathbb{C}P^2$.*

The following property is Petrovsky-Landis generic for fields in $\mathcal{A}_n$: the only integral curve homeomorphic to a sphere with points removed is the line at infinity with the singular points at infinity removed.

Theorem. *If two fields in $\mathcal{A}_n$ with the property above are topologically equivalent, then the sets of characteristic numbers of the singular points at infinity are $\mathbb{R}$ -linearly equivalent (see [56], [84]). The last statement means that there is an $\mathbb{R}$ -linear mapping ${}^{\mathbb{R}}\mathbb{C} \rightarrow {}^{\mathbb{R}}\mathbb{C}$ which carries one set into the other.*

Remark. The proof uses the generalized Poincaré theorem about the rotation number.

Lemma [84]. *Suppose that two topologically equivalent germs of conformal mappings $(\mathbb{C}, 0) \rightarrow (\mathbb{C}, 0)$ have linear parts which are rotations. Then the angle of rotation is the same for both germs (except perhaps for sign).*

For smooth mappings this need not be the case. The lemma is true for mappings for which the boundary of the image of any sufficiently small neighbourhood of zero intersects the boundary of the pre-image.

For admissible direction fields corresponding to homogeneous vector fields in $\mathbb{C}^2$, $\mathbb{R}$ -equivalence of the set of characteristic numbers of the singular points at infinity implies topological equivalence [71].

Theorem [56]. *Almost every field in $\mathcal{A}_n$ has a neighbourhood in $\mathcal{A}_n$ for which there exists a neighbourhood of the identity homeomorphism of $\mathbb{C}P^2$ to itself such that every field in the first neighbourhood which is conjugate to the original field by a homeomorphism in the second neighbourhood is, in fact, affinely equivalent to the original field.*

This property of a field is called *absolute rigidity*.

The number of complex limit cycles of an equation in $\mathcal{A}_n$ (i.e., the number of cycles corresponding to isolated fixed points of the monodromy) is at most countable [88].

Theorem [56]. *Almost all equations in $\mathcal{A}_n$ have an infinite number of complex limit cycles which are homologically independent (in the sense that the cycles on an integral curve are independent as elements of the homology group with compact supports of the curve).*

Theorem [100]. *Density of all but finitely many trajectories, absolute rigidity, and possession of infinitely many homologically independent limit cycles are properties possessed by all fields in $\mathcal{A}_n$ except for those on a real-algebraic submanifold of real codimension one (and not simply a set of measure zero) in the space of coefficients.*

Remark. It appears that these properties are possessed by almost all admissible direction fields in $\mathbb{C}P^d$ generated by polynomial vector fields of degree $n > 1$ in $\mathbb{C}^d$ (cf. [55]).

Chapter 7

Analytic Theory of Differential Equations

This chapter deals with the theory of differential equations without movable critical points and linear equations with complex time.

§ 1. Equations Without Movable Critical Points

The concept of a critical point arises when solutions are studied as functions of a distinguished variable.

1.1. Definitions. 1. Let F be a holomorphic function in a domain of $\mathbb{C}^{n+2}$. By a *solution* of the differential equation

$$F(z, w, w', \dots, w^{(n)}) = 0 \quad (1)$$

we mean a complete analytic continuation of a germ of a holomorphic function $z \mapsto w(z)$ which satisfies the equation. Equation (1) is said to be *algebraic* if F is a polynomial in $w, w', \dots, w^{(n)}$.

2. A *critical point* is a point of ramification of the solution. (The term “critical point of a solution” is synonymous with “ramification point”; we adhere to the traditional terminology.)

3. A differential equation (1) has a *movable critical point* if the critical points of its solutions fill out a domain on the z -axis; points of this domain are called *movable critical points* of the equation.

1.2. Movable Critical Points of a First Order Equation

Example. Consider an equation

$$\frac{dw}{dz} = \frac{P(z, w)}{Q(z, w)}, \quad (2)$$

with a rational right hand side in which P, Q are polynomials. Suppose the curve $Q=0$ projects onto the whole z -axis. Then, all but a finite number of points on the z -axis are movable critical points of equation (2). In fact,

suppose a is a nonsingular point of the vector field $Q \frac{\partial}{\partial z} + P \frac{\partial}{\partial w}$ such that

$Q(a)=0$ and $Q|_{z=z(a)} \not\equiv 0$. Let φ_a be the integral curve of equation (2) which goes through a . The restriction $z|_{\varphi_a}$ is not constant; therefore, for some natural number m , the function $t=(z-z(a))^{1/m}$ is a local parameter on φ_a at the point a ; hence $z(a)$ is an algebraic ramification point of the solution with the initial condition a . Thus the movable critical points of the equation are singularities of the projection of the integral curves to the z -axis; the integral curves themselves are holomorphic (they have no singularities, see Chap. 1, § 1.9).

Painlevé's Theorem. *All movable critical points of an algebraic equation*

$$F(z, w, w') = 0 \quad (3)$$

are algebraic ramification points.

An algebraic equation (3) cannot have movable critical points of any kind other than those possessed by the equation analyzed in the previous example; the proof of Painlevé's theorem is based on this fact.

1.3. The Riccati Equation. The rest of this section is devoted to the investigation of equations without movable critical points (see [63], [43] and the references therein).

Theorem [43]. Equation (2) has no movable critical points if and only if it is the Riccati equation

$$\frac{dw}{dz} = a_0(z)w^2 + a_1(z)w + a_2(z).$$

◀ For, if equation (2) has no critical points, then the corresponding direction field extends analytically to the product $\mathbb{C} \times \mathbb{C}P^1$, $z \in \mathbb{C}$, $w \in \mathbb{C}P^1$, and the extended field is transverse to all but finitely many fibres $\{z\} \times \mathbb{C}P^1$. We lift $\frac{\partial}{\partial z}$ to a vector field v generated by the direction field. Projecting v on to the fibres $\{z \times \mathbb{C}P^1\}$ along the z -axis gives a family of holomorphic vector fields on the projective line. But such fields are given by second degree polynomials. ►

1.4. Implicit Equations. L. Fuchs found necessary and sufficient conditions for an algebraic equation (3) to have no movable critical points.

Suppose the function F is defined on a domain $\Omega \times \mathbb{C}^2$, $z \in \Omega$, $(w, p) \in \mathbb{C}^2$.

We complete the surface E given by the equation $F=0$ by adjoining “points at infinity” to get a closed surface $\bar{E}$ in $\Omega \times \mathbb{C}P^2$. The points at which the projection of the surface E along the p -axis onto the z, w -plane is not a local diffeomorphism are called, as in Chap. 1, § 7, *singular points* of the equation (3). The Fuchs conditions are explicitly prescribed restrictions on the behaviour of the surface of the equation near the points at infinity and near the singular points. A complete list of these conditions is given in the book [63]; in [43] one condition is omitted. Algebraic equations (3) without critical points either reduce to Riccati equations or can be explicitly integrated [43], [80]. In conclusion we cite the following theorem due to S. Malmquist.

Theorem. Let F be a polynomial in three variables. Then an equation (3) with at least one meromorphic, non-rational solution has no movable critical points.

The theory of algebraic equations (3) without movable critical points is presented, in the language of differential algebra, in [80]. A proof of Malmquist’s theorem and its generalizations (apparently the first such proof without gaps) can be found in [39], which also gives an extensive list of references.

1.5. Painlevé Equations. P. Painlevé and B. Gambier classified equations

$$w'' = R(z, w, w')$$

with no movable critical points under the condition that R be a rational function in w and w' with coefficients which are meromorphic functions in z defined on some domain Ω of the z -plane. Equations with these properties are often called *equations of class P*.

Definition. Consider two equations of class P . Let the right hand side of one equation be defined for $z \in \Omega_1$, and the right hand side of the other

for $z \in \Omega_2$. We shall say that the *first equation is induced from the second* if there is a holomorphic mapping $H: \Omega_1 \times \mathbb{C}P^1 \rightarrow \Omega_2 \times \mathbb{C}P^1$ which carries the fibres $z = \text{const}$ into one another by a linear-fractional transformation on each fibre $\{z\} \times \mathbb{C}P^1$, and carries the first equation into the second.

Painlevé and Gambier gave a list of 50 equations such that every equation of class P could be induced from one of the equations on the list [63], [43]. Of the 50 equations, 44 either integrate by quadratures or reduce to algebraic equations of the form (3); the remaining 6 are known as *Painlevé equations*. We enumerate them:

$$\text{I. } w'' = 6w^2 + z;$$

$$\text{II. } w'' = 2w^3 + zw + a;$$

$$\text{III. } w'' = w'^2 w^{-1} + e^z(aw^2 + b) + e^{2z}(cw^3 + dw^{-1}), |b| + |d| \neq 0;$$

$$\text{IV. } w'' = \frac{1}{2}w'^2 w^{-1} + \frac{3}{2}w^3 + 4zw^2 + 2(z^2 - \alpha)w + \beta w^{-1};$$

$$\text{V. } w'' = w'^2 \left(\frac{1}{2w} + \frac{1}{w-1} \right) - \frac{w'}{z} + \frac{(w-1)^2}{z^2} \left(\alpha w + \frac{\beta}{w} \right) + \gamma \frac{w}{z} + \delta \frac{w(w+1)}{w-1};$$

$$\begin{aligned} \text{VI. } w'' = & \frac{w'^2}{2} \left(\frac{1}{w} + \frac{1}{w-1} + \frac{1}{w-z} \right) - w' \left(\frac{1}{z} + \frac{1}{z-1} + \frac{1}{w-z} \right) + \\ & + \frac{w(w-1)(w-z)}{z^2(z-1)^2} \left[\alpha + \beta \frac{z}{w^2} + \gamma \frac{z-1}{(w-1)^2} + \delta \frac{z(z-1)}{(w-z)^2} \right]. \end{aligned}$$

All solutions of the first four Painlevé equations are meromorphic functions. Solutions of the fifth equation have logarithmic ramification points at $z=0$ and $z=\infty$; solutions of the sixth have them at $z=0$, $z=i$, and $z=\infty$. Recently a connection has been discovered between integrable differential equations of mathematical physics and equations of class P . For example, if w is a solution of the second Painlevé equation, the function

$$u(z, t) = t^{-2/3} (w'(z) + w^2(z))$$

is a solution of the Korteweg-de-Vries equation. For further details about Painlevé equations and their applications see [44].

§ 2. Local Theory of Linear Equations with Complex Time

This section deals with the theory of formal and analytic normal forms of linear equations and systems. The coefficients of the equations are the germs at 0 of meromorphic functions. If some coefficient has a pole at the point 0, we say that 0 is a *singular point*.

2.1. Regular and Irregular Singular Points

Definition. The singular point 0 of the system

$$\dot{z} = A(t)z, \quad z \in \mathbb{C}^n \tag{4}$$

is said to be *regular* if there is a (negative) number k such that all solutions of the system, in every sector of the t -plane with vertex at 0, grow no faster than the modulus of t raised to power k as t tends to zero within the sector (i.e., the exponent k is the same for all sectors and all solutions).

Note. It does not make sense to speak about the rate of growth of a many-valued function when the argument tends to a logarithmic ramification point unless we restrict the function to a sector: the function $\ln t$ increases as $|t|^{-1}$ when t tends to 0 along a particular spiral.

A singular point which is not regular is said to be *irregular*. Regularity and irregularity of a singular point 0 of the equation

$$x^{(n)} + a_1(t)x^{(n-1)} + \dots + a_n(t)x = 0 \quad (5)$$

are defined similarly.

Examples. 1. When $n=1$, equation (4) has a regular singular point at 0 if and only if A has a simple pole at 0.

2. If A is a constant matrix, the singular point 0 of the system $\dot{z} = (A/t)z$ is regular. The germ of the fundamental matrix of the solutions is of the form $X(t) = t^A$ where, by definition, $t^A = e^{A \ln t}$. We note that the analytic continuation of this germ will, generally speaking, have a logarithmic ramification point at 0.

3. Let L be a polynomial with complex coefficients. The equation

$$L\left(t \frac{d}{dt}\right)x = 0$$

is known as *Euler's equation*. Let $\lambda_1, \dots, \lambda_m$ be the roots of L and $k_1, \dots, k_m$ their multiplicities, respectively. Then a fundamental system of solutions of Euler's equation has the form

$$t^{\lambda_1}, t^{\lambda_1} \ln t, \dots, t^{\lambda_1} (\ln t)^{k_1-1}, \dots, t^{\lambda_m} (\ln t)^{k_m-1}.$$

It is clear that the singular point 0 of this equation is regular.

Remark. The substitution $t = e^\tau$ transforms Euler's equation into an equation $L\left(\frac{d}{d\tau}\right)x = 0$ with constant coefficients.

Definition. If the matrix function A in system (4) has a simple pole at 0, then the singular point 0 of (4) is said to be *Fuchsian*.

Theorem 1. A Fuchsian singular point of (4) is regular.

Remark. The converse is false. Indeed, the singular point of the Euler equation of order n is regular. We can transform this equation into system (4) by the substitution $z = (x, \dot{x}, \dots, x^{(n-1)})$. The singular point of the resulting system will be regular, and its matrix will, in general, have a pole of order n at 0.

Theorem 2. A singular point 0 of equation (5) (whose coefficients are germs of meromorphic functions at 0) is regular if and only if all functions $t^j a_j$ are holomorphic at 0.

The requirement that the functions $t^j a_j$ be holomorphic at 0 is known as the *Fuchs condition*.

Remark. The substitution $z = \left(x, \left(t \frac{d}{dt} \right) x, \dots, \left(t \frac{d}{dt} \right)^{n-1} x \right)$ transforms an equation (5) which satisfies the Fuchs condition at 0 into a system (4) with a simple pole at 0. This indicates that $\ln t$ is the natural time for equations with a regular singular point.

2.2. Formal, Holomorphic, and Meromorphic Equivalence. In this section, π denotes the projection $\mathbb{C} \times \mathbb{C}^n \rightarrow \mathbb{C}$ along the second factor.

Definition. Two equations (4) are said to be *holomorphically* (resp., *meromorphically*) *equivalent* if there is a holomorphic (resp., meromorphic with a pole at the point 0) mapping of $B \times \mathbb{C}^n$ into itself which is linear in z , preserves the time t , and transforms one equation into the other. Here B is a neighbourhood (resp., punctured neighbourhood) of 0. In other words, there is a holomorphic (resp., meromorphic) mapping $H: B \rightarrow \text{GL}(n, \mathbb{C})$, called a *conjugating map*, such that the substitution $(t, z) \mapsto (t, H(t)z)$ carries one equation into the other. (Conjugating maps are automorphisms of the trivial vector bundle with one-dimensional base B and fibre $\mathbb{C}^n$.)

Two systems $\dot{z} = A(t)z$ and $\dot{w} = B(t)w$ are conjugate by means of the map H if and only if $B = \dot{H}H^{-1} + HAH^{-1}$.

Definition. If H is a formal Taylor (or Laurent) series in t with coefficients in $\text{GL}(n, \mathbb{C})$ which formally satisfies the relation above, then the systems are said to be *formally $\mathcal{O}$ -equivalent* (resp., *formally $\mathcal{F}$ -equivalent*). Here, $\mathcal{O}$ denotes the ring of functions holomorphic at 0 and $\mathcal{F}$ its quotient field. Sometimes, we speak of “formal holomorphic equivalence” and “formal meromorphic equivalence” instead of “ $\mathcal{O}$ -equivalence” or “ $\mathcal{F}$ -equivalence”.

Theorem [28]. *Germs of systems with a regular singular point which are formally $\mathcal{O}$ -equivalent (resp., $\mathcal{F}$ -equivalent) are holomorphically (resp., meromorphically) equivalent and the conjugating formal series converges.*

This does not hold for systems with an irregular singular point.

2.3. Monodromy. Let $t_0 \neq 0$ be a point at which the matrix A of system (4) is defined. By analytic continuation along a loop which starts and ends at t_0 and goes around the point 0 once in the positive direction, the space of germs at t_0 of solutions of (4) maps into itself. The automorphism is linear and does not depend on the choice of loop with the above properties; it is called the *monodromy map*. For example, the monodromy map of the system $\dot{z} = (A/t)z$ is equal to $\exp(2\pi i A)$.

Knowing just the monodromy enables us to write down the solutions of a system (4) with a regular singular point.

Theorem. *A system (4) with a regular singular point is meromorphically equivalent to the system $\dot{z} = (C/t)z$, where $2\pi iC$ is any value of the logarithm of the monodromy map of system (4).*

◀ For let the point t_0 be as above, and let X be the germ of the fundamental matrix of solutions of system (4) at the point t_0 . Consider the analytic continuation of the germ $H = X t^{-C}$ on a punctured neighbourhood of 0. It is single-valued, because both the germs X and t^C are multiplied on the right by the matrix $T = \exp(2\pi iC)$ of the monodromy map upon being analytically continued over a loop circling 0. The continuation of the germ H is meromorphic at 0 because 0 is a regular singular point. Finally, t^C is the fundamental matrix of solutions of the system $\dot{z} = (C/t)z$. ▶

The last two theorems enable us to seek solutions in a neighbourhood of a regular singular point of linear differential equations and systems by seeking formal solutions which are series in powers of t and $\ln t$ without bothering about convergence [28].

The search for formal solutions of (2) starts by solving the “indicial equation”. When equation (5) has regular singular point 0, it can be written in the form

$$\left(t \frac{d}{dt}\right)^n x + b_1(t) \left(t \frac{d}{dt}\right)^{n-1} x + \dots + b_n(t) x = 0,$$

where the b_j are holomorphic functions. In this case the *indicial equation* has the form

$$\lambda^n + b_1(0) \lambda^{n-1} + \dots + b_n(0) = 0.$$

If the roots λ_j of this equation are pairwise distinct, then equation (5) has a fundamental system of solutions $\varphi_j(t) t^{\lambda_j}$, where the φ_j are holomorphic functions and $\varphi(0) \neq 0$ (see, e.g., [28]).

2.4. Formal Theory of Linear Systems with a Fuchsian Singular Point. We can use the general methods of the theory of Poincaré normal forms (Chap. 3) to help with the formal classification of linear systems. Consider the nonautonomous system

$$t^r \dot{z} = A(t)z, \quad A(0) \neq 0, \quad (6)$$

where the matrix-valued function A is holomorphic at 0. We consider the corresponding autonomous system

$$z' = A(t)z, \quad t' = t^r,$$

where τ is the new time, and the prime ' denotes $\frac{d}{d\tau}$.

The phase curves of the new system coincide with the integral curves of the old system off the plane $z=0$. The Poincaré-Dulac theorem (Chap. 3,

§ 3.2) enables us to find a formal normalizing substitution which effects the reduction of the autonomous system to a formal normal form all of whose nonlinear terms are resonant. This theorem can be strengthened for systems linear in z : the normal form and normalizing substitution are linear in z , and the latter preserves the time t .

Definition. An n -tuple $(\lambda_1, \dots, \lambda_n) \in \mathbb{C}^n$ of eigenvalues of $A(0)$ is *resonant* for a system (6) with a fuchsian singular point ($r=1$) if the difference of some pair of the λ_j is an integer.

Theorem. If $t\dot{z} = A(t)z$ is a system with a fuchsian singular point and if the spectrum of the matrix $A(0) = A$ is nonresonant, then the system is formally $\mathcal{O}$ -equivalent, and therefore holomorphically equivalent, to the standard system $t\dot{z} = Az$.

Corollary. Under the conditions of the previous theorem, the fundamental matrix of solutions of the system has the form $\varphi(t)t^A$, where $\varphi: B \rightarrow \text{GL}(n, \mathbb{C})$ is a holomorphic matrix-valued function and B is a neighbourhood of 0.

Proof of the Theorem. Let $(\lambda_1, \dots, \lambda_n)$ be the spectrum of the matrix $A = A(0)$, and suppose the singular point is fuchsian, i.e., $r=1$ in formula (6). The linear part of the corresponding autonomous system has eigenvalues $\lambda_1, \dots, \lambda_n, 1$. The monomial $t^k z_i \frac{\partial}{\partial z_j}$ is resonant if and only if $\lambda_j = \lambda_i + k$. The theorem now follows from the strengthened Poincaré-Dulac theorem. ►

In the resonant case, the expression for the fundamental matrix of solutions is more complicated, but the formal normal form of a linear system with a regular singular point can always be integrated. The Frobenius method is based on this fact; it enables one to integrate equation (5) with a regular singular point using series [28], whether there are resonances or not.

2.5. Formal Theory of Linear Systems with a Non-Fuchsian Singular Point

Definition. An n -tuple $(\lambda_1, \dots, \lambda_n)$ is said to be *resonant* for a system (6) with a non-fuchsian singular point (i.e., $r > 1$) if two of the λ_i are equal and *nonresonant* otherwise. The system (6) is said to be *resonant* or *nonresonant* accordingly.

Theorem. Let $r > 1$ in system (6) and suppose that the spectrum of the matrix $A(0)$ is nonresonant. Then the system is formally $\mathcal{O}$ -equivalent to the system

$$t^r \dot{w} = B(t)w, \quad B(t) = \text{diag}(b_1(t), \dots, b_n(t));$$

where $B(t)$ is a diagonal matrix with polynomials $b_i(t)$ of degree no higher than $r-1$ on the diagonal.

Remark. The latter system is called the *formal normal form* of system (6) or a *normalized system*. It splits into 1-dimensional equations and can be

integrated. Let

$$\int B(t) t^{-r} = D(t^{-1}) + C \ln t,$$

where C is a constant diagonal matrix, and D is a diagonal matrix with polynomials of degree no higher than $r-1$ on the diagonal.

Corollary. *The fundamental matrix of solutions of the system (6) has the form*

$$X(t) = \hat{H}(t) t^C \exp D(t^{-1}),$$

where $\hat{H}$ is a matrix whose entries are formal Taylor series [113], [28].

The proofs of this theorem available in the Russian literature are cumbersome; we therefore sketch a short proof.

We replace system (6) by the autonomous system

$$\frac{dz}{d\tau} = A(t)z, \quad \frac{dt}{d\tau} = t^r.$$

◀ Let $\lambda_1, \dots, \lambda_n$ be the spectrum of the matrix $A(0)$. The linear part of the autonomous system has the spectrum $\lambda_1, \dots, \lambda_n, 0$. We apply the Poincaré-Dulac theorem, as modified in § 2.3, to this system. Since a monomial $t^k z_i \frac{\partial}{\partial z_j}$ is resonant if and only if $\lambda_i = \lambda_j$, there is a formal substitution $(t, z) \mapsto (t, \tilde{H}(t)z)$ which carries the autonomous system to one of the form

$$\frac{dw}{d\tau} = \tilde{B}(t)w, \quad \frac{dt}{d\tau} = t^r,$$

where $\tilde{B}$ is a diagonal matrix with formal Taylor series on the diagonal. We put

$$\tilde{B}(t) = B(t) + t^r B_1(t).$$

Then the fundamental matrix of solutions of the system $t^r \dot{w} = \tilde{B}(t)w$ has the form $W = e^{B_2(t)} t^C \exp D(t^{-1})$, where C and D are as above and $\dot{B}_2 = B_1$. The substitution $(t, z) \mapsto (t, \hat{H}z)$, $\hat{H} = e^{-B_2} \tilde{H}$ is the one needed. ►

We formulate a similar result for resonant systems.

Definition. Two equations

$$\dot{z} = A(t)z, \quad \dot{w} = B(t)W,$$

where A and B are formal series in a fractional power of t , are *formally fractionally meromorphically equivalent* if there exists a formal series H in fractional powers of t , with coefficients in $GL(n, \mathbb{C})$, such that:

$$B = \dot{H}H^{-1} + HAH^{-1}.$$

Here the series A, B, H have only a finite number of terms with negative exponents.

Theorem [10]. Any equation of the form (6) is formally fractionally meromorphically equivalent to the normal form:

$$t^{r_0} \dot{w} = B(t); \quad B = B_{r_1} z^{r_1} + \dots + B_{r_m} z^{r_m} + C z^{r_0-1}$$

where $0 < r_1 < \dots < r_m < r_0 - 1$; $\{r_0, \dots, r_m\} \in q^{-1}\mathbb{Z}$ for some positive integer q ; the operators B_{r_j}, C commute pairwise; and all the B_{r_j} are diagonalizable. The numbers $r_j - r_0$ are called the canonical levels of B and are invariants of the formal fractional meromorphic classification.

2.6. Asymptotic Series and the Stokes Phenomenon. The normalizing substitutions described in the previous section sometimes diverge. For example, the series $\sum k! t^k$ is a formal solution of the equation $t^2 \ddot{x} + (3t-1)\dot{x} + x = 0$ [28]. However, formal substitutions enable us to find so-called “asymptotic series”, which approximate proper solutions in sufficiently narrow sectors with vertex at 0.

Definition. Let f be a holomorphic function in a sector S with vertex at 0. A series $\sum f_k t^k$ is called an *asymptotic series* for f if, for every k , its k th partial sum differs from f by $o(t^k)$ as $t \rightarrow 0$ in S .

Theorem. Under the hypotheses of the theorem in § 2.5, every ray with vertex 0 belongs to some sector S in which the system (6) is analytically equivalent on $S \times \mathbb{C}^n$ to its formal normal form. Moreover, there is a holomorphic map $H: S \rightarrow \text{GL}(n, \mathbb{C})$ effecting the conjugation of the equation to its normal form and such that $\hat{H}$ is an asymptotic series for H in the sector S ; here, $\hat{H}$ is the formal substitution given by the theorem of § 2.5.

This theorem can be sharpened.

Definition. A ray, with vertex at 0, on which $\text{Re}(\lambda_i - \lambda_j) t^{1-r} = 0$, is called a *separating ray* corresponding to the pair (λ_i, λ_j) .

Complement. In the preceding theorem, we may take the sector S to be any closed sector with vertex 0 which satisfies the following condition: S does not contain two separating rays corresponding to any single pair (λ_i, λ_j) .

We now describe the *Stokes phenomenon* for a system with a non-Fuchsian singular point 0. Consider two intersecting sectors S_1 and S_2 with vertex at 0 which each satisfy the preceding requirement but whose union does not. Let H_{S_1} and H_{S_2} be the normalizing mappings (which preserve t) corresponding to the sectors S_1 and S_2 . Let $S = S_1 \cap S_2$. The mapping $H_S = H_{S_1} \circ H_{S_2}^{-1}$ also preserves t and carries one normalized system into the other. Let L_S denote the space of all solutions of the normalized system which are defined in the sector S . The mapping H_S defines a linear automorphism C_S of the space L_S which is called the *Stokes operator*.

We note that the mapping H_S has a trivial asymptotic expansion $H_S \sim E$ since the asymptotic expansions for H_{S_1} and H_{S_2} coincide. That is, $H_S = E + o(t^N)$ as $t \rightarrow 0$ in S for any positive integer N .

2.7. Analytic Classification of Nonresonant Systems in a Neighbourhood of an Irregular Singular Point. We consider the class of nonresonant systems (6) with a non-fuchsian singular point. We suppose that $A(0)$ is diagonal (which can be achieved in the nonresonant case by a linear change of variables). In this section, we consider "strong holomorphic equivalence" of such systems in which it is required that the conjugating substitution differ from the identity by $O(t)$ as $t \rightarrow 0$: $H = E + O(t)$. The Stokes operators are invariants of the strong holomorphic classification of such systems. We describe what sort of operators can arise as the Stokes operators. This description is facilitated by the fact that the normalized system integrates explicitly; we fix this system.

To every sector $S = S_1 \cap S_2$, where S_1, S_2 are as in § 2.6, there corresponds a Stokes operator C_S and a mapping $H_S: S \rightarrow \text{GL}(n, \mathbb{C})$, $H \sim E$. The solutions of the form

$$\varphi_j(t) = a_j(t) \frac{\partial}{\partial w_j}, \quad \text{where } \dot{a}_j = t^{-r} b_j(t) a_j, \quad b_j(0) = \lambda_j,$$

form a basis of the space L_S . We note that

$$a_j(t) = \exp[\lambda_j t^{1-r}(1/(1-r) + O(t))].$$

The mapping H_S and the Stokes operator C_S are related as follows. Let

$$C_S \varphi_i = \psi_i, \quad \psi_i = \sum c_{ij} \varphi_j.$$

Then

$$H_S(t) \frac{\partial}{\partial w_i} = a_i^{-1}(t) \psi_i(t).$$

Since $H_S \sim E$, we obtain $c_{ii} = 1$; $c_{ij} \neq 0$ implies that $\text{Re}(\lambda_i - \lambda_j) t^{1-r} \rightarrow -\infty$ as $t \rightarrow 0$ in S .

Definition. An operator $C_S: L_S \rightarrow L_S$ which satisfies the preceding condition will be called *admissible* for the sector S and the system (7).

Definition. Cover a disc centred at 0 by a finite number of sectors S_j with vertex at the center, and suppose that no sector contains two separating rays corresponding to a single pair (λ_p, λ_q) . To each ordered pair S_i, S_j with a non-empty intersection $S_{ij} = S_i \cap S_j$, we associate the operator C_{ij} which is admissible for this sector and the normalized system (7). These operators satisfy $C_{ij} = C_{ji}$ and $C_{ij} C_{jk} C_{ki} = E$. The set $\{C_{ij}\}$ is called a *Stokes system*.

Theorem [18], [104]. *Every Stokes system can be realized as a set of Stokes operators for a nonresonant system with an irregular singular point formally equivalent to a prescribed normalized system.*

Theorem [104]. *Two formally equivalent nonresonant systems with an irregular singular point and with the same collection of Stokes operators are holomorphically equivalent in a neighbourhood of the singular point.*

Similar theorems also hold in the resonant case, but they then pertain to the fractional meromorphic classification. Moreover, the construction and

description of the Stokes operators is not so explicit. In the original papers, these results are phrased in the language of cohomology theory, a Stokes system being interpreted as a 1-cocycle of a so-called "Stokes bundle" over a circle, and so on (see [18]).

In conclusion we note that the cokernel of the differential operator

$$t^{-r} \frac{d}{dt} - A(t)$$

from the space of germs of vector functions meromorphic at 0 to itself is always finite-dimensional [18].

§ 3. Theory of Linear Equations in the Large

This section deals with the theory of linear equations and systems with regular singular points on $\mathbb{CP}^1$ and its applications to Abelian integrals and Kleinian groups.

3.1. Equations and Systems of the Fuchsian Class. We formulate the "Fuchsian condition at infinity".

Definition. A system $\dot{z} = A(t)z$ has a *Fuchsian singular point* (resp., *nonsingular point*) at infinity if the system obtained by making the substitution $\tau = 1/t$ has a Fuchsian singular point (resp., nonsingular point) at 0. The *Fuchsian condition at infinity* for an equation

$$x^{(n)} + a_1(t)x^{(n-1)} + \dots + a_n(t)x = 0$$

is defined similarly.

An Equivalent Definition. The system $\dot{z} = A(t)z$ (resp., equation (5)) satisfies the Fuchsian condition at infinity if the matrix-function $tA(t)$ (resp., the vector-function $(ta_1(t), \dots, t^n a_n(t))$) is holomorphic at infinity.

Remark. A system $\dot{z} = A(t)z$ has only Fuchsian singular points on the Riemann sphere if and only if it has the form

$$\dot{z} = \sum \frac{A_j}{t - \alpha_j} z_j, \quad \text{where the } A_j \text{ are constant matrices.} \quad (8)$$

Equation (5) has only Fuchsian singular points on the Riemann sphere if and only if $a_j(t) = P_j(t)Q^{-j}(t)$, where $Q(t) = \prod (t - \alpha_k)$ and $\deg P_j \leq (m-1)j$. The matrices A_j in the system (8) are called *matrix-residues*.

The equations and systems described by the theorem are said to be *Fuchsian*.

3.2. Extensibility and Monodromy. In what follows, the universal covering of a domain Ω is denoted by $\hat{\Omega}$. To say that a germ of an analytic function

at a point $p \in \Omega$ extends to the universal covering $\hat{\Omega}$ means that the germ can be continued over any curve $\gamma \subset \Omega$ starting at p and the result is not changed under a continuous deformation of γ in Ω which preserves the initial point and endpoint of the curve.

Theorem. *Every solution of a linear equation or a system with meromorphic coefficients extends to the universal covering of the domain on which the coefficients are holomorphic.*

This theorem is the analogue, in the complex-valued case, of the theorem guaranteeing that solutions of linear equations with real time extend over any interval in which the coefficients are continuous.

By extending all solutions over a loop which does not pass through a pole of the coefficients, the space of germs of solutions at the initial point of the loop maps into itself. This automorphism is linear and is called the *monodromy map*. Making successive circuits around loops corresponds to taking products of the monodromy map. We obtain a homomorphism of the fundamental group of the domain on which the coefficients are holomorphic into a subgroup of $GL(n, \mathbb{C})$. This homomorphism is called the *monodromy* of the equation or system; the linear operator corresponding to the loop γ depends only on the homotopy class of the loop, and is denoted by T_γ . The image of the monodromy homomorphism is called the *monodromy group*.

3.3. The Riemann-Fuchs Theorem. Solutions of fuchsian equations extend to the universal cover of the punctured complex t -axis with the poles of the coefficients deleted. These solutions determine the monodromy group and are regular (in any sector with vertex at a singular point they increase no faster than some power of the distance from the singular point). It turns out that fuchsian equations are the only equations whose solutions have these properties.

Theorem [95]. *Suppose that a germ of a holomorphic vector-function φ extends holomorphically to the universal cover of the Riemann sphere with the points $\alpha_1, \dots, \alpha_m, \infty$ deleted, and suppose that the Wronskian of the extended vector-function (which is also denoted by φ) vanishes nowhere. Suppose that φ defines a monodromy group; that is, extending it over each loop in the punctured Riemann sphere induces a linear automorphism of the linear space generated by the components of the germ. Suppose that the extension is regular; that is, as t tends to a deleted point α within a sector with vertex at α , the modulus of $\varphi(t)$ increases no faster than some power of the distance to α on the Riemann sphere. Then there is a fuchsian equation, for which φ is the germ of a fundamental system of solutions.*

Corollary 1. *Any algebraic function of a single variable satisfies a fuchsian equation.*

◀ Consider the set Ω of nonsingular points of the function and let $p \in \Omega$ be an arbitrary point. Let $\varphi_1, \dots, \varphi_m$ denote the germs of holomorphic functions at p corresponding to different sheets of the algebraic function. Select from among them a maximal linearly independent collection $\varphi_1, \dots, \varphi_n$ (n may be less than m ; e.g., $\sqrt[m]{t}$). The germ $\varphi = (\varphi_1, \dots, \varphi_n)$ extends to the universal cover $\hat{\Omega}$ and generates a monodromy group; the sheets of the algebraic function are permuted under extension along a loop. The Wronskian W of the vector-function φ is multiplied by a constant (equal to the determinant of the monodromy mapping); therefore $W=0$ at a finite number of points $\beta_j \in \Omega$ and these points also have to be deleted. The proof that an algebraic function is regular at singular points is elementary. Hence, we can apply the previous theorem. ▶

Corollary 2. *The Abelian integral depending on a parameter, considered in Chap. 6, § 5.2, satisfies a fuchsian equation (called the Picard-Fuchs equation).*

◀ We give the proof, assuming that H is regular. The ramification points of the integral are the critical values of the Hamiltonian H and infinity; the monodromy is described by the Picard-Lefschetz theorem [31]; the proof of regularity is elementary. ▶

3.4. Analytic Functions of Matrices. I.A. Lappo-Danilevskiy applied the theory of analytic functions of matrices to the calculation of monodromy groups of linear differential equations and to the reconstruction of the equation from the monodromy group.

Lappo-Danilevskiy's investigations relate mainly to Fuchsian systems. Consider the system (8). Fix loops $\gamma_1, \dots, \gamma_m$ with a common initial point, each of which goes once around just one pole of the coefficients. The corresponding monodromy matrices $T_j = T_{\gamma_j}$ generate the monodromy group of the system (8).

For fixed poles α_j the monodromy matrices depend only on the matrix-residues A_j .

Theorem 1. *The matrices T_j are entire functions of the matrix-residues A_j :*

$$T_j = E + 2\pi i A_k + \sum_{1 \leq k, j \leq n} a_{ki}(\alpha) A_k A_j + \dots, \quad \alpha = (\alpha_1, \dots, \alpha_m);$$

where the right hand sides are power series in the non-commuting matrix variables A_j which converge for all values of the variables.

Theorem 2. *If the monodromy matrix is sufficiently close to the identity matrix, then there is a unique fuchsian system with the given monodromy, whose matrix-residues are close to the zero matrix. They are given by the formulas*

$$A_j = \frac{1}{2\pi i} (T_j - E) + \sum_{1 \leq k, j \leq n} b_{kj}(\alpha) (T_k - E)(T_k - E) + \dots;$$

where the right hand sides are power series in $T_1 - E, \dots, T_m - E$ which converge in a neighbourhood of 0.

The latter series are obtained by reversion of the former series. Theorem 2 is a sort of implicit function theorem (for non-commuting variables). Lappo-Danilevskiy calculated the coefficients a_{kj} , b_{kj} , ... as functions of α [73]; see also [62].

3.5. Connection with the Theory of Kleinian Groups. The space of germs of meromorphic and holomorphic solutions of the Riccati equation at a point distinct from a pole of the coefficients and from ∞ is isomorphic to $\mathbb{C}P^1$. Extending the solutions over a loop around a pole of the coefficients, induces a map of the solution space onto itself which is a fractional-linear transformation. The group of all transformations constructed in this way is called the *monodromy group of the Riccati equation*.

Consider a second-order fuchsian differential equation. Its monodromy group is a subgroup of $GL(2, \mathbb{C})$. We map the phase space $\mathbb{C}^2$ to $\mathbb{C}P^1$: $(z_1, z_2) \mapsto (z_1 : z_2)$. The original equation goes over into a Riccati equation; its monodromy group becomes the monodromy group of the Riccati equation. A group of fractional-linear (Möbius) transformations $\mathbb{C}P^1 \rightarrow \mathbb{C}P^1$ is called a *Kleinian group* if there is a domain on which it operates discretely (no orbit of the group accumulates at an interior point of the domain). Every Kleinian group can be realized as the monodromy group of a Riccati equation (this follows from the affirmative solution of the Riemann-Hilbert problem for Fuchsian equations; see §4). This connection enriches both the theory of Kleinian groups and the theory of differential equations [92]. In particular, it is convenient to study deformations of Kleinian groups by deforming the coefficients of the corresponding equations.

3.6 Integrability in Quadratures. Liouville showed that second-order linear equations are not, in general, integrable by quadrature; the solution cannot be expressed in terms of the coefficients by means of arithmetical operations, solving algebraic equations, raising to powers, and integration. (Note: solving algebraic equations in this context does not mean solving by radicals; it is assumed that the set of zeros of each polynomial is known, and that the integral of each function is known.) In particular, the equation $\ddot{x} + tx = 0$ is not integrable by quadrature.

The general theory of integrability by quadrature of linear equations and systems is based on the methods of differential algebra. To every linear equation or system with rational coefficients, one associates a Galois group which is solvable when the corresponding equation or system is integrable (see [65], [67]). In conclusion we formulate the following geometrical result.

Khovanskiy's Theorem [62], [67]. *If the monodromy group of a fuchsian system has a solvable normal divisor of finite index, then the system is integrable in quadratures. If the monodromy group does not have this property, then the system is not even integrable by "generalized quadrature". This means that the solution of the system cannot be expressed in terms of the coefficients*

by solving algebraic equations, integration, and composition with entire functions of any number of variables.

3.7. Special Equations of Mathematical Physics. A second-order fuchsian equation with three singular points on the sphere is called a *Riemann equation*. Its coefficients are uniquely determined by the singular points and the roots of the defining equations corresponding to these points. Second-order equations with more than three singular points and higher order equations do not have this property. The Riemann equation with singular points 0, 1, ∞ is *Gauss's hypergeometric equation*

$$t(t-1)\ddot{x} + [(\alpha + \beta + 1)t - \gamma]\dot{x} + \alpha\beta x = 0.$$

Linear second-order equations are often encountered in mathematical physics. Most of them reduce to particular or limiting cases of the hypergeometric equation. Examples of such equations are:

$$\text{Bessel's equation: } t^2 \ddot{x} + t \dot{x} + (t^2 - \nu^2)x = 0$$

$$\text{Weber's equation: } \ddot{x} + (a - t^2)x = 0$$

$$\text{Legendre's equation: } (t^2 - 1)\ddot{x} + 2t\dot{x} - \nu(\nu + 1)x = 0.$$

General linear second-order equations with linear coefficients reduce to the degenerate hypergeometric equation

$$t\ddot{x} + (b - t)\dot{x} - ax = 0.$$

All the basic ordinary differential equations encountered in mathematical physics are obtained from the following equation with 5 regular singular points

$$\ddot{x} + \frac{1}{2} \sum \dot{x}/(t - a_r) + \frac{3}{16}(A + Bt)t^2 x / \prod (t - a_r) = 0 \quad (1 \leq r \leq 4)$$

(a theorem due to Klein and Bôcher; see [63, Chap. XX]).

3.8. The Monodromy Group of the Gauss Equation. The monodromy group of the hypergeometric equation with real α, β, γ is connected with the triangle bounded by circular arcs meeting at angles $\pi\lambda, \pi\mu$ and $\pi\nu$ where $\lambda^2 = (1 - \gamma)^2$, $\mu^2 = (\gamma - \alpha - \beta)^2$, $\nu^2 = (\alpha - \beta)^2$.

We suppose that $\lambda \in (0, 1)$, $\mu \in (0, 1)$ and $\nu \in (0, 1)$. The group generated by the reflections in the side of the triangle contains a subgroup G of index two consisting of fractional-linear transformations. The standard projection $\mathbb{C}^2 \setminus \{0\} \rightarrow \mathbb{C}P^1$ carries the monodromy group of a hypergeometric equation satisfying the above conditions into the group G . If the sum $\lambda + \mu + \nu$ is less than 1 (resp., equal to 1, greater than 1), then this group is a subgroup of motions of the Lobachevsky plane (resp., Euclidean plane, Riemann sphere). This fact enables one to investigate the integrability of the hypergeometric equation in algebraic functions (H.A. Schwarz [63, p. 393]). The integrable cases are connected with triangles having the angles $(\pi/2, \pi/2, \pi/n)$ – a dihedron, $(\pi/2, \pi/3, \pi/3)$ – a tetrahedron, $(\pi/2, \pi/3, \pi/4)$ – an octahedron,

$(\pi/2, \pi/3, \pi/5)$ – an icosahedron. Theorems about the insolubility of the hypergeometric equation in “generalized quadratures” also follow from these results [67].

§ 4. The Riemann-Hilbert Problem

“To prove that there is a fuchsian linear differential equation with prescribed singular points and a prescribed monodromy group”; this is Hilbert’s 21st problem [48]. It admits various generalizations.

4.1. Formulation of the Problem. (We do not touch here on the many investigations dealing with the calculation of the coefficients of the equation in terms of the monodromy group. Extensive material on this question is available in N.P. Erugin’s book “The Riemann problem”, Minsk, 1982.)

It is required to prove the existence, on the Riemann sphere, of the following objects with prescribed singular points and a prescribed monodromy group:

- A. fuchsian equations,
- B. systems $\dot{z} = A(t)z$ with regular singular points,
- C. fuchsian systems $\dot{z} = \sum A_j(t - \alpha_j)^{-1}z$.

Problem C, which is important for applications [97], has not yet been solved, in spite of the many claims spread throughout the literature ([75], [97], [89, § 85]). We outline below a solution of problem B based on ideas of Birkhoff and H. Röhrl (this clearly implies a solution of problem A). We also sketch a solution of problem C under additional restrictions on the monodromy group (see [20], [41:58]).

4.2. The Riemann-Hilbert Problem for a Disk. If one replaces the sphere by a disk in the statement of the problems above, then the resulting problems can be solved as follows. One constructs a matrix-function on the universal cover of the disk with the singular points deleted with the prescribed monodromy group and regular singular points. Then one verifies that the function satisfies a fuchsian system of equations.

Let $\alpha = (\alpha_1, \dots, \alpha_m)$ be the given set of singular points, let $\gamma_1, \dots, \gamma_m$ be loops which go around these points and have a common initial point (see Fig. 22), and let $T_1, \dots, T_m$ be the monodromy mappings corresponding to these loops. We cover the disk K by domains U_j with piecewise-smooth boundaries in such a way that each domain U_j contains just one point $\alpha_j \in \alpha$ and the intersection $U_j \cap U_{j+1}$ has a piecewise-smooth boundary and is simply connected; $\gamma_j \subset U_j$ (Fig. 22). For any pair of domains U, G with $U \subset G \subset \mathbb{C}$, which contain the point P , we let G_α denote the difference $G \setminus \alpha$, $\hat{G}_\alpha$ the universal cover of G_α with base point P , and $\hat{U}$ the connected component containing P of the domain in $\hat{G}_\alpha$ which lies over U .

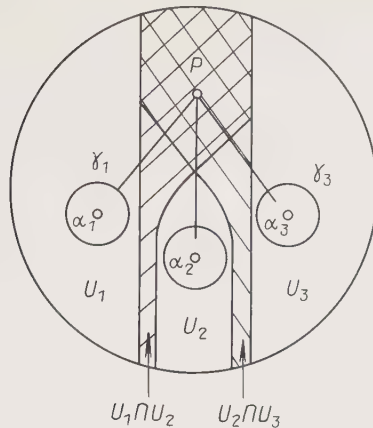


Fig. 22. The covering of the disk for the solution of the Riemann-Hilbert problem

Definition. A matrix-function X on $\hat{U}_j$ has *fuchsian principal part* t^C with monodromy T if $X = \Phi t^C$ where $T = \exp(2\pi i C)$ and Φ is a holomorphic mapping $U_j \rightarrow \text{GL}(n, \mathbb{C})$ (strictly speaking, we ought to write $X(\hat{t}) = \Phi(P\hat{t}) \cdot (P\hat{t})^C$, where $\hat{t} \in \hat{G}_\alpha$ and P is the projection $\hat{G}_\alpha \rightarrow G$; however, for brevity, we simply write Φ instead of $\Phi \circ P$).

We shall seek a matrix-function X on each domain $\hat{U}_j$ of the form $\Phi_j t^{C_j}$, where $2\pi i C_j$ is an arbitrary value of $\ln T_j$ and the mapping $\Phi_j: U_j \rightarrow \text{GL}(n, \mathbb{C})$ is holomorphic. The condition that these expressions coincide on the intersection $\hat{U}_j \cap \hat{U}_{j+1}$ is

$$\Phi_j^{-1} \Phi_{j+1} = t^{C_j} t^{-C_{j+1}}|_{\hat{U}_j \cap \hat{U}_{j+1}}.$$

The problem of finding holomorphic mappings $\Phi_j: U_j \rightarrow \text{GL}(n, \mathbb{C})$ satisfying the preceding equations is soluble for any right-hand sides with values in $\text{GL}(n, \mathbb{C})$. Suppose, for simplicity, that $m=2$. Note that the domain $\hat{U}_j \cap \hat{U}_{j+1}$ is a single sheeted cover of $U_j \cap U_{j+1}$ and can be identified with it.

Lemma (H. Cartan [41:31]). *Let U_1 , U_2 and $U_1 \cap U_2$ be domains with piecewise-smooth boundaries for which the intersection $U_1 \cap U_2$ is connected and simply-connected. Then, for any holomorphic mapping $F: \overline{U_1 \cap U_2} \rightarrow \text{GL}(n, \mathbb{C})$, there exist holomorphic mappings $\Phi_j: \bar{U}_j \rightarrow \text{GL}(n, \mathbb{C})$, $j=1, 2$, such that $\Phi_1^{-1} \Phi_2 = F$. (We recall that a map is holomorphic on a closed set if it is holomorphic on a neighbourhood of the set).*

Thus, when $m=2$, there is a matrix-function on $\hat{U}_1 \cup \hat{U}_2$ of the form $X = \Phi_j t^{C_j}|_{\hat{U}_j}$. Its singular points are obviously regular. When $m>2$ the matrix-function X is constructed in a similar manner by induction on m .

Lemma. *The matrix-function X satisfies a linear equation on K with simple poles.*

◀ Indeed $\dot{X}X^{-1}|_{\dot{U}_j} = (\dot{\Phi}_j + \Phi_j C_j t^{-1}) \Phi_j^{-1}$. ▶

This completes the solution of the Riemann-Hilbert problem on a disk.

4.3. The Riemann-Hilbert Problem for a Sphere. Without loss of generality we can suppose that the set α of singular points contains the point ∞ :

$$\alpha = (\alpha_1, \dots, \alpha_m, \infty).$$

We use the preceding result. Cover the sphere by two disks K^+ and K^- with centres at 0 and ∞ such that K^+ contains all the loops $\gamma_1, \dots, \gamma_m$, and the intersection $U = K^+ \cap K^-$ contains the point P . Let $\gamma_{m+1} \subset U$ be a loop which starts and ends at the point P and goes around the point ∞ in the positive direction (Fig. 23). We choose the loops $\gamma_1, \dots, \gamma_m$ so that the loop $\gamma_1 \dots \gamma_m \gamma_{m+1}$ is homotopic to zero on K_α^+ . Then $T_{m+1} = (T_m \dots T_1)^{-1}$ is the monodromy of the loop γ_{m+1} . Let $2\pi i C_j$ be any value of $\ln T_j$, $j = 1, \dots, m+1$. We construct, as in § 4.2, a matrix-function X_+ on $\hat{K}_\alpha^+$ with Fuchsian principal parts t^{C_j} , $j = 1, \dots, m$. We seek a matrix-function X on the universal cover of $\hat{C} \setminus \{\alpha_1, \dots, \alpha_m, \infty\}$ which is such that:

$$X|_{\hat{K}_\alpha^+} = \Phi_+ X_+, \quad X|_{\hat{K}_\infty} = \Phi_- (1/t)^{C_{m+1}}.$$

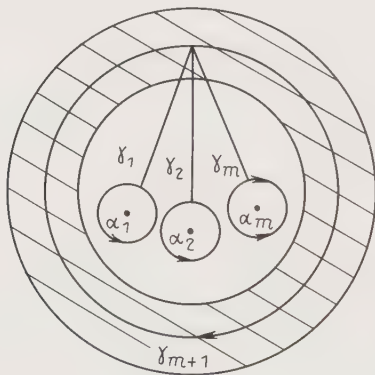


Fig. 23. The disk K^+

This matrix-function exists if the following factorization problem is soluble:

$$\Phi_+^{-1} \Phi_- = F|_U, \quad F = X_+ X_-^{-1}|_U, \quad X_- = (1/t)^{C_{m+1}}.$$

The matrix-functions X_+ and X_- are not single-valued on U , but continuation along the loop γ_{m+1} induces the same monodromy mapping; therefore F is a holomorphic mapping $U \rightarrow \text{GL}(n, \mathbb{C})$.

The analogue of Cartan's lemma for this problem is the following.

Birkhoff's theorem [20]. *A holomorphic mapping of an annulus $F: U \rightarrow \text{GL}(n, \mathbb{C})$ can be expressed as a product $F = \Phi_+ \Phi_-$, where $\Phi_\pm: K^\pm \rightarrow \text{GL}(n, \mathbb{C})$*

is a holomorphic mapping of the disk K^+ , and $\Phi_- : K^- \rightarrow \text{GL}(n, \mathbb{C})$ is a meromorphic mapping with a pole at ∞ .

This completes the solution of problem B. The matrix A of the required equation is equal to $\dot{X}X^{-1}$. It has simple poles at the points α_j and possibly, but not necessarily, a simple pole at ∞ , which is in any case a regular singular point for the equation $\dot{z} = A(t)z$.

4.4. The Riemann-Hilbert Problem for Fuchsian Systems

Theorem. *If one of the operators $T_1, \dots, T_m, M_{m+1} = (T_m \dots T_1)^{-1}$ is diagonalizable, then, for any m -tuple $\alpha = \{\alpha_1, \dots, \alpha_m\}$, there is a fuchsian system with poles belonging to α which is such that the monodromy operator T_j corresponds to a loop around the point α_j .*

Suppose the matrices T_{m+1} and C_{m+1} are diagonal and, as in §4.3, let $X = \Phi_- t^{-C_{m+1}}$ over K^- and $X = \Phi_+ X_+$ over K^+ .

Lemma (A. Souvage [46:1]). *A meromorphic mapping $\Phi_- : K \rightarrow \text{GL}(n, \mathbb{C})$ with a pole at ∞ can be expressed as $\Phi_- = P\Phi t^D$ where D is a diagonal matrix, Φ is a holomorphic mapping $K^- \rightarrow \text{GL}(n, \mathbb{C})$, and P is a polynomial matrix with determinant equal to 1.*

We apply the lemma to the mapping Φ_- in the expression for X . Set $Y = P^{-1}X$. As above, the matrix-function Y has fuchsian principal parts at the points $\alpha_1, \dots, \alpha_m$. Moreover, $Y|_{K^-} = \Phi t^D t^{-C_{m+1}}$. The matrices D and C_{m+1} are diagonal and, hence, commute. Therefore, the product $\Phi t^D t^{-C_{m+1}} = \Phi t^{D-C_{m+1}}$ is a fuchsian principal part. Hence the matrix $\dot{X}X^{-1}$ has simple poles on the sphere; problem C has been solved.

Remark. The arguments in this section follow those in J. Plemelj's book [89]. Plemelj gave the first solution of the Riemann-Hilbert problem and all later investigators make reference to his work. Plemelj carries out the preceding argument without using formulae; he virtually reproves Souvage's lemma; and he uses the fact that the matrices D and C_{m+1} commute without saying so explicitly. For matrices D and C_{m+1} which do not commute, Plemelj's argument does not go through. For example, if $\tau = 1/t$, $X = \tau^D \tau^C$, $D = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$, $C = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, then the matrix-function $A(\tau) = \frac{dX}{d\tau}$, $X^{-1} = \begin{pmatrix} 0 & \tau^{-2} \\ 0 & \tau^{-1} \end{pmatrix}$ has a pole at the origin which is not simple.

4.5. Generalizations for Non-Standard Time. In the contemporary literature, linear systems of differential equations are interpreted as connections on vector bundles. This enables one to solve the Riemann-Hilbert problem for a non-classical "time" (in which t takes values on an arbitrary Riemann surface or manifold [29]), but the Riemann-Hilbert problem for fuchsian systems has still not been completely solved. We mention some applications of the theory of vector bundles on a sphere to the Riemann-Hilbert problem;

the first half of § 4.7 translates results obtained earlier into geometrical language.

4.6. Vector Bundles on the Sphere

Definition. A *vector bundle of rank n on the Riemann sphere* is a triple (E, π, S) which satisfies the following conditions. E is an $(n+1)$ -dimensional complex manifold (called the *total space*), containing a sphere S (called the *base* or the *zero section of the bundle*), and $\pi: E \rightarrow S$ is a holomorphic map which is identity on S (a *retraction*). Each fibre $F_t = \pi^{-1}t$ is biholomorphically equivalent to $\mathbb{C}^n$; so that a linear structure is defined on the fibres. It is further required that the triple should be locally trivial: for every disk K on S there is a biholomorphic mapping $H_K: \pi^{-1}K \rightarrow K \times \mathbb{C}^n$ which carries each fibre F_t linearly onto the fibre $\{t\} \times \mathbb{C}^n$.

Remark. Let K_+ and K_- be discs with centres at 0 and ∞ respectively which cover the spherical surface and set $U = K_+ \cap K_-$. The transition function $(H_K)^{-1} \circ H_{K_-}$ determines a holomorphic mapping $F: U \rightarrow \text{GL}(n, \mathbb{C})$. Conversely, every such mapping determines a holomorphic vector-bundle on the sphere: the total space E is obtained from the union of $K_+ \times \mathbb{C}^n$ and $K_- \times \mathbb{C}^n$ by identifying the points (t, z) and $(t, F(t)z)$ for $t \in U$ and $z \in \mathbb{C}^n$; the projection π is induced by projection along the second factor. We shall call the resulting vector bundle on the sphere the *vector bundle with transition function F* .

Definitions. a) Two vector bundles are said to be *equivalent* if there is a biholomorphic mapping of the total space of one to the space of the other which commutes with the projections and is linear on the fibres.

b) A vector-bundle is said to be *trivial* if it is equivalent to the direct product of the sphere and $\mathbb{C}^n$.

c) A vector-bundle is said to be a *direct sum of line bundles* if it is equivalent to the vector bundle with transition function $F: t \mapsto \text{diag}(t^{\alpha_1}, \dots, t^{\alpha_n})$.

d) The *determinant* of a vector bundle with transition function F is a rank one vector bundle (a line bundle) with the transition function $\det F$.

Remarks. 1. A vector bundle with transition function F is trivial if and only if the equation $F = \Phi_-^{-1} \Phi_+$ has a holomorphic solution (Φ_+, Φ_-) , where $\Phi_+: K_+ \rightarrow \text{GL}(n, \mathbb{C})$, $\Phi_-: K_- \rightarrow \text{GL}(n, \mathbb{C})$.

2. Souvage's Lemma (1886) (§ 4.4) means that every vector bundle on the sphere is equivalent to a direct sum of line bundles.

4.7. Application to the Riemann-Hilbert Problem. Let $\alpha_1, \dots, \alpha_m, \infty$ be prescribed points and $T_1, \dots, T_m, T_{m+1}$ the corresponding monodromy mappings, $T_1 \dots T_{m+1} = E$. Every set of fuchsian principal parts $(t - \alpha_1)^{C_1}, \dots, (1/t)^{C_{m+1}}, \exp(2\pi i C_j) = T_j$, determines a vector bundle on the sphere by the construction described in § 4.2 and § 4.3. If the vector bundle is trivial, then there is a fuchsian system with the given poles and monodromy operators. This bundle is not, however, always trivial. The existence of a linear system with regular

singular points follows from the existence of a meromorphic trivialization of this vector bundle with a pole at ∞ (i.e., there is a reference frame in each fibre which depends holomorphically on $t \in \mathbb{C}$ and meromorphically on t at ∞). Birkhoff's theorem (§4.3) shows that such a trivialization always exists.

Let S^1 be the unit circle.

Theorem (D. Shaun, *Topology*, **12**, No. 4 (1973)). *Let $\mathcal{L}$ be the space of all germs of holomorphic mappings $(\mathbb{C}, S^1) \rightarrow \mathrm{GL}(n, \mathbb{C})$ which determine analytic vector bundles on the sphere with trivial determinant. The subset of mappings corresponding to nontrivial vector bundles is a proper analytic subvariety. This means that if A is any finite-parameter analytic family of maps in $\mathcal{L}$ the subset corresponding to nontrivial vector bundles is a closed analytic subset; there are arbitrarily small deformations of A for which this subset is a proper subset.*

Hence, after some calculation, the following theorem is obtained.

Theorem. (Yu.S. Il'yashenko). *There is a countable union of proper analytic submanifolds of $(\mathrm{GL}(n, \mathbb{C}))^m$ such that every m -tuple $(T_1, \dots, T_m)$ in the complement has the following property: For each set $(t - \alpha_1)^{C_1}, \dots, (1/t)^{C_{m+1}}$ of fuchsian principal parts*

$$\exp(2\pi i C_j) = T_j, \quad j = 1, \dots, m+1, \quad T_{m+1} = (T_1 \dots T_m)^{-1}, \quad \sum_{j=1}^{m+1} \mathrm{tr} C_j = 0$$

there is a fuchsian system whose fundamental matrix of solutions has fuchsian principal parts in the given set.

4.8 Isomonodromic Deformations and the Painlevé Equations. Suppose the coefficients of a linear differential equation or system depend holomorphically on a single complex parameter. Such a family is called a *deformation of the equation*. A deformation is said to be *isomonodromic* if the monodromy group does not change when the parameter changes. L. Schlesinger has found conditions for a deformation to be isomonodromic. For some second-order equations the conditions for isomonodromicity can be written as an equation of class *P*. R. Garnier [43, p. 284] has shown that all the Painlevé equations are obtained as isomonodromicity conditions for deformations of linear equations. Recently, new connections have been discovered between Painlevé equations and isomonodromic deformations. These relations have applications in theoretical physics [97].

Bibliography

We begin with a few comments on some of the books listed below. The textbooks [93], [6], [49] set out the basic theory of ordinary differential equations and discuss its connection with other branches of mathematics

and its applications to various fields of natural science: mechanics, electrical engineering, ecology, etc.

The books [91] and [75] are the original sources of the qualitative theory of differential equations and stability theory; for the most part, the results we have surveyed are outgrowths of investigations begun by H. Poincaré and A.M. Lyapunov.

The books [63], [86], [28], [46] and [7] are general monographs. The books [86], [28] and [46] reflect the state of the qualitative theory of differential equations at the end of the 40s, 50s, and 60s respectively. The books [63] and [43] set out the theory of differential equations in the complex domain. In particular, [63] presents the theory of integral transforms and its application to the solution of linear equations. The fundamentals of the theory of linear equations with complex time are covered in the books [28] and [46].

The book [7] contains a survey of the contemporary state of the theory of ordinary differential equations. It sets out the foundations of Poincaré's method of normal forms and its applications in recent years to the foundations of the theory of smooth dynamical systems and the local theory of bifurcations.

The book [1] gives the first systematic exposition of the foundations of the qualitative theory of differential equations in the plane. It covers the theory of auto-oscillations (limit cycles) and discontinuous (relaxation) oscillations, and analyzes many applications to physics and engineering.

The phenomena of soft and hard loss of stability ("dangerous and safe boundaries of the stability region") are investigated in the book [14]. Many applications are also analyzed.

The book [2] deals with the geometrical theory of differential equations in the plane and on the sphere. It contains a practically complete collection of the topological invariants of analytic vector fields with isolated singular points on a two-dimensional sphere.

The book [23] is devoted to resolution of singularities, the method of normal forms, and applications to the study of analytic differential equations and the problems of mechanics. The book [15] deals with the theory of normal forms, mainly for formal and smooth vector fields and mappings. A geometrical exposition of the method of resolution of singularities is contained in the book [34]. The local theory of invariant manifolds is set out in [50], [77], [28], [34]; in [77] it is applied to the equations of mathematical physics.

The book [82] sets out the local theory of Hamiltonian systems. Because of the presence of an invariant symplectic structure, specific phenomena occur here which are not encountered in generic differential equations.

The book [22] is devoted to perturbation theory which is used to investigate systems which are close to integrable systems.

The foundations of the theory of differential equations on manifolds is set out in the books [6] and [85]. The Frobenius theory is given in [85]; a geometrical exposition is given in [7].

The book [109] deals with the theory of foliations, a part of which is the theory of differential equations with real and complex time. It contains a detailed exposition of differential equations on 2-dimensional and 3-dimensional tori.

The book [5] is devoted to Anosov systems, a natural class of structurally stable differential equations on manifolds. Geodesic flows on compact manifolds belong to this class.

The book [113] is devoted mainly to the asymptotic theory of linear differential equations with complex time.

The Riemann-Hilbert problem is discussed in the books [29], [89], [41] and [97]. [89] deals with the theory of integral equations, by means of which the first solution of the problem was obtained.

The book [41] contains the theory of vector bundles and its application to the Riemann-Hilbert problem for equations on non-compact Riemannian surfaces. In the book [97] the solubility of the Riemann-Hilbert problem is applied to the theory of fields. In the book [29] the theory of linear differential equations is set out in the language of connections on vector bundles; this enables one to formulate and solve the Riemann-Hilbert problem for equations with multidimensional time.

For the convenience of the reader, references to reviews in *Zentralblatt für Mathematik* (Zbl.) and *Jahrbuch über die Fortschritte der Mathematik* (Jrb.) have, as far as possible, been included in this bibliography.

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Index

- | | |
|---|---|
| <ul style="list-style-type: none"> α-limit set of phase curves 29 Abelian integral 109, 127 absolute rigidity 114 absorbing domain 37 adjacent classes of germs 63, 66 <ul style="list-style-type: none"> – diagram 63, 66 admissible direction field in the complex plane 112 – fields 112, 113 – operator 124 algebraic equation 115, 116 <ul style="list-style-type: none"> – of a single variable 126 – insolubility 89 – solubility 51, 66 algebraically soluble 49–51, 102 <ul style="list-style-type: none"> – problems 49–51 almost eigenbasis 25 – resonant n-tuple 73 analytic center manifolds of differential equations in the plane 78 <ul style="list-style-type: none"> – classification of elementary singular points in the complex plane 112 – of nonresonant linear systems near an irregular singular point 124 – dependence of classes of pairs of conformal mappings 93 | <ul style="list-style-type: none"> – function of matrices 127 – invariant manifolds 76 – normal form of a resonant germ 95 <ul style="list-style-type: none"> – forms, geometry 94 – theory of differential equations 114ff. analytically equivalent admissible fields 113 <ul style="list-style-type: none"> – differential equations 47 – germs 92 – homeomorphisms 42 – orbitally equivalent differential equations 47 – soluble problems 49 analyticity of the center manifold 77 angular function 41 Arnold, V.I. 50 asymptotic series 123 asymptotically stable 23, 25 attractors 36, 37 autonomous differential equation 9 |
| | <ul style="list-style-type: none"> Bessel's equation 129 bicritical node 21 Borel subgroup of shifts 70 Bryuno's condition 75 Bryuno's method 87 |

- C^k -equivalent differential equations 47
- C^k -orbitally equivalent differential equations 47
- Cartan subgroups of hyperbolic rotations 70
- center 21, 88–92, 106, 107
 - according to linear terms 89
 - manifold 56, 57
 - of a cycle 102
- centralized chains 69
- characteristic number of a singular point 113
 - trajectory 83, 84, 88, 102
- Chetaev function 24, 25
- classification of linear systems 48
 - of resonant mappings 93
 - of resonant vector fields with generic nonlinearities 93
- complete system of first integrals 12
- complex autonomous equation 13
 - cycle of a holomorphic vector field 98
 - time 13
- complexification of phase space 20
 - of a linear operator 20
 - of an equation 20
- compound cycles 91, 105
- condition A 75
- conjugating mapping 147ff.
- contact plane 33
- convergence of linear operators 19
- correspondence mapping of the hyperbolic sector of a singular point 107
- criminant 33
- criteria for Lyapunov stability (for degeneracies up to codimension 3) 62
 - for stability 58
- critical point 114, 115
- cross-resonances 77
- cycle 26, 41
 - , multiplicity of 27
 - , nondegenerate 28, 41
 - , orbitally asymptotically stable 28
 - , orbitally Lyapunov stable 28
 - , semi-stable 27
 - , stable 27
 - , unstable 27
 - , vanishing 110
- cycles 119ff.
- d -dimensional volume of a finite covering of a compact set 37
- deformation of an equation 135
- degeneracies of finite codimension 50
 - of small codimension 58
- degenerate elementary singular points 94, 96
 - hypergeometric equation 129
 - node 21
- dependence of solutions on initial conditions and on parameters 17
- derivative of a mapping 11
- determinant of a vector bundle 134
- diffeomorphism 10
 - , action on a vector field 12
 - , rectifying 14
- diffeomorphisms, one-parameter group of 10
- differentiable vector field 11
- differential equation corresponding to a vector field 9
 - equations of higher order 13
 - – on a 2-dimensional torus 40ff.
 - – on manifolds 14
 - – on surfaces 43ff.
- direct sum of line bundles 134
- direction field 8
 - of a differential equation 9, 33
- discriminant curve 33
- dissipative equation 22
- distinguished solution of a functional equation 89
- distinguishing between a center and a focus 88–92
- distortion of phase volume 22
- Dulac 92
- Dulac's method 91
- eigenbasis 25
- eigenvalues of a singular point 12
- eigenvector 20
- elementary compound cycle 107
 - singular points 81
- endomorphisms of a circle 46
- equation of class P induced from another 117
- equations of class P 116, 135
 - of Fuchsian class on the Riemann sphere 130
 - on a torus with irrational rotation numbers 43
 - with periodic right hand side 103
 - without movable critical points 115
- equilibrium points 12
- equivalence of formal vector fields 52
- equivalent equations 33
 - pairs of conformal mappings 93
 - vector bundles 134
- Euler polygonal curve 16
- Euler's equation 118
- existence and uniqueness theorem 14

- exponential of a linear operator 19
- extended phase space 9
- extensibility 125
- extension of a nilpotent operator 70
 - of a N -jet 51
 - of a solution 17
- f -lowerable vector field 30
- families of differential equations on a torus 46
- fiber space 134
- field of planes 15
- finitely smooth equivalence of germs of vector fields 68
- finiteness theorems 43, 107
- first focus number 91
 - integral of a differential equation 12
 - integrals 12, 19
- flat germ 106
 - vector field 80
- focus 21, 88, 90–91
- fold points 33
- folded foci 34, 35
 - nodes 34, 35
 - saddles 34, 35
 - singular points 35
- folding 33
- foliations with k -dimensional leaves on an n -dimensional manifold 97
- formal center manifold 77
 - classification of germs of vector fields 52
 - equivalence 119
 - first integral 83
 - holomorphic equivalence 119
 - meromorphic equivalence 119
 - normal form 121
 - – – , three applications 54
 - substitutions with fixed point 0 52
 - Taylor series 52
 - theory of linear systems with a fuchsian singular point 120
 - theory of linear systems with a non-fuchsian singular point 121
 - vector field 52
- formally equivalent vector fields 53
 - $\mathcal{F}$ -equivalent system 119
 - finitely determined formal vector field 55
 - N -determined formal vector field 55
 - $\mathcal{O}$ -equivalent 119
 - orbitally equivalent germs of holomorphic vector fields 94
- freely homotopic paths 98
- Frobenius method 121
 - theorem for existence of integral surfaces 15
- Fuchs 116
 - condition 119
- fuchsian class of equations and systems 125
 - condition at infinity 125
 - nonsingular point at infinity 125
 - principal part of a matrix-function 131
 - singular point 118, 120, 124
 - – – at infinity 125
- function invariant 93
- fundamental alternative 84
 - system of solutions 20, 21
- Γ -nondegenerate vector fields 87
- Galois group 128
- Gambier 116, 117
- Gauss's hypergeometric equation 129
- generalized first focus number 91
 - quadratures 128, 130
- generating cycle 109
 - field 10
- generation of complex limit-cycles 109
 - of real limit-cycles 109
- generic field having property A 113
- generic field in the Petrovsky-Landis sense 113
 - holomorphic vector field 53
 - singular points 25
- germ of a vector field at a singular point 50
 - of a vector field with a nonresonant linear part 96
- germs of conformal mappings with the identity as linear part 92
 - of diffeomorphisms 50
 - of vector fields with symmetries 67
- good blow-ups 81
- Grobman-Hartman theorem 21
- group of all symmetries of a differential equation 30
 - of quasihomogeneous dilations 32
 - , one-parameter, of diffeomorphisms 10
- Hadamard-Perron theorem 21
- Hausdorff dimension of a compactum 37, 38
- Hilbert's 21st problem 130
- higher order differential equation 13
- holomorphic direction-field 13
 - equivalence 119
 - foliation 97
 - vector field 13
- holonomy group of a foliation 98
- homogeneous equations 31
 - vector field 31
- hyperbolic cycle 102
 - fixed point of a diffeomorphism 100
 - linear vector field 79

- sector 85, 107
- singular point 40, 56
- – points, topological classification 47
- implicit differential equations 33, 116
- incommensurable in the Bryuno sense 73
- indicial equation 120
- integrability in quadratures 128
- integral curve 9
 - – of a differential equation 9
 - – of a holomorphic direction field 13
- interval of continuity of coefficients 22
- invariant manifold of a differential equation 56
 - manifolds of a cycle 101
 - – of germs of diffeomorphisms 101
 - – of a vector field 56
- involution 35
- irreducible representation of the Lie algebra $\mathfrak{sl}(2)$ 70, 71
- irregular singular point 118, 119, 121
- isomonodromic deformation 135
- jet negative (positive, neutral) as regards a given property 49
- jets of a function 48, 49
 - of vector fields 49
- Jordan chain of operators 20
 - node 21
- $K[[x]]$ 52
- k -compressive system 38
- k -resonant n -tuple 55
- Kleinian groups 128
- Korteweg-de-Vries equation 117
- Lappo-Danilevsky 127
- Legendre's equation 129
- lemma, Cartan 131
 - Sauvage 133
- limit cycle 27
 - cycles of polynomial vector fields in the plane 104ff.
 - – of systems close to Hamiltonian systems 108ff.
 - sets 29
- linear differential equations 19
 - – – of higher order 22
 - normal forms 72, 99, 103
 - part of a vector field 12
 - vector field 79
 - – – , complex-hyperbolic, weakly hyperbolic 79
- linearization of a differential equation at a singular point 12
 - Liouville-Ostrogadsky formula 21
 - Lipschitz condition 15
 - local phase flow 18
 - problems of analysis 48
 - theory of diffeomorphisms 99
 - – of linear equations with complex time 117
 - Lojasiewicz condition 82, 91, 106, 108
 - Lotka-Volterra model 112
 - Lyapunov focus numbers 89, 91
 - function 24, 25
 - stability 48
 - stable 23
 - Lyapunov's theorem on stability by linearization 24
- matrix-residues 125, 127
- meromorphic equivalence 119
- method of successive approximations 16
- monodromic singular point 84, 90
- monodromy 97, 119, 125
 - group 98, 126
 - – of a foliation 98
 - – of the Gauss equation 129
 - – of the Riccati equation 128
 - mapping 25, 97, 119, 126
 - – of a closed phase curve 25, 26
 - – of a complex cycle 98
 - – of a compound cycle 105
 - – of a periodic equation 103
- matrices 127
- of the path 97
- operator 103
- $\triangleright$ transformation of a singular point 84
- motions of the Euclidean plane, Lobachevsky plane, Riemann sphere 129
- movable critical points 114, 115
 - – – of a 1st-order equation 115
- multiplicative resonant monomial 100
 - type (C, v) 99
- multiplicity of a cycle 27, 41
 - of a singular point 28
- NFIM class 76
- N -jet of a smooth function 48, 49
- N -jets of vector fields 49
- negative semi-trajectory 29
- Newton diagram of a vector field 87
- nilpotent Jordan block 68, 69
- node 21
- nonautonomous system 9
- nondegenerate cycle 28, 41
 - singular point 12, 39
- non-fuchsian singular point 153ff.

- non-monodromic singular point with no characteristic trajectory 85
- non-removable terms 69
- nonresonant system 121
- normal form of folded singular point 35
 - of linear equation with periodic coefficients 103
 - of vector fields 68
 - on the invariant manifold 76
 - forms, analytic case 83
 - in smooth case 83
 - Poincaré-Dulac form 104
- normalized N -jets 59
- normalizing series 72
- , divergence of 73, 74
- ω -limit set of phase curves 29
- orbital topological classification of elementary singular points in the complex plane 96
- orbitally asymptotically stable 28
 - equivalent 35
 - Lyapunov stable 28
 - topologically equivalent differential equations 47
 - – germs 50
- Painlevé equations 116, 117, 135
- period of a trajectory 41
- periodic point of a diffeomorphism 41
- phase curve 9
 - of a differential equation 9, 26
 - flow 10, 19
 - , local 11
- phase-portrait of quotient systems 61
 - space 9
 - velocity field 10
- Picard approximations 16
- Picard-Fuchs linear differential equations 111, 127
- Picard-Lefschetz theorem 127
- plane field 15
- plane field of the hyperbolic variables 57
- Plemelj 133
- Poincaré 26
 - domain 72, 99, 104
 - normal form 120
 - theorem 53, 99
 - type of vector field 72
- Poincaré-Bendixson theorem 29
- Poincaré-Dulac normal form 53
 - theorem 53, 100, 120, 122
- point at infinity 116
 - of bifurcation 115
- polar blow-up in the plane 80
- polynomial differential equation in the complex plane 112
 - fields 113
 - normal forms 55
 - vector fields 91
 - of the degree two 108
- positive semi-trajectory 29
- principal part of a vector field 105
- problem of finiteness 104
- punctured Riemann sphere 126
- quadratic fields 108
- quasihomogeneous dilatations 32
 - function 32
 - vector field of degree r 32
- quasi-hyperbolicity 68
- quotient system 31, 61
- realization 93, 98
- rectifiability of a vector field 14
- rectification of a vector field 15
- rectifying diffeomorphism 14
- reduction principle 57, 101
 - theorem 56, 57
- regular polynomial of degree $n+1$ in 2 complex variables 110
 - singular point 33, 117–119
- relation between formal and analytic classification 74
 - – and smooth classification 66
- representatives of a germ 50
- residual 44
- resolution of singularities 80
- resonance zone 45, 46
- resonances 53
- resonant linear vector field 53
 - monomial 104
 - n -tuple 53
 - for system with a fuchsian singular point 121
 - for system with a non-fuchsian singular point 121
 - normal forms 104
 - set of eigenvalues 72
 - of a periodic equation 104
 - linear system 121
 - term 53
- restriction of equation to a formal invariant manifold 76
- retraction 134
- Riccati equation 115, 128
- Riemann equation 129
 - surface 13
- Riemann-Fuchs theorem 126
- Riemann-Hilbert problem 128, 130, 134

- – for a disc 130
- – for a sphere 132
- – for Fuchsian systems 133
- rotation number of a differential equation 42, 43
- – of a homeomorphism of a circle 42
- set 46
- – of an endomorphism 46
- Runge-Kutta method 16
- saddle 21
- saddle-node 81
- saddle-resonant vector fields 93, 96
- sectorial normalization 95
- self oscillations 112
- semi-algebraic manifold 49
 - manifold 49
- semi-regular germ 106
 - mapping 105, 106
- semi-stable cycle 27
- semi-transversal 105
- separating ray 123
- separatrix, (incoming, outgoing) 40
- Siegel domain 72, 99
- σ -process in the plane 80
- singular point 83, 117
 - – at infinity 113
 - –, multiplicity of 28
 - – of a differential equation 11
 - – of a vector field 11
 - points, elementary 81
 - –, generic 25
 - – of a differential equation in a higher dimensional complex phase space 72
 - – of a differential equation in a higher dimensional real phase space 47
 - – of vector fields 80
 - – without exclusive directions 90
- small denominators 72
- smooth arc 84
 - classification of germs of vectors fields 66
 - orbital classification of elementary singular points in the plane 82
- smoothly equivalent germs 50
- smoothness 47
- solution of a differential equation 9
 - – – by power series 16
- space of 1-jets 33
 - of invariants 92, 93
- stability 23
 - by linearization 24
 - criteria 62
 - of motion 24
- stable center manifold 57
 - cycle 27
 - manifold of a cycle 101
 - singular point 24
- standard elliptic sector 85
 - hyperbolic sector 85
 - parabolic sector 85
 - representation of the algebra $\mathfrak{sl}(2)$ 70
 - – of the group $SL(2)$ 70
 - saddle 58
- stochastic attractor 37
- Stokes bundle 125
 - operator 123
 - phenomenon 123
 - system 124
- strange attractor 37
- strictly Siegel n -tuple 72
 - – type of vector field 72
- structurally stable diffeomorphism 42
 - – differential equations 39
 - – – equations on a torus 42
 - – system 39
 - – – on a 2-dimensional sphere 40
- structure of stability criteria 58
- support of a vector monomial 87
 - of an analytic vector field 87
- surface of an equation 33
- swallowtail surface 36
- symmetries, use of, to reduce the order 31
- symmetry group of a differential equation 30
 - of a direction field 30
 - of a vector field 30
- systems with symmetries 30
- Theorem, Belitsky 54
 - Bendixson 81
 - Berezovskaya 87
 - Bibikov 76
 - Birkhoff 132
 - Bryuno 100
 - Camacho-Sad 96
 - Chen 101
 - Cibrario 33
 - Davydov 35
 - Denjoy 43
 - Dulac 106
 - Dumortier 82
 - Elizarov 96
 - Floquet 103
 - Galkin 46
 - Grobman-Hartman 21, 100, 102
 - Guckenheimer 79
 - Hadamard-Perron 21, 56, 101
 - Il'yashenko 66, 79, 93, 135
 - Jakobson 43
 - Khazin-Shnol 50

- Khovansky 128
- Theorem, Klein-Bôcher 129
- Ladis 79
- Theorem, Malmquist 116
- Painlevé 115
- Petrov 111
- Picard-Lefschetz 127
- Poincaré 53, 72, 99, 104
- Poincaré-Bendixson 29
- Poincaré-Dulac 54, 100, 121
- Riemann-Fuchs 126
- Seidenberg 82
- Sell 68
- Shaun 135
- Shoshitaishvili 57
- Siegel 73, 99, 104
- Sternberg 100
- Sue 98
- Takens 51
- Varchenko 111
- Voronin 93
- theory of linear equations in the large 125
- topological classification 48
 - of analytic vector fields with singular points on a sphere 86
 - of compound singular points with a characteristic trajectory 83
 - of differential equations in the plane near a singular point 85
 - of germs of smooth vector fields 58
 - of hyperbolic singular points 47
 - of singular points 58
 - of singular points in the complex domain 79, 96
 - finite determinacy 87
- topologically equivalent admissible fields 113
 - differential equations 47
 - germs 50
 - homeomorphisms 42
 - phaseflows 11
 - orbitally equivalent differential equations 39
 - unstabilizable jets 51
- trace of an operator 21
- trajectory 41
 - of a point under the action of a diffeomorphism 101
- trivial vector bundle 134
- two-dimensional torus 40
- unstable center manifold 57
 - cycle 27
 - manifold of a cycle 101
 - singular point 23, 25
- vanishing cycle 110
- variation of monodromy 109
- variational equation 17
- vector bundle of rank n on the Riemann sphere 134
 - with the transition function F 134
 - bundles, equivalent 134
 - on the Riemann sphere 134
 - fields 9
- weak Hilbert conjecture 110
- weakly hyperbolic linear vector field 79
- Weber's equation 129
- Whitney pleat 33, 35
- Wronskian 22
- zero of order N of a formal power series 55
 - section of a fibre bundle 134

II. Smooth Dynamical Systems

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Translated from the Russian
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Contents

Preface (D.V. Anosov)	151
Chapter 1. Basic Concepts (D.V. Anosov)	154
§ 1. The Concept of a Dynamical System	154
1.1. Flows and Cascades	154
1.2. Random Processes as Dynamical Systems. Symbolic Dynamics	156
1.3. Trajectories, Motions, Invariant Sets	159
1.4. Reversal and Change of Time	160
1.5. Morphisms of Dynamical Systems	161
1.6. Various Remarks	162
§ 2. Smooth Dynamical Systems	164
2.1. Smooth Flows	164
2.2. The Variational Equations	166
2.3. First-Return Mapping	168
2.4. Equilibrium Points and Periodic Trajectories	170
2.5. The Morse Index	173
Chapter 2. Elementary Theory (D.V. Anosov)	174
§ 1. Introduction	174
1.1. Contents of the Chapter	174
1.2. Genericity and Structural Stability	175
§ 2. The Kronecker-Poincaré Index and Related Questions	177
2.1. The Kronecker-Poincaré Index	177

2.2. Summary of Topological Results About Fixed Points	178
2.3. The Fuller Index	181
2.4. Zeta-Functions	183
§ 3. Morse-Smale Systems	183
3.1. General Information About Morse-Smale Systems	183
3.2. The Structure of Phase Manifolds of Morse-Smale Systems	187
3.3. Existence of Morse-Smale Systems whose Generating Diffeomorphism (or Vector Field) has Prescribed Topological Properties and whose Periodic Trajectories Have Prescribed Properties	192
3.4. Other Questions	195
Chapter 3. Topological Dynamics (D.V. Anosov, I.U. Bronshtein)	197
§ 1. Introduction	197
§ 2. Attractors, Morse Decompositions, Filtrations, and Chain Recurrence	198
2.1. Attractors and Morse Decompositions	198
2.2. Chain Recurrence	200
2.3. Lyapunov Functions	202
§ 3. Indices of Isolated Invariant Sets	203
3.1. Isolated Invariant Sets	203
3.2. Isolating Blocks and Index Pairs	204
3.3. Homological and Homotopic Indices	206
3.4. The Morse-Conley Index	207
§ 4. "Repetitive" Motions	211
4.1. Nonwandering Points. The Centre	211
4.2. Variants of the Nonwandering Concept. Prolongations	212
4.3. Minimal Sets	213
4.4. Distality and Some Types of Extensions of Minimal Sets	214
§ 5. Extensions of Dynamical Systems and Nonautonomous Differential Equations	216
5.1. Nonautonomous Differential Equations	216
5.2. Linear Extensions	217
Chapter 4. Flows on Two-Dimensional Manifolds (S.Kh. Aranson, V.Z. Grines)	219
§ 1. Singular Trajectories	219
§ 2. The Poincaré Rotation Number. Transitive and Singular Flows on a Torus	222
§ 3. The Homotopy Rotation Class. Classification of Transitive Flows and Nontrivial Minimal Sets of Flows on Surfaces	223
§ 4. Generic Properties of Flows on Surfaces	226
Bibliography	227
Index	230

Preface

The *qualitative theory of ordinary differential equations* (QTDE) and the theory of *dynamical systems* (TDS) arose within the theory of differential equations; in time, the TDS attained a definite autonomy, and it can now be regarded as an independent branch of mathematics, which continues to develop intensively. It retains a close connection with the theory of differential equations, and the boundary between them is not particularly sharp (unfortunately, there are differences in terminology in the region where the two theories overlap). At the same time, the TDS has established new connections with other branches of mathematics which appear even more essential for certain questions in the TDS. Even the concept of a *dynamical system* (DS) has itself evolved considerably.

A dynamical system may be smooth, topological (see the definition in Chap. 1, § 1), or measurable. Measurable DSs are studied in ergodic theory or in the metric theory of DSs (where “metric” refers to the existence of a measure, not a metric). Volume 2 deals with this topic; the basic object, here, is a DS with an invariant measure (and the properties of the DS are examined relative to this measure). The theory of topological DSs is often called topological dynamics: we mostly present only the parts of this theory which are of interest in connection with the theory of smooth DSs, which is sometimes called differentiable dynamics. This distinction is largely a matter of convention: where, for example, do the metric properties of smooth DSs belong? They have, in fact, been put into volume 2; mainly with the idea of distributing the material evenly. In the same way, volume 3 and part of volume 4 have been devoted to specific properties of DSs related to those considered in analytical mechanics and volume 5 is devoted to bifurcations, although the systems considered in these volumes are smooth.

In an article of this size it is not possible to throw light on all aspects of smooth DSs. Since this article precedes, to some extent, the articles on the TDS in other volumes of this encyclopedia, it will be necessary to summarize the frequently used concepts, detailing various shades of meaning, variants of definitions and terminology, and the like. This is done in the introductory Chap. 1 and, on occasion, in the other chapters. In Chaps. 2 and 3, we examine various questions connected in one way or another with topology (those in Chap. 2 are connected with algebraic topology; those in Chap. 3, apart from § 3, are connected with point set topology). Their common characteristics are given at the beginning of these chapters. Chapter 4 deals with flows on surfaces; this theory (apart from the particular cases of structurally stable systems and flows in a plane) is little known and has not yet received a systematic exposition.

After that, it would have been natural to turn to the hyperbolic theory, *i.e.*, the theory of DSs whose trajectories exhibit hyperbolic behaviour. In the sixties, as the theory of smooth DSs was becoming an independent discipline, hyperbolic theory played a predominant rôle; today the hyperbolic

theory continues to develop, and its concepts, methods, and results are used in other areas. It is not surprising, therefore, that some textbooks and surveys with titles such as "The theory of smooth dynamical systems" deal mainly with the hyperbolic theory; around which are grouped all other topics (if any). Since the most important other trends are discussed in other parts of this series (see below), the hyperbolic theory would also have occupied a central position here, if this article had been several times longer.

However, many important questions of hyperbolic theory are well presented in various textbooks and semi-expository monographs; they are partly touched on in volume 2. It is true that the content of these books and of part of volume 2 does not exhaust the hyperbolic theory, much less the topics closely associated with it. But, in view of the limited space in the present article, it would have been impossible to say much more. Instead of presenting yet another treatment of the same material, which would inevitably have been short and formal, it seemed more sensible to me to deal here in detail with other questions, some of which have not been adequately presented in the literature. Apart from their intrinsic interest, these topics "ought" to precede the hyperbolic theory. In cases where they are not logically necessary, they are useful for orientation. Perhaps a special volume of this series will be devoted to the questions of the hyperbolic theory and related issues which were not included in volume 2.

Essays on local and non-local bifurcations and on the application of the TDS to mathematical questions in hydrodynamics would be closely related to the material of the present volume and to the parts of hyperbolic theory included in volume 2. Presumably, such articles will be included in the later volumes (some are already included in volume 5). As regards other aspects of the theory of smooth DSs, I remark that analytic questions, including small denominators, are touched on in the first article of this volume and in volumes 2 and 3. Variational methods and integrable systems are dealt with in volume 3 (see also volume 4 regarding the latter). Concerning less general topics, I mention questions relating to maps of an annulus (see volume 3) whose study involves the most varied types of arguments (analytic, geometric, variational) and (once again) questions relating to ergodic properties of smooth DSs, where there are issues which relate to neither hyperbolicity nor small denominators.

The phase space (the "scene of action") in the theory of smooth DSs is a smooth manifold, usually different from the Euclidean space $\mathbb{R}^n$. So some knowledge of manifolds is required. For the most part, a general background will suffice: the material which, in S. Lang's well-known description, belongs to the no-man's-land lying between advanced calculus and the three great differential theories—differential topology, differential geometry, and the theory of differential equations. The *key-words* are: manifold, chart, tangent vector, tangent bundle, vector field, vector bundle, smooth mapping, diffeomorphism, Riemannian metric, submanifold, transversality. In all modern courses on the theory of differential equations, a geometrical treatment

is given of autonomous systems on $\mathbb{R}^n$. If the reader is familiar with this (and feels at home in “no-man’s-land”), then passage to DSs on manifolds is not difficult. From time to time, specific topological concepts are encountered: the degree of a mapping, the rotation index of a vector field, homology, the fundamental group, coverings and covering transformations, homotopies and homotopy equivalence, isotopy, handle decomposition. The formulation of the results will not go outside the framework provided by these concepts (though this does not apply to the proofs). The concepts from point set topology we need do not, in fact, go beyond complete metric spaces and metric compacta.

A few words about notation: $\mathbb{R}, \mathbb{R}^+, \mathbb{C}, \mathbb{Z}, \mathbb{N}$ are the sets of real numbers, non-negative real numbers, complex numbers, integers, and natural numbers (non-negative integers), considered with the usual structure (algebraic operations, topology). S^n denotes an n -dimensional sphere: sometimes the “standard” sphere $|x|=1$ in $\mathbb{R}^{n+1}$ and sometimes a manifold homeomorphic to a sphere; $\mathbb{D}^n$ is the “standard” n -dimensional closed ball $|x| \leq 1$ in $\mathbb{R}^n$. The restriction of a mapping $f: A \rightarrow B$ to $C \subset A$ is denoted by $f|_C$. Composition of mappings is sometimes denoted by putting a small circle between them, sometimes by juxtaposition. 1_A is the identity mapping of a set A . We let TM denote the tangent bundle of a smooth manifold M and $T_x M$ the tangent space to M at the point x . A smooth mapping $f: M \rightarrow N$ of smooth manifolds induces a map $Tf: TM \rightarrow TN$ (the “tangent mapping”, “differential”, “derivative”); we write $Tf(x) = Tf|_{T_x M}$. The derivative with respect to the independent variable t (“time”) is denoted by a dot above the variable. An overbar denotes closure and $\oplus$ a direct sum. Other notation will be introduced in the text.

In conclusion I make some remarks about the literature of smooth DSs. Some basic textbooks (slanted mainly towards the hyperbolic theory) are: [2], [62], [37], [63], [71]. Some surveys of a general character are: [53], [74]. These books and surveys contain information about the history of the questions under consideration and extensive bibliographies. The Mathematical Encyclopedia [55] can also be used as a source-book; articles from it are cited in the format: [ME, “name of article”]. Finally, the theory of smooth dynamical systems has been discussed at all recent international mathematical congresses, lectures (usually surveys) can be found in the published proceedings.

The bibliography to this article is quite small, and the history is hardly touched upon. On occasion, only the most recent article of several on the same topic is listed; it can be consulted for references to earlier and, possibly more important, articles. (Accordingly, there is no reference here to the classical works of the founders of the TDS; but a number of such references can be found in the bibliography to the first part of this volume.)

D.V. Anosov

Chapter 1

Basic Concepts

D.V. Anosov

§ 1. The Concept of a Dynamical System

1.1. Flows and Cascades. We shall consider two kinds of DSs: flows and cascades. (For a broader notion of DS, consult [ME, “dynamical system”, “topological dynamical system”] and volume 2 of this series.)

A *flow* is a one-parameter group or semigroup of transformations acting on a set M (typically, this set will be endowed with an additional structure which the transformations will be required to respect, see below). M is called the *phase space of the flow*. In other words, associated to each $t \in \mathbb{R}$ or $t \in \mathbb{R}^+$, there is a mapping $g^t: M \rightarrow M$ such that the *group property* holds, i.e.,

$$g_0 = \mathbf{1}_M, \quad (1)$$

$$g^{t+s} = g^t g^s \quad \text{for all } t, s \text{ under consideration.} \quad (2)$$

For illustrative purposes, an analogy is often drawn with the steady flow of a liquid or gas, where a similar family of transformations arises. After time t a fluid particle moves from a point x to the point $g^t x$. This explains the use of the word “flow”. But it should be noted that the “hydrodynamical” analogy is quite superficial; the imaginary “phase fluid”, “flowing” in the phase space, differs from a real continuous medium in that there is no interaction between neighbouring particles.

A *cascade* differs from a flow in that the maps g^t are only defined for $t \in \mathbb{Z}$ or $t \in \mathbb{N}$. In this case, we often use k and nearby letters of the alphabet instead of $t, s, \dots$. The notation g^k denotes the k^{th} iterate of the map $g = g^1$ for $k > 0$ and the k^{th} iterate of g^{-1} when $k < 0$. The name “cascade” is used to contrast it to a “flow”.

We mentioned earlier that M will be endowed with an additional structure. This will either be the structure of a topological space or of a manifold (always smooth, Hausdorff, with a countable basis, and, unless otherwise stated, connected and without boundary). In the first case, it is required that $g^t x$ be continuous with respect to (t, x) (in which case, we have a *topological* DS, flow, cascade; for a cascade this reduces to the requirement that g be a homeomorphism if the cascade is defined for all t in $\mathbb{Z}$, or a continuous mapping if it is defined for t in $\mathbb{N}$). In the second case *smooth* DSs are defined. A cascade g^k is *smooth* if all the g^k are smooth mappings; this reduces to the fact that g is a diffeomorphism (if $k \in \mathbb{Z}$) or a smooth mapping (if $k \in \mathbb{N}$). A *flow* is *smooth* if $g^t x$ depends smoothly on (t, x) and

$$\mathbf{v}(x) = \left. \frac{d}{dt} \right|_{t=0} g^t x$$

is a smooth vector field on M (the *phase velocity field*). The latter completely determines (“generates”) the flow: for a fixed x_0 and variable t , $g^t x_0$ is a solution of the differential equation

$$\dot{x} = \mathbf{v}(x) \quad (3)$$

with initial condition x_0 ; that is,

$$g^t x_0 = x(t), \quad \text{where } \dot{x}(t) = \mathbf{v}(x(t)) \quad \text{and} \quad x(0) = x_0. \quad (4)$$

(For $M = \mathbb{R}^n$, see article 1, Chap. 1, §§ 1, 2; for arbitrary M , see § 2.)

In applications, the points of M (the *phase points*) often represent the states of some isolated (closed) physical system, or, more abstractly, these states themselves are regarded as elements of a phase space (*i.e.*, the set of states is regarded as having the corresponding topological or smooth structure), and the DS describes the evolution of the system; *i.e.*, the dynamical system describes the change in the states with time: the system moves from the state x to the state $g^t x$ after time t . (Accordingly, the variable t in the TDS is called the time, although other applications are possible in which the physical meaning of t is different.) The “isolation” of a system is manifested by the fact that the transition from the state $g^s x$ to $g^{s+t} x$ as the time changes from s to $s+t$ takes place in the same way as the transition when the time changes from 0 to t , a fact which is reflected in (2).

Originally a DS was understood to be an isolated mechanical system with a finite number of degrees of freedom (the name DS originated in this way). The evolution of such a system is described by a smooth flow in the phase manifold M , *i.e.*, by an equation of the form (3). (We often have $M = \mathbb{R}^n$, but this is not the case for, say, a circular pendulum or a rigid body with a fixed point. Moreover, it can happen that the flow g^t is defined in $\mathbb{R}^n$, but has an invariant manifold M (*i.e.*, an invariant set which is a manifold; see § 1.3) and we are interested only in the restriction $\{g^t|_M\}$ of our flow to M .) However, many of the associated geometrical (or, more precisely, kinematical) ideas – the motion of a point in M along a phase curve, etc. – do not depend on whether equation (3) describes a *mechanical* system. The term DS therefore began to be applied more widely to describe what was earlier called a smooth flow. At the same time, the QTDE began to be applied to physical systems which are not mechanical (electrical engineering, ecological, ...), and this lent support to the notion that it is not inevitably attached to mechanics. Flows are most often encountered in applications, but cascades also appear. For example, in ecology one might want to study changes in a population with non-overlapping adult generations. Here, the generations play the rôle of discrete time. Nevertheless, the main significance of cascades lies in the fact that they are usually technically somewhat simpler than flows; at the same time, the essence of the matter may be the same in both cases. Thus, results obtained for cascades frequently carry over to flows, often not by way of a formal reduction, but by some modification of the proofs.

If we wish to emphasize the fact that some cascade or flow is only a semigroup, we will use the prefix “semi-”: semi-cascade, semi-flow. We will also use the words irreversibility, irreversible systems, etc., to refer to the semigroup case and reversibility, and so on, to refer to the group case although the term “reversible system” has an entirely different meaning in questions related to mechanics (see Article I, Chap. 4, § 3.2).

A cascade is also called a *DS with discrete time*, or a *discrete DS*, and a flow is called a *DS with continuous time*, or a *continuous DS*; but these terms are ambiguous. Thus, for different, but equally justifiable, reasons, systems with distributed parameters, described by partial differential equations, might be called “continuous”. Besides, when DSs are understood in their widest sense as an action of a (semi-)group (*cf.* the articles in ME cited above), then it is natural to interpret “continuity” or “discreteness” as referring to the topology of the (semi-)group.

1.2. Random Processes as Dynamical Systems. Symbolic Dynamics. Up to now, we have considered systems in which changes of state are deterministic (randomness can enter only if there is an initial probability distribution on the phase space). The concept of a “random process” in probability theory formalizes the notion of a classical (non-quantum) system whose evolution is not deterministic but has definite probabilistic characteristics. We shall not attempt to cite all the relevant formulations completely, but only indicate here what is needed for the TDS.

In the general case a *random process* is defined as a family of measurable mappings ξ_t of a space Ω of elementary events in a measurable space A (called the *phase space*¹ of the process; the points of A are called the *states* of the process); the mappings ξ_t are parametrized by the parameter t (the “time”), which runs over $\mathbb{R}$, $\mathbb{R}^+$, $\mathbb{Z}$, or $\mathbb{N}$ in cases of interest to us. We will be concerned with (strictly) stationary processes (the stationarity condition is the analogue of isolatedness or autonomy). Examples of such processes are obtained as follows: if $\{g^t\}$ is a measurable DS on Ω which preserves the measure defining the probability and $f: \Omega \rightarrow A$ is a measurable map, then $\xi_t(\omega) = f(g^t \omega)$. It might seem that this gives a rather special class of stationary random processes, but it turns out that, under fairly mild assumptions, a stationary random process is equivalent in some sense to a process of the type above. This equivalence is effected by replacing the original space of elementary events by a new space M whose elements are realizations (sample functions, trajectories) of the original process, *i.e.*, the functions $t \mapsto \xi_t(\omega)$ with all possible ω . The space of realizations can be naturally endowed with the structure of a normed measure space. Take f to be the mapping which associates to a function $x(t)$ its value when $t=0$, and define

¹ The phase space is mentioned only when it is nontrivial in some sense. For the frequently encountered processes with numerical values, A is $\mathbb{R}$ with the standard structure, and it need not be specially mentioned.

g' to be the shift which translates the argument of any function by t :

$$(g^t x)(s) = x(t + s).$$

It turns out that the shifts preserve the measure on the space of realizations. This is the so-called *natural representation* of a random process². The construction above can always be carried out, but the resulting dynamical system need not be measurable unless it satisfies certain definite (albeit rather weak) assumptions (strictly speaking, it may be necessary to “touch up” the original process $\xi_t(\omega)$ by changing its values for each t on a set of measure zero). Incidentally, when the time is discrete, difficulties with measurability do not arise. In this case, M is a subset of either $A^{\mathbb{Z}}$ or $A^{\mathbb{N}}$; the elements of M are two-sided or one-sided infinite sequences $\{x_n\}$ according as $n \in \mathbb{Z}$ or $n \in \mathbb{N}$.

Thus, stationary random processes often reduce to measurable DSs with an invariant normalized measure. (It must be borne in mind that the phase space M of the DS does not coincide with the phase space A of the process; the meanings of the words “trajectory” and “state” are also different). For probability theory this is perhaps not very important (although applying Birkhoff’s ergodic theorem immediately gives the strong law of large numbers). On the other hand, it does provide the TDS with interesting examples. Some DSs of probabilistic origin are examined in volume 2. Here, however, we consider the topological analogue of such DSs.

A *topological Bernoulli automorphism* or *topological two-sided shift* (resp., *endomorphism* or *one-sided shift*) σ , and the *topological Bernoulli cascade* (resp., *semi-cascade*) $\{\sigma^k\}$ obtained by iterating it, act on the space Ω_n of infinite two-sided (resp., one-sided) sequences of symbols from some finite “alphabet” $A = \{a_1, \dots, a_n\}$. The space Ω_n is equipped with the topology it inherits as a direct product of an infinite number of copies of A each of which has the discrete topology. Hence, it is compact and metrizable. Of the various metrics on Ω_n which determine this topology, the metric most generally used is

$$\rho(\{x_i\}, \{y_i\}) = \alpha \exp(-\beta \min\{|i|: x_i \neq y_i\})$$

with fixed $\alpha, \beta > 0$. The shift σ acts by shifting sequences, whose elements are written from left to right in the order of increasing subscript, one step to the left. In the case of one-sided sequences, the element with zero subscript drops out:

$$\sigma(\{x_i\}) = \{x'_i\}, \quad \text{where } x'_i = x_{i+1}.$$

This cascade has many normalized invariant measures. Here are some of them.

Let p be a normalized measure on A , i.e., assign each a_i a probability $p_i \geq 0$ such that $\sum p_i = 1$ and, for $B \subset A$, set

$$p(B) = \sum \{p_i, a_i \in B\}.$$

² This is a word for word translation of a standard Russian term. Apparently, there is no standard term in English. Sometimes one speaks of “a process of function-space type”.

Let μ be the measure (compatible with the topology) on Ω_n obtained as the direct product of the measures p . (A “cylinder” set

$$\{\{x_i\}: x_{i_1} \in B_1, \dots, x_{i_r} \in B_r\}, \quad \text{where } B_j \subset A,$$

is assigned the measure $p(B_1) \dots p(B_r)$. One extends this in the standard way from cylinder sets to all Borel sets, and then to a complete measure.) Relative to μ , the cascade $\{\sigma^k\}$ can be regarded as a random process (with phase space A) which describes the sequence of independent trials which all have outcomes $a_i \in A$ and the same probability distribution p . This explains the reference to Bernoulli in the name. The mapping σ , considered together with μ , is called a *metric Bernoulli automorphism* (resp., *endomorphism*).

Obvious generalizations are possible when A is an infinite set equipped with a topology or measure, and the “time” runs over discrete (semi-)groups other than $\mathbb{Z}$ or $\mathbb{N}$. We shall not need these.

The next random process in order of complexity is a homogeneous Markov chain with a finite number of states $a_1, \dots, a_n$ and discrete time. Its realizations are also two-sided or one-sided sequences of symbols in A and, hence, the natural representation of this random process also reduces to introducing a normalized measure μ (into Ω_n) which is invariant under σ and is compatible with the topology. I shall not describe the construction (see volume 2), but mention only that, if the probability of passing from the state a_i to a_j is 0, then the set of all sequences in which a_j follows a_i at least once has measure zero. This suggests the following topological analogue.

Suppose certain pairs of symbols in A are “admissible”. The set of all sequences $\{x_k\}$ for which the pairs (x_i, x_{i+1}) are admissible for all i is a closed subset $\Omega' \subset \Omega_n$ which is σ -invariant, i.e., $\sigma\Omega' = \Omega'$. (This subset may be empty for an “unsuccessful” choice of “admissible” pairs; we will assume that the set of admissible pairs has been chosen “successfully”, i.e., that $\Omega' \neq \emptyset$.) This is the most important example of a closed, σ -invariant subset of Ω_n . The dynamical system on Ω' generated by the shift $\sigma|_{\Omega'}$ is called a *topological Markov chain*. The set of admissible pairs can be given by a matrix $B = (b_{ij})$, with $b_{ij} = 1$ if the pair (a_i, a_j) is admissible, and $b_{ij} = 0$ otherwise. (We then write Ω_B instead of Ω' .) It is also possible to give the set of admissible pairs by an oriented graph with n vertices, denoted by $a_1, \dots, a_n$, in which there is an oriented edge going from a_i to a_j if and only if the pair (a_i, a_j) is admissible. The vertices of the graph correspond to the states of a quasi-random process (the prefix “quasi-” is used because there is no concept of probability in our “topological” variant); a state $\{x_i\}$ of the DS corresponds to an infinite path in the graph, going along the edges in the positive direction (i.e., the direction from x_i to x_{i+1}).

The study of topological Bernoulli cascades, their invariant measures, the closed σ -invariant subsets, etc., is called *symbolic dynamics*. More precisely, this is symbolic dynamics in the strict sense of the words; in a broader sense, we understand symbolic dynamics to also include the application of

symbolic dynamics in the strict sense to the study of dynamical systems (for example, smooth systems) which are defined quite independently of Ω_n and σ .

Suppose that we are interested in more than the fact that a current state of a quasi-random process changes, from a_i to a_j : let us suppose, say, that the transition from a_i to a_j could take place in several different ways and we want to distinguish among them. Such a situation can be described by a matrix $B=(b_{ij})$ in which b_{ij} is equal to the number of different ways of going from a_i to a_j ; this (the situation and the matrix) can also be represented by an oriented (multi-)graph with vertices $a_1, \dots, a_n$ in which b_{ij} edges lead from a_i to a_j . As above, a state of the DS (which is also called a topological Markov chain) corresponds to an infinite path in the graph along the oriented edges; this differs from the previous case in that a path is not merely defined by the vertices $\{x_i\}$ passed in moving it, but it is also necessary to give the edges which are traversed. A “purely symbolic” formulation of what we have said is obvious.

Actually, there is no real generalization here. For, we can pass from a topological Markov chain in the latter sense to a topological Markov chain in the first sense by considering infinite sequences of edges and regarding a pair of edges as admissible if the end of the first edge coincides with the initial point of the second edge. In practice, however, when using symbolic dynamics in the TDS, topological Markov chains in which there are several transitions between two states of the chain often arise directly.

The monographs [1], [22] are devoted to symbolic dynamics. Consequently this subject is barely touched on in volumes 1 and 2, although it was used to obtain some of the results cited in volume 2.

1.3. Trajectories, Motions, Invariant Sets. The following few definitions relate to both flows and cascades. We let $\{g^t\}$ denote a DS on a phase space M in which t runs over $\mathbb{R}$, $\mathbb{R}^+$, $\mathbb{Z}$ or $\mathbb{N}$.

The function $t \mapsto g^t x$ is called the *motion* of the point x (any such function will also be called a motion of the DS). For a smooth flow this is the same as the solution of the differential equation (3), but in other situations it is not appropriate to speak of a solution (a solution of what?). The (*phase*) *trajectory* of a point x or a motion $t \mapsto g^t x$ is the set $\{g^t x\}$. Any such set is called a trajectory of the DS. For a topological flow this is a curve, except in the case where x is an *equilibrium point*, i.e., $g^t x = x$ for all t . (Equilibrium points are also called *rest points*. In article I, trajectories are called phase curves, a term not commonly used in the TDS, and equilibrium points are called singular points, which is appropriate in the smooth situation if it is borne in mind that such points are the zeros of the phase-velocity vector field, and so no direction is defined at them.) A related concept in the smooth case is that of an integral curve of a direction field (Article I, Chap. 1, § 1.1).

The words “trajectory” and “integral curve” are sometimes used in the

literature with different meanings, one of which is what we are calling here a "motion".

A trajectory $\{g^t x\}$ is said to be *periodic* if it is the trajectory of a *periodic motion*, i.e., if $g^{t+T}x = g^t x$ for all t and some $T \neq 0$. All such T are called *periods* of the motion and (especially in the case of a cascade) of the point x . This point is then called a *periodic point* (of the system and, in the case of a cascade, of the map g). According to the definition, periodicity also includes the case where $g^t x$ does not depend on t ; in the case of a cascade, x is said to be a *fixed point* (of the cascade and of the mapping which generates it) and, in the case of a flow, x is an equilibrium point. But, in speaking of a periodic trajectory of a flow, one often tacitly excludes equilibrium points. A periodic trajectory of a flow is called a *closed trajectory*; because it is a closed curve if $g^t x$ depends continuously on t . (This may not be so in the measurable case, where individual trajectories are usually neglected.) Under the same assumptions, all the periods are integral multiples of a certain *minimal period* $T_0 > 0$; this is clearly true for a periodic trajectory of a cascade as well. Often the word "period" is understood to be the minimal period.

Periodic trajectories are sometimes called *cycles*.

The set $\{g^t x, t \geq t_0\}$ is called a positive semi-trajectory; $\{g^t x, t \leq t_0\}$ is called a negative semi-trajectory (this is only considered in the reversible case). We call $x_0 = g^{t_0} x$ the initial point (or origin) of both these semi-trajectories. Alternatively, we sometimes say that the semi-trajectories are semitrajectories "of x_0 " and of the system. For a given initial point x_0 of a semi-trajectory we can always suppose that $t_0 = 0$ upon replacing x by x_0 . All semi-trajectories of periodic trajectories (including equilibrium points of a flow) coincide. In the reversible case, any other trajectory is divided by any of its points x_0 into a positive and a negative semi-trajectory with x_0 as initial point.

Another general concept is that of an *invariant set*. In the group case, an invariant set is a set $A \subset M$ consisting of entire trajectories; in other words, $g^t A = A$ for all t . For a cascade $\{g^t, k \in \mathbb{Z}\}$, this is equivalent to requiring that $g^{-1} A = A$ or $g A = A$. In the semigroup case, we require that one of the following hold for all t ,

$$1) (g^t)^{-1} A = A; \quad 2) g^t A = A; \quad 3) g^t A \subset A.$$

For a semi-cascade $\{g^k\}$, this is equivalent to requiring that g satisfy the relations above. All three variants are used. If we need to distinguish among them, we will speak of two-sided invariance if 1) holds and semi-invariance, positive invariance, or invariance in the positive direction if 3) holds. For a cascade, 2) is usually called *g-invariance* (compare with Ω' in § 1.2). Positive invariance also has an obvious meaning in the group case.

1.4. Reversal and Change of Time. In the case when t runs over a group, we can define a new DS $\{g_-^t\}$ from $\{g^t\}$ by setting $g_-^t = g^{-t}$. We say that

$\{g_-^t\}$ is obtained from $\{g^t\}$ by *time reversal*. If $\{g^t\}$ is a smooth flow with phase velocity $\mathbf{v}$, then $\{g_-^t\}$ is a smooth flow with phase velocity $-\mathbf{v}$.

Given a flow $\{g^t\}$, we can pass to a flow $\{f^t\}$ which has the same trajectories and direction of motion as $\{g^t\}$ but which flows with a different velocity, *i.e.*, in passing to $\{f^t\}$, the time taken to traverse some arc of a trajectory changes. We say that $\{f^t\}$ is obtained from $\{g^t\}$ by a *change of time*. For smooth flows a smooth change of time is equivalent to multiplying the phase-velocity vector field by a smooth, positive, scalar function.

1.5. Morphisms of Dynamical Systems. The modern reader is accustomed to expect that, where there are objects, there will also be morphisms and isomorphisms, and that classifying isomorphism classes will be one of the natural problems of the theory. Although this problem may turn out to be insoluble (the reader is probably also accustomed to this), partial classification, various invariants, etc., may be useful.

This is generally true also in the TDS, although the degree of explicit attention paid to morphisms and classification varies among its subdivisions. The most explicit treatments occur in the more abstract subdivisions. In the local QTDE, the isomorphisms appear in the guise of coordinate changes. Not only do morphisms appear under other names, but they can also be present “implicitly”. Thus, the restriction of a dynamical system to an invariant set A is, in an obvious sense, a “subsystem”, and the topological embedding of A into the phase space is a monomorphism of systems. Here are a few more definitions related to morphisms.

Let $\{f^t\}$ and $\{g^t\}$ be topological DSs with phase spaces M and N . A continuous mapping $h: M \rightarrow N$ is called a *homomorphism* of DSs (more precisely, a homomorphism from the first DS to the second) if $hf^t = g^t h$ for all t . (If it is not assumed that the motions are defined for all t (see §§ 1.6, 2.1), then it is necessary to specify whether one requires that $f^t x$ be defined for all t for which $g^t h x$ is defined.) For (semi-)cascades $\{f^k\}$ and $\{g^k\}$ it suffices to require that $hf = gh$.

Sometimes it is said that such a homomorphism effects a *semi-conjugation* of the DS (sometimes this phrasing is reserved for surjective homeomorphisms). The disadvantage of this nomenclature is that it creates the impression that semi-conjugacy of DSs is a symmetric relation.

When the homomorphism h above is surjective (*i.e.*, $hM = N$), the first DS is called a (*topological*) *extension* of the second, and the second is called a (*topological*) *factor* of the first (or an extension or factor under the mapping h , if it is necessary to be more precise). If h is a surjective homeomorphism, then h is called an *isomorphism* (of the DSs), and the DSs are said to be (*topologically*) *isomorphic* or (*topologically*) *conjugate*. For (semi-)cascades $\{f^k\}$, $\{g^k\}$, we also say that the mappings f and g are (*topologically*) *conjugate* in this case. If the DSs are smooth, and if h is a diffeomorphism, then we speak of *smooth conjugacy* (or *C^r -conjugacy* if h is a C^r -diffeomorphism).

Topological flows $\{f^t\}$ and $\{g^t\}$ with phase spaces M and N , respectively, are said to be (*topologically*) *equivalent* if there is a (surjective) homeomorphism $h: M \rightarrow N$ which carries the trajectories of one flow to the trajectories of the other while preserving the direction of motion. (In article I, Chap. 2, § 1.1 topological equivalence is called topological orbital equivalence.) In other words, the first flow is conjugate to a flow obtained from the second by a change of time (§ 1.4). For the sake of uniformity in formulating some results, we sometimes say that cascades are equivalent, having in mind that they are conjugate. In the theory of smooth DSs it turns out that C^r -conjugacy, where r depends on the actual situation, is adequate in some situations (see, for example, article I, Chap. 2, § 4; Chap. 3, § 1.1, § 6.4; Chap. 6, § 2.1; in article I, C^∞ -conjugacy and C^ω -conjugacy (that is, analytic conjugacy) also appear). In other contexts, topological equivalence is more appropriate (even for smooth DSs).

In ergodic theory, there are also concepts which play the rôle of morphisms (see volume 2). The basic concept is that of a metric isomorphism of DSs with invariant measures. Unlike topological isomorphism, a metric isomorphism is not a homeomorphism, but an isomorphism of measure spaces.

1.6. Various Remarks. The questions in the TDS and the QTDE are connected in one way or another with the qualitative disposition of trajectories in phase space and the qualitative properties of motions. They may concern the qualitative picture and the behaviour in the whole phase space (the *global* theory, study, picture, ...) or in some part of it (the *local* theory, etc.). With due respect to the fuzziness of boundaries and shades of meaning, it may be pointed out that in the QTDE the trajectories of a flow (viewed as non-parametrized curves) often play a greater rôle than the motions, while motions play the chief role in the TDS, especially in its more abstract subdivisions. In global questions in the TDS, it is usually assumed that the motion of a phase point is defined for all $t \in \mathbb{R}$, $\mathbb{R}^+$, $\mathbb{Z}$, or $\mathbb{N}$ (as was reflected in § 1.1), but this assumption is regarded as less essential in the QTDE. Besides if we are concerned only with the disposition of the trajectories of a flow, and not with the motions along them, then it is always possible to change time (§ 1.4) so that every motion will be defined for all t (but, for cascades, such a “trick” is impossible). Finally, in local questions one is usually not concerned with the fate of trajectories after they leave the domain of phase space under consideration and, hence, no suppositions are made about the corresponding motions. Formalizing the situation for flows in terms of the motions (which is necessary in the non-smooth case, where it is impossible to reduce everything to specifying the phase-velocity field on the domain of interest) is rather cumbersome, although really quite trivial. The idea will become clear from what will be said in § 2 in connection with the smooth case. (For cascades the analogous situation is easily formalized; one does not assume that the map which generates the cascade is defined everywhere, but only on some open set.) Besides, one can often assume that there is

a “true” DS in the sense of § 1.1 on some phase space M and one is studying only what takes place in some part of M . It must also be said that, although the TDS considerably extends the range of objects investigated by the QTDE, the possibilities for a meaningful study of local questions are restricted in the more abstract situation. Thus, the local theory, if it goes outside the framework of the QTDE in some respects, nevertheless remains close to it.

Individual trajectories are usually of interest in the QTDE and the TDS when they have properties which can considerably affect, even if only locally, the qualitative picture. In particular, such trajectories include the periodic trajectories (including equilibrium points of flows). The investigation of neighbourhoods of such trajectories is the “most local” part of the local theory, belonging essentially to QTDE. The parts of the local theory relating to other sorts of domains may be called *semilocal*. They are newer, more complicated, and have been less studied. They are not always classified as parts of the QTDE: there are no precise boundaries, and the classification often depends simply on the traditions of the various schools.

In the TDS, a large part is played by various properties and considerations which are connected in some way or another with the compactness of various objects. In view of the small size of the present article, the phase space will be usually assumed to be a compact metric space or a closed manifold. Consequently, a number of properties connected with compactness hold automatically. In a more complete exposition, it would be necessary to sometimes assume, sometimes prove, that these properties hold, and to analyze what happens when they are absent.

For example, the property that a (semi-)trajectory be relatively compact often appears in accounts of the theory (see, e.g., article I, Chap. 1, § 5.5). Poincaré described such a (semi-)trajectory as *Lagrange stable*. At that time there was no general concept of compactness, and it seems superfluous today to keep a special (and lengthier) name for the particular case in question.

Sometimes it is convenient to pass from a noncompact phase space to a compact one in some way (making a change of time, if necessary). Thus, it is sometimes convenient to pass from $\mathbb{R}^2$ to S^2 (a one-point compactification), or to the projective plane $\mathbb{R}P^2$, or to the Poincaré (hemi-)sphere [18], [46]. There are instructive nontrivial examples of compactification in [20], [59].

On the other hand, the phase space may be neither compact nor locally compact. Examples of such phase spaces are the infinite-dimensional function spaces which arise in studying DSs described by partial differential or functional-differential equations, as well as DSs corresponding to certain (but not all, cf. § 1.2) random processes.

The latter case is fairly specific and is touched on in volume 2. The former case is closer to the QTDE and the theory of smooth DS. I remark that a choice of suitable function space for such DSs (physically they are systems with distributed parameters) is not usually self-evident. Requiring that $g'x$

be continuous with respect to (t, x) turns out to be too restrictive; it is usually replaced by the condition that the semi-group $\{g^t\}$ be strongly continuous. Although the phase space is noncompact, the best results relate to cases in which the motions become more concentrated along a compact set as time progresses. Fortunately, such cases are actually encountered, and the corresponding problems are definitely interesting (compare this with the rôle played by compact operators in the theory of operator equations). However, the application of the ideas and methods of the QTDE and the TDS in this domain has really only just begun. An article devoted to such applications in the field of hydrodynamics is planned for one of the later volumes of this series. A question relevant to this area has already been touched on in the present volume (article I, Chap. 1, § 8). Relevant material can be found in [54] and [40].

§ 2. Smooth Dynamical Systems

In this section we explain how certain concepts and facts pertaining to smooth DSs on $\mathbb{R}^n$ carry over to the more general case of smooth DSs on manifolds. Basically this is done by paraphrasing formulations in an “invariant” manner, *i.e.*, so that they do not depend on (nor, perhaps, even require appeal to) the actual choice of local coordinates.

2.1. Smooth Flows. Let $\mathbf{v}$ be a smooth vector field on a manifold M (*i.e.*, a map which assigns, to each point $x \in M$, a vector $\mathbf{v}(x) \in T_x M$ which depends smoothly on x in a natural sense). We consider the differential equation (3): $\dot{x} = \mathbf{v}(x)$. If $x(t)$ is a smooth function of a scalar argument t with values in M , the derivative $\dot{x}(t) \in T_{x(t)} M$ is well-defined. The function $x(t)$ is a solution of (3) if $\dot{x}(t) = \mathbf{v}(x(t))$ for all t in the interval of definition of $x(t)$. As in the case $M = \mathbb{R}^n$, this is connected to the intuitive idea of a phase point moving in M (“among” the fixed phase points). The motion takes place in such a way that, at each moment of time t , the velocity vector $\dot{x}(t)$ is equal to the vector $\mathbf{v}(x(t))$ associated to the point of phase space at which the moving point is at that moment. Like the concept of a solution itself, this kinematic representation does not depend on local coordinates.

At the same time, equation (3) can be written in terms of the local coordinates as a system of the form

$$\dot{x}_i = v_i(x_1, \dots, x_n), \quad i = 1, \dots, n, \quad (5)$$

and $x(t)$ viewed as a solution of this system. To be precise, let (U, ψ) be a chart of M , *i.e.*, $U \subset M$ is a coordinate neighbourhood and $\psi: U \rightarrow \mathbb{R}^n$ a coordinate mapping associating with a point $x \in U$ its local coordinates $(x_1, \dots, x_n)$. As long as the point $x(t)$ remains in U , we can speak of its local coordinates

$$\psi(x(t)) = (x_1(t), \dots, x_n(t)).$$

The set of functions $x_i(t)$ is a solution of the system (5), in which

$$(v_1(x_1, \dots, x_n), \dots, v_n(x_1, \dots, x_n))$$

are the components of the vector $\mathbf{v}(x)$ in the natural basis of the space $T_x M$ associated to the chart (U, ψ) (if the tangent vectors are regarded as linear, homogeneous, first-order differential operators, this basis consists of the vectors $\frac{\partial}{\partial x_i}$).

Conversion to local coordinates shows at once that various local assertions (about the existence of a solution, and so on) carry over immediately from equations in $\mathbb{R}^n$ to equations on manifolds. If $x(t)$, $y(t)$ are two solutions, and if $x(t_0) = y(t_0 + a)$, then $x(t)$ and $y(t + a)$ coincide on their common domain. This enables us (cf. article I, Chap. 1, § 2.5) to extend a solution with the initial value $x(0) = x_0$ in the usual way to the maximum interval on which the solution exists; we denote this interval by

$$J(x_0) = (t^-(x_0), t^+(x_0))$$

(where $t^\pm$ may even be equal to $\pm\infty$). The extended $x(t)$ can go through various coordinate neighbourhoods and is described in terms of the local coordinates by various sets of functions which are solutions of the corresponding systems of the form (5) (and if $x(t)$ returns to a coordinate region where it had been at some previous time, then it will, in general, go along a different curve from that along which it moved the first time). We introduce the mapping g^t (the shift along trajectories by time t) as in (4). Generally speaking, this is only a partial transformation, i.e., the domain of g^t is only a part of M (perhaps empty). The set

$$D = \{(x, t) \in M \times \mathbb{R} : t \in J(x)\}$$

is open, and the mapping

$$D \rightarrow M \quad (x, t) \mapsto g^t x \tag{6}$$

is smooth (of the class C^r if $\mathbf{v} \in C^r$; actually the degree of smoothness with respect to t differs somewhat from that with respect to x ; I shall not go into this here). Relations (1) and (2) hold, where if $D \neq M \times \mathbb{R}$, we interpret (2) with the following reservations: the domain of definition of the right hand side of (2) is contained in that of the left hand side (the two need not coincide, although they will if t and s have the same sign), and the right hand side is the restriction of the left hand side to the smaller domain. If $|t^\pm(x)| < \infty$, then, for any compact $K \subset M$, we shall have $g^t x \notin K$ for all t sufficiently close to $t^\pm(x)$. If the trajectories $\{g^t x\}$ are relatively compact in M , then $J(x) = \mathbb{R}$. In particular, this is the case if M is compact.

According to § 1, we must demand that $D = M \times \mathbb{R}$ in order to speak of a flow $\{g^t\}$. But often one speaks of a flow even when $D \neq M \times \mathbb{R}$. For the

most part, we leave the reader to decide when one can (through a small modification of the text) use the word "flow" in this context.

It is clear that a diffeomorphism of two manifolds allows us to "transfer" a smooth flow from one to the other. Here is a somewhat more general fact.

Let $h: M \rightarrow N$ be a smooth mapping of manifolds and let $\mathbf{v}$ and $\mathbf{w}$ be smooth vector fields on M and N defining the flows $\{f^t\}$ and $\{g^t\}$. The vector fields $\mathbf{v}$ and $\mathbf{w}$ are said to be *h-related* if

$$Th(x) \cdot \mathbf{v}(x) = \mathbf{w}(h(x)) \quad \text{for all } x \in M,$$

i.e.,

$$Th \circ \mathbf{v} = \mathbf{w} \circ h \quad (7)$$

(the latter formula uses the fact that the vector fields $\mathbf{v}$ and $\mathbf{w}$ are mappings $M \rightarrow TM$ and $N \rightarrow TN$). In this case, h is a homomorphism of the flows $\{f^t\}$ and $\{g^t\}$ in the sense of § 1.5:

$$h \circ f^t = g^t \circ h \quad \text{for all } t \quad (8)$$

(with the obvious reservations if $D \neq M \times \mathbb{R}$ where D is defined for $\{f^t\}$ as in (6)). Conversely, (8) implies (7).

2.2. The Variational Equations. In the case $M = \mathbb{R}^n$, when the same everywhere defined coordinates $(x_1, \dots, x_n)$ are used for all time, we have the following well-known result about the differentiability of the solutions $g^t x$ of (3) with respect to the initial data (cf. article I, Chap. 1, § 2.7). Suppose the initial data depend smoothly on a parameter c . To be precise, let $\gamma(c)$ be a smooth function with values in $\mathbb{R}^n$ such that $\gamma(c_0) = x_0$, and consider the solution $g^t \gamma(c)$ of the system (3). Then the vector

$$\delta x(t) = \left. \frac{\partial g^t \gamma(c)}{\partial c} \right|_{c=c_0}, \quad (9)$$

traditionally called the *variation of the solution* $x(t) = g^t x_0$ of the system (3), is a solution of the so-called *variational equations* along the trajectories $x(t)$, i.e., of the system

$$\delta \dot{x}(t) = \mathbf{v}'(x(t)) \delta x(t), \quad (10)$$

where $\delta \dot{x} = \frac{d}{dt} \delta x$, and $\mathbf{v}'$ is the matrix $\begin{pmatrix} \partial v_i \\ \partial x_j \end{pmatrix}$. Moreover, the initial condition uniquely distinguishing $\delta x(t)$ from other solutions of system (10) is:

$$\delta x(0) = \frac{d\gamma(c_0)}{dc}. \quad (11)$$

We now consider a smooth flow $\{g^t\}$ on a manifold M defined by equation (3). It is easy to see that the mappings Tg^t define a flow on TM . This flow "projects" onto $\{g^t\}$ by the natural projection $p: TM \rightarrow M$ (carrying $T_x M$

into x):

$$p \circ Tg^t = g^t \circ p.$$

The discussion above really concerned the flow $\{Tg^t\}$ when $M = \mathbb{R}^n$. For if $\delta x(t)$ is defined in accordance with (9) (which, incidentally, makes sense even if $M \neq \mathbb{R}^n$), then it is the tangent vector to M at the point $g^t x_0$, and $\delta x(t) = Tg^t \cdot \delta x(0)$, where $\delta x(0)$ is determined from (11). Thus the differential equations for $\delta x(t)$ are precisely those obtained by representing the flow $\{Tg^t\}$ by means of the corresponding phase-velocity vector field. In invariant form (which is also suitable in the case $M \neq \mathbb{R}^n$, without even appealing to local coordinates), the latter field is a vector field on TM (i.e., a map $TM \rightarrow TTM$). It turns out that this is the mapping $s \cdot T\mathbf{v}$. Here $\mathbf{v}$ is viewed as a map $M \rightarrow TM$ and s is the involution in TTM defined in terms of local coordinates by $(x_i, \dot{x}_i, \delta x_i, \delta \dot{x}_i) \mapsto (x_i, \delta x_i, \dot{x}_i, \delta \dot{x}_i)$.

In an obvious notation, the equations for the flow $\{Tg^t\}$ in local coordinates are

$$\dot{x}_i = v_i(x), \quad \delta \dot{x}_i = \sum \frac{\partial v_i(x)}{\partial x_j} \delta x_j. \quad (12)$$

These differ from (10) because a vector $\delta x \in TM$ has $2n$ coordinates (where $\dim M = n$): viz., the n coordinates $(x_1, \dots, x_n)$ of the point $x = p\delta x$ on M , and a further n coordinates $(\delta x_1, \dots, \delta x_n)$ of the vector δx in the space $T_x M$. The classical variational equations for the trajectory $g^t x_0$ of the flow $\{g^t\}$ describe $\{Tg^t\}$ over the trajectory, but (12) describes $\{Tg^t\}$ over all trajectories in the coordinate neighbourhood.

The flow $\{Tg^t\}$ is C^{r-1} if $\mathbf{v}$ is C^r . Moreover, when $\mathbf{v}$ is C^1 the special character of the flow $\{Tg^t\}$ (it is, in an obvious sense, linear over every trajectory of the flow $\{g^t\}$), ensures that it is uniquely determined by its phase-velocity field even though it is not smooth.

A well-known classical proposition asserts that the phase velocity of the motion is a solution of the corresponding system of variational equations. This may be rephrased in invariant terms as

$$\text{or as} \quad Tg^t(x) \cdot \mathbf{v}(x) = \mathbf{v}(g^t x) \quad (13)$$

$$\mathbf{v}(M) \subset TM \quad \text{is an invariant manifold of the flow } \{Tg^t\}. \quad (14)$$

A cascade $\{g^k\}$ induces the cascade $\{Tg^k\}$ on TM . It is clear that the latter “projects” onto $\{g^k\}$, i.e. $p \circ Tg^k = g^k \circ p$, and, in an obvious sense, is “linear” over each trajectory $\{g^k x\}$. The rôle of the variational equation is played by the representation of the mapping $Tg^k(x)$ as a composition

$$Tg(g^{k-1}x) \circ \dots \circ Tg(gx) \circ Tg(x).$$

There is no generally accepted name for $\{Tg^t\}$ and $\{Tg^k\}$. In the language of topological dynamics they are linear extensions of the systems $\{g^t\}$ and $\{g^k\}$. But there are many different linear extensions, and a name is needed for precisely $\{Tg^t\}$ and $\{Tg^k\}$. I venture to call them the *tangential linear*

extensions (“*differential*” extensions would also be an acceptable name, but not *variational*, because the latter adjective, in contrast to the noun “*variation*”, has a very specific meaning connected with the calculus of variations). Strictly speaking, $\{Tg^t\}$ and $\{Tg^k\}$ fall under the general concept of the “*first prolongation*” of a smooth group of mappings, which goes back to S. Lie. But, in the TDS the term “*prolongation*” has a different meaning (Chap. 3, § 4.2).

2.3. First-Return Mapping. Suppose that the n -dimensional phase manifold M^n of a smooth flow $\{g^t\}$ contains a hypersurface V (a manifold of codimension one) transverse to the phase-velocity field (*i.e.*, the latter is nowhere tangent to V). V is called a *cross-section* (or a hypersurface section, or a transversal (usually when $n=2$), or, but only when $n=2$, a nontangential arc). One may further specify whether the transversal (*etc.*) is *local* or *global*; in the latter case, any trajectory intersects V , and V is a closed subset of M . To each point $v \in V$, we associate the first (with respect to time) point of intersection after v of the trajectory $\{g^t v\}$ with V if there is such an intersection. We obtain a mapping F of an open subset $V_0 \subset V$ into V . In other words, $Fv = g^{s(v)}v$, where $s(v)$ is the *time of first return of v to V* , *i.e.*, $s = s(v)$ if

$$s > 0, \quad g^s v \in V \quad \text{and} \quad g^t v \notin V \quad \text{when } 0 < t < s.$$

(It may happen, of course, that F is nowhere defined, *i.e.*, no trajectory intersects V twice.) F is called the *first-return mapping* (or the *monodromy operator*, or the *Poincaré mapping*, or, in the Russian literature, the *successor mapping*). It is used both in theoretical investigations and to visually represent the results of computer calculations of the trajectories of three-dimensional systems.

In the latter (three-dimensional) case, one calculates the solution over a very long time to find the successive points of intersection of a trajectory with a suitable transversal, *i.e.*, the points $F^k v$ for a fixed $v \in V$. By collecting a sufficient number of such points (in a typical numerical experiment the number of points might be of the order of 100 or 1000), it is possible use their disposition to say something about the behaviour of the trajectory $\{g^t v\}$. If it lies on an invariant surface, then the points $F^k v$ will lie on a curve; if $\{g^t v\}$ fills a domain densely, then the points $F^k v$ densely fill a domain on V ; if $g^t v$ approaches a set A , getting arbitrarily close as time increases (*i.e.*, A is an ω -limit set of the trajectory; see article I, Chap. 1, § 5.5), then the points $F^k v$ accumulate near the set $A \cap V$ and reproduce its structure to some extent. Presenting results this way often proves to be more convenient than trying to draw the trajectories (by projecting to two planes or by axonometric means) — it is easy to take in at a glance hundreds or thousands of points, but a similar number of “*loops*” made by a trajectory leaving and returning to V would look intricate in any complicated situation, no matter what the method of representation.

The first-return mapping is more often used to study (semi-)local questions than global questions. This is because a global transversal need not exist and, when it does, there is no guarantee that $V_0 = V$. An “extreme” example is the flow

$$\dot{x} = \frac{1}{1+y^2}, \quad \dot{y} = 1 \quad (15)$$

on the cylinder $M = S^1 \times \mathbb{R} = \{(x, y)\}$, where x is taken mod π . Here $V = 0 \times \mathbb{R}$ is a transversal for which $V_0 = \emptyset$. This is connected with the noncompactness of M .

It can be shown that if M is closed, then $V = V_0$ for a global transversal (we remark that the condition that V be a closed subset of M further implies, in this case, that V is also a closed manifold). Generally speaking, the time $s(v)$ of first return is not constant. But, by a smooth change of time, we can arrange that it becomes constant and equal to one, say. In this case M and $\{g^t\}$ can be “recovered” (up to a diffeomorphism commuting with g^t) from V and F by means of the following construction, known as the *Smale suspension* of the diffeomorphism $F: V \rightarrow V$ (and of the cascade $\{F^k\}$). M is obtained from the direct product $V \times [0, 1]$ by identifying the “bottom” $V \times 0$ and the “top” $V \times 1$ by the map which sends $(v, 1)$ to $(Fv, 0)$. The motion takes place along the “interval” $v \times [0, 1]$ in such a way that the second coordinate increases at a unit rate; and when the moving point hits the position $(v, 1) = (Fv, 0)$ it begins to move along the “interval” $Fv \times [0, 1]$, and so on. This construction is applicable in the more general case when F is a homeomorphism of a topological space V (and even when $F: V \rightarrow V$ is a continuous mapping, in which case $\{g^t\}$ will be a semi-flow). In our case it enables a given V and F to be obtained as a global transversal and first-return mapping of a smooth flow on a smooth manifold. (At first glance it might seem that “breaks” might appear in pasting the ends together. To convince ourselves of the smoothness, it suffices to rephrase the construction in terms of a suitable identification in $V \times \mathbb{R}$.)

A periodic nonautonomous system (article I, Chap. 6, § 3)

$$\dot{x}_i = v_i(x_1, \dots, x_n, t) \quad i = 1, \dots, n \quad (16)$$

can be interpreted as an autonomous system in the cylinder $\mathbb{R}^n \times S^1$, where $S^1 = \mathbb{R}/(\tau\mathbb{Z})$ and τ is the period of the right hand side of (16) in t . If the solutions of (16) are defined for all t , then the trajectories periodically return to the global transversal $V = \mathbb{R}^n \times s$, $s \in S^1$ and, in this case, $V_0 = V$.

In the general case, the intersection of a closed trajectory L of the flow $\{g^t\}$ on M with the transversal V is a periodic trajectory of the first-return mapping; the period m of this trajectory is in no way connected with the period of L . In the preceding example, the number $m\tau$ was the period of L (but not necessarily the minimal period of the corresponding solution of the system (16)). On the other hand, given any point of a closed trajectory L , there is a local transversal which intersects L only at the point and which is, therefore, a fixed point of the first-return mapping. It is clear that the

study of the behaviour of trajectories of the flow in a neighbourhood of a closed trajectory and the study of iterates of the first-return mapping in the neighbourhood of its fixed point or of a periodic trajectory are almost one and the same topic.

In the case of iterates of the first-return mapping it can be seen that it is sometimes expedient to speak, in contrast to § 1, of a cascade $\{g^n\}$ in which the transformations g^n are partial. This is the analogue of a flow in which $D \neq M \times \mathbb{R}$ in (6).

2.4. Equilibrium Points and Periodic Trajectories. Replacing $\mathbb{R}^n$ by a phase manifold M introduces nothing new into the local study of periodic trajectories and equilibrium points. Except in the case when M is nonorientable and we consider a closed trajectory L along which the orientation changes, such a treatment does not even formally increase the generality, for local coordinates can be introduced along the trajectory and everything may be regarded as taking place in $\mathbb{R}^n$. Even in the excluded case, where a neighbourhood of L is the direct product of a Möbius strip and $\mathbb{R}^{n-2}$, the first-return mapping can still be regarded as a (local) transformation on $\mathbb{R}^{n-1}$. The arbitrariness in the choice of local coordinates is of no consequence, because the definitions and results of article I that concern us can be formulated (in case they were not already so formulated) without using coordinates and, hence, are immediately applicable to smooth DSs on M . This is obvious for concepts such as stability and invariant manifolds. Concepts related to "linear approximation" are easily rephrased invariantly in terms of $\{Tg^t\}$ and $\{Tg^k\}$ described above.

For example, if x_0 is an equilibrium point of the flow (3), then the linear flow $\{Tg^t(x_0)\}$ on $T_{x_0}M$ is determined by a linear vector field on $T_{x_0}M$ (the phase-velocity field). We let $\mathbf{v}'(x_0)$ denote the linear map which gives the latter. In terms of local coordinates it is described by the matrix $\left(\frac{\partial v_i(x_0)}{\partial x_j}\right)$ (which, therefore, in the case when $\mathbf{v}(x_0)=0$, can be regarded as a matrix of components of a tensor; when $\mathbf{v}(x_0) \neq 0$ this is not so). The *eigenvalues of the equilibrium point* x_0 are the eigenvalues of this map (multiplicity taken into account); the equilibrium point is *hyperbolic* if all the eigenvalues lie off the imaginary axis, and it is *nondegenerate* if they are all nonzero. The *multipliers* of a closed trajectory $L=\{g^t x_0\}$ of period τ are the eigenvalues (with multiplicity taken into account) of the linear mapping $\{Tg^t(x_0)\}$; they do not depend on the choice of $x_0 \in L$. The period, here, is usually understood to be the minimal period τ_0 . It is sometimes useful to consider other periods as well; then the multipliers corresponding to each period $k\tau_0$ are k th powers of the multipliers corresponding to τ_0 . In any case, in view of periodicity and (13),

$$Tg^\tau(x_0) \cdot \mathbf{v}(x_0) = \mathbf{v}(x_0),$$

so that there is always a multiplier equal to one; in article I it was discarded. The *eigenvalues of a fixed point* x_0 of a cascade $\{g^k\}$ are the eigenvalues

of the operator $Tg(x_0)$. The *eigenvalues of a periodic trajectory* $l = \{g^k x_0\}$ with period m are the eigenvalues of x_0 as a fixed point of the cascade $\{g^{mk}\}$. They do not depend on the choice of $x_0 \in l$. If the flow has a transversal V with first-return mapping F and if a closed trajectory L intersects V , then the eigenvalues of the periodic trajectory $L \cap V$ of the cascade $\{F^k\}$ are also multipliers of L ; in addition to these, L still has the “superfluous” multiplier equal to one. A closed trajectory of the flow is said to be *hyperbolic* (resp., *nondegenerate*) if it has only one multiplier on the unit circle (resp., equal to 1). A periodic trajectory of a cascade and each of its periodic points is *hyperbolic* (resp., *nondegenerate*) if it has no eigenvalues on the unit circle (resp., equal to 1).

If L_0 is a closed trajectory with period $k\tau_0$ and minimal period τ_0 , and L_0 is the underlying trajectory (pointwise) with period τ_0 , then L can be viewed geometrically as the result of making k circuits around L_0 . We denote this symbolically by $L = L_0^k$. L will be called a *k-circuit trajectory* or a *k-fold repetition* of L_0 and L_0 a *single-circuit trajectory* (the expressions “multiple trajectory” and “multiplicity”, which suggest themselves, must be rejected because they have a different meaning – see article I, Chap. 1, § 5.3).

An equilibrium point is said to be *isolated* if it is an isolated zero of the phase-velocity vector field, i.e., if some neighbourhood of it contains no other equilibrium points. In other cases, a periodic trajectory of a flow or cascade (and the corresponding periodic point of a cascade) with (not necessarily minimal) period τ is said to be *isolated* if it has a neighbourhood containing no other periodic trajectories with period close to τ (equal to τ in the case of a cascade). It may happen that a periodic trajectory is nondegenerate or isolated as a single-circuit trajectory, but not as a k -circuit trajectory (hyperbolicity, however, does not depend on the number of circuits). A nondegenerate periodic trajectory (including an equilibrium point of a flow) is isolated and preserved (in a natural sense) under small perturbations. If it is hyperbolic, it remains so after a small perturbation. Smallness means C^1 -small.

The results about asymptotic stability and invariant manifolds of periodic trajectories (including equilibrium points) cited in article I (Chap. 3, § 4.1 and Chap. 6, §§ 2.3, 2.4) automatically carry over to our case. We fix the relevant notation. The stable and unstable manifolds of a hyperbolic, periodic (with period p) point x of a reversible cascade $\{g^k\}$ are denoted by $W^s(x)$ and $W^u(x)$ where the superscripts “s” and “u” indicate “stable” and “unstable”, respectively. The stable and unstable manifolds of a periodic trajectory $L = \{g^i x\}$ are

$$W^s(L) = \bigcup_{i=0}^{p-1} W^s(g^i x), \quad W^u(L) = \bigcup_{i=0}^{p-1} W^u(g^i x).$$

($W^s(x)$ and $W^u(x)$ coincide with the stable and unstable invariant manifolds of x as a fixed point of the map g^p and are first constructed locally and then extended by the action of iterates of g^{-p} and g^p .) The stable and unstable

manifolds of a hyperbolic equilibrium point of a flow are denoted and constructed in a similar way. Finally, if L is a closed hyperbolic trajectory of a flow $\{g^t\}$, with period τ , we have to distinguish between

1) the stable and unstable invariant manifolds $W^s(L)$ and $W^u(L)$ of the trajectory, which are also denoted by $W^s(x)$, $W^u(x)$ for any $x \in L$, and

2) the stable and unstable manifolds of a point $x \in L$, denoted by $W^{ss}(x)$ and $W^{uu}(x)$.

Sometimes $W^s(x)$ is called the weak stable manifold of x , but sometimes $W^s(x)$ is called the stable manifold of x and $W^{ss}(x)$ the strong stable manifold. Regardless of names,

$$\begin{aligned} W^s(L) &= \{y: \rho(g^t y, L) \rightarrow 0 \text{ as } t \rightarrow \infty\}, \\ W^{ss}(x) &= \{y: \rho(g^t y, g^t x) \rightarrow 0 \text{ as } t \rightarrow \infty\}, \end{aligned}$$

where ρ is any fixed metric. The right hand sides of these relations define sets in more general situations, and even in the general context of topological dynamics. In our case, these sets turn out to be injectively immersed manifolds: the tangent space $E^{ss}(x) = T_x W^{ss}(x)$ is the invariant subspace of the operator $Tg^t(x)$ corresponding to the multipliers λ with $|\lambda| < 1$, and $E^s(x) = T_x W^s(x)$ is the direct sum of $E^{ss}(x)$ and $\mathbb{R}v(x)$ (where v is the phase velocity, as before).

It is clear that

$$g^t W^{ss}(x) = W^{ss}(g^t x), \quad W^s(L) = \bigcup_{0 \leq t \leq \tau} W^{ss}(g^t x).$$

If the trajectory $L \ni x$ is completely unstable (i.e., if it is asymptotically stable as $t \rightarrow -\infty$), then we set $W^s(L) = L$ and $W^{ss}(x) = x$. What was stated above trivially remains in force in this case too. Furthermore, upon making the obvious changes, all that we have said holds for $W^u(L)$ and $W^{uu}(x)$. The corresponding tangent spaces are denoted by $E^u(x)$ and $E^{uu}(x)$. For an equilibrium point x of a flow, or a periodic point x of a cascade, $W^s(x)$ and $W^u(x)$ are also immersed manifolds. The tangent spaces

$$E^s(x) = T_x W^s(x), \quad E^u(x) = T_x W^u(x)$$

can also be described in terms of the linear approximation $\{Tg^t(x)\}$. If x is completely unstable, then we set $W^s(x) = x$; if it is stable, then $W^u(x) = x$. (There is some inconsistency in this (traditional) nomenclature: in the case of a cascade, the manifold denoted by $W^s(x)$ is the analogue of $W^{ss}(x)$.)

It is worth noting that when the dimension of the phase manifold is greater than two, a neighbourhood of a closed trajectory can contain trajectories which exhibit many different types of behaviour. It seems pointless in this case to speak of "limit cycles" (just as one does not speak of "limiting equilibrium points" in connection with equilibrium points on the plane). One already has the notion of asymptotic stability; a broader notion could encompass types of behaviour qualitatively different from one another, and there is no justification for lumping such together.

2.5. The Morse Index. The *Morse index*³ of a hyperbolic periodic trajectory (including an equilibrium point) is defined to be the dimension $\dim W^u(L)$. In the case of a cascade, the index of a point $x \in L$ is the same as the index of L . (Some authors define the Morse index of a closed trajectory to be one less than that defined here.) We denote the Morse index by u , $u(x)$, or $u(L)$. For an equilibrium point of a flow or periodic trajectory of a cascade, the index is equal to the number of the corresponding eigenvalues λ with $\operatorname{Re} \lambda > 0$ for $|\lambda| > 1$; for a closed trajectory it is equal to one plus the number of the multipliers λ with $|\lambda| > 1$. The formulation in terms of λ also defines a number in the non-hyperbolic case which we shall not need.

Morse defined the index of a critical (stationary) point x of a smooth (C^2) function $f: M \rightarrow \mathbb{R}$ as the negative index of inertia of the “second differential” $d^2 f(x)$, i.e., in terms of local coordinates $(x^1, \dots, x^n)$, as the number of negative eigenvalues of the matrix $\left(\frac{\partial^2 f}{\partial x^i \partial x^j} (x) \right)$. Consider a gradient flow

$$\dot{x} = -\operatorname{grad} f(x), \quad (17)$$

where the gradient is taken with respect to some Riemannian metric. (More explicitly, the metric allows one to transform the covariant vector df into the contravariant vector $\operatorname{grad} f$ by standard “index manipulation”: in terms of local coordinates, the i th component of the gradient is

$$(\operatorname{grad} f)^i = \sum_j g^{ij} \frac{\partial f}{\partial x_j},$$

where the matrix (g^{ij}) is the inverse of the matrix of coefficients of the metric tensor.) Its equilibrium points are precisely the critical points of the function f , and hyperbolicity of an equilibrium point is equivalent to nondegeneracy of the critical point (the rank of $d^2 f(x)$ is equal to the dimension of M). In the latter case, the Morse index of x as a critical point of f is the same as its Morse index as an equilibrium point (but, in general, the first index coincides with the number we spoke of at the end of the previous paragraph).

In the calculus of variations in the large there is a variant of the concept of the Morse index, relating to certain sets of critical points (“nondegenerate critical manifolds”). This would correspond to the concept of Morse index for a set of periodic trajectories (satisfying definite conditions). But such a variant is less essential for the TDS, and I restrict myself here to the simplest, but most important, variant relating to separate trajectories.

³ In mathematics there are many magnitudes and objects which bear the name “index” (not counting the indices which appear as superscripts of a symbol). In this article, too, several different types of indices will be encountered. Hence, various qualifying words have to be added unless it is clear from the context what index is under discussion; in that case, as is often done in the literature, we simply speak of “the index”.

Chapter 2

Elementary Theory

D.V. Anosov

§ 1. Introduction

1.1. Contents of the Chapter. The title of this chapter needs explanation. Its contents are elementary in the sense that it is not necessary (even in the omitted proofs) to seriously consider possible complicated limiting behaviour of motions – either because the questions we examine do not call for such considerations or because the dynamical systems we investigate have motions with simple limiting behaviour. In other cases, not considered in this chapter, complex limiting behaviour has to be dealt with seriously.

On first acquaintance with the QTDE, one is apt to be greatly impressed by the Poincaré-Bendixson theory, which establishes the possible types of limiting behaviour of trajectories of flows on the plane and on the two-dimensional sphere. (The fundamental case of a relatively compact semi-trajectory of a smooth flow with isolated equilibrium points was considered in article I, Chap. 1, § 5.5. There is a rather complete description in other cases, as well; see the references in [ME, “Poincaré-Bendixson theory”].) One can scarcely help but think that if the Jordan curve theorem (“a closed curve separates a plane”) can be used so effectively in this theory, then the use of more powerful topological tools should give strong results in the higher dimensional case.

However, the most distinctive feature of flows in the two-dimensional case is that the trajectories locally separate the phase space. This fact is true for spheres and other closed surfaces; it is also true for reversible cascades on a circle S^1 . A substantial general theory is obtained for the corresponding DSs; see Chap. 4 and article I, Chap. 2, § 2, § 4. (Only cascades on S^1 generated by orientation-preserving homeomorphisms are considered in article I. But if a homeomorphism reverses orientation, then it has a fixed point (more precisely, at least two such points), and everything reduces to the simpler case of iterations of a homeomorphism of an interval.) But when the trajectories do not separate the phase space, the situation becomes fundamentally more complicated, and no amount of topology can help in the general case.

There are well-defined, comparatively restricted classes of DSs for which one can obtain sufficiently complete information about the character of the phase portraits and its connection with the topology of the phase space. The actual selection of the classes of DSs necessarily reflects the limits of our knowledge, but is neither arbitrary nor random. The part of the theory

considered here is closely connected with the general concepts of structural stability and genericity, which we shall discuss in § 1.2. This provides a basis for the belief that, although new (wider or different) classes of DSs may be introduced and studied in the future, those currently known will continue to be important. The simplest such class is considered in § 3.

In § 2, we present a few results of a topological nature which do not depend on whether the dynamical system belongs to one of these special classes. These results relate to periodic trajectories and are connected with some of their local properties. They say nothing about the behaviour of other types of trajectories and, so, the information they provide about the phase portrait (if any) is, in principle, incomplete. (In Chap. 3, § 3, we cite further results of a topological nature, which also hold for dynamical systems which do not belong to the classes mentioned above. These relate to certain invariant sets which need not consist of periodic trajectories and also touch only on certain aspects of the phase portrait.)

1.2. Genericity and Structural Stability. Consider the space of all reversible C^1 DSs on a closed n -dimensional manifold M . (Cascades are assumed reversible mostly for the sake of definiteness; the changes required in the formulation for the non-reversible case are insignificant and obvious.) In the case of continuous time, this is the space of all C^1 generating vector fields on M and, in the case of discrete time, it is the space of all C^1 -diffeomorphisms $M \rightarrow M$. In either case, we provide the space with the C^1 -topology.

We shall say that a property of a DS is *structurally stable*, *exceptional*, or *generic* according as the DSs with this property form an open, first category (the union of a countable number of nowhere-dense sets), or a second category subset of the space of all C^1 DSs. (A second category subset is understood in the narrow sense as a set complementary to a set of first category, i.e., as a set containing a dense G_δ .) DSs which have an exceptional or generic property are themselves said to be *exceptional* or *generic* (this is clearly an abuse of language, but one we allow ourselves when the criterion by which the DSs are distinguished is clear from context; a similar convention regarding the term “structural stability” is not admissible, because the term “structurally stable system” has a different meaning; see below). In an obvious sense, we will speak of the structural stability, exceptionality, or genericity of a given situation or case. (We remark that a property which is not generic need not be exceptional.)

We could also consider the space of all C^r , $1 \leq r \leq \infty$, dynamical systems equipped with the C^r -topology, and define genericity, exceptionality, and structural stability in the C^r sense, by replacing C^1 by C^r in the preceding paragraph. We will also speak of C^r -generic (i.e., generic in the C^r sense) and similarly of C^r -exceptional, and C^r -structural stability. (What we have just said also applies to the next section.) Throughout this article, unless the contrary is stated, we take $r=1$; the space of C^1 dynamical systems,

with the corresponding topology, is adequate for most of the questions we consider.

A DS is said to be *structurally stable* if it is topologically equivalent to any sufficiently close DS by a homeomorphism $h: M \rightarrow M$ sufficiently close to $\mathbb{1}_M$ in the C^0 -sense. (More formally, for any neighbourhood $\mathcal{V}$ of the homeomorphism $\mathbb{1}_M$ in the space of all homeomorphisms with the C^0 -topology, there is a C^1 -neighbourhood $\mathcal{U}$ of the DS in question such that the DS is equivalent to any DS in $\mathcal{U}$ by means of some $h \in \mathcal{V}$. Note that closeness of DSs is understood in the C^1 -sense, but the closeness of h to $\mathbb{1}_M$ is understood in the C^0 -sense, and that h effects equivalence, not conjugacy.) I shall not dwell overmuch on structurally stable systems, since I wrote a separate survey of them (*Trudy Mat. Inst. Akad. Nauk*, vol. 169, Moscow; Nauka, 1985, pp. 59–93).

The notion of genericity does not relate exclusively to the TDS. It plays a significant rôle in many branches of mathematics. Instead of genericity, one often speaks of the “generic or general position case”. For the generic case, the picture may prove to be simpler than for exceptional cases; at the same time, it is precisely the generic case which merits primary attention.

When properties of points in $\mathbb{R}^n$ are being discussed, one can base the notion of genericity on Lebesgue measure instead of sets of second category. The metric and category points of view are similar in many respects, but do not coincide; in fact a first-category set may have total measure. In infinite-dimensional function-space there is no particular natural measure, and so the metric variant drops out. But, the metric point of view can be adopted for a family of DSs depending on a finite number of parameters $(\lambda_1, \dots, \lambda_n) \in \mathbb{R}^n$. There are interesting problems in which the distinction between this point of view and the category point of view manifests itself ([12], § 11, K). Moreover, smooth (of some class or other) families of smooth DSs themselves form a function space, so in this space the category approach may be applied: then the metric approach can be applied to a generic (in the sense of category) family. So far the metric point of view has played a rôle in questions connected with small denominators (volume 1, article I; volume 3).

Asking whether a given property of a smooth DS is generic is a clear-cut question, requiring an answer “yes” or “no”. Some questions of this kind have been solved, others remain open. According to the Kupka-Smale theorem, for a generic, smooth DS all periodic trajectories (including the equilibrium points in the case of a flow) are hyperbolic, and their stable and unstable manifolds intersect transversally. DSs (flows, cascades) with these properties are called *Kupka-Smale DSs* (*flows, cascades*); a diffeomorphism which generates a Kupka-Smale cascade is called a *Kupka-Smale diffeomorphism*. In connection with genericity, see § 3.1 and the references cited in the preface.

(Digressing for a moment, I formulate part of the Kupka-Smale theorem for irreversible cascades. In the space of smooth mappings of M to itself

considered with the C^1 -topology, the generic mappings (*i.e.*, the mappings which form a second-category set) are those whose periodic points do not have zero eigenvalues and are hyperbolic (in the same sense as before).)

A broader problem is to find a set of conditions which are satisfied for generic DSs and which, at the same time, significantly delimit their possible properties, thereby making the situation more or less transparent. This problem is not as precise as the preceding. However, it is certain that this problem has been solved in the case of phase spaces of small dimension (§ 3.1), and is unsolved in the general case.

Questions as to whether particular properties are generic as well as the more general problem, can be posed for narrower classes of DSs defined by some special property, but I shall not dwell on this.

As regards structurally stable systems, the basic problem is to find necessary and sufficient conditions for structural stability in terms of the qualitative properties of the behaviour of the trajectories in phase space. This has been solved only for cascades (R. Mañé, *Publ. Math. I.H.E.S.*).

§ 2. The Kronecker-Poincaré Index and Related Questions

2.1. The Kronecker-Poincaré Index. The Kronecker-Poincaré index is defined for isolated periodic trajectories, including equilibrium points of flows. (We could then define the index of an invariant set which consists of several periodic trajectories as the sum of their indices, and use approximation techniques to introduce the Kronecker-Poincaré index for certain invariant sets consisting of periodic trajectories. We will not need this and, so, will not formulate conditions under which it can be done.) The definition is connected with the concept of the rotation index of a nowhere vanishing vector field $\mathbf{v}$ on a sphere $S^{m-1} \subset \mathbb{R}^m$, *i.e.*, the definition involves the degree of the mapping $x \rightarrow \frac{\mathbf{v}(x)}{|\mathbf{v}(x)|}$ to the unit sphere. From now on, all mappings and vector fields are understood to be continuous.

The set of fixed points of a mapping f is denoted by $\text{Fix } f$. The number of elements of a set A is denoted by $\#A$.

For an equilibrium point x , the *Kronecker-Poincaré index* is equal to the rotation index of the phase-velocity field on a small sphere containing x (the field and the sphere are carried into $\mathbb{R}^n$ by means of local coordinates). In topology, one speaks in this case of the index of the zero of a vector field. The index $\text{ind}(a, f)$ of an isolated fixed point a of a continuous (not necessarily smooth) mapping f is defined to be equal, in terms of local coordinates, to the index of the corresponding zero of the displacement field $f(x) - x$. (Topologists often take the index of the vector field $x - f(x)$; the factor $(-1)^n$ in the Lefschetz formula then drops out; see below.) The index of a periodic point a (with period l) of a mapping f is equal to $\text{ind}(a, f^l)$. It turns out that all points $f^i a$ have the same index, and so the index can be assigned

to the corresponding trajectory. (This is obvious if f is a local diffeomorphism at the points of this trajectory. In the general case one can use approximation considerations, combining the Kupka-Smale theorem for irreversible cascades with proposition a) in § 2.2.) For a closed, l -circuit trajectory L (see Chap. 1, § 2.4) of a flow we construct a local transversal at a point $x \in L$ and we define $\text{ind } L$ to be $\text{ind } (x, F^l)$, where F is the first-return mapping. In all the cases, it can be shown that the index does not depend on the choices used to define it.

A small (in the C^0 -sense) perturbation of a DS which has a trajectory of one of the types considered above (equilibrium point, *etc.*) with nonzero index results in a perturbed system which also has trajectories of the same type (equilibrium point, *etc.*) located close to the original ones and, if we are not speaking about equilibrium points, with nearly the same period (in the case of cascades, the *same* period). (See also § 2.2, e.)

In the nondegenerate case (Chap. 1, § 2.4), the index is determined by the linear approximation: it is equal to $(-1)^i$, where i is the number of real eigenvalues (or multipliers) which are less than 0 in the case of an equilibrium point, or are less than 1 in the remaining cases. For a nondegenerate linear mapping A of a vector space V on to itself, we shall let $\varepsilon(A)$ be 1 or -1 according as A preserves or reverses the orientation of V (this does not depend on the choice of the orientation of V). Then, for a hyperbolic, periodic (with period l) point x of a cascade $\{f^k\}$, we have $\text{ind } (x, f^l) = (-1)^{u+n} \Delta$, where $u = u(x)$ (the Morse index, Chap. 1, § 2.5) and $\Delta = \varepsilon(Tf^l|E^u(x))$. We call Δ the *orientation type* of the point x and its trajectory. For a closed trajectory L of a flow $\{g^t\}$ with period τ , we have $\text{ind } L = (-1)^{u+n} \Delta$, where $u = u(L)$ and $\Delta = \varepsilon(Tg^\tau|E^{uu}(x))$. When $\Delta = 1$ we say that L is *not twisted* and, when $\Delta = -1$, that L is *twisted*. These names, clearly, do not apply to L as a curve in M , but to the vector bundle over L with the fibre $E^{uu}(x)$; they reflect whether the latter is orientable (a “cylinder”) or not (a “Möbius strip”).

From the topological point of view, the properties of a periodic point x of a reversible cascade $\{f^k\}$ are characterized by the quadruple (l, u, Δ, δ) , where $\delta = \varepsilon(Tf^l(x))$, and the local properties of the closed trajectory L of the flow $\{g^t\}$ are characterized by the triple (u, Δ, δ) , where $\delta = \varepsilon(Tg^\tau(x))$, i.e., δ is equal to -1 or 1 depending on whether a circuit around L does or does not change orientation. In the case of a cascade on an orientable manifold M , δ depends on whether l is even or odd, and whether the diffeomorphism f preserves the orientation of M . The latter depends only on the homotopy type of f , and so, for a given homotopy type, the quadruple (l, u, Δ, δ) reduces to the triple (l, u, Δ) . For a closed trajectory L of a flow, the invariant δ depends only on the homotopy type of L as a closed curve and on whether or not M is orientable. When M is orientable, (u, Δ, δ) reduces to (u, Δ) .

2.2. Summary of Topological Results About Fixed Points. There are a number of topological results connected with the Kronecker-Poincaré index.

a) Let $V \subset \mathbb{R}^n$ be a compact domain with smooth boundary ∂V . Suppose that $\mathbf{v}$ is a (continuous) vector field with isolated zeros on V which does not vanish anywhere on ∂V , so that the rotation index of $\mathbf{v}$ is defined on ∂V . Then the rotation index is equal to the sum of the indexes of the zeros of $\mathbf{v}$ in V .

This fact is often used in the QTDE. Thus, if we wish to prove the existence of a periodic solution of a periodic nonautonomous system (Chap. 1, § 2.3), we consider the rotation index of the displacement field of the corresponding first-return mapping on sufficiently large spheres.

A corollary of a) is the Bohl-Brouwer theorem: a continuous mapping $f: D \rightarrow D$, where D is homeomorphic to $\mathbb{D}^n$, has a fixed point. In this connection I mention a similar theorem (which is proved differently).

b) Browder's theorem. Suppose that $D \subset \mathbb{R}^n$ is a closed domain homeomorphic to $\mathbb{D}^n$, and suppose that D is contained in a larger open domain $W \subset \mathbb{R}^n$. Let $f: W \rightarrow \mathbb{R}^n$ be a continuous map such that all the images $f^i(D) \subset W$ (in particular, they are all well-defined). Suppose that, after some point, all images are contained in the interior of D . Then f has a fixed point in D .

This theorem can be used to establish the existence of periodic solutions of a periodic nonautonomous system when the system is dissipative at infinity, i.e., if, in time, all solutions enter a fixed bounded domain and remain there forever.

c) The Poincaré-Hopf theorem. If a vector field on a closed manifold M has only isolated zeros, then the sum $\chi(M)$ of the indices of the zeroes is equal to the Euler characteristic of M . If $\chi(M)=0$, then there is a vector field on M without zeroes.

d) The Lefschetz formula. If M is closed n -dimensional manifold, the Lefschetz number of a continuous map $f: M \rightarrow M$ is defined to be

$$L(f) = \sum_{i=0}^n (-1)^i \operatorname{Tr} f_{i*},$$

where $\operatorname{Tr} f_{i*}$ is the trace of the linear map

$$f_{i*}: H_i(M; \mathbb{R}) \rightarrow H_i(M; \mathbb{R}) \quad (1)$$

on homology induced by f . If the fixed points of f are isolated, then the sum of the indices of the fixed points is equal to $(-1)^n L(f)$. If M is simply-connected and $L(f)=0$, then f is homotopic to a map with no fixed points.

e) In addition to explicit expressions for the sum of the indices, a), c), and d) implicitly contain an equally important proposition about the conservation of the sum of the indices under continuous changes of the vector field or mapping, viz.:

In a), c) and d), suppose the vector field $\mathbf{v} = \mathbf{v}_\theta$ (respectively, the mapping $f = f_\theta$) depends continuously on a parameter θ , $0 \leq \theta \leq 1$ (i.e., $\mathbf{v}_\theta(x)$ and $f_\theta(x)$ are continuous in (θ, x)). The zeros (resp., fixed points) are assumed to be isolated only when $\theta=0$ and $\theta=1$. In case a), we also suppose that $\mathbf{v}_\theta$ does

not vanish at any point of ∂V for any value of θ . Then the sum of the indices of the zeros (resp., fixed points) is the same for both $\theta=0$ and $\theta=1$. In particular, if the sum is not zero when $\theta=0$, then the field (resp., the map) must have a zero (resp., a fixed point) when $\theta=1$.

f) The requirement of simple-connectivity at the end of d) is essential. If the manifold is not simply-connected, we make the following refinement.

We say that two points $x, y \in \text{Fix } f$ are in the same *Nielsen class* if they can be joined by a path γ for which γ and $f\gamma$ are homotopic as paths with fixed endpoints. It can be proved that this actually defines a decomposition of $\text{Fix } f$ into classes which are closed subsets of M at a positive distance from one another. In particular, the number of classes is finite. If $\# \text{Fix } f < \infty$, then we define the *index of a Nielsen class* to be the sum of the indices of the fixed points in the class. Define an *essential class* to be a class with nonzero index. We define the *Nielsen number* $N(f)$ to be equal to the number of essential classes. If $f_t: M \rightarrow M$, $0 \leq t \leq 1$, is a homotopy, and if $\# \text{Fix } f_0 < \infty$ and $\# \text{Fix } f_1 < \infty$, then $N(f_0) = N(f_1)$. (This also enables us to define $N(f)$ when $\# \text{Fix } f = \infty$.) In fact, a homotopy establishes a bijective relation between the Nielsen classes for f_0 and f_1 : a class N_0 of f_0 corresponds to a class N_1 of f_1 if some (and, therefore, any) $x_0 \in N_0$ and $x_1 \in N_1$ can be joined by a path x_t homotopic (with fixed ends) to the path $f_t x_t$. (In general, this correspondence depends on the homotopy between f_0 and f_1 , although it does not, in a sense the reader will be able to make precise, change if the homotopy undergoes a continuous deformation.) The indices of N_0 and N_1 turn out to be the same.

Any map $f: M^n \rightarrow M^n$ has at least $N(f)$ fixed points. If $n \neq 2$, or if $\chi(M) \geq 0$, then, for any f , there is a g which is homotopic to f and such that $\# \text{Fix } g = N(g) = N(f)$. When $n=2$ and $\chi(M) < 0$, this is true for homeomorphisms. In the former case, g can even be taken to be smooth; in the latter, the corresponding question has not yet been finally answered.

Unfortunately, it is much more difficult to calculate the Nielsen numbers than pure homotopy invariants like $L(f)$. In [34], an "abelianized" variant of the Nielsen theory is advanced, which may turn out to be more convenient for calculations (but passing to it involves losing some information). If M is a torus (of any dimension), then $N(f) = |L(f)|$ [25]. The same is true for nilmanifolds.

In topology, there are variants of d), e) and f) which apply not just to manifolds, but to finite polyhedra, and sometimes to more general spaces.

g) Suppose that $\sup_k |L(f^k)| = \infty$. Does it follow that the cascade $\{f^k\}$ has an infinite number of periodic trajectories? We may assume that the periodic trajectories are isolated (otherwise the answer is obvious) and try to use the Lefschetz formula. The question then arises of whether there can exist a periodic point x with minimal period p for which the sequence

$$i_k = \text{ind}(x, f^{kp})$$

is unbounded. It turns out that this is possible in the continuous case, but $\{i_k\}$ is bounded, and even periodic [27], in the smooth case. This fact and related circumstances may prove useful for other purposes. Meanwhile, I remark that it has been suggested that this fact could profitably be used in conjunction with the φ -index of a single-circuit trajectory L which is defined as the mean value (over i) of $\text{ind } L^i$ [27].

2.3. The Fuller Index. The Fuller index is defined for an isolated l -circuit closed trajectory L of a flow. It is equal to $\frac{1}{l} \text{ind } L$, where $\text{ind } L$ is the Kronecker-Poincaré index [35]. (As in § 2.1, I shall not discuss the possibility of defining the index for infinite sets of closed trajectories.) This definition stems from the attempt to obtain analogues of the results in § 2.2, for closed trajectories of flows. The following theorem (due to F.B. Fuller) is an analogue of the key assertion e).

Consider a flow (*i.e.*, a phase-velocity field) on a closed manifold M which depends continuously on a parameter θ , $0 \leq \theta \leq 1$. Suppose that for $\theta=0$ and $\theta=1$, all closed trajectories whose (not necessarily minimal) periods lie in the interval $[\alpha, \beta]$, where $\alpha > 0$ and $\beta < \infty$, are isolated. Suppose, further, there is no θ for which the flow has closed trajectories with the periods α and β . Then the sum of the Fuller indices of the periodic trajectories with periods $\tau \in [\alpha, \beta]$ is the same for $\theta=0$ and for $\theta=1$.

Moreover, if we want the index of a closed single-circuit trajectory to be the same as its Kronecker-Poincaré index and demand that the previous proposition hold, then the index has to be defined precisely as above.

Suppose that all trajectories of a smooth flow $\{f^t\}$ on a closed manifold M are periodic with the same minimal period τ . For each trajectory, we identify all points on the trajectory to obtain a new manifold N and a projection $p: M \rightarrow N$. (In the general case, a non-Hausdorff space might be obtained.) The following theorem (due to H. Seifert and G. Reeb) is easily proved using Fuller's theorem. If the Euler characteristic of N is nonzero, $\chi(N) \neq 0$, then any flow on M which is sufficiently close (in the C^0 -sense) to $\{f^t\}$ (*i.e.*, any flow whose phase-velocity field is C^0 -close) has a closed trajectory.

We digress slightly to discuss a possible approach to the proof of this theorem using perturbation theory. If the perturbation of the original DS $\{f^t\}$ is small, the motion in the perturbed DS basically takes place along the closed trajectories of the flow $\{f^t\}$ (*i.e.*, along the curves $p^{-1}x$, $x \in N$), on which a small transverse drift is superimposed. The effect of the latter is to cause a small change of x : after one circuit along $p^{-1}x$, there is a certain drift, which can be calculated approximately by the averaging method. A geometrically invariant treatment of the latter gives us a vector field v on N which describes the mean drift after one revolution. The closed trajectories of the perturbed flow, which have period approximately equal to τ , will be

located where the mean deflection is zero, *i.e.*, they are situated near the curves $p^{-1}a$ for which $\mathbf{v}(a)=0$. (In this connection, the rôle of the condition $\chi(N)\neq 0$ is obvious.)

However, this argument needs further justification. The simplest variant is as follows: if the perturbation is small in the C^1 -sense, and if a is a nondegenerate equilibrium point of the averaged flow on N , then it is easily proved that there must indeed be a closed trajectory of the perturbed flow near $p^{-1}a$, which closes after one revolution around $p^{-1}a$. But, we do not, in fact, exclude cases in which the equilibrium points are degenerate or even non-isolated. Such cases could also, of course, be investigated by perturbation theory, but it is not clear *a priori* what the results of such an investigation would be and whether it would be possible to handle all the cases which arise in a uniform way. Nor is it clear what conditions must be imposed on the smallness of the perturbations: C^0 , C^1 , or some C^n ? In summary, perturbation theory provides effective computational procedures in a specific situation, but is less effective than topological considerations in studying the qualitative behaviour in the general case. The Seifert-Reeb theorem can be obtained by more analytic considerations, but these differ from the usual perturbation theory [21].

The condition that the flow be close to $\{f^t\}$ cannot be dropped: on any M^n with $n>2$ and $\chi(M)=0$, there are flows without closed trajectories or equilibrium points [78]. The results for closed trajectories are inevitably weaker than those for periodic points (see also [14]). The reason for this, in the final analysis, is connected with the fact that the period of a closed trajectory may change under a perturbation, while it cannot for a periodic point (as long as this point is, in an obvious sense, preserved under the perturbation). As a result of this fact, the following phenomena, which have no analogues for cascades, can occur for flows which are arbitrarily smooth and depend arbitrarily smoothly on a parameter θ .

1) For $\theta < \theta_0$ the flow has a closed trajectory L_θ which depends continuously on θ , is isolated, has a nonzero index, and is even hyperbolic (so that once the trajectory exists for some θ , it must exist for all nearby values of θ), but the length of the trajectory increases without bound as $\theta \rightarrow \theta_0$. As a result, when $\theta \rightarrow \theta_0$ this trajectory, together with its index, "vanishes without trace into the blue sky"; this phenomenon has therefore been called the "*blue-sky catastrophe*". (This name was probably first used as a joke, but later it became adopted as standard terminology. "Catastrophe" is merely a synonym for "bifurcation", *i.e.*, a qualitative change which occurs when the parameter passes through a certain "critical" value, but in the present case the emotional nuance evoked by the word "catastrophe" is to some extent justified.)

2) When $0 \leq \theta \leq \theta_2$, the flow has a closed trajectory L_θ and for some $\theta = \theta_1 \in (0, \theta_2)$ a new closed trajectory L'_θ branches off from the "twice traversed" L_θ (*i.e.*, from L_θ^2), and this new L'_θ merges when $\theta = \theta_2$ with the "once traversed" L_θ (so that as $\theta \rightarrow \theta_2$ their minimal periods approach one another),

after which both trajectories disappear. In their domains of definition, both trajectories depend continuously on θ ; L_θ , as a trajectory with minimal period, is isolated and has a non-zero index when $\theta < \theta_2$; the same is true for L'_θ when $\theta_1 < \theta < \theta_2$; there are no other closed trajectories in a neighbourhood of these trajectories [3].

(Of course, a closed trajectory can disappear by merging with an equilibrium point or tending to a “separatrix contour” consisting of equilibrium points and the trajectories joining them. This, too, is not quite like the bifurcation of periodic trajectories of cascades, but it is also not very complicated.)

2.4. Zeta-Functions. There is yet another characteristic of DSs connected with periodic trajectories.

One can associate various functions with a sequence of numbers $a = \{a_i\}$. An example is the generating function $\sum a_i t^i$, but other functions are also used. We shall need the zeta-function

$$Z_a(t) = \exp \left(\sum_{i=1}^{\infty} \frac{1}{i} a_i t^i \right)$$

(as can be seen, the numbers a_i are assumed to be given for $i \geq 1$; they must not increase faster than exponentially in i).

(The name “zeta-function” stems from an analogy with the zeta-function in algebraic geometry, which, in turn, on being expressed in terms of another variable, is similar to the Riemann zeta-function; see [M.E. “zeta-function”].)

The zeta-function of the sequence $a_i = \# \text{Fix } f^i$ is written $Z_f(t)$ and called the *zeta-function of the mapping f* or of the *cascade* $\{f^k\}$. The definition does not require smoothness. We can also consider $Z_{f|A}$, where A is an invariant set of a cascade $\{f^k\}$. Although the definition is quite general, the zeta-function is, in fact, considered in rather more concrete situations. It is not difficult to calculate it for a Bernoulli topological shift or for a Markov chain; in other cases, one can sometimes show that it is well-defined and determine whether or not it is rational.

If we take a_i to be the sum of the Kronecker-Poincaré indices of the periodic points with period i , then a *homological zeta-function* is obtained, which is sometimes called the *false zeta-function*. It is easily calculated by means of the Lefschetz formula in terms of the mappings (1) [74].

It is doubtful whether it is possible to define a “good” analogue of the zeta-function for flows of general type, although there is an analogue in certain special cases (which can be formally carried over to the general case, but it is not clear what it gives [74]).

§ 3. Morse-Smale Systems

3.1. General Information About Morse-Smale Systems. A smooth DS (flow or cascade) is called a *Morse-Smale system* (flow or cascade), or “M-S system”,

if all its motions tend (in both time directions) to periodic trajectories (including equilibrium points in the case of flows), if the number of periodic trajectories is finite, if they are all hyperbolic, and if their stable and unstable manifolds intersect transversally. A *M-S diffeomorphism* is a diffeomorphism whose iterates generate a M-S cascade.

Morse-Smale dynamical systems may quite justifiably be regarded as systems with “simple” phase portraits and, for this reason, merit attention.

When the dimension n of the phase manifold is small, $n=1, 2$ for flows and $n=1$ for cascades, M-S systems play a particularly important role. The results described in article I, Chap. 2, § 1 and § 3, can be formulated as follows: the structurally stable flows on S^2 and the structurally stable cascades on S^1 are precisely the M-S systems; moreover, these systems are generic (in an even stronger sense than in § 1.2: they form an everywhere dense, open set). This is also true for flows on any closed surface, *i.e.*, on closed two-dimensional manifolds (see the references in Chap. 4). In the low-dimensional case, the three concepts, genericity, structural stability and simplicity merge. But for larger values of n there are strict inclusions instead of coincidence: M-S systems are structurally stable (and the property that a system be a M-S system is also structurally stable), but there are structurally stable DSs which are not M-S systems (the behaviour of their trajectories is more “complicated”); also structurally stable systems are not everywhere dense in the space of smooth DSs. (This assertion about the trajectories is not a theorem, since the word “complicated” is not a precise term. From one point of view [1], DSs with positive topological entropy should be regarded as “complicated” (see volume 2 or *ME* for the concept of “topological entropy”). The topological entropy is positive in all known examples of structurally stable DSs which are not M-S systems, but it has not been proved that this is necessarily the case for flows (for cascades it is).)

For general DSs, the stable and unstable manifolds, $W^s(x)$, $W^{ss}(x)$, *etc.*, are injectively immersed submanifolds, but the topology of such a manifold W may not coincide with its topology as a subset of M (the manifold may even be everywhere dense). In the case of M-S systems, these topologies coincide: in other words, the identity inclusion $i: W \rightarrow M$ is an embedding in the topological sense. This can be restated as follows. Define the limit set L_f of a continuous mapping $f: W \rightarrow M$ to be the set

$$L_f = \bigcap \{f(\overline{W \setminus K}) : K \subset W \text{ compact}\},$$

i.e., the set of points of the form $\lim f(x_n)$ as $x_n \rightarrow \infty$ in W (*i.e.*, for any compact $K \subset W$, $x_n \notin K$ for sufficiently large n). (The definition of L_f does not require that W be one of $W^s(x)$, *etc.*, or, even, that W be a manifold; it is natural when W is a noncompact, locally compact topological space with a countable basis.) Then, for i , we shall have $L_i \cap W = \emptyset$, $L_i = \overline{W} \setminus W$. This L_i is sometimes denoted below by ∂W (which does not conflict with the use of the symbol ∂ for the boundary in the compact case).

For a gradient flow ((17) in Chap. 1), all motions approach the set of equilibrium points arbitrarily closely as $t \rightarrow \pm \infty$. If all the critical points of the function f are nondegenerate, then the flow satisfies the conditions defining M-S systems except, perhaps, the transversality condition on the intersections. But this condition is satisfied in the generic case: the M-S flows are an open, everywhere dense set in the space of all gradient flows. (In particular, on any M there is an M-S flow without closed trajectories, and on any M there exists a structurally stable flow.) The phase portraits of M-S gradient flows are very simple.

Let x, y be periodic points of a M-S system (flow or cascade) with Morse indices $u(x)$ and $u(y)$, which lie on different trajectories and suppose that $W^u(x) \cap W^s(y) \neq \emptyset$. It follows from the transversality condition on the intersections that $u(x) \geq u(y)$ and $\partial W^u(x) \supset W^u(y)$; if the DS is a flow and y is an equilibrium point, then $u(x) > u(y)$. Points in $W^u(x) \cap W^s(y)$ and their trajectories are said to be *heterclinic* if $u(x) = u(y)$. If there are such points, on approaching y , the manifold $W^u(x)$ begins to oscillate strongly being pressed onto $W^u(y)$. When the dimensions of $W^u(x)$ and $W^u(y)$ coincide, the relation $\partial W^u(x) \supset W^u(y)$ now involves a set-theoretic phenomenon beyond the scope of naive geometry. Because of this the phase portrait does not appear all that simple. (It is true that nothing peculiar takes place in the case of a flow in which $u(x) = u(y) = 1$. In this case x is an equilibrium point, and $W^u(x) \setminus x$ consists of two trajectories ("whiskers") and y lies on an asymptotically stable, closed trajectory into which the whisker spirals.) Even worse, if there is a periodic point z such that $W^u(x) \cap W^s(z) \neq \emptyset$ and $W^u(z) \cap W^s(y) \neq \emptyset$, then $W^u(x) \cap W^s(y)$ consists of an infinite number of trajectories (and if there is no such z , it consists of a finite number of trajectories). If there are no heteroclinic points, the M-S system (flow, cascade, diffeomorphism) is said to be *gradient-like*. (The terminology here has not been completely established, and this term can also be used in a different sense.)

Digressing for a moment, I mention that even more significant complications of the phase portrait are encountered when there are so-called "homoclinic" points. A transversal *homoclinic point* x in a smooth DS is a point of transversal intersection of the invariant manifolds $W^u(L)$ and $W^s(L)$ of a hyperbolic periodic trajectory L . The entire trajectory of x consists of transversal homoclinic points and is called a (transversal) *homoclinic trajectory*.⁴

⁴ Trajectories which lie in $(W^u(L) \cap W^s(L)) \setminus L$ are also called *doubly asymptotic* (to L), since they tend to L when $t \rightarrow \pm \infty$. Here, no conditions are imposed on the nature of the intersection, so a closed separatrix of a flow on the plane is doubly asymptotic to the corresponding saddle; its stable and unstable "whiskers" (the "halves" of W^s and W^u) coincide instead of intersecting transversally. One speaks of a, possibly non-transverse, homoclinic point (or trajectory), only when no similar fusion of $W^u(L)$ and $W^s(L)$ takes place near the point. To date, nontransversal homoclinic points have actually been considered for cascades in the two-dimensional case when $u(L) = 1$ (i.e., when W^u and W^s are curves and the appropriate precise formulations are reasonably obvious, including the characterization of the degree of "degeneracy" of a homoclinic point as the order of tangency of the curves) and for flows in the three-dimensional case, where the use of a suitable transversal reduces us to the previous case.

In the case of a flow, it follows from the latter fact and the condition that W^u and W^s intersect transversally that

$$\dim W^u(L) + \dim W^s(L) \geq n + 1,$$

and, hence, that L must be a periodic trajectory. A neighbourhood of the closure of a homoclinic trajectory necessarily contains a nontrivial ("complicated") hyperbolic set (see Volume 2) and, in particular, an infinite number of periodic trajectories. This last circumstance implies that homoclinic points do not occur in M-S systems. For a periodic trajectory L of an M-S system

$$W^u(L) \cap W^s(L) = L.$$

There is another respect in which a M-S flow $\{g^t\}$ is similar to a gradient flow, namely, there is a smooth (C^∞) function $f: M \rightarrow \mathbb{R}$ such that:

1. The critical points of f are periodic points of the flow. Equilibrium points correspond to nondegenerate critical points with the same indices, but critical points on a closed trajectory L are inevitably degenerate (if $x \in L$, then $T_x L$ is an eigenspace corresponding to the eigenvalue zero of the linear map given by the quadratic form $d^2 f(x)$), but this degeneracy is minimal (the rank of $d^2 f(x)$ equals $n-1$).

2. Outside the periodic trajectories, we have

$$\mathbf{v}f = (df, \mathbf{v}) = \frac{d}{dt} \bigg|_{t=0} f \circ g^t < 0,$$

and its order near a periodic trajectory is that of the square of the distance to the trajectory. For an M-S cascade $\{g^k\}$ there is also a C^∞ function f which has critical points (which are nondegenerate and of the same index) at the periodic points of g , which decreases along all other trajectories, and which is such that the order of the difference $f(gx) - f(x)$ near a periodic point a is of the order of the square of distance to a .

Because of the obvious similarity to the Lyapunov function in stability theory (article I, Chap. 1, § 4.3), f is called the *Lyapunov function* of the M-S system (and also the *energy function*) [56].

We introduce a partial order on the set of periodic trajectories of a M-S system by setting $L \geq L'$ if $W^u(L) \cap W^s(L') \neq \emptyset$, i.e., either $L = L'$, or there are trajectories going from L to L' . Among other things, it can be proved that

$$\partial W^u(L) = \bigcup \{W^u(L') : L' < L\}.$$

It turns out that $\geq$ is indeed an ordering. A finite partially ordered set can be represented in an obvious way as a graph with oriented edges (the arrows pointing in the "decreasing" direction). We "label" the graph by associating to each vertex the local characteristic of the corresponding periodic trajectory (§ 2.1; in the case of a flow, of course, a vertex corresponds to an equilibrium point or a closed trajectory). The labelled, oriented graph obtained in this way is called the *phase diagram* of the M-S system. It is

clear that it contains considerable information about the DS. But this information is not complete: two M-S systems on the same manifold M with the same phase diagram need not be equivalent. This gives rise to two questions:

- 1) What phase diagrams occur for M-S systems on a given manifold M ?
- 2) What data must be added to a phase diagram to uniquely determine the M-S system up to equivalence?

These questions can be naturally modified in various ways. For example, it is possible to ignore the edges in the phase diagram, *i.e.*, to retain only the set of local topological data about the periodic trajectories (the periodic data). When question 1 is modified in this way, quite a lot is known (§ 3.3). Question 2 has been investigated only for some particular classes of M-S systems (§ 3.4 and Chap. 4).

3.2. The Structure of Phase Manifolds of Morse-Smale Systems. Certain decompositions of the phase manifold M^n are associated with a M-S system. These enable one to establish a connection between the topology of the phase manifold and the properties of the phase portrait.

1) In the case of M-S flows on a surface (a closed, two-dimensional manifold) M^2 , the phase portraits have a simple direct description: they amount to a decomposition of M into cells filled with trajectories with the same type of behaviour.⁵ These cells are connected domains obtained from M by removing the singular trajectories, the periodic trajectories (including equilibrium points) and the separatrices of saddles. The cells, the lines bounding them, and the equilibrium points on the boundaries can be regarded as connected components of the intersection of all possible (not just one-dimensional) stable and unstable manifolds of periodic trajectories. Such a decomposition admits a “finite” description which sharpens the phase diagram and turns out to determine the system up to equivalence.

But all this is connected with the special simplicity of the two-dimensional case for flows. In this case it is possible to obtain significant (albeit not so exhaustive) information about the possible types of phase portraits of flows which are not even M-S, but have a finite number of equilibrium points and separatrices. In the more general case, there is a longer list of singular trajectories and the connected components of the set obtained by removing the singular trajectories from M can be more complicated (in particular, they are no longer necessarily domains). See Chap. 4.

2) In the case of higher dimensions (and, for cascades, already in dimension two) the decomposition of M into the connected components of the intersections $W^u(L) \cap W^s(L)$ becomes too complicated. For example, these intersections, even the n -dimensional ones, can have an infinite number of connected components. We can consider the coarser decomposition, into either stable or unstable manifolds.

⁵ The word “cell” is not used here in the topological sense.

In fact, one uses (especially in precise formulations) not so much the decomposition $\{W^u(L)\}$ (for definiteness, we restrict our attention to the unstable manifolds), but a certain filtration connected with it. Recall that a *filtration* of a topological space X is a finite, increasing system of closed sets

$$\phi = X_0 \subset X_1 \subset \dots \subset X_r = X.$$

In our case we start with $A_0 = \emptyset$. Then we consider those L such that $\partial W^u(L) = \emptyset$, i.e. such that $u(L) = 1$ if L is a closed trajectory of a flow, and $u(L) = 0$ if L is a periodic trajectory of a cascade or an equilibrium point of a flow. These L are the minimal elements in the set of periodic trajectories with respect to the ordering introduced earlier. Speaking less formally, they are *sinks*, i.e., asymptotically stable periodic trajectories. Taking $W^u(L)$ for all such L , or for some of them, we obtain A_1 . Once we have constructed A_i , we consider the L for which $\partial W^u(L) \subset A_i$, i.e., all trajectories "emanating from" L approach A_i arbitrarily closely. In the phase diagram, these L correspond to vertices which are connected by edges to vertices corresponding to the periodic trajectories in A_i . We obtain A_{i+1} by adding some or all of the $W^u(L)$ with the indicated L to A_i . Generally speaking, this process allows a certain freedom at each step. Sooner or later, it terminates and we obtain $A_j = M$. Thus, the A_i form a filtration of M (it is obvious that the A_i are closed). Despite the complications which arise when there are heteroclinic points (where the A_i may not be locally connected), the filtration $\{A_i\}$ was successfully used in the first paper in the field to prove the Morse-Smale inequalities cited below (because of the complications mentioned, Čech cohomology was used to deal with the "bad" sets). Since this filtration is directly connected with the behaviour of the trajectories, it will be necessary, as can be imagined, to return to it from time to time.

3) It is often convenient to represent phase space in another form which coincides with a handlebody decomposition, well-known in topology, or something similar (when there are closed trajectories). The relevant topological theory has been set out in detail in the literature (see [M.E., "handle theory"]), and so I shall mention only the simplest information about it (omitting in sections a) and b) below the necessary refinements connected with smoothness). I shall then indicate how to modify it to investigate M-S systems.

a) An n -dimensional handle of index i is a pair (H, h) (or simply H) where H is the homeomorphic image of $\mathbb{D}^i \times \mathbb{D}^{n-i}$ under a *characteristic homeomorphism* $h: \mathbb{D}^i \times \mathbb{D}^{n-i} \rightarrow H$ (which is regarded as fixed). Here, $\mathbb{D}^m$ denotes the standard m -dimensional ball with center at 0. Furthermore,

$$h(0, 0), \quad h(\mathbb{D}^i \times 0), \quad h(0 \times \mathbb{D}^{n-i}), \quad h(\partial \mathbb{D}^i \times \mathbb{D}^{n-i})$$

are called respectively the *center*, the *core*, the *co-core*, and the *attaching tube* of the handle.

b) A manifold N' (with boundary) is said to have been obtained from a manifold N (with boundary) by (simultaneously) attaching handles if $\overline{N' \setminus N}$ is a disjoint union of handles whose attaching tubes lie in ∂N .

c) A handle of index 0 has no attaching tube, and so it can be "attached" to the empty set. More generally, attaching such a handle means that one takes a disjoint union of N with the handle.

For handles of index 1, the attaching tube consists of two connected components. Attaching them to different components of N results in the two lying in one component of N' ; thus, the number of connected components (of N' compared with N) is reduced by one. We call such a handle of index one a *connecting handle*. On the other hand, if we attach a handle of index one so that the number of connected components does not change, then we obtain a new one-dimensional cycle which goes along the handle and closes in the component of N to which the handle is connected. Attaching handles of higher index does not change the connectedness and does not create new one-dimensional cycles.

d) A *handlebody decomposition of a manifold M* is a filtration of M

$$\phi = M_0 \subset M_1 \subset \dots \subset M_r = M \quad (2)$$

in which the M_j are smooth manifolds with boundaries and M_{j+1} is obtained from M_j by attaching handles.

e) Let $f: M \rightarrow \mathbb{R}$ be a C^2 function with only nondegenerate critical points. We can associate a handlebody decomposition (2) with f so that the vector field $(-\text{grad } f)$ points strictly into the interior of M_i on ∂M_i and each critical point of f on $M_i \setminus \overline{M_{i-1}}$ is at the center of a handle with the same index as the critical point. Conversely, given a handlebody decomposition, we can construct a function f whose associated handlebody decomposition is the given one.

f) Suppose that f has exactly m_i critical points of index i , and let b_i be the i -dimensional Betti number of the manifold M ; in particular, b_0 is the number of components of M . In order to obtain b_0 components of M from m_0 handles of index 0 by attaching handles, we need $m_0 - b_0$ connecting handles (the number of components cannot be greater than m_0). The remaining $m_1 - (m_0 - b_0)$ handles of index one give one-dimensional cycles which generate the one-dimensional homology group of M (but which become dependent, perhaps, when handles of index 2 are attached). Hence

$$m_0 \geq b_0, \quad m_1 - m_0 + b_0 \geq b_1.$$

These are the first two of the Morse inequalities

$$\sum_{j=0}^i (-1)^j m_{i-j} \geq \sum_{j=0}^i (-1)^j b_{i-j}, \quad (3)$$

the last of which (when $i=n$) is an equality. The derivation of the inequalities (3) depends on taking into account the effect of attaching handles not only

on the connectivity, *i.e.*, on the zero-dimensional homology, but also on the higher dimensional homology; this may seem different technically, but the idea is essentially the same. The relation (3) can be strengthened slightly, by replacing b_i by $b_i + c_i + c_{i-1}$, where c_i is the number of i -dimensional torsion coefficients ($c_{-1} = c_0 = c_n = 0$).

It can be supposed that the handle decomposition is connected with a gradient flow ((17) in Chap. 1) or with a M-S flow which approximates it. One can associate a similar handle decomposition to a M-S system which has no closed trajectories (or to a M-S cascade). In the case of a flow, the phase-velocity field on ∂M_i , as before, points strictly into M_i . In the case of a cascade $\{g^k\}$, this corresponds the fact that gM_i lies strictly within M_i . We express the last property by saying that: the DS (or diffeomorphism) preserves the filtration (2), or that (2) is a filtration for the given DS (or diffeomorphism). Moreover, we can suppose that if a periodic point a is a center of a handle H , then $W^u(a) \cap H$ is the core and $W^s(a) \cap H$ as the co-core. In the case of a cascade, it is necessary that the handles corresponding to the points of a single periodic trajectory be attached simultaneously if we want the cascade to preserve the filtration, as is required in § 3.3. (But to obtain the inequalities (3), it suffices to have a handlebody decomposition in which the number of handles of each index i is equal to the number of periodic points of index i .) In other respects, the order in which the handles are attached allows a certain degree of arbitrariness, similar to that which occurs in the construction of the filtration $\{A_{ij}\}$; it is necessary only that, when $L \geq L$, the handles corresponding to points of L should be attached later than the handles corresponding to points of L . The construction of such a handlebody decomposition and the construction of a Lyapunov function are essentially the same thing.

In the case of a M-S flow with closed trajectories, the only change is that a closed trajectory of L corresponds to a so-called *round handle* H of index $i = u(L)$. This is a fibre bundle over L (*i.e.*, from the topological point of view, a fiber bundle over a circle) homeomorphic to a Whitney sum⁶ $E_1 \oplus E_2$ of bundles E_1 and E_2 with standard fibres $\mathbb{D}^{i-1}$ and $\mathbb{D}^{n-i}$, which are unit disk subbundles of vector bundles over L (which can, if desired, be taken to be the bundles with the fibres $E^{uu}(x)$ and $E^{ss}(x)$). We can suppose that the characteristic homeomorphism $h: E_1 \oplus E_2 \rightarrow H$ carries the fibre of the bundle E_1 over the point $x \in L$ into $W^{uu}(x) \cap H$, and the fibre of E_2 over x into $W^{ss}(x) \cap H$. The attaching tube of the round handle is $h(\partial E_1 \oplus E_2)$.

In the context of M-S systems, inequalities (3) are called the *Morse-Smale inequalities*. If there are closed trajectories, these continue to hold with the following modifications: m_i is the sum of the number of equilibrium points of index i and the number of closed trajectories of index i or $i+1$; if there are twisted or orientation-reversing closed trajectories, the Betti numbers

⁶ The Whitney sum of two fibre bundles $p_i: E_i \rightarrow B$, $i=1, 2$, over the same base is the subset of the direct product $E_1 \times E_2$ formed by points (e_1, e_2) with $p_1 e_1 = p_2 e_2$, regarded as a fibre bundle over B in the obvious manner.

b_i must be taken over a field of characteristic 2. But if there are no closed trajectories, then inequalities (3) and their sharpened version (with $b_i + c_i + c_{i-1}$) hold exactly (where m_i is now the number of periodic points of index i).

Time reversal (Chap. 1, § 1.4) in an M-S system corresponds to changing to the dual handlebody decomposition: the filtration (2) is replaced by the filtration

$$\phi = \overline{M \setminus M_r} \subset \overline{M \setminus M_{r-1}} \subset \dots \subset \overline{M \setminus M_1} \subset \overline{M \setminus M_0} = M.$$

Here $\overline{M \setminus M_j}$ is obtained from $\overline{M \setminus M_{j+1}}$ by attaching precisely the handles which were attached to M_j to give M_{j+1} , except that a handle (or round handle) of index i is now regarded as a handle (or round handle) of index $n-i$ (resp., $n-i+1$), because its attaching tube is the part of its boundary which is attached to $\partial \overline{M \setminus M_{j+1}}$, i.e., $h(\mathbb{D}^i \times \partial \mathbb{D}^{n-1})$ (resp., $h(E_1 \oplus \partial E_2)$). Thus, in the case of an ordinary handle, $\mathbb{D}^i$ and $\mathbb{D}^{n-i}$ interchange roles; this can be described formally as changing from (H, h) to (H, h') , where $h'(x, y) = h(y, x)$. Similarly, in the case of a round handle, the roles of E_1 and E_2 are interchanged.

In contrast to the decomposition into $\{W^u(L)\}$, the construction of a handlebody decomposition entails an element of arbitrariness. Essentially this is the same arbitrariness which occurs in topology when we say that a torus is a sphere with a handle (where, exactly, is the handle?). Perhaps it is because of this that a certain psychological barrier has to be overcome when speaking of handles. But this is not difficult.

Only the simplest facts pertaining to handlebody theory have been given in this section. As is well known, this theory includes a much more complicated part, an example of which is the proof of the Poincaré conjecture in dimensions greater than or equal to five. Of an allied nature, is the use of handles (and other decompositions of M) to investigate certain questions in the following section. There, we shall merely state the various results.

4) Finally, given a handlebody decomposition of M , it is possible to construct a cell complex K homotopy equivalent to M . The starting point is the obvious remark that a handle $H = h(\mathbb{D}^i \times \mathbb{D}^{n-i})$ can be contracted on to its core by means of the homotopy

$$\varphi_t h(x, y) = h(x, (1-t)y), \quad 0 \leq t \leq 1.$$

a) If N' is obtained from N by simultaneously attaching handles $H_1, \dots, H_s$ with indices $i_1, \dots, i_s$, then, upon contracting each H_i to its core, we find that N' is homotopy equivalent to a topological space obtained by attaching cells of dimensions $i_1, \dots, i_s$ to N .

b) If N is then deformed (within itself) into a cell complex K , then the boundaries of the cells attached to N are "carried along" under the deformation (speaking more precisely, the deformation determines a homotopy of the corresponding attaching maps). This determines attaching maps of the cells to K .

c) By a further homotopy of the attaching maps, we can arrange that a cell of dimension i_j is attached to the (i_j-1) -dimensional skeleton of K . As a result, we find that N' is homotopy equivalent to the cell decomposition K' obtained by attaching the corresponding cells to K .

d) The operation of attaching a round handle of index i can be replaced by successively attaching two ordinary handles of index $i-1$ and i . Geometrically the first corresponds to the fibre of the fibre bundle $E_1 \oplus E_2$ over a fixed point of L , and the second to the remaining part of the round handle.

Applying these considerations to the decomposition of M into handles and round handles, we obtain a finite cell complex K , whose cells correspond to the handles and round handles in accord with [31]. It must be borne in mind that the relation of K to M and the DS needs some discussion. The point is that the homotopies of the attaching maps determine K only as an "abstract" cell complex (which lies, figuratively speaking, "somewhere to the side of M "); although one still obtains a homotopy equivalence $M \rightarrow K$. See [31] for a discussion (mostly in the case of M-S flows without closed trajectories, a case which is definitely "better" in some respects than the general case). Time reversal, *i.e.*, passing to the dual handle decomposition, corresponds to passing to the dual cell complex $\tilde{K}$. In topology, this observation is related to a particularly transparent proof of Poincaré duality and, thus, the passage to $\tilde{K}$ has been analyzed in sufficient detail in the literature.

3.3. Existence of Morse-Smale Systems whose Generating Diffeomorphism (or Vector Field) Has Prescribed Topological Properties and whose Periodic Trajectories Have Prescribed Properties

1) It was shown in § 3.1 that, on any manifold M , there is a M-S flow without closed trajectories. At the other extreme are M-S flows without equilibrium points. In order that such a flow exist on M it is necessary that the Euler characteristic of M vanish: $\chi(M)=0$. It can be proved by using round handles that this condition is also sufficient when $n \neq 3$. Moreover, any homotopy class of nowhere vanishing vector fields on M contains a field which defines an M-S flow without equilibrium points [13]. There exist *three-dimensional* M (with $\chi(M)=0$) on which there are no M-S flows without equilibrium points [58].

2) Let $\{g^t\}$ be an M-S flow on M . Then g^1 is not an M-S diffeomorphism, but there are arbitrarily close (in the C^1 -topology) M-S diffeomorphisms. Thus, there are M-S diffeomorphisms and cascades on any manifold M . But, the M-S diffeomorphisms obtained in this way are clearly isotopic to $\mathbb{1}_M$. It is natural to ask whether there exist M-S diffeomorphisms in a given isotopy class $\mathcal{A}$ of diffeomorphisms [76].⁷ This question is more complicated. I shall just say a few words about one approach to it and cite the simplest concepts and results pertaining to it.

⁷ Here, we have in mind the following papers of M. Shub and D. Sullivan: *Bull. Amer. Math. Soc.*, 80, no. 1 (1974) 27–41; *Topology*, 14 (1975) 109–139.

In addition to the handle decomposition and the cell decomposition K associated in § 3.2 to a M-S cascade $\{g^k\}$, we also need to take into account the action of g on these objects. We can suppose that the filtration (2) is such that each $\overline{M_{j+1}} \setminus \overline{M_j}$ consists of handles $H_1^j, \dots, H_l^j$ corresponding to one periodic trajectory. Then

$$gH_i^j \subset H_{\sigma(i)}^j \cup M_j,$$

where σ is a permutation of the numbers $1, \dots, l$. Furthermore, the image of the core of H_i^j contains the core of $H_{\sigma(i)}^j$, and the map is a homeomorphism on the inverse image of the latter. It easily follows that some cellular map g' , homotopic to g , induces a map $g'_\#$ of the group $C(K)$ of chains generated by the cells of K and that this map has the following form in any dimension: it has a triangular block matrix of coefficients, each block of which corresponds to a permutation of parts of the elements of the basis, possibly with a change of sign. Such matrices are called "virtual permutation matrices" (no meaning is attached to the phase "virtual permutation" by itself).

We now consider chains with real coefficients. The map $g_* = g'_*$ induced by g on homology is obtained by restricting $g'_\#$ to the invariant space of cycles and taking the quotient with respect to the invariant space of the boundaries. So the eigenvalues of g_* are contained among the eigenvalues of $g'_\#$ and are roots of unity.

Thus, in order that a M-S diffeomorphism exist in $\mathcal{J}$, it is necessary that all eigenvalues of the linear maps on homology induced by diffeomorphisms in $\mathcal{J}$ be roots of unity. It turns out that if M is simply-connected, $n > 5$, and $\mathcal{J}$ satisfies this condition, then some iterates of the diffeomorphisms in $\mathcal{J}$ are isotopic to M-S diffeomorphisms.

One first proves that some iterate $f^m = h$ of a diffeomorphism $f \in \mathcal{J}$ admits the following "chain model". There exists a finitely generated free chain complex $C = \{C_i\}_{i=0}^n$ and an endomorphism $\varphi = \{\varphi_i\}$ such that the coefficient matrices of φ_i are virtual permutation matrices, and the pair (C, φ) is chain homotopy equivalent to $(C(K), h'_\#)$, where K is a cell complex of M , and h' is a cellular map homotopic to h . One then proves that a diffeomorphism h which admits a "chain model" is isotopic to a M-S diffeomorphism g (under the above conditions on M and under the additional condition, which can always be arranged, that $C_1 = C_{n-1} = 0$). The "complicated" part of handle theory establishes that, under certain conditions, a given finitely generated free chain complex C , which is chain homotopy equivalent to $C(K)$ can be realized by a handle decomposition of M , i.e., C is isomorphic to $C(K_1)$ where the cell complex K_1 is obtained by contracting handles to their cores. The construction of g is connected with a supplement to the theory which deals not only with the realization of the chain complex C , but also with the realization of the given endomorphism $\varphi: C \rightarrow C$. The additional properties of the realization ensured by this construction are enough to guarantee that g is a M-S diffeomorphism.

It is not clear whether we can take the M-S diffeomorphism to actually belong to $\mathcal{J}$; the corresponding obstruction has been described using algebraic K -theory [33]. See [50] concerning the non-simply-connected case. In the particular case of a torus (of any dimension) the condition cited above is sufficient [41].

3) One can further ask whether there exist M-S systems with a given number of periodic trajectories with prescribed local topological properties.

The simplest such question is: are there M-S flows on M without closed trajectories which have exactly m_i equilibrium points with Morse index i , $0 \leq i \leq n$, where the m_i are prescribed numbers? An equivalent question is: are there functions f on M with nondegenerate critical points, exactly m_i of which have the index i ? The necessary conditions are given in § 3.2. From topology, it is known that these are also sufficient if $n=2$ (this is easy) or if $n>5$ and M is simply-connected (this is actually the same result from the "complicated" part of handle theory which was mentioned in 2); see [73]. Obvious considerations when $n=2$ or direct reference to [73] gives f with the minimal possible m_i : namely, with $m_i = b_i + c_i + c_{i-1}$. But it is easy to change f so as to add two new critical points with indices i and $i+1$. By making such changes to f it is possible to go from an f with the minimum possible m_i to an f with any prescribed m_i satisfying the corresponding inequalities).

4) We now turn to manifolds M with Euler characteristic zero. It is natural to ask whether there is a M-S flow on M without equilibrium points which has exactly r closed trajectories L_i prescribed, in the sense of § 2.1, by giving (u_i, Δ_i) or $(u_i, \Delta_i, \delta_i)$. Of the results related to this question, I mention only the simplest and most complete one in which $M=S^3$. The principal role is played by the untwisted L_i , i.e., the (u_i, Δ_i) with $\Delta_i=1$. In order that there exist a M-S flow on S^3 without equilibrium points and with exactly a_j untwisted closed trajectories with Morse index $u=j$, $1 \leq j \leq 3$, it is necessary and sufficient that the inequalities

$$a_1 \geq 1, \quad a_3 \geq 1, \quad a_2 \geq a_1 - 1, \quad a_2 \geq a_3 - 1 \quad (4)$$

hold. Moreover, the flow has no twisted closed trajectories with $u(L)=1, 3$ (this is obvious), and the number of twisted trajectories with $u(L)=2$ can be arbitrary.

In this case the inequalities (3) reduce to the first two in (4) and they express the trivial fact that some periodic trajectories of an M-S system must be (asymptotically) stable, the so-called *sinks*, and some completely unstable (i.e., asymptotically stable as $t \rightarrow -\infty$), the so-called *sources*. Moreover, these are untwisted on S^3 . The inequality $a_2 \geq a_1 - 1$ is proved by arguments similar to those given in § 3.2, 3), f); reversing time, we obtain $a_2 \geq a_3 - 1$. The sufficiency of the conditions (4) and the assertion about twisted L with $u(L)=2$ are proved by direct construction.

As can be seen, inequalities (3) are too coarse when there are closed trajectories. Their derivation does not take account of the fact that attaching round

handles of indices i and $i+1$ affects the i -dimensional homology differently. A sharpening of the M-S inequalities has been obtained in [30] for M-S flows without equilibrium points. Only untwisted trajectories are involved in the sharpened inequalities, *i.e.*, they are written in terms of the a_i and the (ordinary) Betti numbers b_i . For simply-connected, torsion-free M^n with $\chi(M)=0$ and $n>5$, the sharpened inequalities turn out to be also sufficient; if the numbers $a_i \geq 0$ satisfy these inequalities, then there is an M-S flow on M without equilibrium points which has exactly a_i closed untwisted trajectories of index i and no closed twisted trajectories.

5) Finally, one can ask whether a given isotopy class $\mathcal{J}$ of diffeomorphisms on M contains a M-S diffeomorphism for which a prescribed set $\{(l_i, u_i, \Delta_i)\}_{i=1}^r$ or $\{(l_i, u_i, \Delta_i, \delta_i)\}_{i=1}^r$ is the set of data for all periodic trajectories. This question has been investigated mainly for two-dimensional, orientable M (for the higher-dimensional case see [32]). Some necessary conditions are obvious. Some of the periodic trajectories of a M-S system must be sinks and sources. For a sink, in our case, $u_i=0$, $\Delta_i=1$, and for a source $u_i=2$, $\Delta_i=\varepsilon^l$, where $\varepsilon=1$ or -1 depending on whether the diffeomorphism preserves or reverses orientation. The homological zeta-function calculated from $\{(l_i, u_i, \Delta_i)\}$ must be the same as that determined by the action on homology induced by the diffeomorphisms of $\mathcal{J}$. These conditions are sufficient if $\mathcal{J} \ni \mathbf{1}_M$ (in which case M need not be orientable) or if $M=T^2$ (torus) and $\varepsilon=1$. An unexpected complication arises when $\varepsilon=-1$. It turns out that an orientation reversing homeomorphism of a surface of genus g which has more than $g+1$ periodic trajectories with mutually distinct, odd, minimal periods has positive topological entropy [42], while M-S systems have entropy equal to zero (for "topological entropy" see Volume 2 or M.E.). When M is S^2 or T^2 the question has been completely elucidated: when $\varepsilon=-1$ certain other conditions are imposed, which together with the previous conditions turn out to be sufficient [16], [17].

3.4. Other Questions. In this section, I shall examine conditions for M-S systems to be equivalent. There are only a few papers on this question, and they relate mainly to flows. For the two-dimensional case, see Chap. 4. A system of data has been found which uniquely characterizes an M-S flow (up to equivalence) on an orientable three-dimensional manifold M provided that each intersection $W^u(L) \cap W^s(L')$ with two-dimensional W^u and W^s consists of a finite number of trajectories. (This is equivalent to saying that, if $L > L'$ and $u(L)=u(L')=2$, then all the trajectories in $W^u(L)$ tend to a sink.) Unfortunately, the result is awkward to formulate even when there are no closed trajectories [80].

It is proved in [65] that if a M-S flow on M^n , $n \geq 3$, without closed trajectories is uniquely determined by its phase diagram, then the diagram must have the following property:

- a) If $x > y$, then either $u(x)=0$ or $u(y)=0$.

This is not possible on every M^n : if there exists an M-S flow on M without closed trajectories which satisfies a), then it follows that M must be torsion-free in dimensions up to and including $n-2$ (so if M is orientable, there is no torsion at all). In fact, a) implies that all cells in the underlying cell complex K (§ 3.2, (4)), except the one-dimensional and n -dimensional ones, determine cycles in $C(K)$.

Suppose that there is an M-S flow $\{g^t\}$ on S^n , $n \geq 3$, without closed trajectories which satisfies a). It turns out that the flow is then uniquely determined by its phase diagram [66] and, furthermore, the flow and the diagram have the following special properties:

b) $u(x)$ takes only the values 0, 1, $n-1$, n (otherwise S^n would have "superfluous" homology).

c) If $u(x)=n-1$, then $\partial W^u(x)$ reduces to a single sink. In other words, there is only one y with $y < x$; for it, $u(y)=0$. (In view of a), if $y < x$, then necessarily $u(y)=0$ and $\partial W^u(x)$ is connected.)

d) The 1-dimensional skeleton K^1 of the complex K (see § 3.2, (4)) is a connected tree.

We remark that if there are no closed trajectories, then the structure of K^1 can be completely recovered from the phase diagram. In addition, K^1 can be realized as the subset

$$P = \cup \{ \overline{W^u(y_j)} : u(y_j) = 1 \} \subset S^n.$$

e) It follows from d) and the Morse-Smale inequalities that the number of sources (x with $u(x)=n$) is one larger than the number of y for which $u(y)=n-1$. If such y exist, then, in view of c), the corresponding $\overline{W^u(y)}$ are $(n-1)$ -dimensional spheres, separating S^n into domains. In each such domain there is exactly one source x . The part of P which lies inside a domain is connected and consists of the vertices and edges which correspond to z with $z < x$ and $u(z)=0, 1$. Apart from the given inequality, such a z satisfies only those inequalities directly connected with K^1 (a z which corresponds to an edge is greater than those t which correspond to the vertices). But if z lies on the boundary of a domain, i.e., if $z \in P \cap \partial W^u(y)$ for some y with $u(y)=n-1$, then $z < y$ for all such y and $z < x$ for all those sources x which lie in domains bounded by the corresponding $\overline{W^u(y)}$. Apart from these inequalities, z satisfies only those inequalities which are connected with K^1 (moreover, z corresponds to a vertex of K^1).

f) The flow $\{g^t\}$ obtained from $\{g^t\}$ by time reversal, and the phase diagram (which is obtained in an obvious way from the phase diagram of the flow $\{g^t\}$) also satisfy the conditions a)–e).

Clearly, conditions d)–f), like a)–c), can be formulated in terms of the phase diagram alone, although this is quite awkward for e). It turns out that, if a given phase diagram (a marked, oriented graph, which does not give closed trajectories) satisfies conditions a)–f), then it is the phase diagram of a M-S flow on S^n [66].

Chapter 3

Topological Dynamics

(§§ 1–3 were written by D.V. Anosov, §§ 4–5 by I.U. Bronshtein.)

§ 1. Introduction

This chapter deals with topological DSs (Chap. 1, § 1.1). We consider flows and cascades on a complete metric space M with metric ρ . Unless otherwise stated, M is assumed to be compact. Actually, many of the definitions make sense and many results remain valid in other cases also. More complete and general treatments can be found in the literature (see below for references; see also “topological dynamics” and “topological dynamical systems” in M.E.).

Until recently, the main focus of attention in topological dynamics has been the study of motions and the corresponding classes of invariant sets which exhibit some sort of “repetitiveness”. It has been tacitly assumed (although not stated explicitly) that these are the motions which “ought” to be studied because all motions tend in time to “repetitive” motions in the compact case (see § 4.1 for more precise details). But for Morse-Smale systems, for example, the “repetitive” motions are the periodic motions, and the intrinsic structure of the corresponding invariant set is trivial. On turning our attention to other motions we arrive at a meaningful theory (which, among other things, relates the properties of “repetitive” motions to the topology of M , something which is impossible to discover from the analysis of “repetitive” motions alone). Finally, the influence of the QTDE and the theory of smooth DSs on topological dynamics resulted in the consideration of questions which go beyond the scope of the earlier themes. Simultaneously (under the same influence), interest turned to finding structurally stable properties of DSs, *i.e.*, properties which are preserved under small perturbations. It must be borne in mind, however, that the concept of a C^1 -structurally stable property is adequate for the theory of smooth DSs; systems with such properties form an open set in the space of C^1 DSs with the corresponding topology, as defined in Chap. 2, § 1.2 (sometimes one speaks of C^r -structurally stable properties with $r > 1$; it is clear what this means). For topological DSs one speaks of perturbations which are C^0 -small and, hence, of C^0 -structurally stable properties. The C^0 -topology on the space of topological DSs on M is defined by means of the metric

$$\rho(\{f^t\}, \{g^t\}) = \max \{ \rho(f^t x, g^t x) : |t| \leq 1, x \in M \}.$$

Many C^1 -structurally stable properties of smooth DSs (for example, the property of being a Morse-Smale system) are not C^0 -structurally stable. But when a property turns out to be not only C^1 -structurally stable, but also C^0 -structurally stable, the fact is worth noting. (We are speaking only of

C^0 -structurally stable properties and not of C^0 -structurally stable systems. There is no concept of a C^0 -structurally stable system. Furthermore, by analogy with Chap. 2, § 1.2, we can speak of C^0 -generic properties. For references, see “general position” and “topological dynamics” in M.E. It is not clear, however, how useful this notion is for the theory of smooth DSs).

Since questions involving the concepts above are not traditional in topological dynamics, and are at present less well known and less extensively dealt with in the literature, we shall discuss them in more detail (§§ 2, 3) than the more traditional questions relating to “repetitive” motions (§ 4).

In the theory of DSs, a noteworthy (although usually auxiliary) role is played by extensions (Chap. 1, § 1.5). In this chapter they are encountered in §§ 4, 5. In § 4, they crop up in connection with minimal sets; this is a development of a traditional theme. We saw in Chap. 1, § 2.2, that an “invariant” treatment “uniform with respect to all trajectories” of the variational equations of a smooth DS leads to extensions of a special type. By abstracting from this example, we arrive at the general concept of a linear extension (§ 5). The concept of a hyperbolic linear extension is very important. The treatment of nonautonomous differential equations from the point of view of topological dynamics also leads to extensions, and, in particular, to linear extensions of linear systems. This, too, is discussed in § 5.

We conclude with a word about the literature. The first systematic treatment of questions relating to “repetitiveness” of motions was given by G.D. Birkhoff, who can, for this reason, be regarded as the founder of topological dynamics. However, Birkhoff considered smooth DSs. The fact that this is not essential and that it is natural to develop the theory for topological DSs (at first, for flows only) was noticed by a group of Soviet mathematicians at the beginning of the thirties. They added much new material during the thirties and forties. This stage is reflected in [60] and [72]; see also [19]. For further developments of the theory of minimal sets, see [23], [29], [36] and [81]. In [26], this theory is set out together with ergodic theory, and both are combined in the analysis of examples of algebraic origin. In connection with similar questions, see also [M.E.: “distal dynamical systems”, “quasidiscrete systems”, and “nilflows”]. In connection with § 2, see [24], [28], [71] and [75]; in connection with § 3 see [28] and [75]; and in connection with § 5 see [24], [47], [57], [69] and [70]. In the articles in the M.E. cited above, there are references to other topics which are not covered or even mentioned here.

§ 2. Attractors, Morse Decompositions, Filtrations, and Chain Recurrence

2.1. Attractors and Morse Decompositions. A closed invariant set A of a topological DS $\{g^t\}$ is called an *attractor*, or an *asymptotically stable set* if it has the following properties:

a) It is *Lyapunov stable*, i.e., for every neighbourhood U of A there is a neighbourhood V of A such that any positive semi-trajectory which starts in V is entirely contained in U ;

b) if x is any point sufficiently close to A , then $\rho(g^t x, A) \rightarrow 0$ as $t \rightarrow \infty$.

When A is an equilibrium point, these definitions of Lyapunov stability and asymptotic stability are equivalent to those in Article I, Chap. 1, § 4.1. When A is a closed trajectory, (asymptotic) stability of A coincides with the property which was called orbital asymptotic stability in Article I, Chap. 1, § 5.4 (in contrast to Article I, we speak here about the stability of closed invariant sets only and not about the stability of motions. The difference between these notions already appears in the case of a closed trajectory and a periodic motion. It is even sharper in the hyperbolic theory: an attractor can consist of trajectories which correspond to unstable motions). The definition above of an attractor is equivalent to that given in Article I, Chap. 1, § 8.1.

The property that a DS have an attractor in a fixed open set $U \subset M$ is C^0 -structurally stable.

Let $\omega(x)$ and $\alpha(x)$ be the ω -limit and α -limit sets of the trajectory through the point x (the definitions are word for word the same as in Article I, Chap. 1, § 5.5). If B is a set, then the sets

$$W^s(B) = \{x \in M : \omega(x) \subset B\}, \quad W^u(B) = \{x \in M : \alpha(x) \subset B\}$$

are called respectively the *basin of attraction* and the *zone of repulsion*, respectively, of the set B (compare the definitions of W^s and W^u in Chap. 2. Although not formally required by the definition, W^s and W^u will be used only in the case that B is a closed invariant set).

If A is an attractor, then $A^* = M \setminus W^s(A)$ is a “repeller”, i.e., an attractor for the DS $\{g_t^-\}$ obtained by reversing time (Chap. 1, § 1.4). In this section, asterisks will be used only to denote repellers connected with attractors in this way. The intersection of a finite number of attractors is an attractor, and the intersection of a finite number of repellers is a repeller. The intersection of an attractor and a repeller is called a *Morse set*.

A finite, ordered (by index) system $\{A_1, \dots, A_r\}$ of mutually non-intersecting, closed, invariant subsets is called a *Morse decomposition*⁸ if:

1) for any $x \in M$ there are i, j such that

$$i \leq j, \quad \alpha(x) \subset A_j, \quad \omega(x) \subset A_i;$$

2) if $\alpha(x) \subset A_i$ and $\omega(x) \subset A_i$, then $x \in A_i$

(compare with the collection of periodic trajectories of a Morse-Smale DS. In this example, as in the general case, the Morse decomposition is only partially ordered). In particular, if A is an attractor, the pair (A, A^*) is a Morse decomposition.

⁸ This is the standard nomenclature in English, but since this system is not actually a decomposition of M , it would have been better to use the neutral word “collection”, as is done in Russian.

There is a bijective correspondence between Morse decompositions and filtrations of M (Chap. 2, § 3.2, 2)) by attractors. If

$$\phi = A_0 \subset A_1 \subset \dots \subset A_r = M$$

is such a filtration, set $A_i = A_i \cap A_{i-1}^*$. Conversely, if $\{A_1, \dots, A_r\}$ is a Morse decomposition, set $A_i = W^u(A_1 \cup \dots \cup A_i)$. (It is instructive to examine the example of a Morse-Smale DS with A_i as above.) In particular, a Morse decomposition consists of Morse sets. (Conversely, every Morse set $A \cap B^*$, where A, B are attractors, is contained in a Morse decomposition, namely

$$\{A \cap B, A \cap B^*, A^* \cap B, A^* \cap B^*\}.$$

The following relation holds:

$$A_1 \cup \dots \cup A_r = \bigcap_{i=1}^r (A_i \cup A_i^*). \quad (1)$$

The property that a DS have a Morse decomposition $\{A_1, \dots, A_r\}$ with $A_i \subset U_i$ where the $U_i \subset M$ are fixed open sets, is a C^0 -structurally stable property.

We now generalize (and weaken) the concept, introduced in Chap. 2, § 3.2, 3), of a filtration of a DS. A homeomorphism g and the cascade $\{g^k\}$ are said to preserve the filtration

$$\Phi = \{M_i\}, \quad \phi = M_0 \subset M_1 \subset \dots \subset M_r = M$$

if gM_i lies strictly within M_i . A flow $\{g^t\}$ preserves Φ if every g^t with $t > 0$ preserves Φ . Φ is also said to be a *filtration for the DS* (or *homeomorphism*). We can associate the Morse decomposition

$$A_i = \bigcap_t g^t(M_i \setminus M_{i-1}) \quad (2)$$

to Φ . The corresponding A_i are given by $A_i = \bigcap_{t>0} g^t M_i$. (Here t is understood

to take integral values in the case of a cascade, and real values in the case of a flow.) Of course, it is not excluded that some $A_i = \phi$; this happens when $g^t M_i \subset M_{i+2}$ for sufficiently large t .

The property that a cascade preserve a given filtration Φ is a C^0 -structurally stable property. But a flow $\{f^t\}$ obtained by a small perturbation of a flow preserving Φ need not preserve Φ ; all one can say is that the sets $f^t M_i$ will lie strictly within M_i for, say, $t \geq 1$. The A_i for $\{f^t\}$ will remain within small neighbourhoods of the A_i of the original flow.

2.2. Chain Recurrence. An ε -trajectory (an ε -motion would be a better name) of a cascade $\{g^k\}$ is a (finite or infinite) sequence $\{x_k\}$ such that $\rho(x_{k+1}, g x_k) < \varepsilon$ for all k (for which the expression makes sense). An ε -trajec-

tory of a flow $\{f^t\}$ is a parametrized, possibly discontinuous⁹, curve $x(t)$ defined on a finite or infinite time interval such that

$$\rho(x(t+\tau), g^\tau x(t)) < \varepsilon \quad (3)$$

for $0 \leq \tau \leq 1$ and for all t (for which this makes sense). One could require that (3) hold for $0 \leq \tau \leq a$ for some fixed a . If we tentatively call such an object an “ (ε, a) -trajectory”, then the assertion that the actual choice of a is not important can be explicitly interpreted as follows: for any flow $\{g^t\}$ and any ε , a and b , there exists δ tending to 0 as $\varepsilon \rightarrow 0$ (for fixed a , b and $\{g^t\}$) such that a (δ, a) -trajectory is an (ε, b) -trajectory.

A related concept is that of a (δ, s) -chain. This is a sequence of intervals of actual trajectories of time length greater than s with the property that the initial point of each interval is within a δ -neighbourhood of the preceding interval. The initial point of a (δ, s) -chain is the initial point of its first interval, and its endpoint is the endpoint of its last interval (if the number of intervals is finite).¹⁰

A point $x \in M$ has the *chain recurrence* property and is said to be *chain recurrent* if, for any $\varepsilon, T > 0$, there is an ε -trajectory starting from x and returning to x at some time after T . Equivalently, for any $\delta, s > 0$, there is a (δ, s) -chain with initial point and endpoint at x . All points of a trajectory either are or are not chain recurrent, so that we can speak of the chain recurrence of a trajectory. Chain recurrence is the widest (and therefore the weakest) of the various “repetitiveness” properties which are considered in topological dynamics. Speaking loosely, chain recurrence does not mean that a motion “repeats”, rather that the DS “does not interfere with” the repetition of “approximate motions”.

The set of all chain recurrent points is a closed, invariant set. We denote it by $\mathcal{R}$ or, if necessary, by $\mathcal{R}(\{g^t\})$ (or $\mathcal{R}(g)$ in the case of a cascade $\{g^k\}$). This set coincides both with the intersection of the sets $A \cup A^*$ where A runs over all attractors of the DS and (see (1)) with the intersection of all sets $\bigcup_{i=1}^r A_i$ corresponding to the possible Morse decompositions $\{A_1, \dots, A_r\}$.

Thus Morse decompositions “approximate $\mathcal{R}$ from above”. For the restric-

⁹ If the phase space is a manifold, one can restrict attention to continuous ε -trajectories, but this is not possible in the general context of topological dynamics. An ε -trajectory of a smooth flow $\dot{x} = v(x)$ can be taken to be a smooth (or piecewise smooth) curve $x(t)$ for which $|\dot{x}(t) - v(x(t))| < \varepsilon$ for all t (except perhaps for some isolated values corresponding to discontinuities of $\dot{x}$). Such an ε -trajectory will be a $C\varepsilon$ -trajectory in the previous sense, where C depends on v . If we restrict a smooth flow to an invariant set A which is not a manifold, the availability of a similar modification depends on whether we require that the ε -trajectory actually be in A or merely very close to A .

¹⁰ It is clear that something like ε -trajectories or (δ, s) -chains can be defined in a case where there is a uniform structure on M instead of a metric. A uniform neighbourhood system then plays the role of small ε and δ . This is essentially done in [28] where the compact set M is not initially assumed to be metrizable. Here, however, the author uses a uniform covering system instead of a uniform neighbourhood system.

tion $\{g^t | \mathcal{R}(\{g^t\})\}$ of the DS $\{g^t\}$ to $\mathcal{R}(\{g^t\})$, we have

$$\mathcal{R}(\{g^t | \mathcal{R}(\{g^t\})\}) = \mathcal{R}(\{g^t\}). \quad (4)$$

The dependence of $\mathcal{R}(\{g^t\})$ on $\{g^t\}$ is upper semicontinuous in the sense used for multivalued mappings: if $\{f^t\}$ is sufficiently close to $\{g^t\}$, then the set $\mathcal{R}(\{f^t\})$ lies within a small neighbourhood of the set $\mathcal{R}(\{g^t\})$ ($\mathcal{R}$ cannot “increase sharply” under a small perturbation, but $\mathcal{R}$ can “decrease sharply”: it suffices to consider the example where $g^t x \equiv x$).

2.3. Lyapunov Functions. Simple examples suggest that the behaviour of trajectories outside the set of “repetitive” motions is reminiscent of the behaviour of the trajectories in the gradient system. However, the way to pass from this observation to an exact formulation is neither obvious nor unique.

We shall say that a continuous function $f: M \rightarrow R$ (for brevity, we simply say “function”) is *non-increasing* along trajectories if

$$f(g^t x) \leq f(x) \quad \text{for all } x \in M, \quad t > 0,$$

and that f *decreases along a trajectory* L if

$$f(g^t x) < f(x) \quad \text{for all } x \in L, \quad t > 0.$$

(This is something like a Lyapunov function, but we will reserve the term for more specialized f .)

Consider, first, all possible functions which are non-increasing along trajectories. For each such f the set of x for which $f(g^t x)$ does not depend on t is a closed invariant set. Take the intersection A of all such sets for all possible f . It is the smallest closed invariant set outside of which there exists a resemblance to a gradient system. It can be shown that there is a function f which decreases along any trajectory not in A . In fact, the set A can be characterized by other means which are not related to non-increasing functions [15]. (The alternate description of A uses a modification of prolongations; see §4.2 concerning the latter.) However, the set A has so far not been used outside the narrow limits of one of the branches of topological dynamics (concerned with the analysis of various aspects of the concept of stability). This, perhaps, is connected with the fact that neither description of A is convenient enough for other branches of the theory of dynamical systems. Generally speaking, A is smaller than $\mathcal{R}$, but greater than the set of nonwandering points (§4.1).

Since it is impossible to “catch” anything smaller than A by functions which do not increase along trajectories, and since the only known, “sensibly defined”, larger set exhibiting some sort of “repetitiveness” is $\mathcal{R}$, it is natural to try to connect $\mathcal{R}$ with functions which do not increase along trajectories. It turns out that $\mathcal{R}$ is indeed “caught” by narrowing the class of functions.

We limit ourselves to the case of a flow. Consider the functions which decrease on certain trajectories and are constant on others, and which map

the union of the latter into a nowhere-dense subset of $\mathbb{R}$. It turns out that any such function is constant on the connected components of $\mathcal{H}$ (and, in particular, on any trajectories entering $\mathcal{H}$) and that there exists one such function f which takes different values on different components of $\mathcal{H}$ and which decreases on every trajectory not in $\mathcal{H}$. This function f does the most which can be done by functions of the type considered, for which reason it is called the *complete Lyapunov function* in [28]. It is not difficult to check that the assertion that it exists entails the assertion that $\mathcal{H}$ is “caught” by means of a filtration.

If M is connected, then the following assertions are equivalent:

- 1) $\mathcal{H} = M$;
- 2) the DS has no proper attractors (i.e., no attractors other than $\emptyset$ and M);
- 3) if U is a nonempty, open set and if $t > 0$, then $g^t \bar{U} \not\subset U$ (the property of weak incompressibility).

In general, $\mathcal{H}$ is the union of all closed invariant subsets $A \subset M$ on which the restricted DS (i.e., the DS $\{g^t|_A\}$) is weakly incompressible [82], but $\mathcal{H}$ itself may not have this property.

§ 3. Indices of Isolated Invariant Sets

3.1. Isolated Invariant Sets. In this section, invariant sets are assumed to be compact, unless otherwise stated. An invariant set A is said to be *isolated* or *locally maximal* if there is a neighbourhood U of A in which there is no larger (possibly noncompact) invariant set. In other words, A is the intersection of all $g^t U$; any trajectory entirely contained in U lies in A . We say that U is an *isolating neighbourhood* for A . But if we are primarily interested in U and not A , then U is said to be an isolating neighbourhood and A , which is uniquely determined, is not specified. We do not exclude the case $A = \emptyset$; in terms of U , this means that no single trajectory is entirely contained in U .

An isolated equilibrium point, a closed trajectory of a flow, or a periodic trajectory of a cascade (Chap. 1, § 2.4) may fail to be isolated closed sets (for example, an equilibrium point of the type of a center). But if they are hyperbolic, then they are also isolated as invariant sets.

The intersection of a finite number of isolated invariant sets is an isolated invariant set, but their union need not be. Attractors and repellers (and, hence, Morse sets, § 2.1) are isolated invariant sets. A set of chain-recurrent points (§ 2.2) need not be isolated. If A is an isolated invariant set of a DS $\{g^t\}$, then any DS $\{f^t\}$ sufficiently close to $\{g^t\}$ has an isolated invariant set contained in a small neighbourhood of A .

The concept of an isolated invariant set is often encountered in the investigation of various topics: travelling waves, dynamical systems whose trajecto-

ries exhibit hyperbolic behaviour, the three-body problem, and iterations of one-dimensional maps. This does not mean that it is necessary to try to construct a general theory of isolated invariant sets; rather the reverse, for their properties may be much too diverse. However, there are two clearly distinguishable groups of topics which can legitimately be called theories pertaining to isolated sets. These are the theory of hyperbolic isolated sets (these form a much narrower class than the class of all isolated sets) and “index” theory (the class of isolated invariant sets is not restricted, but only a special group of properties is considered).

Up to now, index theory has only been developed for flows. So, for the rest of this section, we shall speak only about flows.

In some cases “index” considerations enable one to draw conclusions about the existence of an isolated invariant set A in a given domain U by the studying the behaviour of the trajectories along the boundary of U . Naturally, the internal structure of A cannot be determined in these circumstances. Nevertheless, one obtains some information about A and this can be useful (cf. the end of § 3.3); sometimes something can be said about trajectories which tend to A as $t \rightarrow \infty$ or $t \rightarrow -\infty$.

The indices considered in this section differ essentially from those considered in Chap. 2, § 2, although both types are of a topological nature and are connected in some way with the Morse indices (Chap. 1, § 2.5). Moreover, the theory below can be regarded as a development of another tradition, also old, but which does not usually appear in the foreground. Even in the QTDE on the plane, considerations of the following sort can be used in investigating an equilibrium point a : if some trajectories “turn away” from a in one direction, and others turn away from a in another direction, then there must be trajectories “in between” which go into a . This is essentially a topological argument, although the topology is trivial in this case. In the higher dimensional case, P. Bohl used similar arguments and, for this purpose, proved the theorem which now bears his name. (See Chap. 2, § 2.2; thus, Bohl’s use of his theorem differed from the later, more common applications, which were mentioned in Chap. 2, § 2.2.) T. Ważewski’s principle was formulated later; this enables one to use the study of the behaviour of the trajectories along the boundary of U to draw conclusions about the existence of a semi-trajectory L (and, thereupon, an entire trajectory L , which is a limiting trajectory of L) that lies entirely within U [43]. Indices of isolated invariant sets enable one to draw conclusions of a similar sort. In this respect, they are formally somewhat weaker than Ważewski’s principle (see, e.g., [28]), but their more algebraic character makes them more flexible; moreover, the additional information they provide about A and the trajectories tending to A can be valuable.

3.2. Isolating Blocks and Index Pairs. In this section, as a rule, we shall be dealing with compact, as opposed to open, neighbourhoods. When is a compact set N an isolating neighbourhood? Clearly a necessary and suffi-

cient condition is that no boundary point of N (in M) should lie on a trajectory entirely contained in N . This is the case, for example, if any boundary point x is such that the positive or negative semi-trajectory starting at x immediately leaves N . (More formally, there is an $\varepsilon = \varepsilon(x) > 0$ such that $g^t x \notin N$ for all $t \in (-\varepsilon, 0)$ or all $t \in (0, \varepsilon)$. In the latter case x is called an *exit point*; the set of all exit points is denoted by N_+). Such N are called *isolating blocks*.

Various definitions of isolating blocks occur in the literature. They played a large role in the early research in the field, when smooth flows were considered. At that time, their definition involved a number of other conditions which were used in various arguments; it was proved that an isolating invariant set of a smooth flow contains an isolating neighbourhood N which is an isolating block (satisfying these additional conditions). In order to study topological flows, the definition had to be modified and the theory lost its former elegance. Ultimately, isolating blocks were replaced by *index pairs* in the general theory (see below). However, in applications of the theory to smooth flows, it is still convenient to use isolating blocks. In practice, these are understood in the sense defined in the previous paragraph, and N is an n -dimensional submanifold (with boundary) of the manifold M , and only the boundary may have "corners".

For example, for a saddle $(\lambda, \mu > 0)$

$$\dot{x} = \lambda x + o(|x| + |y|), \quad \dot{y} = -\mu y + o(|x| + |y|) \quad (5)$$

on the plane, the small square $|x|, |y| \leq \varepsilon$ can serve as an isolating block; its vertical sides form N_+ . A small disc would also serve, but the square is a more obvious choice.

Let N be a compact isolating neighbourhood of an isolated invariant set A . An (ordered) pair $\langle N_1, N_2 \rangle$ of compact subsets of N is called the *index pair* (for A) relative to N if A lies strictly within $N_1 \setminus N_2$ and:

- 1) if a positive semi-trajectory starting at $x \in N_1$ eventually leaves N_1 , then it simultaneously leaves N . Symbolically, if $g^t x \in N$ for $0 \leq t \leq t_0$, then $g^t x \in N_1$ for the same t . (Here, "exit" is understood in a weaker sense than before.)
- 2) if a positive semi-trajectory starting at $x \in N_1$ eventually leaves N_1 (or what, in view of 1), is the same thing: if it leaves N , then it must first have entered N_2 ("exit from N_1 takes place only through N_2 ").

It has been proved that there exists an index pair $\langle N_1, N_2 \rangle$ for any isolated invariant set A and any compact isolating neighbourhood N of A . If N' is an isolating block (for A), then $\langle N', N'_+ \rangle$ is an index pair (for A relative to N').

Another example: for the saddle (5) we can take

$$N = \{(x, y): |x|, |y| \leq \varepsilon\};$$

$$N_1 = \{(x, y): |x| \leq \varepsilon, |y| \leq \varepsilon/2\}; \quad N_2 = \{(x, y): 0 \leq 1 - |x| \leq \varepsilon/2, |y| \leq \varepsilon\}.$$

In the general case, N_2 serves as a "coarse approximation" (sufficient for our purposes) of N'_+ , and N' plays two roles, one of which (mainly, a control

on how close to A everything takes place) goes to N , and the other (the main one) to N_1 .

3.3. Homological and Homotopic Indices. The space $N_1/(N_1 \cap N_2)$ obtained from N_1 by identifying all points of $N_1 \cap N_2$ with one another is called the *index space* of A . It is viewed as a space with a distinguished point (as topologists say, a “pointed” space, the distinguished point being the point x_0 obtained from $N_1 \cap N_2$). Formally, this means that we are considering the pair $(N_1/(N_1 \cap N_2), x_0)$; to be even more precise, the second term should be written as $\{x_0\}$. If $N_2 = \phi$, the index space N_1/ϕ is the disjoint union of N_1 and the single point x_0 (more formally, the single point space; but more picturesquely, a point lying somewhere to the side of N_1). We can regard this as a special convention regarding the meaning of N_1/ϕ ; alternatively, it is possible to give to define N_1/B with $B \subset N_1$ so that this is obtained automatically when $B = \phi$.

It turns out that all the index spaces for A have the same homotopy type as pointed spaces. (For two spaces (X, x_0) and (Y, y_0) , this means that there are continuous mappings $f: X \rightarrow Y$ and $g: Y \rightarrow X$ such that gf is homotopic to $\mathbf{1}_X$ and fg is homotopic to $\mathbf{1}_Y$, where all maps under consideration, f, g and the mappings $X \rightarrow X$ and $Y \rightarrow Y$ which arise under the homotopies, carry one distinguished point into the other.) Roughly speaking, the homotopy equivalence is constructed by shifting suitably along the trajectories. This allows us to define the *homotopy index* $h(A)$ to be the homotopy type of the index space and the *(co-)homological index* as the total (co-)homology, of the index space modulo the distinguished point. The latter index is uniquely determined by $h(A)$ and, as $h(A)$ is also more accessible, I shall restrict myself to using this.

When working with the homotopy type $[X]$ of a space X , *i.e.*, the set of spaces homotopy equivalent to X , it is not necessary to visualize all the elements in some way. On the other hand, it is often possible to easily visualize some sufficiently simple representative $Y \in [X]$; we think of X as having been continuously deformed into Y . If, say, X is homotopy equivalent to S^n , then S^n will naturally be taken as this representative. The corresponding homotopy type is denoted by S^n . (Any point of S^n can be taken as the distinguished point.)

In example (5), upon shrinking both vertical sides of the square into a single point, it is easy to see that we obtain a space homotopy equivalent to a circle. In general, a hyperbolic equilibrium point with Morse index u has a homotopy index S^u . (When $u=0$, it is worthwhile checking that the convention regarding N_1/ϕ “works”.)

In some cases $h(A)$ can be expressed in terms of simpler homotopy types by means of the operations of *sum* or *wedge* $\vee$ and *product* $\wedge$. These operations are defined on homotopy types by first defining them on the pointed spaces which represent the homotopy types as follows:

$$(X, x_0) \vee (Y, y_0) = (X \cup Y) / \{x_0, y_0\}$$

(the union is disjoint; when dealing with spaces instead of homotopy types, we speak more often of wedge rather than sum);

$$(X, x_0) \wedge (Y, y_0) = (X \times Y) / ((X \times y_0) \cup (x_0 \times Y)).$$

These operations are associative and commutative; moreover, multiplication distributes over sums; the homotopy types

$$\hat{0} = [(x_0, x_0)], \quad \hat{1} = \Sigma^0 = [(\{x_1, x_0\}, x_0)], \quad \text{where } x_1 \neq x_0,$$

play the rôle of zero and unity. A sum is equal to $\hat{0}$ if and only if the summands are each equal to $\hat{0}$.

If an isolated invariant set A is a disjoint union of isolated invariant sets A_1 and A_2 , then $h(A) = h(A_1) \vee h(A_2)$. The homotopy index of a hyperbolic, untwisted, closed trajectory L with Morse index $u > 1$ is equal to $\Sigma^u \vee \Sigma^{u-1}$. But when $u(L) = 1$, then $(S^1 \cup x_0, x_0)$, where $x_0 \notin S^1$, is a representative of $h(L)$. (This differs from the particular case of the previous formula with $u = 1$: $\Sigma^1 \vee \Sigma^0$ has representative $(S^1 \cup x_0, x_1)$, where $x_1 \in S^1$.) A hyperbolic, twisted, closed trajectory L with Morse index u (which is necessary at least two) has homotopy index

$$h(A) = [RP^2] \wedge \Sigma^{u-2},$$

where RP^2 is the projective plane; the choice of distinguished point on it does not matter.

There are many papers applying index theory to various concrete problems. Many of them are devoted to problems connected with travelling waves. As an example of the general sort of argument used, we sketch a typical application of these ideas.

Imagine that we are given an autonomous system in $\mathbb{R}^n$ and that we want to show that it has a trajectory going from one equilibrium point a to another b . The points a and b themselves are easily investigated. Suppose they prove to be hyperbolic. It is easy to construct an isolating block N containing a and b . If $[N/N_+] = \hat{0}$, it follows that the maximal, closed, invariant set A contained in N (or what is the same thing, for which N is an isolating neighbourhood) does not reduce to $\{a, b\}$; otherwise we would have $\Sigma^{u(a)} \vee \Sigma^{u(b)} = \hat{0}$. After this point, index considerations play no further role. By using other special features of the system under consideration, one proves that a trajectory which is entirely contained in N and different from a and b must go from a to b . (In fact, for these special systems the behaviour of the trajectories within N is the same as in a gradient system.)

There are also other approaches to problems of this type. But index methods have repeatedly been applied to such problems and (if only because of the number of such applications) some of the problems were first solved by these methods. In this connection, an exposition of index theory is included in the book [75] on mathematical questions in the theory of shock waves.

3.4. The Morse-Conley Index. In this section, we consider a related problem which also, in the final analysis, is actually connected with travelling waves,

but which is discussed in [28] in an abstract form divorced from its origins.

Imagine that we are given an autonomous system in $\mathbb{R}^2$ which depends on a parameter $\lambda \in [\lambda_0, \lambda_1]$ and which is such that there is an isolating block N , which is suitable for all λ and which contains the equilibrium points a and b . Suppose, further, the latter are hyperbolic with $u=1$ (saddles) for all λ and they have small isolating blocks N_a and N_b , also suitable for all λ . When $\lambda = \lambda_0$ and $\lambda = \lambda_1$ there are no other trajectories lying entirely in N ; but, the invariant manifolds $W^u(a)$ and $W^u(b)$ leave ∂N when $\lambda = \lambda_0$ differently from the way they do when $\lambda = \lambda_1$: the same whisker meets different components of N_+ . (This assertion is meaningful, since the whiskers can be extended along the parameter as long as the equilibrium point remains hyperbolic.) By using this fact together with other properties, we want to prove that there must be a trajectory going from a to b for some λ .

In this case index theory is used to establish that the maximal invariant set A contained in N does not reduce to $\{a, b\}$ for all λ . As in the previous example, the fact that the trajectory lying wholly in N must go from a to b is proved using other actual special features of the system in question. We outline the index-theoretic argument as follows.

We argue by contradiction and assume that $A_\lambda = \{a, b\}$ for all λ . Shifting along the trajectories determines a homotopy equivalence

$$(N_a/N_{a+}) \vee (N_b/N_{b+}) \rightleftarrows N/N_+,$$

and this equivalence depends continuously on λ . But when $\lambda = \lambda_0$ and $\lambda = \lambda_1$, the shifts along the trajectories determine mappings

$$N_a/N_{a+} \rightarrow N/N_+ \tag{6}$$

which are not homotopic.

Filling in the details of this outline requires arguments of the same sort as those involved, say, in the proof of the fact that the index spaces of a fixed A are homotopy equivalent, or that $h(A) = h(A_1) \vee h(A_2)$ where A is a disjoint union of A_1 and A_2 . It is natural to wish that the result of these arguments could be neatly formulated once and for all in the general theory. The statement would have to involve some sort of mapping of indices $h(A) \rightarrow h(A_\lambda)$ arising from shifts along trajectories. But for homotopy types, it is impossible to define morphisms in a reasonable way. The point is that saying X and Y are homotopy equivalent does not say which homotopy equivalence $X \rightarrow Y$ is to be used (the equivalences need not be homotopic, e.g., $X = Y = S^1$).

The way out of this situation is to restrict the objects we consider. Instead of arbitrary spaces homotopy equivalent to the index spaces, we only take the actual index spaces themselves and instead of arbitrary mappings of one into the other, which are homotopy equivalences, we consider only those which are obtained by shifting along trajectories.

A *connected simple system* is defined to be a set of spaces and continuous mappings between them such that:

- a) a composition of mappings belonging to the system belongs to the system;
- b) the identity maps of the spaces in the system belong to the system;
- c) for any two spaces X, Y in the system, the set of mappings $X \rightarrow Y$ belonging to the system is not empty and consists of homotopy equivalences which are homotopic to one another.

(More succinctly, a system is a category of topological spaces and continuous mappings which satisfies c).)

The *Morse-Conley index* $I(A)$ of an isolated invariant set A is a connected simple system consisting of the index spaces of A and mappings of the index spaces which are either shifts along the trajectories or mappings homotopic to such shifts. (A complete formulation would have to make precise exactly how the mappings are defined by means of shifts along the trajectories; I omit this.) For Morse-Conley indices, as well as for connected simple systems in general, we can introduce morphisms and, in particular, the notion of homotopy equivalences. (As is well-known, for (co-)homology there is still another useful type of mapping: the connecting homomorphisms in the exact sequence of a pair. The analogue of this sequence in homotopy theory is the Puppe exact sequence. In our case, if A is an isolated invariant set of a flow $\{g^t\}$ and if $A_1 \subset A$ is an attractor of the flow $\{g^t|A\}$, then it is possible to construct an analogue of this sequence under certain additional assumptions. I mention only that the first connecting map c in the sequence characterizes in some sense the behaviour of the trajectories in $A \setminus A_1$ which tend to A_1 and the arcs of nearby trajectories in M .)

Generally speaking, $I(A)$ does not contain representatives which are as simple as those in $h(A)$. It is therefore expedient to combine different types of indices in an argument.

If $\{g_\lambda^t\}$ is a flow, which depends continuously on a parameter λ taking values in a topological space A , we let $\mathcal{S}_\lambda$ be the set of all isolated invariant sets of the flow $\{g_\lambda^t\}$, and $\mathcal{S}$ be the disjoint union of all the $\mathcal{S}_\lambda$, $\lambda \in A$. We provide $\mathcal{S}$ with a topology which is especially suitable for discussing extension of isolated invariant sets with respect to the parameter. (This topology does not coincide with the topology which seems most natural at first sight: the topology on $\mathcal{S}$ induced as a subset of $F(M) \times A$, where $F(M)$ is the set of all closed subsets of M equipped with the topology associated to the Hausdorff metric.) For any compact $N \subset M$, we let

$$A(N) = \{\lambda: N \text{ is an isolating neighbourhood for } \{g_\lambda^t\}\}.$$

When $\lambda \in A(N)$, we let $\sigma_N(\lambda)$ be the maximal invariant set of the flow $\{g_\lambda^t\}$ contained in N . The sets $\sigma_N(U)$ for all possible compacts $N \subset M$ and all open sets $U \subset A(N)$ are taken as a prebasis of the topology on $\mathcal{S}$ (i.e., their

intersections form a basis). With this topology and the natural projection

$$\mathcal{S} \rightarrow A, \quad \mathcal{S}_\lambda \mapsto \lambda,$$

$\mathcal{S}$ turns out to be (the total space of) a sheaf of sets over A . If two isolated invariant sets, regarded as points of $\mathcal{S}$ can be joined by a continuous path, then their Morse-Conley indices are homotopy equivalent.

As is well-known, the sheaf topology often possesses rather unusual properties; and the present case is no exception. Apart from these, other peculiarities are also present. These, however, reflect definite properties of isolated invariant sets and, hence, can scarcely be regarded as defects of the definition, which might perhaps be removed by modifying the latter. Thus, the mapping

$$A(N) \rightarrow \mathcal{S}, \quad \lambda \mapsto \sigma_N(\lambda)$$

is continuous (it is a section of the sheaf), but geometrically the set $\sigma_N(\lambda)$ can “jump” as λ changes. To make the example at the beginning of this section more specific, let us suppose that one of the whiskers goes from a to b for some λ and that A_λ consists of a, b and this whisker. But A_λ only consists of a and b for other λ . If N is the original isolating neighbourhood, then $A(N) = [\lambda_0, \lambda_1]$ and $\sigma_N(\lambda) = A_\lambda$. But, in this connection, we can assert that the properties of A_λ which we have discussed, such as $h(A_\lambda)$ and, in a sense $I(A_\lambda)$, do not actually change when λ changes, although A_λ itself “jumps”. Furthermore,

$$A(N_a \cup N_b) = [\lambda_0, \lambda_1], \quad \sigma_{N_a \cup N_b}(\lambda) = \{a, b\},$$

and so two continuous sections of $\mathcal{S}$ coincide everywhere except for one value of λ ; the topology of $\mathcal{S}$ is not Hausdorff. But this reflects the fact that the definition of the extension of an isolated invariant set with parameter depends on the choice of the isolating neighbourhood.

The theory in §§ 3.3 and 3.4 can sometimes be applied to objects which fail to be flows because the trajectories are not unique or the solutions are defined only when $t \geq 0$ or, perhaps, only for small $t \geq 0$. In all these cases it is possible to regard the set of all possible solutions as a subset Φ of the space of parametrized curves on M which have the property of *local positive invariance* relative to shifts of the argument: that is, if $\varphi \in \Phi$, there is a neighbourhood $U(\varphi)$ of φ in Φ and an $\varepsilon(\varphi) > 0$ such that the images of $U(\varphi)$ under shifts of the argument on $t \in [0, \varepsilon(\varphi))$ do not go outside Φ . More important is the fact that the same methods make it possible to consider, within the framework of this theory, questions relating to equations with time delays, partial differential equations, and integral equations. In the literature, it has therefore become customary to start with a locally positive invariant Φ . In connection with the material of this section, see [45, 68] in addition to [28, 75] and the references therein.

Although all the refinements in this section serve a purpose, the simpler concepts and results of § 3.3 are adequate for most applications to date. It is too early to decide whether the more complicated theory will be fruitful.

I wish to point out, however, that the Morse-Smale inequalities (Chap. 2, § 3.2) admit a natural, general treatment within the context of this theory [28].

§ 4. “Repetitive” Motions

4.1. Nonwandering Points. The Centre. A point $x \in M$ is said to be *wandering* if there is a neighbourhood U of x and a $t_0 > 0$ such that $g^t U \cap U = \emptyset$ for all $t > t_0$. A point which is not wandering is called a *nonwandering point*. Together with x , all points $g^t x$ are either simultaneously wandering or nonwandering, and so we can speak of *wandering* or *nonwandering trajectories*. Time reversal or change in time (Chap. 1, § 1.4) does not alter the property of a point being wandering or nonwandering.

The set of all nonwandering points of a DS $\{g^t\}$ is denoted by $NW(\{g^t\})$, or $NW(g)$ for a cascade. The Greek letter Ω is often used instead of NW . It is a closed invariant set; it contains all ω -limit and α -limit sets of all trajectories (see Article I, Chap. 1, § 5.5 for the definitions), and is a (in general, proper) subset of $\mathcal{R}(\{g^t\})$ (see (§ 2.2)).

The analogue of (4) with $\mathcal{R}$ replaced by NW does not hold. In this connection, we set

$$\Omega_1 = NW(\{g^t\}), \dots, \Omega_{i+1} = NW(\{g^t | \Omega_i\}), \dots, \\ \Omega_\omega = \bigcap_{i < \omega} \Omega_i, \quad \Omega_{\omega+1} = NW(\{g^t | \Omega_\omega\}), \dots$$

(transfinite induction). This process stabilizes at some (transfinite) number α of countable order, i.e.,

$$\Omega_\alpha = \Omega_{\alpha+1} = \dots$$

This Ω_α is called a *center* of the DS $\{g^t\}$. The center is the largest closed invariant set A all of whose points are nonwandering for the DS $\{g^t | A\}$. With regard to the *depth of the center*, i.e., the number of steps in the transfinite process necessary to reach it, see [ME, “center”].

A positive (resp., negative) semi-trajectory is said to be *Poisson-stable* if it is contained in its ω -limit set (resp., α -limit set). For trajectories we have to distinguish between Poisson-stability in the positive direction, i.e., stability for some (and, therefore, any) positive semi-trajectory, Poisson-stability in the negative direction (i.e., the similar property for negative semi-trajectories), and simply Poisson-stability, meaning that both these properties hold. (If it is necessary to emphasize the difference, we use the term two-sided Poisson-stability to refer to the latter case.) A point x is (positive, negative, or two-sided) Poisson-stable if its trajectory has the same property; in other words, if $g^t x$ approaches x arbitrarily closely as $t \rightarrow \infty$, or as $t \rightarrow -\infty$, or as both $t \rightarrow \infty$ and $t \rightarrow -\infty$. The closure of the set of all Poisson-stable points coincides with the center.

Let U be a neighbourhood of the set NW . Then there are $m, T > 0$ such that, for any x , the set $\{t: g^t x \notin U\}$ is contained in no more than m intervals

of length T . In this sense, it can be said that any motion approximates the set of recurrent motions in time. In a weaker sense, it can be asserted that they concentrate around the center. That is, for any neighbourhood U of the center and for any x , the proportion of those t in the interval $[0, T]$ for which $g^t x \in U$, tends to 1 as $T \rightarrow \infty$. (However, the smallest closed invariant set with the same property, the *minimal center of attraction*, is, generally speaking, only a part of the center [60], [72].)

Examining the dependence of $NW(\{g^t\})$ on the DS $\{g^t\}$, we encounter a phenomenon which is new compared to the situation for $\mathcal{R}$ (see the end of § 2.2). The DS $\{g^t\}$ is said to undergo a $C^0\Omega$ -“explosion” if there is a neighbourhood U of $NW(\{g^t\})$ for which there exist DSs $\{f^t\}$, arbitrarily close to $\{g^t\}$, for which $NW(\{f^t\}) \not\subset U$. Closeness is understood in the sense adopted for topological DSs (§ 1), but when considering smooth DSs, we also speak, in a self-explanatory sense, of $C^r\Omega$ -explosions. If $NW(\{g^t\}) = \mathcal{R}(\{g^t\})$, then $\{g^t\}$ cannot undergo a $C^0\Omega$ -explosion; the converse is true when M is a closed manifold [71] (but not when M is arbitrary; a flow on a “figure eight” with a single equilibrium point is an example). An interesting question (not only in its own right, but because of its relation to other problems) is whether the same is true for a $C^1\Omega$ -explosion.

4.2. Variants of the Nonwandering Concept. Prolongations. A point $x \in M$ is said to be *weakly nonwandering*¹¹ for the DS $\{g^t\}$ (A.N. Sharkovskiy) if, for any neighbourhood U of x and for any number $t_0 > 0$, there exists a number $t > t_0$ and a DS $\{f^t\}$ arbitrarily close to $\{g^t\}$ for which $f^t U \cap U \neq \emptyset$. The set of weakly nonwandering points is denoted by NW_w . It is clear that $NW \subset NW_w \subset \mathcal{R}$. If M is a manifold, then $NW_w = \mathcal{R}$, but this is not necessarily so in general (as can be seen from the “figure eight” example at the end of § 4.1).

The *prolongation of a point x with respect to the initial data* is defined to be the set

$$D(x) = \{y: \text{there are } x_n \rightarrow x, t_n \geq 0 \text{ such that } g^{t_n} x_n \rightarrow y\}$$

and the *limiting prolongation* is the subset $D'(x)$ defined by requiring that $t_n \rightarrow \infty$ above. It is clear that $D(x)$ is the union of $D'(x)$ and the positive semi-trajectory of x ; it is also clear that $D'(x) = \bigcap_{t > 0} g^t D(x)$. $D'(x)$ is also said

to be *prolongation of the trajectory of the point x* .

For example, the prolongation of the stable whisker of a saddle a contains $W^u(a)$ (this example was given by Poincaré and Bendixson). A point x is a nonwandering point if and only if $x \in D'(x)$.

Various modifications of prolongations exist. One of these (*prolongation with respect to the initial conditions and the DS*) is obtained by varying not

¹¹ The term “weakly nonwandering set” is also encountered in ergodic theory, but with a different meaning; see [ME, “invariant measure”].

only the initial conditions, but also the DS:

$$P(x) = \{y: \text{there exist } \{f_n^t\} \rightarrow \{g^t\} \text{ and } x_n \rightarrow x, t_n \geq 0, \text{ such that } f_n^{t_n} x_n \rightarrow y\}.$$

The subset $P'(x)$ is defined similarly with the additional condition that $t_n \rightarrow \infty$.

P and P' are related in the same way as D and D' . Weakly nonwandering points are the points x for which $x \in P'(x)$.

It is possible to vary only the DS (*prolongation with respect to the DS*). On a manifold, this is the same as the previous prolongation, i.e., P . (Authors who are dealing with DSs on manifolds often give formulations special to this case.)

Apart from the questions touched on here, prolongations (and also ε -trajectories) are used in investigating various aspects of the stability concept in the general context of topological dynamics. To date, however, the nuances thus revealed have not been necessary in the theory of smooth DSs.

4.3. Minimal Sets. A set $A \subset M$ is said to be *minimal* if it is nonempty, closed and invariant, and has no proper subsets which have these three properties. Every DS (with a compact phase space, as usual) contains at least one minimal set. A point $x \in M$ is said to be *recurrent* (in the Birkhoff sense) if, for any $\varepsilon > 0$, there exists $L > 0$ such that the entire trajectory of x is contained in an ε -neighbourhood of any of its intervals of time length L . Together with x , all points of its trajectory are either simultaneously recurrent or non-recurrent; so one can speak of a recurrent trajectory. This property is stronger than Poisson-stability. Every point of a compact minimal set is recurrent; and the closure of the trajectory of a recurrent point x is a compact minimal set (whether M is compact or not). It follows that recurrence is preserved by a continuous change, or reversal, of time. (For the closure of a trajectory to be minimal, *almost-recurrence* is sufficient: this means that, for any neighbourhood U of the point x , the set of t for which $g^t x \in U$ is relatively dense, i.e., there exists $L > 0$ such that, in each interval $[s, s+L]$, there is at least one point of the set. When the closure of the trajectory is compact, almost-recurrence is the same as recurrence.)

Frequently, the different terminology suggested by W.H. Gottschalk and G.A. Hedlund is used. They call Poisson-stability *recurrence*. What we call recurrence (more exactly, almost-recurrence, but we shall not dwell on the refinements necessary in the noncompact case), they call *almost-periodicity of the point x* or *almost-periodicity of the DS at the point*. They also speak of *almost-periodicity of a DS on the trajectory of a point x* ; this means that the motion $t \mapsto g^t x$ is almost-periodic in the usual sense (see below). This use of the words “almost-periodicity” is unusual and, so, for recurrence in the Birkhoff sense, the term *uniform recurrence* was proposed.

A function $\varphi: \mathbb{R} \rightarrow M$ or $\varphi: \mathbb{Z} \rightarrow M$ is *almost-periodic* if, for every $\varepsilon > 0$, there is a relatively dense set of ε -almost-periods, i.e., a relatively dense set of numbers τ such that $\rho(\varphi(t+\tau), \varphi(t)) < \varepsilon$, for all t . In particular, one can speak of an almost-periodic motion $t \mapsto g^t x$. An almost-periodic motion can

cease to be almost-periodic after a change of time; so the expression “almost-periodic trajectory”, which is often encountered, is not to be taken literally; one is actually thinking of the trajectory as equipped with a parametrization, *i.e.*, one is actually speaking about the motion.

Examples of DSs with almost-periodic motions can be obtained by the following algebraic construction. Let G be a topological group. Choose an element $a \in G$ and a one-parameter subgroup $a(t)$ of G . For any $x \in G$, we set

$$g^k x = a^k x, \quad g^t x = a(t)x. \quad (7)$$

The resulting DS (cascade or flow) on G is said to be a *DS of group shifts* (more precisely, left shifts; one could also take right shifts). When G is a compact metric space, all motions of this DS are almost-periodic.

If the subgroup $\{a^k\}$ or $\{a(t)\}$ is dense in G , then all trajectories will also be dense in G . A group with a subgroup of this type is said to be *monothetic* (in the case of $\{a^k\}$) or *solenoidal* (in the case of $\{a(t)\}$); such a group is, of course, commutative. We shall only need to consider compact, metrizable, commutative G . Such a group is solenoidal if and only if it is connected (this is in [60]); and monotheticity is equivalent to the condition that its character group be isomorphic to a countable subset of the group $\mathbb{R}/\mathbb{Z}$, considered as an abstract group (without a topology).

If a DS (on a complete metric space) has an almost-periodic motion, then its restriction to the closure of the trajectory of the latter is topologically isomorphic (Chap. 1, § 1.5) to a DS of group shifts (and the corresponding G and $\{a^k\}$ or $\{a(t)\}$ have the properties indicated above). Here is another equivalent property of such a motion: the closure A of its trajectory is compact and the DS $\{g^t|A\}$ is *equicontinuous* (as a family of transformations depending on the parameter t), *i.e.*, for any $\varepsilon > 0$, there is a $\delta > 0$ such that, for any $t \in \mathbb{R}$, $x \in A$ and $y \in A$, the distance $\rho(g^t x, g^t y) < \varepsilon$ whenever $\rho(x, y) < \delta$.

A *minimal DS* is a DS whose phase space is a minimal set. Minimality must not be confused with topological transitivity. *Topological transitivity* is the property that some (but not necessarily every) trajectory has M as its ω -limit set. If, at each point, M has dimension greater than 0 in the case of a cascade, and greater than one in the case of a flow, then topological transitivity is equivalent to the existence of an everywhere-dense trajectory on M . (For other definitions of topological transitivity, not always equivalent, see [ME, “topological transitivity”].)

4.4. Distality and Some Types of Extensions of Minimal Sets. Points x and y of M are said to be *proximal* if

$$\inf\{\rho(g^t x, g^t y) : t \in \mathbb{R}\} = 0,$$

and *distal* otherwise. (This is a property of a pair of points.) A DS is said to be *distal* if any two distinct points of the phase space are distal.

A DS of group shifts is distal. An example of a distal DS of a different type is generated by the following map g of a two-dimensional torus $\mathbb{R}^2/\mathbb{Z}^2$:

$$g(x, y) = (x + \alpha, y + x),$$

where x, y are cyclic (modulo 1) coordinates on a circle, and α is a fixed number. The cascade $\{g^k\}$ is an extension (Chap. 1, § 1.5) of the cascade $\{f^k\}$ on the circle generated by $f(x) = x + \alpha$. This extension is obtained for the map h of the torus to the circle, where $h(x, y) = x$. Under the action of g , the distance between two points of the “fibre” $h^{-1}x$ of the extension (in the natural metric on the fibres) does not change. But, in general, the cascade $\{g^k\}$ is not equicontinuous and, in particular, does not preserve the metric on the torus. When α is irrational, the cascade $\{g^k\}$ is minimal. When α is rational, the cascade is not minimal, but all its trajectories are recurrent: some are periodic, others are almost-periodic.

In the general case, too, all trajectories of a distal DS are recurrent. So, in studying such DSs, it is natural to concentrate on DSs which are minimal. In 1963 H. Furstenberg proved that every minimal, distal DS can be obtained from the trivial DS, whose phase space is a single point, by a transfinite extension process, in which each extension is, in a sense, isometric: the corresponding DS preserves the metric on the fibres. (The precise formulation will not be given here.) If we want a formulation not connected with a particular metric, then it can be said that every extension is equicontinuous in the sense defined below.

Let $\{f^t\}$ be a DS on M which is an extension of the DS $\{g^t\}$ on N under the mapping $h: M \rightarrow N$. The extension is said to be:

1) *equicontinuous* if the family of mappings

$$\{f^t | h^{-1}u: t \in \mathbb{R}, u \in \mathbb{N}\}$$

is equicontinuous;

2) *proximal* if any two points on any fibre $h^{-1}u$ are proximal;

3) *distal* if any two distinct points on any fibre are distal;

4) *weakly mixing* if the “coordinate-wise” action of $\{f^t\}$ on

$$\{(x, y): x, y \in M, hx = hy\} \quad (f^t(x, y) = (f^t x, f^t y))$$

defines a topologically transitive DS;

5) a *PI-extension* if h can be decomposed into a projective transfinite sequence of extensions of the first two types. In the case when N is a point, 1)–3) reduce to properties of DSs which we already know, and 4) signifies the topological transitivity of the “Cartesian square” of the DS $\{f^t\}$ (i.e., the DS on $M \times M$ defined by the coordinate-wise action of $\{f^t\}$). The name “weak mixing” stems from the analogy with ergodic theory when topological transitivity is regarded as an analogue of ergodicity. It is well known that ergodicity of the Cartesian square of a DS is equivalent to the property that the original DS be weak mixing (Volume 2, Chap. 1, § 3).

The following theorem about the structure of minimal sets holds [81].

Let $\{f^t\}$ be a minimal DS with a compact phase-space M which is an extension of a (necessarily minimal) DS $\{g^t\}$ under the mapping $h: M \rightarrow N$. Then there are canonically defined minimal DSs $\{f_*^t\}$ and $\{g_*^t\}$ on compact

phase spaces M_* and N_* , and homeomorphisms

such that: $\tau: M_* \rightarrow M$, $\sigma: N_* \rightarrow N$, $h_*: M_* \rightarrow N_*$

- 1) $h \circ \tau = \sigma \circ h_*$;
- 2) τ is a proximal extension;
- 3) σ is a PI-extension;
- 4) h_* is a weakly mixing extension.

The problem of describing the structure of weakly mixing extension remains unsolved. We also remind readers of the old problem: what conditions must a space M satisfy in order that there exist a minimal flow on M ? (In particular, can S^3 be the minimal set of a flow?)

§ 5. Extensions of Dynamical Systems and Nonautonomous Differential Equations

5.1. Nonautonomous Differential Equations. Nonautonomous systems of differential equations

$$\dot{x} = \mathbf{u}(x, t) \quad (8)$$

often arise in the following situation. There is an autonomous system in "triangular" form

$$\dot{x} = \mathbf{v}(x, y), \quad \dot{y} = \mathbf{w}(y). \quad (9)$$

If $y(t)$ is a solution of the system

$$\dot{y} = \mathbf{w}(y), \quad (10)$$

then

$$\dot{x} = \mathbf{v}(x, y(t)) \equiv \mathbf{u}(x, t). \quad (11)$$

The flow described by the system (9) is an extension of the flow (10), and (8) with $\mathbf{u}$ of the form (11) describes the behaviour of the solutions of (9) which lie over the trajectories of the flow (10).

In the discussion of the variational equation (Chap. 1, § 2.2), we encountered a similar situation, except that the phase space of the extension was not a direct product, as it is for (9), but the total space of a bundle.

Finally, it can happen that system (8) is given irrespective of any extensions, but that it is possible to associate an extension with it so that it will describe the motions of this extension over a trajectory of a factor. A simple example is the interpretation of a periodic, nonautonomous system as a DS on a cylinder (Chap. 1, § 2.3). In that case a circle serves as the factor, the motion along the circle taking place with constant speed.

Formally, (8) can be changed into the autonomous system

$$\dot{x} = \mathbf{u}(x, y), \quad y \in \mathbb{R}, \quad \dot{y} = 1, \quad (12)$$

which is an extension over the flow $\dot{y} = 1$ on $\mathbb{R}$. But this is not helpful because all the trajectories of (12) are noncompact ($y(t) \rightarrow \infty$) and they behave, roughly

speaking, as a family of parallel lines; the application of the concepts and results of the TDS in this case is fruitless. A different picture emerges if the phase space of the factor ("where y moves") is compact, as in the example of a periodic system. In this example, the factor is a smooth manifold. In more general cases, it can happen that (8) is obtained from an extension in which the phase space of the factor is not a manifold. The advantages associated with compactness may outweigh the inconveniences associated with passing from differential equations to a topological flow.

In order not to clutter the exposition with detail, we describe the main idea of this method in the special, but important, case where (8) is a linear homogeneous system:

$$\dot{x} = A(t)x, \quad x \in \mathbb{R}^n. \quad (13)$$

Suppose that the matrix $A(t)$ is uniformly bounded and uniformly continuous as a function of t . We associate to (13) all systems of the form

$$\dot{x} = \tilde{A}(t)x \quad (14)$$

where the matrix function $\tilde{A}$ is a shift of A by some time, *i.e.*, a function of the form $t \mapsto A(t + t_0)$, or a limit of such shifts in the compact-open topology, *i.e.*, a function $t \mapsto \lim_{k \rightarrow \infty} A(t + t_k)$, where the limit notation indicates uniform convergence on finite time intervals. The set of all $\tilde{A}$ is, in a natural way, a compact metric space M on which a flow of shifts operates:

$$(\sigma^t \tilde{A})(s) = \tilde{A}(t + s)$$

(compare Chap. 1, § 1.2). The system (14) can be regarded as defining a flow $\{g^t\}$ in $M \times \mathbb{R}^n$: if $x(t)$ is a solution of (14) with initial condition $x(0) = x$, then $g^t(\tilde{A}, x) = (\sigma^t \tilde{A}, x(t))$. $\{g^t\}$ is an extension of $\{\sigma^t\}$ under the map $(\tilde{A}, x) \mapsto \tilde{A}$, and the x -components of the motions, which are carried into $\{\sigma^t A\}$ by this map, change with time in accordance with the original system (13).

This method enables us to apply various arguments of the TDS to investigate systems (14) which are limiting, in the sense described, for (13). (At first the method was applied to the systems (13) with almost-periodic coefficients; in this case, the convergence of $\sigma^{t_k} A \rightarrow \tilde{A}$ can be understood in another way, *viz.*, as uniform convergence on the whole axis.)

5.2. Linear Extensions. Let (E, p, B) be an n -dimensional real vector bundle with a compact base B and let $\{f^t\}$ and $\{g^t\}$ be DSSs in E and B such that the first is an extension of the second under the projection $p: E \rightarrow B$. Suppose that the mapping

$$f^t|_{E_b}: E_b \rightarrow E_{g^t b}$$

is linear for all $b \in B$ (here $E_b = p^{-1}b$ is the fibre over b). We then speak of $\{f^t\}$ as a *linear extension*.

A linear extension is said to be *hyperbolic* if there are vector sub-bundles E^s and E^u , and numbers $d > 0$ and $\alpha > 0$ such that

$$E = E^s \oplus E^u$$

(where " $\oplus$ " denotes the Whitney sum; see the footnote in Chap. 2, § 3.2, 3)) and

$$\begin{aligned} |f^t x| &\leq d|x| e^{-\alpha t} \quad \text{for all } x \in E^s \quad \text{and } t > 0, \\ |f^{-t} x| &\leq d|x| e^{-\alpha t} \quad \text{for all } x \in E^u \quad \text{and } t > 0. \end{aligned}$$

Here $|\cdot|$ is a Riemannian metric on (E, p, B) ; since B is compact, the actual choice of metric is immaterial.

Here are two weaker conditions. Let

$$E_b^s = \{x \in E_b : \lim_{t \rightarrow \infty} |f^t x| = 0\}, \quad E_b^u = \{x \in E_b : \lim_{t \rightarrow -\infty} |f^t x| = 0\}$$

(again the choice of metric is unimportant). The condition

$$E_b = E_b^s + E_b^u \quad \text{for all } b \in B$$

is called the *transversality condition for linear extensions*. (Here, it is not assumed that the dimensions of E_b^s and E_b^u are complementary.) If $|x| = 0$ whenever $\sup_t |f^t x| < \infty$, then we say that *there are no nontrivial bounded motions in the linear extension*. It turns out that if $\mathcal{R}(\{g^t\}) = B$ (see § 2.2), then a linear extension is hyperbolic if and only if there are no nontrivial bounded motions on it.

In the theory of linear systems, a large role is played by the concept of an adjoint system. For (13), the adjoint system is

$$\dot{\xi} = -A^*(t)\xi, \quad (15)$$

where the asterisk denotes the conjugate matrix. Let $X(t)$ and $\Xi(t)$ be the matrixants of the systems (13), (15), i.e., solutions of the matrix equations

$$\dot{X} = A(t)X, \quad X(0) = \mathbf{1}_{\mathbb{R}^n}; \quad \dot{\Xi} = -A^*(t)\Xi, \quad \Xi(0) = \mathbf{1}_{\mathbb{R}^n},$$

so that the solutions of systems (13) and (15) with initial conditions $x(0) = x$ and $\xi(0) = \xi$ are

$$x(t) = X(t)x, \quad \xi(t) = \Xi(t)\xi. \quad (16)$$

Putting $(\xi, x) = \sum \xi_i x_i$, we view ξ as a linear homogeneous functional of x . Then the relation between systems (13) and (15) can be characterized by noting that their solutions (16) satisfy

$$(\xi(t), x(t)) = (\xi, x) = \text{const.}$$

Therefore, $\Xi = (X^*)^{-1}$, which can also be expressed as:

$$(\Xi(t)\xi, x) = (\xi, X^{-1}(t)x) \quad \text{for all } \xi, x.$$

The concept of the adjoint linear extension plays a similar role for linear extensions. In a natural notation, let the original extension be

$$p: (E, \{f^t\}) \rightarrow (B, \{g^t\}). \quad (17)$$

Then the phase space of the *adjoint extension*

$$p^*: (E^*, \{f_*^t\}) \rightarrow (B, \{g^t\}) \quad (18)$$

is the total space of the vector bundle E^* dual to E (the fibre $E_b^* = (p^*)^{-1}b$ is the dual space of E_b i.e., it consists of homogeneous linear functionals $x \mapsto (\xi, x)$ on E_b ; and the topology on E^* is the natural one), and

$$(f_*^t \xi, x) = (\xi, f^{-1}x) \quad \text{for all } \xi \in E_b^*, x \in E_{g^t b}, b \text{ and } t.$$

The following five conditions are equivalent:

- 1) (17) satisfies the transversality condition;
- 2) (18) has no nontrivial bounded motions;
- 3) if the nonlinear extension

$$p: (E, \{\varphi^t\}) \rightarrow (B, \{g^t\}) \quad (19)$$

is sufficiently close to (17) in the Lipschitz sense, then the DSs (17) and (19) are isomorphic by a fibered homeomorphism $E \rightarrow E$;

- 4) there is a Green's function for the invariant section problem;
- 5) there exists a continuous function $\Phi: E^* \rightarrow \mathbb{R}$, which is a quadratic form on each fibre E_b^* and which is such that

$$\Phi(f_*^t \xi) > \Phi(\xi) \quad \text{for all } \xi \in E^*, |\xi| \neq 0 \quad \text{and } t > 0.$$

If the linear extension (17) satisfies the transversality condition, then the set

$$A(\{f^t\}) = \{b \in B: E_b^s \oplus E_b^u = E_b\}$$

is closed, each of the subsets

$$A_k = \{b \in A(\{f^t\}): \dim E_b^u = k\} \quad (k = 0, 1, \dots, n)$$

is closed and invariant, and $\{A_0, A_1, \dots, A_n\}$ is a Morse decomposition. Moreover, $A(\{f^t\}) = A(\{f_*^t\})$, and (17) is a hyperbolic extension over $A(\{f^t\}) \supset \mathcal{R}(\{g^t\})$.

In the theory of smooth dynamical systems, there are applications of the theory of linear extensions to questions concerning structural stability. The generalization of the theory of linear extensions to the case of Banach extensions is of great interest.

Chapter 4

Flows on Two-Dimensional Manifolds

S.Kh. Aranson, V.Z. Grines

§ 1. Singular Trajectories

Let M be a connected, closed, orientable or non-orientable, two-dimensional manifold (a surface) of genus p , and let $\{g^t\}$ be a topological flow on M . To describe the properties of $\{g^t\}$ it is natural to begin by picking out

the singular trajectories since their character and mutual arrangement completely determine the qualitative structure of the phase portrait.

A *separatrix* is defined to be a trajectory $\{g^t x\}$ which approaches an equilibrium point a as $t \rightarrow \infty$ or $t \rightarrow -\infty$ and is such that there are trajectories arbitrarily close to it which first come close to a , "running alongside the trajectory $\{g^t x\}$ ", as it were, and then move away from a by a finite distance. Formally, this means that there is a neighbourhood U of a , a sequence $x_n \rightarrow x$, and sequences of numbers s_n and t_n such that $s_n \rightarrow \infty$ (resp., $s_n \rightarrow -\infty$), $g^{s_n} x_n \rightarrow a$, $g^{t_n} x_n \notin U$ and $t_n > s_n$ (resp., $t_n < s_n$).

The following trajectories on M are called *singular trajectories*:

- 1) equilibrium points;
- 2) separatrices;
- 3) closed trajectories for which the first-return map (Chap. 1, § 2.3) differs from the identity;
- 4) non-closed trajectories L which are Poisson-stable (Chap. 3, § 2.1) and which have the property that, for any point $a \in L$, there is a neighbourhood $U \ni a$ and an open arc $\lambda \subset L$, $\lambda \ni a$, which splits U into two semi-neighbourhoods one of which is such that there are no points which lie on non-closed, Poisson-stable trajectories.

On a sphere, a projective plane, or a Klein bottle, there can be no non-closed, Poisson-stable trajectories [7], [51].

Theorem 1.1. Suppose $\{g^t\}$ has only a finite number of singular trajectories of types 1) and 2) (no assumption is made about singular trajectories of types 3) and 4)). Then, if all the singular trajectories are removed from M , the remaining set breaks up into connected components of the following types:

1) components filled with non-closed trajectories which are not Poisson-stable in the positive and negative directions. These components are planar domains (i.e., they are homeomorphic to domains in $\mathbb{R}^2$) which are either simply-connected or doubly-connected, and all the trajectories in the same domain have the same ω -limit points and the same α -limit points (Article I, Chap. 1, § 5.5). Components, whose accessible from inside boundary contains the non-closed, Poisson-stable semi-trajectories, are always simply-connected (see [ME, "Accessible boundary point"]);

2) components filled with closed trajectories. Such a component is either a doubly-connected planar domain, or it coincides with M , which is then a torus;

3) components filled with non-closed, Poisson-stable trajectories. Such a component either coincides with M , which is then a torus, or it is a connected set which is not path connected and which is a second category set in its closure. The number of such components is not greater than p if M is orientable, and not greater than $[(p-1)/2]$ (where $[\cdot]$ denotes the integer part) if M is not orientable (see Fig. 1).

(For $M = S^2$, see [4] and [18]; for orientable M , see [49]; for non-orientable M , see [7], [51] and [52].)

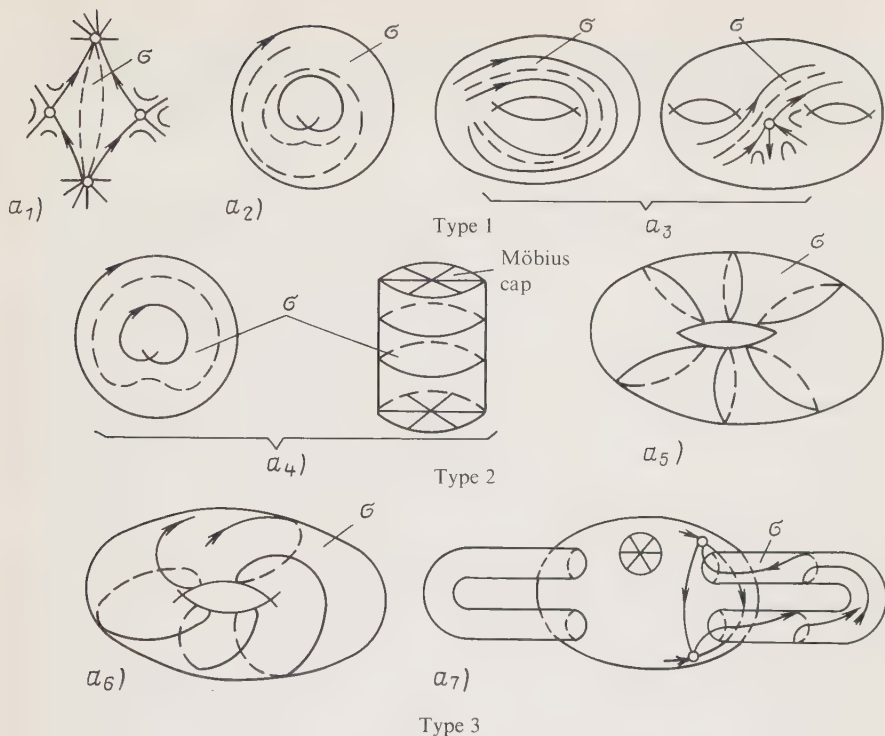


Fig. 1. Types of components σ possible on two-dimensional manifolds. Type 1: a_1) a simply-connected component, filled with non-closed trajectories which are not Poisson-stable; a_2) a doubly-connected component, filled with non-closed trajectories which are not Poisson-stable; a_3) a simply-connected component whose accessible from inside boundary contains the non-closed Poisson-stable semi-trajectories; Type 2: a_4) a doubly-connected component filled with closed trajectories; a_5) a component which is a torus, filled with closed trajectories; Type 3: a_6) a component which is a torus, filled with non-closed, Poisson-stable trajectories; a_7) a component filled with non-closed Poisson-stable trajectories, which is a connected, but not path connected, point set.

For flows on the sphere which have a finite number of singular trajectories, we define a *flow scheme*, which includes the following information: the number and nature of the equilibrium points, the number and mutual arrangement of the limiting continua (in particular, the limit cycles), and the behaviour of the separatrices. The scheme is a complete topological invariant of a flow with a finite number of singular trajectories on a sphere; two such flows are topologically equivalent if and only if their flow schemes are isomorphic in a natural sense [4], [18].

For other surfaces, the fundamental group plays a big part in flows which have non-closed, Poisson-stable trajectories. If there are no such trajectories,

a complete topological description can be given using topological invariants similar to the flow scheme of a sphere. In particular, a complete topological invariant of Morse-Smale flows is the distinguished graph of the flow [64].

§ 2. The Poincaré Rotation Number. Transitive and Singular Flows on a Torus

Let $M = \mathbb{R}^2 / \mathbb{Z}^2$ be a torus, and let $\pi: \mathbb{R}^2 \rightarrow M$ be the natural projection. A flow $\{g^t\}$ on M "lifts" in a natural way to $\mathbb{R}^2$, i.e., one obtains a covering flow $\{f^t\}$ on $\mathbb{R}^2$ which is an extension of $\{g^t\}$ under the mapping π ; it is uniquely determined. Let L be a positive semi-trajectory of $\{g^t\}$, and let $l = \{(x(t), y(t))\}$ be any semi-trajectory of $\{f^t\}$ which is an inverse image of L in $\mathbb{R}^2$. If $x^2 + y^2 \rightarrow \infty$ as $t \rightarrow \infty$, then there exists a finite or infinite limit $v(L) = \lim_{t \rightarrow \infty} y(t)/x(t)$ which does not depend on the choice of l in $\pi^{-1}(L)$; it

is called the *rotation number* of the semi-trajectory L .

If $\omega(L)$ (i.e., $\omega(x)$ with $x \in L$, see Chap. 3, § 2.1) contains a closed trajectory which is not homotopic to zero, or a closed separatrix contour (which consists of equilibrium points and separatrices) which is not homotopic to zero, then $v(L)$ exists and is either rational or equal to $\pm\infty$. If $\omega(L)$ contains a non-periodic, Poisson-stable trajectory, then $v(L)$ exists and is irrational. If $\{g^t\}$ has a global transversal (Chap. 1, § 2.3; when the flow is only continuous, a global transversal is a simple closed curve which all trajectories intersect, passing locally from one side to the other), then $v(L)$ is the same for all L . When the global transversal is homotopic to the zero meridian, $v(L) \bmod 1$ coincides with the classical Poincaré rotation number (Article I, Chap. 2) of the corresponding first-return mapping. In particular, this is the case for a topologically transitive flow without equilibrium points.

Theorem 2.1. *Two topologically transitive (Chap. 3, end of § 4.3) flows without equilibrium points on a torus are topologically equivalent if and only if their rotation numbers v and $\tilde{v}$ are commensurable, i.e., there is a unimodular matrix*

$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ *with integer entries such that*

$$\tilde{v} = \frac{c + dv}{a + bv} \quad [9], [60], [62], [11].$$

A flow on a torus is said to be *singular* if it has no equilibrium points, no closed trajectories, and its minimal set (Chap. 3, § 4.3) is nowhere dense in the torus. For singular flows on a torus, the rotation number (up to commensurability), the number of components, and their locations are a complete topological invariant [11]. Singular flows on the torus cannot be smoother than C^1 (a consequence of Denjoy's theorem, see Article I, Chap. 2). See also [ME, "Differential equations on a torus", and "Kneser's theorem"].

§ 3. The Homotopy Rotation Class. Classification of Transitive Flows and Nontrivial Minimal Sets of Flows on Surfaces

Recent progress in studying of flows on orientable surfaces M of genus $p \geq 2$ began with the systematic investigation of the asymptotic behaviour of lifts of trajectories of flows on the universal cover $\tilde{M}$ (the Lobachevsky plane in Poincaré's realization as the disc $|z| < 1$ in the complex z -plane). That is, one studies the behaviour of the trajectories of the covering flow. This idea, in a particular case, was advanced by A. Weil in [83] (which, however, is mostly devoted to developing the analogous approach for flows on a torus) and, in general, by D.V. Anosov in the mid-sixties. The fundamental topological invariant obtained here is the homotopy rotation class which generalizes the classical Poincaré rotation number [9].

We express M as a quotient space $\tilde{M}/\Gamma$, where Γ is a discrete group of fractional linear transformations isomorphic to the fundamental group of M . We let π denote the natural projection $\tilde{M} \rightarrow M$. Each element $\gamma \in \Gamma$, other than the identity, has exactly two fixed points; an attracting point γ^+ and a repelling point γ^- ; they lie on the "absolute" $|z| = 1$. The set of all fixed points of elements of Γ other than the identity is called the *set of rational points*. It is countable and is everywhere dense on the absolute. The complement in the absolute of the set of rational points is called the *set of irrational points*. (An exposition of the basic facts relating to this representation of M is given in [61].)

Let L be a positive semi-trajectory of the flow $\{g^t\}$ on M . We consider any inverse image l of L on $\tilde{M}$, i.e., a semi-trajectory $l = \{f^t z\}$ of the covering flow $\{f^t\}$ which projects to L . D.V. Anosov showed that, under quite general assumptions on $\{g^t\}$ (for example, if $\{g^t\}$ has only a finite number of equilibrium points), l is either contained in a bounded domain of $\tilde{M}$ (in which case, its possible limiting behaviour is then described by the usual Poincaré-Bendixson theory in the plane and, so, the same is true for L), or "it goes to infinity", i.e., $|f^t z| \rightarrow 1$. In the second case, $f^t z$ tends to a certain point on the absolute. (For a generalization of the last assertion, see [67]. The reader should be warned that Theorem 2 in [67] is not true.) Independent of the assumptions on $\{g^t\}$, if $\omega(L)$ contains a closed trajectory which is not homotopic to zero or a closed separatrix contour which is not homotopic to zero, then l tends to a rational point on the absolute; if $\omega(L)$ contains a non-periodic, Poisson-stable trajectory, then l tends to an irrational point on the absolute [9].

In the general case, let $\sigma(l)$ denote the set of limit points of l which belong to the absolute. The set

$$\mu(L) = \bigcup_{\gamma \in \Gamma} \gamma(\sigma(l))$$

is called the *homotopy rotation class* of the semi-trajectory L of the flow $\{g^t\}$ on M . Every automorphism τ of the group Γ uniquely induces a homeomorphism τ^* of the absolute. The homotopy rotation classes $\mu(L)$ and $\mu(\tilde{L})$ of semi-trajectories L and $\tilde{L}$ of flows $\{g^t\}$ and $\{\tilde{g}^t\}$ on M are said to be *commensurable* if there is an automorphism τ of Γ such that $\mu(\tilde{L}) = \tau^*(\mu(L))$.

On orientable surfaces of genus $p \geq 2$, the natural analogue of a topologically transitive flow on a torus without equilibrium points is a topologically transitive flow which has a finite number of singular trajectories of types 1) and 2) (see § 1) and no separatrix going from one equilibrium point to another. Such flows will be called *class T flows*.

Theorem 3.1. *Two class T flows which are defined on M and have no equilibrium points with exactly two separatrices¹ are topologically equivalent if and only if there exist two Poisson-stable semi-trajectories of separatrices of these flows which have commensurable homotopy rotation classes [9].*

The main element in using the homotopy rotation class to prove Theorem 3.1 and in further considerations is the construction of a special fundamental domain on $\tilde{M}$ (Fig. 2).

Let $\{g^t\}$ be a topological flow on M which has a nontrivial (i.e., different from an equilibrium point or a closed trajectory) minimal set Ω .² It is known that Ω is nowhere dense on M . The set $M \setminus \Omega$ is the union of a finite or countable set of connected components which are domains. Trajectories in Ω accessible from the interior of these domains are called *boundary trajectories*. A pair of boundary trajectories in Ω is called a *special pair* if they form the boundary (accessible from within) of a simply-connected domain of $M \setminus \Omega$. The trajectories which form a special pair are themselves called *special*.

Deviating slightly from the terminology of Chap. 1, § 1.5, we shall say that minimal sets Ω and $\tilde{\Omega}$ of flows $\{g^t\}$ and $\{\tilde{g}^t\}$ on M are *topologically equivalent* if there is a homeomorphism $h: M \rightarrow M$ whose restriction to Ω makes the DSs $\{g^t|_{\Omega}\}$ and $\{\tilde{g}^t|_{\tilde{\Omega}}\}$ topologically equivalent in the sense of Chap. 1, § 1.5.

Theorem 3.2. *Nontrivial sets Ω and $\tilde{\Omega}$ of the flows $\{g^t\}$ and $\{\tilde{g}^t\}$ which do not contain special trajectories are topologically equivalent if and only if there are boundary semi-trajectories $L \subset \Omega$ and $\tilde{L} \subset \tilde{\Omega}$ which have commensurable homotopy rotation classes [8].*

Theorem 3.3. *(The existence of a standard representative.) Let Ω be a non-trivial minimal set of a flow $\{g^t\}$ which does not contain special trajectories.*

¹ An example of such an equilibrium point is the origin of the local flow $\dot{x} = x^2 + y^2$, $\dot{y} = 0$ on $\mathbb{R}^2$. The separatrix entering $(0, 0)$ is the negative semi-axis of the x -axis, and the outgoing separatrix is the positive semi-axis; together they form a trajectory which "comes to a halt", as it were. These separatrices do not affect the behaviour of neighbouring trajectories and do not actually depend on other particular features of the phase-portrait.

² For C^2 flows, this is possible only when $\Omega = M^2$ a torus [43]. (In this section the genus of M is greater than one.)

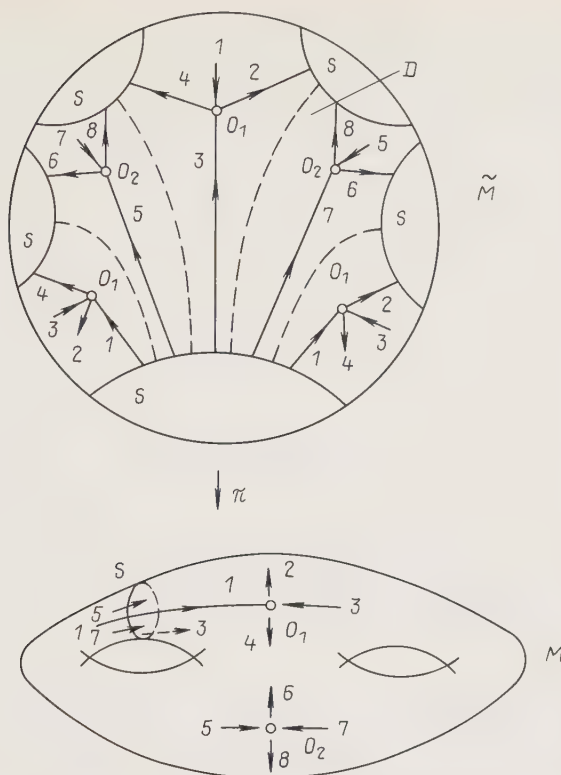


Fig. 2. The special fundamental domain D on $\tilde{M}$ for a class T flow $\{g^t\}$ containing two saddles O_1 and O_2 on a manifold M of genus $p=2$. S denotes a transversal for the flow $\{g^t\}$ on M . The separatrices of the saddle O_1 are denoted by the numbers 1, 2, 3, 4, and the separatrices of O_2 by 5, 6, 7, 8. The same letters and numbers are used on $\tilde{M}$ to denote the lifts of S , the saddles, and their separatrices. For convenience, congruent points and curves are also denoted by the same letters. $\tilde{M}$ denotes the universal covering of M .

Then, there is a flow g_0^t on M , generated by a Lipschitz vector-field such that:

- 1) $\{g_0^t\}$ has a nontrivial minimal set Ω_0 whose trajectories are geodesic curves in a metric of constant negative curvature;
- 2) the minimal sets Ω and Ω_0 of the flows $\{g^t\}$ and $\{g_0^t\}$ are topologically equivalent by a homeomorphism homotopic to the identity [10].

All class T flows (up to topological equivalence) without equilibrium points with exactly two separatrices can be obtained by taking a quotient of flows on M , each of which contains exactly one standard nontrivial minimal set Ω_0 for which $M \setminus \Omega_0$ separates into a finite number of simply-connected domains.

The solution of the problem of classifying class T flows and the classification of nontrivial minimal sets of flows on M enables us to find a complete topo-

logical invariant for a wide class of flows on M . This invariant is similar to the flow scheme on a sphere, but takes into account the asymptotic behaviour of non-closed, Poisson-stable trajectories.

For foliations with singularities on surfaces, certain generalizations are obtained in [48]. For M of genus $p \geq 0$ S.Kh. Aranson has recently established results which are analogous to Theorems 3.1–3.3 for the foliations with singularities of a saddle type including needles.

§ 4. Generic Properties of Flows on Surfaces

The main result, here, is that structurally stable flows on a closed surface are generic, and are the Morse-Smale flows (Chap. 2, § 3). There are some additions and refinements connected with modifications of the concepts of structural stability and genericity. In this case, structural stability is the same as Peixoto structural stability. The definition of the latter differs from the definition in Chap. 2, § 1.2, in that the homeomorphism which maps the unperturbed system into the perturbed system is not required to be close to the identity. When M is orientable or nonorientable of genus $p = 1, 2$ or 3 , the C^r -structurally stable systems with $r > 1$ are also Morse-Smale systems (and are, therefore, structurally stable systems in the usual sense) and are C^r generic.

These results are due mainly to Peixoto. A simplification of the proof and some corrections can be found in [38], [39], [6], and [63].

Digressing for a moment, we remark that the situation is more complicated on surfaces which are not closed. Even in the definition of structural stability, one has to specify how the C^1 -topology in the space of flows and the C^0 -topology in the space of homeomorphisms are to be understood (it turns out that they must be understood differently – the first as the fine topology (the Whitney topology) and the second as the compact-open topology). Under certain restrictions on M , necessary and sufficient conditions have been found for structural stability in terms of the phase portrait, but structurally stable systems are no longer dense in the space of C^1 -flows, and the different variants of structural stability are not equivalent [44].

We now return to the compact case. Passage from one structurally stable flow to another can (and, if they are not equivalent, must) take place through flows which are not structurally stable. The simplest such flows are flows of the *first degree of nonstability* (they are defined as being, in a natural sense, structurally stable in the set of flows which are not structurally stable). For an orientable M , necessary and sufficient conditions which distinguish such flows are known [6]. As regards the question of genericity of flows which are not structurally stable, the following proposition holds.

Theorem 4.1. *Let $X^r(M)$ be the space of all C^r , $r \geq 1$, flows on M provided with the C^r -topology and let $\Sigma^r \subset X^r(M)$ be the set of flows which are not structurally stable. Let M be orientable and $r \geq 4$. Then there is a dense subset*

Σ_1^r of Σ^r which is an immersed (in $X^r(M)$) C^{r-1} -submanifold of codimension one and which is such that:

- 1) Σ_1^r contains all flows of the first degree of nonstability; the latter form a codimension one submanifold embedded in $X^r(M)$;
- 2) at each point of Σ_1^r there is a neighbourhood (in the intrinsic topology) consisting of topologically equivalent flows [77], [79].

Remark. For S^2 , flows of the first degree of nonstability are dense in Σ^r when $r \geq 3$. For the torus, they are dense in the part of Σ^r , $r \geq 2$, consisting of flows without equilibrium points. Let M be closed and orientable. If there are equilibrium points and if the genus of M is greater than or equal to one, flows of the first degree of nonstability are not dense in Σ^r , $r \geq 3$. See also volume 5.

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Index

- | | |
|--|--|
| adjoint linear extension 218
almost-periodicity of a DS on a trajectory 213
– of a function 213
– of a point 213
almost-recurrence 213
attaching tube of a handle 188
attractor 198
autonomous system of differential equations in “triangular form” 216

basin of attraction 199
Bernoulli topological automorphism (two-sided shift) 157 | – topological cascade 157
– – endomorphism (one-sided shift) 157
– metric automorphism (endomorphism) 158
Birkhoff-recurrence 213
blue-sky catastrophe 182
Bohl-Brouwer theorem 179, 204
boundary trajectories (of the minimal set of a flow on a surface) 224
– –, special pair 224
Browder’s theorem 179

C^r -exceptionality 175
C^r generic 175 |
|--|--|

- C^r structural stability 175
- C^r structurally stable property 197
- cascade 154
- center of a DS 211
 - , depth of center 211
- chain 201
 - recurrence 200
- characteristic homeomorphism 188
- closed trajectory 160
 - –, hyperbolic 171
 - –, nondegenerate 171
- commensurable homotopy rotation
 - classes 224
 - rotation numbers 222
- components possible on 2-dimensional manifolds, diagrams 221
- connected simple system 209
- connecting handle 189
- continuous time 156
- cross-section 168
 - , global 168
 - , local 168
- cycle 160
- depth of center 211
- discrete time 156
- distal extension 215
- distality 214
- doubly-asymptotic trajectory 185
- DS = dynamical system(s)
- DS, smooth 154
- DS, topological 154
- DS of group shifts 214
- ε -trajectory 200
- eigenvalues of an equilibrium point 170
 - of a fixed point of a cascade 170
 - of a periodic trajectory 171
- energy function 186
- equicontinuity 214
- equicontinuous extension 215
- equilibrium point 159, 170, 220
 - –, hyperbolic 170, 171
 - –, isolated 171
 - –, Kronecker-Poincaré index 177
 - –, nondegenerate 170
- essential Nielsen class 180
- exceptional property 175
- exit point 205
- explosion (C^r - Ω explosion) 212
- extension, distal 215
 - , equicontinuous 215
 - , PI -extension 215
 - , proximal
 - extensions and nonautonomous differential equations 216
- factor 161, 216
- false zeta-function 183
- filtration 188
 - for a Morse-Smale system 200
- first degree of nonstability 226
- first-return map 168
- fixed point 160
- flow 154
 - of class T 224
 - plan (scheme) 221
- flows on two-dimensional manifolds 219ff.
- Fuller index 181
- Fuller's theorem 181
- function not increasing on a trajectory 202
- Furstenberg, H. 215
- generic case 176
 - properties of flows on surfaces 226
- global 162
- gradient-like M-S system 185
- group property 154
 - shifts 214
- handle 188
 - , attaching tube 188
 - , center 188
 - , connecting 189
 - , core 188
 - , co-core 188
 - decomposition 189
- heteroclinic point 185
 - trajectory 185
- homoclinic point 185
 - trajectory 185
- homological index 206
 - zeta function 183
- homomorphism of DS 161
- homotopic rotation class 223
- homotopy index 206
- hyperbolic closed trajectory 171
 - equilibrium point 170
 - linear extension 217
 - periodic trajectory, Morse index 173
- index, φ -index 181
 - , Fuller 181
 - , (co-)homological 206
 - , homotopy 206
 - , Kronecker-Poincaré 177
 - , Morse 173
 - , Morse-Conley 209
 - pair 205
 - space 206

- indices of isolated invariant sets 203
- invariance 160
 - , g - 160
 - , positive 160
 - , semi- 160
- invariant set 160
- irreversibility 156
- isolated equilibrium point 171
 - invariant set 203
 - invariant sets, indices of 203
 - periodic trajectory 171
 - – –, Kronecker-Poincaré index 177
- isolating block 205
 - neighbourhood 203
- isomorphism of DSs 161
- Kronecker-Poincaré index 177
- Kupka-Smale cascade, diffeomorphism, flow, system 176
- Kupka-Smale theorem 176
- Lagrange-stability 163
- Lefschetz number 179
- limiting prolongation 212
- linear extensions 217
- local 162
 - positive invariance 210
- Lyapunov function, complete 203
 - function for M-S system 186
- Lyapunov-stability 199
- matrix, virtual permutation 193
- minimal center of attraction 212
 - DS 214
 - period 160
 - sets 213
- monothetic group 214
- morphisms of DSs 161
- Morse-Conley index 209
- Morse decomposition 199
 - index 173
 - set 199
- M-S = Morse-Smale
- M-S cascade, diffeomorphism, flow, system 183ff.
- M-S diffeomorphisms 184
- M-S inequalities 190
- motion 160
- multiple repetition of a closed trajectory 171
- multipliers of a closed trajectory 170
- nonautonomous DEs 216
- non-closed, Poisson-stable trajectories 220
- nondegenerate closed trajectory 171
 - equilibrium point 170
- nonwandering point, trajectory 211
- orientation types 178
- period 160
- periodic motion 160
 - point 160
 - of trajectory, hyperbolic, isolated, or nondegenerate 171
 - trajectories 160
 - trajectory, stable manifold and unstable manifold of 171
- phase diagram of a M-S system 186
 - point 155
 - space 152, 154
 - – of a random process 156
 - trajectory 159
 - velocity 155
- PI-extension 215
- Poincaré rotation number 222
- Poincaré-Hopf theorem 179
- pointed space 206
- Poisson-stability 211
- product of pointed spaces 206
- prolongation of a trajectory of a point 212
 - with respect to a DS and initial data 212
 - – – to the initial data 212
- property, exceptional 175
 - , generic 175
 - , structurally stable 175
- proximal extension 215
- proximality 214
- QTDE = qualitative theory of differential equations
- random process 156
 - –, natural representation 157
- recurrent trajectory 213
- repeller 199
- repetitive motions 211
- repetitiveness 197, 201, 202
- rest point 159
- reversibility 156
- rotation number 222
- Seifert-Reeb theorem 181
- semi-conjugation of DS 161
- semigroup property 156
- separatrix 220
 - contour 183

- set of irrational points of the absolute 223
- of rational points of the absolute 223
- singular flow on a torus 222
- trajectory 219
- sink 188, 194
- Smale suspension 169
- smooth conjugacy 161
- DS 164
- flows 164
- solenoidal group 214
- source 194
- special pair of trajectories of minimal set of a flow on a surface 224
- trajectories of minimal set of a flow on a surface 224
- stability, asymptotic 199
- stable manifold of a periodic trajectory 171
- stationary processes 156
- structural stability 175
- structurally stable DS 176
- – properties of a system 175
- sum of pointed spaces 206
- symbolic dynamics 158
- tangential linear extensions 167
- TDS=theory of dynamical systems *passim*
- time change 161
- , continuous 155
- , discrete 155
- of return 168
- reversal 161
- topological dynamics 197ff.
- Bernoulli automorphism 157
- equivalence of flows 162
- – of minimal sets on a surface 224
- extension of a DS 161
- factor of a DS 161
- Markov chain 158
- transitivity 214
- trajectory 159
- , closed, nondegenerate 171
- trajectory, ε - 200
- , l -circuit 171
- , single-circuit 171
- , twisted 178
- , untwisted 178
- transversal 168
- , local 168
- , global 168
- transversality condition for linear extensions 218
- twisted trajectory 178
- unstable manifold of a periodic trajectory 171
- untwisted trajectory 178
- variation of solution 166
- variational equations 166
- wandering point 211
- Ważewski's principle 204
- weakly mixing extension 215
- nonwandering point 212
- whiskers 185, 212
- Whitney sum of fibre bundles 190
- zeta-function 13
- , homological 183
- of a mapping or cascade 183
- zone of repulsion 199

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